EXHALE v1.10: physical equations and numerical algorithms

Contents

1. Introduction and conventions **3**

1.1 What the code solves . . . 3

1.2 Units . . . 4

1.3 Notation . . . 5

1.4 How to read this document . . . 5

2. Hydrodynamics **6**

2.1 Conserved variables and the conservation laws . . . 6

2.2 The discrete divergence . . . 6

2.3 The spherical pressure term and gravity . . . 7

2.4 The gravitational potential . . . 8

2.5 Equation of state . . . 8

2.6 The well-balanced option . . . 9

2.7 Low-Mach dissipation of the contact mode . . . 10

2.8 Viscosity and heat conduction . . . 10

2.9 The two-dimensional approximation . . . 12

3. The stellar radiation field and its absorption **13**

3.1 The incident spectrum . . . 13

3.2 Photoionization cross sections . . . 16

3.3 Transfer along the radial ray . . . 17

3.4 Photoionization and photoheating rates . . . 18

3.5 Secondary ionization by photoelectrons . . . 19

3.6 Reprocessed and non-stellar fields . . . 21

4. Ionization and chemical state **22**

4.1 The species carried . . . 22

4.2 The cell system . . . 23

4.3 The rate coefficients . . . 26

4.4 Solving one cell . . . 29

4.5 Ionization carried by the flow . . . 33

4.6 Excited hydrogen and the Lyα field . . . 36

5. The thermal source: heating and cooling **38**

5.1 The heating channels . . . 38

5.2 The cooling channels . . . 41

5.3 Advancing the temperature . . . 45

6. The molecular layer, the trace metals and element transport 46

6.1 The H_2/He reaction network . . . 47
6.2 H_2 photoabsorption resolved into channels . . . 48
6.3 Lyman-Werner photodissociation . . . 49
6.4 The caloric equation of state . . . 50
6.5 Carrier transport . . . 50
6.6 The oxygen option . . . 52
6.7 The lower-atmosphere handoff . . . 52
6.8 Trace metals . . . 57
6.9 Binary element transport . . . 57
6.10 The interdiffusion enthalpy flux . . . 60

7. Numerical scheme 62

7.1 The radial grid . . . 62
7.2 Reconstruction . . . 63
7.3 Approximate Riemann solvers . . . 66
7.4 Positivity repair of a stage . . . 68
7.5 Time integration and the time step . . . 68
7.6 Operator splitting of one step . . . 69
7.7 The marching convergence measure and the stop conditions . . . 71

8. Boundary and initial conditions 73

8.1 The lower boundary . . . 73
8.2 The upper boundary . . . 80
8.3 Initial conditions . . . 81
8.4 Restart . . . 82

9. The stationary state 84

9.1 The stationary residual . . . 85
9.2 The acceptance test . . . 86
9.3 Certification . . . 86
9.4 The Jacobian-free Newton-Krylov solve . . . 90
9.5 Pseudo-transient continuation . . . 92
9.6 The partitioned outer iteration . . . 93

10. v1.10 regional stabilization and stationary closure 96

10.1 Finite-duration regional condition . . . 96
10.2 Lower-boundary thermochemical closure in the stationary problem . . . 97

11. Output and post-processing **97**

11.1 The advection-corrected profiles 97

11.2 The mass-loss rate 101

11.3 Output files 101

11.4 The transmission spectrum 103

11.5 A model transit line against an observed one: the width-matching kernel 104

12. The input keys that select physics **105**

1. Introduction and conventions

1.1 What the code solves

EXHALE integrates the one-dimensional, spherically symmetric equations of radiation hydrodynamics for the escaping upper atmosphere of an irradiated planet. The radial domain runs from a base near the lower atmosphere, which a run may place at the planet radius or at a stated pressure level (the 1 μbar level of the Koskinen et al. (2022) handoff is the one the molecular configurations use), out to the Roche-lobe (L1) radius of the planet–star system or, in the spherical domain mode, to a radius the user states. The unknowns advanced in time are the conserved hydrodynamic state $u = (\rho, \rho v, \mathcal{E})$ on a finite-volume radial grid, together with the ionization and chemical state of the gas: hydrogen and helium always, the metastable He 2^3S level, a set of trace metals, and a molecular network carrying H_2, H_2^+, H_3^+, HeH^+ and, optionally, the oxygen carriers OH, H_2O and CO. The radiative source term couples the two: the stellar spectrum is attenuated along the radial column, photoionizes and heats the gas, and the gas cools through recombination, collisional and line processes whose rates depend on the composition the ionization solve returns. By default the ionization state of each cell is the local balance of its rates; on request the ionization stages of hydrogen and helium (`Ionization transport`) and the He 2^3S level (`He 2^3S transport`) are carried by the flow on their elements' nucleus fluxes instead, beside the transported molecular carriers and the diffusive separation of helium from hydrogen. Heat conduction and viscosity add the Navier-Stokes terms, conduction exchanging heat with the lower atmosphere through the base face. The lower atmosphere itself is not solved by the code: it is handed over as scalars or as a profile, and an offline closure (`src/utils/element_flux_closure.py`) iterates the wind against a Photochem column until the elemental fluxes agree, with the option (`energy_coupling`, default off) of passing the heat the wind conducts into the column back to the column's steady energy balance (`photochem_to_lower_profile.py --conducted-heat-flux`, solved by `src/utils/conducted_heat_column_balance.py`), which makes the coupling two-way in energy as well as in the elements.

The standard route advances the time-dependent equations and assesses the resulting profiles over a stated physical-time interval. A stationary Jacobian-free Newton–Krylov (JFNK) solution may be sought separately, but its absence does not invalidate a marching solution that meets the stated profile criterion. The discrete marching stop and stationary certificate are distinct diagnostics (Secs. 7.7 and 9.).

The code started from ATES v2.0 (Caldiroli et al., 2021; Biassoni et al., 2024) and is by now essentially a new code.

The finite-volume discretization, the three-stage strong-stability-preserving Runge-Kutta integration and the choice of approximate Riemann solvers are recognizably those of ATES; the equation of state, the lower boundary condition, the reconstruction, the steady-state solvers and their certification, the transported species, the heat conduction, the transit synthesis and essentially all of the chemistry were written for EXHALE. Every physics extension is selected at run time by the presence of a key in `input.inp` or of a companion file (`metals.inp`, `opacity.inp`, `base.inp`). Every option that adds a species, a transport process or an external field defaults to off, so that a configuration that states none of them solves the atomic H/He problem; the keys that default to on are parts of the physics of what is already selected, which a key can only turn off for comparison (among them the He 2^3S level, the He/H charge exchange, secondary ionization, the recombination radiation absorbed on the spot, the collisional de-excitation heat of H(n=2), the rovibrational caloric equation of state of H_2, the molecular reaction heat, the resolved H_2 photoabsorption channels, and the interdiffusion enthalpy flux of a run whose elements, molecular carriers or ionization stages move relative to the gas; Table 2).

1.2 Units

The equations are solved in dimensionless form. The normalization is fixed once, in `input_read.f90`, from the planet radius R_0, the base temperature T_0, and the base number density n_0:

$$v_0 = \sqrt{\frac{k_B T_0}{m_{\rm H}}}, \qquad t_s = \frac{R_0}{v_0}, \qquad p_0 = n_0 m_{\rm H} v_0^2, \qquad q_0 = \frac{n_0 m_{\rm H} v_0^3}{R_0}, \tag{1}$$

where $m_{\rm H} = 1.67353284 \times 10^{-24}$ g is the mass of the hydrogen *atom*, not the atomic mass unit. Radii are in units of R_0, so the base level sits at $r = 1$, which is the lower face of the grid (Sec. 7.1); velocities in v_0; mass densities in $n_0 m_{\rm H}$; pressures in p_0; temperatures in T_0; and volumetric energy rates in q_0. The mass unit being the hydrogen atom, the density is the number of hydrogen-atom masses per unit volume, and a species contributes $m_{\rm species}/m_{\rm H}$ to it; the species table therefore carries m(He I) = 3.9715259 and not 4.

The gravitational potential is measured in v_0^2, so it is fixed by the single dimensionless group

$$b_0 = \frac{G M_p m_{\rm H}}{k_B T_0 R_0}, \tag{2}$$

the Jeans escape parameter at the base. Only three further dimensionless numbers enter the potential and the domain: $\tilde{a} = a_{\rm orb}/R_0$, the mass ratio $M_\star/M_p$, and the outer radius.

The base level itself may be stated in three ways: `Log10 lower boundary number density` gives n_0 directly; `Base BC: pressure` states the base pressure in microbar and n_0 follows from $n_0 = p_{\rm base}/(k_B T_0 \hat{n}_{\rm bc})$; and a lower-atmosphere handoff (`base.inp` or a profile file) states $p_{\rm base}$ in bar and n_0 follows the same way. A run that states none is stopped at startup, and so is one whose handoff level disagrees with `Base BC: pressure` (by more than a part in 10^6) or with the density key (by more than 1 percent in the pressure it implies). Without a handoff, `Base BC: pressure` takes precedence over a density key stated beside it, and n_0 is the one the pressure gives; the conflict is not silent: the run prints a WARNING naming both levels (n_0 and the pressure of each) and the one used, and the setup report repeats it under the base level (`input_read.f90`, `write_setup_report.f90`). Here $\hat{n}_{\rm bc}$ is the base particle count per H+He nucleus, which the composition module derives from the element ratios and, with a molecular base, from the H nuclei bound into H_2 (Sec. 2.5).

1.3 Notation

One symbol stands for one quantity throughout this document. The table below fixes the symbols that appear in more than one section. Symbols introduced inside a published fit and defined where they appear (the x of a Verner-type cross section, the λ of a recombination-cooling fit, the α of a spectral index) are local to that fit and are not in the table.

symbol	quantity	note
u	conserved state $(\rho, \rho v, \mathcal{E})$	the unknown of the hydrodynamics, Sec. 2.1
W	primitive state (ρ, v, p)	what the reconstruction works on
ρ, v, p, T	mass density, radial velocity, pressure, temperature	
$\mathcal{E}$, ρe, e, ε	total energy density, internal energy density, internal energy per unit mass, internal energy per particle	$\mathcal{E} = \frac{1}{2}\rho v^2 + \rho e$
Φ	gravitational potential	Φ_i, Φ_c at interfaces and cell centers
γ_{eff}, c_s	effective adiabatic index, sound speed	from the caloric equation of state (14)
m_{H}	mass of the hydrogen atom	the code's mass unit, (1)
μ, κ	dynamic viscosity, heat conductivity	(21) only
n_X, n_{tot}, n_e	number density of species X, of all heavy particles, of free electrons	
$x_{X^{q+}}$, f_i	ionization fraction of a stage, mixing ratio n_i/n_{tot}	fractions of the element's own nuclei
E, $h\nu$	photon energy	always subscripted where a particular energy is meant (E_k, E_{th})
F_E, $\mathcal{F}_k$	incident and attenuated flux per unit photon energy	(35)
σ, τ	photoionization cross section, optical depth	
Γ, α, β	photoionization rate, recombination coefficient, electron-impact ionization coefficient	Sec. 4.2
$\mathcal{H}$, Λ	volumetric heating rate, volumetric cooling rate	erg cm^{-3} s^{-1}; $Q = \mathcal{H} - \Lambda$
ξ_{day}	dayside dilution of every stellar beam	Table 1
J_i, J	diffusive number flux of a carrier, diffusive mass flux of an element	Secs. 6.5, 6.9
dF, S, R	discrete flux difference, source, stationary residual	(6)–(11), (140)
$A_\pm$, ΔV_j, Δr_j	face areas, cell volume, cell width	(5)

1.4 How to read this document

Every equation below is the discrete operator the code applies, taken from the source and not from a derivation. Where the code's expression differs from the textbook form (the spherical pressure term under PLM, the gravitational work term folded into the energy flux, the mean molecular weight that is never formed), the code's form is given and the difference is stated. Routines are named with the file they live in, for instance `RK_rhs.f90`, so that a statement here can be checked against one place in the tree.

Section 2. states the conservation laws, the equation of state and the source terms. The sections that follow it state the radiation field, the ionization and chemical network, the thermal source and the molecular and metal chemistry.

Section 7. then gives the grid, the reconstruction, the Riemann solvers, the time integration and the operator splitting; Sec. 8. the boundary and initial conditions; and Sec. 9. the stationary residual, its acceptance test and the solvers that drive it to zero. A section on the post-processing tools closes the physics account, and Sec. 12. tabulates the input keys that select physics, with the default each of them takes when the key is absent.

2. Hydrodynamics

2.1 Conserved variables and the conservation laws

The conserved state of a cell is the triple

$$u = \left(\rho,\ \rho v,\ \mathcal{E}\right)^{\top}, \qquad \mathcal{E} = \tfrac{1}{2}\rho v^2 + \rho e, \tag{3}$$

with ρe the internal energy density. The map between u and the primitive triple $W = (\rho, v, p)$ is `UW_conversions.f90`: the density and the momentum are trivial, and the pressure follows from the caloric equation of state of Sec. 2.5, $p = p(\rho, \rho e)$, evaluated with the composition of the cell the state belongs to. For a reconstructed face state that cell is the cell the state was extrapolated from, because the composition ratios the equation of state needs are ratios of number densities and are therefore independent of the density.

The equations integrated are the Euler equations in spherical symmetry with gravity and a radiative source,

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \rho v\right) &= 0, \\ \frac{\partial (\rho v)}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \rho v^2\right) + \frac{\partial p}{\partial r} &= -\rho \frac{\partial \Phi}{\partial r} + S_\mu, \\ \frac{\partial \mathcal{E}}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 (\mathcal{E} + p) v\right) &= -\rho v \frac{\partial \Phi}{\partial r} + Q + S_{\mathcal{E}} - \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 q_d\right), \end{aligned} \tag{4}$$

where $Q = \mathcal{H} - \Lambda$ is the net radiative heating minus cooling per unit volume (Sec. 5.) and S_μ, $S_{\mathcal{E}}$ are the viscous and conductive sources of Sec. 2.8, both off unless a run asks for them. $q_d = \sum_s h_s J_s$ is the interdiffusion enthalpy flux of a mixture whose elements or transported carriers move relative to the mass-averaged velocity v (Cook, 2009, eqs. 11–13): it exists where `He_diffusion` moves helium against hydrogen, where a molecular carrier is transported and where `Ionization transport` carries the ionization stages, and is written out in Sec. 6.10: the element part is `interdiffusion_enthalpy_face_flux` (`binary_element_diffusion.f90`), the carrier and stage parts are `molecular_carrier_enthalpy_face_flux` and `ionization_stage_enthalpy_face_flux` (`diffusive_photochemistry.f90`), and their divergences enter the stationary energy row (`steady_residual.f90`), the marching update after an accepted element or carrier step (`EXHALE_main.f90`) and the advected-profile energy solve (Sec. 11.1). The factor 4π that would appear in the face areas and the cell volume cancels between them and is simply not carried.

2.2 The discrete divergence

`RK_rhs.f90` assembles the right-hand side of (4) as a flux difference `dF` and a source `S`, so that one forward Euler stage of the marching loop is $u \leftarrow u - \Delta t\,(dF - S)$ and the stationary residual is $R = dF - S$ with the radiative source subtracted

from its third row. With

$$A_{\pm} = r_{j\pm1/2}^2, \qquad \Delta V_j = \frac{r_{j+1/2}^3 - r_{j-1/2}^3}{3}, \qquad \Delta r_j = r_{j+1/2} - r_{j-1/2}, \tag{5}$$

and $F_{\pm}$ the numerical flux at the two faces of cell j (Sec. 7.3), the three rows are

$$dF_1(j) = \frac{A_+ F_+^{\rho} - A_- F_-^{\rho}}{\Delta V_j}, \tag{6}$$

$$dF_2(j) = \frac{A_+ F_+^{\rho v} - A_- F_-^{\rho v}}{\Delta V_j} + w_p \frac{p_R - p_L}{\Delta r_j}, \tag{7}$$

$$dF_3(j) = \frac{A_+ F_+^{\mathcal{E}} - A_- F_-^{\mathcal{E}} + \delta F_3}{\Delta V_j}, \tag{8}$$

with $F^{\rho} = \rho v$, $F^{\mathcal{E}} = v(\mathcal{E} + p)$, and p_L, p_R the face pressures the Riemann solver returns at the lower and upper face. The weight w_p and the content of $F^{\rho v}$ depend on the reconstruction, because the spherical pressure term is split differently in the two schemes (Sec. 2.3).

The term δF_3 in (8) is the gravitational work,

$$\delta F_3 = A_+ F_+^{\rho} \left[\Phi_i(j) - \Phi_c(j)\right] - A_- F_-^{\rho} \left[\Phi_i(j-1) - \Phi_c(j)\right], \tag{9}$$

with Φ_i the potential at the cell interfaces and Φ_c at the cell centers (`set_gravity_grid.f90`). Expanding the bracket separates (8) into the energy flux of Caldiroli et al. (2021), their Eq. (14), $F^{\mathcal{E}} + F^{\rho}\Phi$ taken at the interface potential, minus $\Phi_c(j)$ times the mass-flux divergence, which is their energy source $\Phi_j L_j^{\rho}$. The two forms are algebraically the same operator; the code carries it as one term so that the mass flux the work term rides on is the very flux the continuity row differenced.

2.3 The spherical pressure term and gravity

The pressure gradient in spherical symmetry is

$$\frac{\partial p}{\partial r} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 p\right) - \frac{2p}{r}, \tag{10}$$

and the two reconstructions use the two sides of this identity. Under PLM, `Phys_flux` adds p to the momentum flux, so the first term of (10) is inside dF_2 and the geometric correction is carried by the source; $w_p = 0$. Under WENO3 the pressure is left out of the flux and the whole gradient enters as the single face difference $(p_R - p_L)/\Delta r_j$; $w_p = 1$. `Source.f90` accordingly returns

$$S_1 = 0, \qquad S_2 = -\frac{\rho_L + \rho_R}{2}\frac{\Phi_i(j) - \Phi_i(j-1)}{\Delta r_j} + \underbrace{\frac{A_+ - A_-}{\Delta V_j} p_c}_{\text{PLM only}}, \qquad S_3 = 0, \tag{11}$$

where ρ_L and ρ_R are the two reconstructed face densities that bound the cell and p_c is the cell's own pressure from the caloric equation of state. Because $(A_+ - A_-)/\Delta V_j \to 2/r$ as the cell narrows, the PLM branch of (11) is the second term of (10).

In a subsonic HLLC star branch, the effective numerical pressure is $P = p_K + S_K \rho_K^*(S_* - v_K)$ on the selected side K. At a supersonic face it is the upstream pressure, with no star increment. The complementary ram part is the complete

Cartesian HLLC momentum flux minus P, so their sum retains that flux at a moving contact, at a sonic clamp, and in supersonic flow. The spherical pressure term uses this same effective pressure in the momentum split.

The three physical terms of the momentum equation, the ram divergence, the spherical pressure gradient and the weight $\rho\partial\Phi/\partial r$, are gathered out of these pieces by `momentum_row_terms_of_cell` (`RK_rhs.f90`) and stored. They are not dF_2 and S_2: under PLM each of those holds a part of order $2p/r$ that cancels against the other, so a magnitude built from them alone reads $2p/r$ on a uniform pressure at rest, where the physical force is zero. The gathered terms are what the stationary residual scales the momentum row by (Sec. 9.1).

2.4 The gravitational potential

`grav_field.f90` carries two potentials, selected by `Domain mode`. The default is the full Roche potential of a synchronously rotating planet,

$$\Phi(r) = -\frac{b_0}{r} - \frac{b_0\mathcal{M}}{\tilde{a}-r} - \frac{b_0(1+\mathcal{M})}{2\tilde{a}^3}\left(\frac{\tilde{a}\mathcal{M}}{1+\mathcal{M}} - r\right)^2, \qquad \mathcal{M} = \frac{M_\star}{M_p}, \quad \tilde{a} = \frac{a_{\rm orb}}{R_0}, \tag{12}$$

in code units, which is Eq. (2) of Caldiroli et al. (2021): the planetary term, the stellar tidal term and the centrifugal term about the center of mass. With `Domain mode: Spherical` the last two terms are dropped and $\Phi = -b_0/r$. Either potential can be carried past the L1 radius with `Outer radius`; past the ridge of (12) the potential falls outward, which the initial condition and the outer ghost take into account (Secs. 8.2 and 8.3). The analytic derivative $d\Phi/dr$ is provided beside it and is used by the initial-condition sonic-point search; the flux assembly itself never differentiates Φ, it differences the tabulated Φ_i and Φ_c, which are evaluated once at startup at the true face and center radii.

The default outer radius in Roche mode is the L1 (Hill) radius $r_{\rm max} = (3\mathcal{M})^{-1/3}\tilde{a}$; `Outer radius` overrides it and keeps the full Roche potential, as the extended domains of Huang et al. (2023) and Yan et al. (2022) do, and a transonic wind needs it past the sonic point (Sec. 8.2).

2.5 Equation of state

The thermal equation of state is that of an ideal gas of all particles including free electrons,

$$p = (n_{\rm tot} + n_e)\,k_B T \qquad \Longleftrightarrow \qquad T = \frac{p}{n_{\rm tot} + n_e} \ \text{ in code units,} \tag{13}$$

which is `comp_T_from_p` and `comp_p_from_T` of `composition.f90`. The particle and electron counts are built in one place, `get_species_densities` in the same file, from the density and the species mass fractions: `calc_ntot` adds one particle for every species the run carries (H I, H II, He I, He II, He III, every metal ion stage, and each of H_2, H_2^+, H_3^+, HeH^+, OH, H_2O, CO), and `calc_ne` adds the free electrons weighted by the net charge of each species. The He 2^3S column is not added by either: it is an excited level of He I and its particle and its nucleus are already counted there. Trace metals enter both sums only under the `eos_metals` policy of Sec. 6.8, on by default, so that the legacy H/He-only budget can be reproduced exactly.

The code never forms a mean molecular weight for the equation of state. The mean molecular weight of Caldiroli et al. (2021), their footnote 1, is a derived diagnostic here; (13) with the explicit particle count is what the hydrodynamics uses, so the temperature of a cell is a function of its composition and never of a fitted composition ratio.

The *caloric* half of the equation of state is `caloric_eos.f90`. For an atomic gas it is the monatomic relation $\rho e = p/(\gamma_{\rm ad}-1)$ with $\gamma_{\rm ad} = 5/3$, exactly as in ATES. This is not valid in a molecular layer: at the 400 to 1300 K of the H_2 base the rotational ladder of H_2 is populated (the $J = 1$ level lies at 170 K), so an H_2 molecule stores about $\frac{5}{2}k_BT$ and the vibrational ladder adds to that above ~ 800 K. Where the gas is 80% H_2 the heat capacity is then wrong by a factor 1.67. The mixture relation used instead is

$$\rho e = \tfrac{3}{2}(n_{\rm tot}+n_e)k_BT + n(\mathrm{H_2})\,k_B\,u_{\rm rv}(T), \qquad \frac{C_V}{k_B} = \tfrac{3}{2}(n_{\rm tot}+n_e) + n(\mathrm{H_2})\,c_{\rm rv}(T), \tag{14}$$

so that the exponent making $\rho e = p/(\gamma_{\rm eff}-1)$ exact is $\gamma_{\rm eff} = 1+(n_{\rm tot}+n_e)k_B/C_V$, a function of both the composition and the temperature. The mean rovibrational energy $u_{\rm rv}$ and the heat capacity $c_{\rm rv}$ are evaluated from the complete bound rovibrational ladder of $H_2\,X^1\Sigma_g^+$, 302 levels to 51966 K, that is to the 4.478 eV dissociation limit (Roueff et al., 2019, Table 2 through VizieR); the same partition function is the internal partition function of the $H+H \leftrightarrow H_2$ equilibrium constant (87), so the energy and the chemistry cannot disagree. H_2^+, H_3^+, HeH^+ and the oxygen carriers are counted as monatomic, a stated approximation valid while they stay trace species (their largest ratio to $n(H_2)$ in the molecular layer of the reference configuration is 1.7×10^{-7}). Dissociation is not a heat capacity and is not in (14); it is a chemical source term. The branch between the two forms is structural, not a limit: a run with no molecules and a cell with $n(H_2)$ exactly zero both take the constant-$\gamma_{\rm ad}$ expression verbatim. The key `Caloric EOS: monatomic` forces the monatomic form everywhere.

The mapping $\rho e \to p$ in the molecular branch is the inverse of (14) and costs one safeguarded Newton iteration per cell; the mapping $p \to \rho e$ is direct. The sound speed used by the Riemann solvers and by the time step is $c_s = \sqrt{\gamma_{\rm eff}p/\rho}$ evaluated with the cell's own composition.

Two base composition scalars follow from the same module. The gas mass per hydrogen nucleus is $m_{\text{H-nuc}} = m(\mathrm{H\,I}) + m(\mathrm{He\,I})\,y + \sum_e A_eX_e$ with $y = n_{\rm He}/n_{\rm H}$; the base particle count per H+He nucleus is

$$\hat{n}_{\rm bc} = \frac{1+y+\sum_e X_e}{1+y} - \frac{1}{2}\,\frac{x_{\rm H_2}}{1+y}, \qquad x_{\rm H_2} = \frac{2q_{\rm H_2}(1+y)}{1+q_{\rm H_2}}, \tag{15}$$

where $x_{\rm H_2}$ is the fraction of base hydrogen nuclei bound into H_2 and the last term is present only with `Molecular base: True`. The mixing ratio $q_{\rm H_2}$ is the one a lower-atmosphere handoff states, or otherwise the chemical-equilibrium fit (93) at $(p_{\rm base}, T_0)$ (Sec. 6.7). The same $x_{\rm H_2}$ is imposed on the inflowing ghost composition, so the equation of state and the species state describe one gas.

2.6 The well-balanced option

`Well balanced: True` (default off) changes the coordinate the pressure is carried in. The local hydrostatic equilibrium of cell j is the constant-density one through its own (ρ_j, p_j),

$$p_{{\rm eq},j}(r) = p_j - \rho_j\left[\Phi(r)-\Phi(r_j)\right], \tag{16}$$

(Eq. 16 of Käppeli & Mishra 2016), whose two face values are $P_{\rm up}(j) = p_j - \rho_j[\Phi_i(j)-\Phi_c(j)]$ and $P_{\rm dn}(j) = p_j + \rho_j[\Phi_c(j)-\Phi_i(j-1)]$. The reconstruction then works on the departure from (16) rather than on the pressure itself, the Riemann pressure jump is formed as $\Delta p = \Delta p_{\rm eq} + q_R - q_L$ from quantities that are all of the size of the departure, and the

momentum row is assembled from the face pressure measured against each side's own equilibrium. The two identities that make the substitution exact are

$$\begin{aligned} A_+P_{\rm up}-A_-P_{\rm dn}-(A_+-A_-)p_j &= -\rho_j\{A_+[\Phi_i(j)-\Phi_c(j)]+A_-[\Phi_c(j)-\Phi_i(j-1)]\},\\ P_{\rm up}-P_{\rm dn} &= -\rho_j\left[\Phi_i(j)-\Phi_i(j-1)\right], \end{aligned} \tag{17}$$

the first for the PLM assembly and the second for WENO3. Their left-hand sides are the equilibrium part of the pressure terms the row carries and their right-hand sides are the discrete gravitational source, so the two cancel in the algebra, `Source.f90` returns $S_2 = 0$ and neither side is ever evaluated. This is the momentum source of Käppeli & Mishra (2014), their Eq. (2.26), in spherical geometry. What cancels is still the physics the row balances, so its magnitude,

$$\mathcal{W}_j = \frac{|\rho_j\{A_+[\Phi_i(j)-\Phi_c(j)]+A_-[\Phi_c(j)-\Phi_i(j-1)]\}|}{\Delta V_j}\quad\text{(PLM)},\qquad \mathcal{W}_j = \frac{|\rho_j[\Phi_i(j)-\Phi_i(j-1)]|}{\Delta r_j}\quad\text{(WENO3)}, \tag{18}$$

is formed separately and is the term the momentum residual is read against under this option.

Under this option the lower boundary shares the same equilibrium: the interior pressure its outgoing characteristic reads is $P_{\rm dn}(1)$, and the two ghost cells continue (16) below the level through the face state (Sec. 8.1), so a column at rest in the discrete equilibrium and meeting the reservoir's pressure at the level carries no flux through the base face either.

2.7 Low-Mach dissipation of the contact mode

The HLLC flux resolves the contact wave exactly, so the dissipation it applies to the entropy family is proportional to the contact speed $S_* \simeq v$ and vanishes as the flow stalls; the acoustic families keep theirs at $|v| \pm c_s$. A stagnant layer can then carry a $2\Delta r$ pattern in v and T that nothing in the scheme removes. `Low-Mach damping:` (default off, `low_mach_dissipation.f90`) adds a gated fourth-difference stress (Jameson et al., 1981) to the numerical momentum flux,

$$D_p^{j+1/2} = \epsilon_4\, g(M)\,\rho_f\,\lambda_f\left(v_{j+2}-3v_{j+1}+3v_j-v_{j-1}\right),\qquad D_{\mathcal{E}}^{j+1/2} = v_f\, D_p^{j+1/2}, \tag{19}$$

with ρ_f, v_f the face averages and $\lambda_f = \frac{1}{2}[(|v|+c_s)_j+(|v|+c_s)_{j+1}]$ the face spectral radius. The gate $g = [\max(0, 1 - M_f^2/M_{\rm th}^2)]^2$, written in M^2 so that it is smooth where the velocity changes sign, is unity at rest and identically zero above $M_{\rm th}$ (default 10^{-3}). The pair $(D_p, D_{\mathcal{E}})$ conserves total energy exactly and converts kinetic energy into heat and never the reverse; the mass flux is untouched. The term is added inside `RK_rhs.f90`, so the marching right-hand side and the stationary residual solve the same equation, which is what distinguishes it from the Shapiro filter of Sec. 7.6.

2.8 Viscosity and heat conduction

`Viscosity: True` and `Conduction: True` (both default off, `viscous_conduction.f90`) add the Navier-Stokes level of the same equations, the level the CETIMB code (the time-dependent upper-atmosphere model of Koskinen and collaborators, Koskinen et al. 2013, 2022) carries. For the Newtonian stress in spherical symmetry the only nonzero components are $\tau_{rr} = \frac{4}{3}\mu(v' - v/r)$ and $\tau_{\theta\theta} = \tau_{\varphi\varphi} = -\frac{1}{2}\tau_{rr}$, so exactly

$$S_\mu = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\tau_{rr}\right)+\frac{\tau_{rr}}{r},\qquad q_\mu = \frac{4}{3}\mu\left(\frac{\partial v}{\partial r}-\frac{v}{r}\right)^2,\qquad S_{\mathcal{E}} = vS_\mu + q_\mu + \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\kappa\frac{\partial T}{\partial r}\right). \tag{20}$$

These differ from the expressions printed as Eqs. (B5) and (B6) of Koskinen et al. (2022): those carry a different gradient term ($\mu' v'$ against $\frac{4}{3}\mu' v/r$), a geometric coefficient $16/3$ against $8/3$, and a dissipation that is not sign definite,

and the printed pair does not satisfy $vS_\mu + q_\mu = \nabla\cdot(\tau\cdot v)$. Physical correctness being the criterion, (20) is what is implemented and the discrepancy is recorded in the source.

Neither Koskinen paper prints the transport coefficients. The heat conduction coefficient is the number-weighted sum of the Banks & Kockarts (1973) coefficients of the constituents, as Salz et al. (2015) (their Eqs. 24–26, electron–ion and neutral hydrogen) and Sutton et al. (2015) (their Eq. 5, with helium) write it,

$$\kappa = \frac{1}{n}\left[n_e\, 1.2\times10^{-6}\, T^{5/2} + n_{\mathrm{HI}}\, 379\, T^{0.69} + n_{\mathrm{HeI}}\, 299\, T^{0.69} + n_{\mathrm{H_2}}\, k_{\mathrm{H_2}}(T)\right] \ \mathrm{erg\,cm^{-1}\,s^{-1}\,K^{-1}}, \tag{21}$$

n the number density of all particles and T in K; ions contribute through n only. The hydrogen term is the atomic hydrogen conductivity of Watson et al. (1981), $4.45\times10^4 (T/1000\,\mathrm{K})^{0.7}$, to within 2 percent between 300 and 7000 K; at He/H = 9.7 the helium term decides κ (the coefficients stand in the ratio $299/379 = 0.79$; the mixture value is 0.81 of the hydrogen one on the LHS 1140 b He/H = 9.7 state, MEASURED, update log stage 3 section 41), and it agrees with the tabulated helium conductivity of Incropera et al. (2007) (Table A.4) to 1 percent at 300 and 1000 K. None of these sources gives an H_2 coefficient: $k_{\mathrm{H_2}}(T) = A\,T^s\,[1 + c\,E(\theta_v/T)]$, $E(x) = x^2e^x/(e^x-1)^2$, $A = 272.523$, $s = 0.735652$, $c = 0.370446$, $\theta_v = 5987$ K (the $v = 0 \to 1$ spacing of H_2 in the Roueff et al. (2019) ladder), an Eucken form fitted to the tabulated conductivity of Incropera et al. (2007), Table A.4, over 200–2000 K (largest deviation 2.5 percent, rms 1.05 percent); it is valid there and an extrapolation outside (below 200 K, where the rotational heat capacity freezes out, it overestimates, by 20 percent at 100 K). The Einstein function is evaluated as $x^2e^{-x}/(1-e^{-x})^2$, which cannot overflow at low T, by its Taylor series $1 - x^2/12 + x^4/240 - x^6/6048$ below $x = 0.05$, where $1-e^{-x}$ loses digits, and as zero above $x = 700$ (`einstein_heat_capacity`; the e^x form returned 0 below 16.9 K and NaN at 8.55–8.59 K until 2026-10-01); the H_2 term is formed only where $n_{\mathrm{H_2}} > 0$. The molecular ions enter through n and n_e only. A continuation factor $0 \le s \le 1$ on κ (`EXHALE_CONDUCTION_SCALE`, `conduction_scale`; 1 when unset, any other value stops the run at startup) multiplies the conductivity in the stationary row and the marching update alike; a state solved at $s < 1$ is a step of a continuation and not a prediction of the model. The factor is part of the state identity: wherever $s \neq 1$ the `cond` token of the options field carries s itself to seventeen significant digits (`factor_token_text`, `load_IC.f90`), so two factors are two states. The viscosity keeps the hydrogen value tied to the Watson conductivity by the Chapman–Enskog/Eucken relation for a monatomic gas, Prandtl number $2/3$, $\mu(T) = \frac{4}{15}\frac{m_{\mathrm{H}}}{k_B}\,4.45\times10^4 (T/1000\,\mathrm{K})^{0.7}$. In code units $\mu_{\mathrm{code}} = \mu_{\mathrm{cgs}}/(n_0 m_{\mathrm{H}} v_0 R_0)$, the inverse Reynolds number, and $\kappa_{\mathrm{code}} = \kappa_{\mathrm{cgs}} T_0/(n_0 m_{\mathrm{H}} v_0^3 R_0)$ (equal to $\frac{15}{4}\mu_{\mathrm{code}}$ only for the Watson value the viscosity is tied to, not for the mixture conductivity).

Both operators are assembled once as tridiagonal coefficient triplets on the same faces and volumes as (5) (`viscous_momentum_coeffs`, `thermal_conduction_coeffs`), and the same triplets serve as the source entering the stationary residual (`viscous_conduction_sources`, called by `steady_residual.f90`, so every stationary solver of Sec. 9. sees the terms) and as the matrix of a Crank-Nicolson update in the marching loop (`viscous_conduction_step`, an operator-split stage of each step), so the marched fixed point and the zero of the residual are the same state. The face coefficient is the cell coefficient interpolated linearly in r to the face, the same on both sides of a face, so the operator is in flux form and its column sum telescopes to the two boundary fluxes.

The outer boundary carries zero diffusive flux, which is the condition an extrapolated outflow ghost admits. At the inner boundary the viscous stress is taken against the velocity of the anchored ghost, held fixed during the implicit step. The heat flux through the base face is taken against a prescribed bath, not against the temperature the advective

ghost happens to carry: the lower atmosphere below the level is a heat bath on the time scale of a stationary solution, whichever way the contact is upwinded. The bath temperature is `conduction_base_level_T`, the lower atmosphere's own temperature at the ghost center $r(0)$ (`base_reservoir_temperature_at`, `base_boundary.f90`): where a lower-atmosphere profile is handed over (Sec. 6.7) it is the profile's temperature at that radius, linear in r between its levels, and otherwise the reservoir's hydrostatic isentrope continued from the level (Sec. 8.1); a reservoir that returns no positive temperature leaves the ghost's own value. The conductivity of the ghost is evaluated at the same bath temperature, so the coefficient and the temperature it acts on belong to one boundary condition. The two choices agree at the level itself; one ghost cell down they differed by 3.5 K on the LHS 1140 b photochemical profiles (188.89 against 185.42 K), which moved the conductive flux into the level by about 4.6 per cent at a fixed cell-1 temperature (MEASURED, update log stage 3 section 70).

The conduction operator is therefore conservative up to one named boundary flux, the heat exchanged with the lower atmosphere through the base face,

$$q_{\mathrm{base}} = \kappa_{1/2}\,\frac{T_1 - T_{\mathrm{bath}}}{r_1 - r_0} \quad \text{(positive into the lower atmosphere)}, \tag{22}$$

the base face's own term of the first row of the triplets, sign reversed. `heat_conducted_into_lower_atmosphere` forms it on the state the certification measures (Sec. 9.3), that is on the state about to be written and not on a perturbed Newton iterate, and `write_setup_report.f90` writes it to `EXHALE_resolved.out` whenever conduction is on: `base_conduction_state` (`unmeasured` before a certification has measured it, then `measured`), `base_conductive_flux_cgs` in $\mathrm{erg\,cm^{-2}\,s^{-1}}$, and the two temperatures of (22), `base_conduction_T_bath_K` and `base_conduction_T_cell1_K`. The energy coupling of the elemental-flux closure with the lower atmosphere reads this flux (Sec. 6.7). On the LHS 1140 b states it carried 35 to 52 percent of the wind's absorbed heat (MEASURED, update log stage 3 section 76).

The Crank-Nicolson stage returns a state only when both tridiagonal eliminations completed, every solved value is finite and every solved temperature is at or above the floor of the equation of state (the floor of the implicit energy step, Sec. 5.3). Otherwise it fails with a named status and reason and returns no state (`conduction_report`). The marching loop then refuses the whole attempted step, in both run modes, exactly as it refuses a failed energy update: the checkpoint is restored and the step is retaken at half the interval, up to the retry budget of the attempted-step controller (eight halvings, `n_step_retry_max`); a stage that still fails at $\Delta t/2^8$ stops the run with exit status 2 and the retry history (Sec. 7.5). A shorter step is the remedy for the failure the large step produces: a mode of the conduction operator with decay rate $\lambda/\Delta t$ is multiplied by $(1-\lambda/2)/(1+\lambda/2)$, which is negative for $\lambda > 2$, so over a step longer than twice the diffusion time of a cell the update overshoots, and halving Δt halves λ. The floor value is never returned, because returning it would add energy with no source behind it.

2.9 The two-dimensional approximation

A one-dimensional radial model of an irradiated sphere has to state how the stellar flux and the mass loss are shared over the planet. The mandatory key `2D approximate method` carries that statement, in the four forms of Table 1. The dilution factor ξ is applied to every stellar beam at the point where the beam is used, and in one place only (`dayside_dilution` in `parameters.f90`): the XUV photon grid, the stellar Ly-α beam and the FUV bands all read it.

The stated fluxes therefore keep their meaning as the flux at the planet's orbit, and the correction to the mass-loss rate is applied once, at the end of the run (148).

Table 1: The `2D approximate method` options, as `input_read.f90` normalizes them, `dayside_dilution` applies ξ and `EXHALE_main.f90` applies the $\dot{M}$ factor.

Key word	Stored as	ξ	$\dot{M}$ factor
`Mdot/4`	`Mdot/4`	1	$1/4$
`Rate/2`	`Rate/2 + Mdot/2`	$1/2$	$1/2$
`Rate/4`	`Rate/4 + Mdot`	$1/4$	1
`alpha`	`alpha`	1	1

The first row is the recipe of Salz et al. (2016) that the ATES paper adopts: the undiluted substellar rates with the resulting mass-loss rate divided by four. The second is the tidally locked reading, dayside averaged rates with the outflow leaving one hemisphere; the third the rapidly rotating one, rates reduced by four with the outflow leaving the whole surface. These are the three recipes of Caldiroli et al. (2021), their Sec. 2.4.

The fourth is different in kind. With `alpha` the run states a coefficient α and the attenuated field carries the further factor $1/(1+\alpha\tau)$ inside the column integral (Sekiya et al., 1980; Odert et al., 2020), so that the cell-mean attenuation becomes

$$\langle e^{-\tau} \rangle_{\mathrm{cell}} = \int_0^1 \frac{e^{-\tau_{\mathrm{out}} - s\,\delta\tau}}{1+\alpha(\tau_{\mathrm{out}} + s\,\delta\tau)}\, ds, \tag{23}$$

which is the cell-mean attenuation (34) with the further factor inside the integral; it has no closed form and is integrated by composite three-point Gauss-Legendre on segments of optical depth at most 0.25, at most 256 of them (`cell_mean_attenuation`, `utilities.f90`). With $\alpha = 0$, the default and the only value for which the field is a pure exponential, it reduces to (34) itself.

The range of validity is stated at the routine: $\xi = 1/2$ is the illuminated hemisphere seen by an *optically thin* absorber. Where a band is shielded the true shell average is lower, the slant columns away from the substellar point being longer; it is 0.20 to 0.33 for the Lyman-Werner band over a molecular layer. No key asks for that refinement, so a shielded band is diluted by the optically thin factor.

3. The stellar radiation field and its absorption

3.1 The incident spectrum

The star is represented by a plane beam that falls radially onto the substellar column. Its spectral shape is chosen once, by the `Spectrum type:` key of `input.inp`, and that single choice fills *every* band of the photon grid: the extreme ultraviolet and X-ray bands, and also the part below the hydrogen Lyman edge where the He 2^3S metastable, the low-ionization-potential metals and hydrogen in $n = 2$ absorb. The three types are implemented in `J_inc.f90` (`stellar_flux_eV`) and `sed_read.f90`:

Power law (`is_PL_sed`). Two power laws of one common index $\alpha \equiv$ `PLind`, normalized separately on the EUV interval $[E_{\rm low}, E_{\rm mid}]$ and on the X-ray interval $[E_{\rm mid}, E_{\rm top}]$. With $L_{\rm X}$ and $L_{\rm EUV}$ the two band luminosities, a the orbital distance and $\mathcal{R} = 10^{L_{\rm X}-L_{\rm EUV}}$,

$$J_{\rm XUV} = \frac{10^{L_{\rm X}} + 10^{L_{\rm EUV}}}{4\pi a^2}, \qquad J_{\rm X} = \frac{\mathcal{R}}{1+\mathcal{R}} J_{\rm XUV}, \qquad J_{\rm EUV} = \frac{1}{1+\mathcal{R}} J_{\rm XUV}, \tag{24}$$

and, for $\alpha \neq -1$ and $P_{\rm l} = \alpha + 1$,

$$J_{\rm inc}(E) = \begin{cases} \dfrac{P_{\rm l} J_{\rm EUV}}{E_{\rm mid}^{P_{\rm l}} - E_{\rm low}^{P_{\rm l}}} E^{\alpha}, & E < E_{\rm mid}, \\ \dfrac{P_{\rm l} J_{\rm X}}{E_{\rm top}^{P_{\rm l}} - E_{\rm mid}^{P_{\rm l}}} E^{\alpha}, & E \geq E_{\rm mid}, \end{cases} \tag{25}$$

the logarithmic form replacing the ratio at $\alpha = -1$, and the X-ray normalization set to zero when `thereis_Xray` is false. Below $E_{\rm low}$ the same law is extrapolated, which is what the type means rather than an accident.

Planck (`spectrum_is_planck`). The photospheric blackbody seen at the orbit,

$$F_\nu = \pi B_\nu(T_{\rm eff}) \left(\frac{R_\star}{a}\right)^2, \qquad B_\nu = \frac{2h\nu^3/c^2}{e^{h\nu/kT_{\rm eff}} - 1}, \tag{26}$$

converted to a flux per unit photon energy by $F_E = F_\nu \, {\rm d}\nu/{\rm d}E$ with ${\rm d}\nu/{\rm d}E = 2.417989242 \times 10^{14}\,{\rm Hz\,eV^{-1}}$. For $h\nu/kT_{\rm eff} > 700$ the Wien form replaces $1/(e^x - 1)$ so that no exponential overflows. The absolute scale comes from $T_{\rm eff}$ and $R_\star$, not from the two luminosities, so the grid-integrated flux generally differs from $J_{\rm XUV}$; `write_setup_report.f90` states both.

Load (`do_read_sed`). A measured spectral energy distribution read by `read_sed` (`sed_read.f90`): two columns, bin central wavelength in Å and flux at the planet in erg cm^{-2} s^{-1} Å^{-1}, ascending in wavelength. Between two rows the field is interpolated log-linearly in $\log E$, exact for the power-law segments a tabulated continuum is made of; outside the band it returns zero, and `spectrum_covers_eV` must be asked first so that the run stops rather than integrating a zero the input never stated.

A monochromatic option (`is_monochr`) puts the whole $10^{L_{\rm EUV}}/4\pi a^2$ at a single energy $E_{\rm low}$.

Bands of the photon grid. `set_energy_vectors.f90` lays the grid out over at most six bands,

$$[E_{\rm floor}, E_{\rm abs}],\ [E_{\rm abs}, 13.598\,{\rm eV}],\ [13.598, 24.587],\ [24.587, 54.418],\ [54.418, E_{\rm mid}],\ [E_{\rm mid}, E_{\rm top}],$$

with fixed bin budgets $(20, 89, 50, 50, 50, 50)$. $E_{\rm low}$, $E_{\rm mid}$ and $E_{\rm top}$ are stated by the `[E_low,E_mid,E_high]` line of `input.inp` (the tutorial run uses 13.60, 123.98 and 1.24×10^3 eV; with `Use only EUV? True` the top is held at 1.24×10^3 eV and the X-ray normalization is zero). $E_{\rm abs}$ is `sub_lyman_absorber_threshold_eV()`, the lowest ionization threshold of an absorber of the bulk gas below the hydrogen edge: the He 2^3S metastable at 4.768 eV when it is tracked, or the neutral stage of an active low-IP metal (K I 4.341, Na I 5.139, Ca I 6.113, Mg I 7.646, Fe I 7.902, Si I 8.152, S I 10.36 eV) when one is lower. $E_{\rm floor}$ is `photon_grid_floor_eV()`, which lowers that further to $E_{\rm th}({\rm H}, n{=}2) = E_{\rm th}({\rm H})/4 = 3.3996$ eV, the head of the Balmer continuum, whenever the excited-hydrogen coupling is armed. The interval $[E_{\rm floor}, E_{\rm abs}]$ carries no bulk-gas absorber: it exists so that the H(n=2) band is stated, that rate being a closed-form integral over it (Sec. 4.6) and not a sum over these bins.

Every ionization threshold is a bin edge. A photoionization rate is the rectangle sum $\sum_k F_k \sigma_k \Delta E_k / E_k$, so a bin that *contains* a threshold would charge the absorber its cross section, read at the bin center, over the part of the bin below the threshold where it cannot absorb at all. `ionization_thresholds_active` therefore collects every threshold the run carries (H I, He I, He II, He 2^3S, H_2 and its two channel thresholds, and every photoionizable stage of an element with nonzero abundance together with its subshell turn-ons); each band is cut at those falling strictly inside it, and its budget is shared over the subintervals in proportion to their logarithmic widths (largest remainder, at least one bin each). The bins are geometric and E_k is the *geometric* mean of the edges, the more accurate abscissa of a rectangle rule for a power-law integrand. A grid whose points sat *on* the thresholds instead overstates the exact $P_{\rm HI}$, $P_{\rm HeI}$ and $P_{\rm HeII}$ by factors 1.095, 1.016 and 1.030.

Dayside dilution. Every stellar beam of the code is multiplied by the dilution $\xi_{\rm day}$ of Table 1: the XUV grid in `set_energy_vectors`, $F_{\rm XUV} \to \xi_{\rm day} F_{\rm XUV}$; the stellar Ly$\alpha$ beam in `lya_rt.f90` and `excited_hydrogen.f90`; and the four far-ultraviolet bands through `fuv_band_flux`.

Bands stated by their integrated flux. Four far-ultraviolet bands lie almost entirely below the photon grid and enter as band-integrated energy fluxes at the planet's orbit rather than through the grid, in the band order of `water_photolysis.f90`: LW (912–1201 Å), B2 (the H I Lyα line at 1215.67 Å, band 1202–1230 Å), B3 (1231–1450 Å) and B4 (1451–2304 Å). The edges are fixed by the physics of the absorbers and not chosen: 912 Å is the Lyman edge, 1201 Å the red end of the H_2 Lyman and Werner line list, and the B2/B3/B4 edges follow the H_2O branching ratios (Sec. 6.6). Each band flux has one of two sources. A key in `input.inp` states it (`Stellar LW flux`, `Stellar Lya flux`, `Stellar FUV B3 flux`, `Stellar FUV B4 flux`), and a stated value, zero included, always stands. Where no key is given and the spectrum is a loaded file (`Spectrum type: Load`), the three continuum bands are integrated from that file by `sed_band_fluxes` (`sed_read.f90`),

$$F_{\rm band} = \int_{\lambda_{\rm lo}}^{\lambda_{\rm hi}} F_\lambda \, d\lambda, \tag{27}$$

the exact integral of the piecewise-linear table, the flux interpolated at each clipped band edge before the trapezoid (a constant or a linear spectrum integrates exactly whatever the nodes), the file being already at the planet and read once for the three bands: LW on any molecular run, B3 and B4 when the oxygen chemistry is on. A file that does not reach both edges of a band the run needs stops the run and asks for the key. The two analytic spectrum types cannot supply these bands (a power law is the XUV fit extrapolated downward, a Planck curve the photosphere, while the stellar far-ultraviolet is chromospheric), and the ionizing grid retained by `read_sed` ends at the lowest active threshold, so the file is re-read for them. Band B2 is never integrated: a Lyα line flux reconstructed from observations is a better number than a trapezoid over whatever rows the file has across 30 Å, so it is always a key. `fuv_band_flux` returns $\xi_{\rm day} F_{\rm band}$, the dilution being applied where the beam is used so that the stated or integrated values keep their meaning, the band flux at the planet's orbit; the setup report states all four and names the source of each.

3.2 Photoionization cross sections

All cross sections are carried in units of 10^{-18} cm^2 (`cross_sec.f90`) and are evaluated once, at the grid centers, into the vectors `s_hi`, `s_hei`, `s_heii`, `s_heiTR`, `s_h2` and the table `sigma_tab`.

Hydrogenic ions (H I, He II), function `sigma`. With $\epsilon = \sqrt{E/E_{\rm th} - 1}$,

$$\sigma(E) = \frac{6.3}{Z^2} \left(\frac{E_{\rm th}}{E}\right)^4 \frac{\exp\left[4 - 4\arctan(\epsilon)/\epsilon\right]}{1 - \exp(-2\pi/\epsilon)}, \qquad E > E_{\rm th}, \tag{28}$$

and $6.3/Z^2$ exactly at threshold, zero below it. $E_{\rm th}$ is the *measured* potential of the ion, the same energy the grid uses as a band edge and the photoelectron is charged against.

He I ground singlet The Verner et al. (1996) single-shell fit,

$$x = \frac{E}{E_0} - y_0, \quad z = \sqrt{x^2 + y_1^2}, \quad Q = 5.5 - P/2, \qquad \sigma = \sigma_0 \left[(x-1)^2 + y_w^2\right] z^{-Q} \left(1 + \sqrt{z/y_a}\right)^{-P}, \tag{29}$$

with the published parameters $(E_0, \sigma_0, y_a, P, y_w, y_0, y_1) = (13.61, 949.2, 1.469, 3.188, 2.039, 0.4434, 2.136)$ and $E_{\rm th}$ taken as the global He I threshold (it only gates the fit). The legacy ATES two-term fit $0.6935/[(E/100)^{1.82} + (E/100)^{3.23}]$ is restored by `ates_photoion_rate` (default off).

He 2^3S metastable Two forms of (29) joined by a log-linear bridge: wing A from the 4.768 eV threshold to the Cooper minimum at 34.70 eV, wing B from the resonance-averaged bump at 45.59 eV upward, and a power law joining them continuously in between. Each wing is a fit at the few-percent level to the Norcross (1971) and TOPbase background.

Metal ions `metal_photoion_sigma` returns the *shell sum* over the seven subshells of Verner et al. (1996) and Verner & Yakovlev (1995); the 1995 form is (29) with $y_0 = y_1 = 0$ and $Q = 5.5 + \ell - P/2$. Above the $E_{\rm max}$ of the 1996 outer-shell table the handover to the 1995 subshell fit is made, as in Verner's own `phfit2`; extrapolating the outer-shell fit is not conservative (the Fe I row rises as E^{2-Q} and reaches 24.9 Mb at 1240 eV against a true 0.51 Mb). An inner-shell absorption is charged to one ionization of the absorbing stage with a photoelectron of $h\nu - E_{\rm th}$(outer), which approximates the Auger cascade and undercounts the charge state by at least one stage; this is the approximation the published exoplanet photochemistry models make (Cecchi-Pestellini et al., 2009; Locci et al., 2022). Fluorescence is neglected: with $E_{\rm top} = 1240$ eV only C, N, O and Na I can have a K hole, and their K yields (Krause, 1979) are 0.28, 0.52, 0.83 and 2.3 percent.

H_2 `sigma_H2` is a piecewise *measured* total photoabsorption: $k_{\rm bx}\times$ Backx et al. (1976) Table 1 over 15.4–18 eV with $k_{\rm bx} = 1.016415$, Samson & Haddad (1994) Table 1 over 18–300 eV, and $k_{\rm hi}\times$ the Yan et al. (1998) eq. (19) $E^{-7/2}$ sum-rule tail above 300 eV with $k_{\rm hi} = 0.885805$, all interpolated in $\log\sigma$ against $\log E$. No piece of the Yan fit survives below 300 eV: its three published pieces do not meet at their own join energies, stepping by a factor 1.719 at 18 eV and 0.619 at 85 eV.

Opacity models `opacity_models.f90` dispatches the cross sections through a model selected in the optional `opacity.inp`: `A` analytic (the curves above, the default), `C` constant above threshold, `P` constant with the Robinson & Catling (2012) pressure broadening $f(p) = 1 + a(p/p_{\rm piv})^n$, and `T` a two-column table. In model `P` the multiplier

$f(p)$ of a cell is stored in `opa_pf(j)` and carried identically in the optical depth and in the local absorption, so that the photons a cell removes from the beam are the photons it absorbs.

3.3 Transfer along the radial ray

The continuous form. What the code evaluates is the frequency-resolved absorption of the incident beam along the substellar radial ray. With $F_E(a)$ the incident flux per unit photon energy at the orbital distance a (for the power law, $J_{\rm XUV} = L_{\rm XUV}/4\pi a^2$ of (24); for the Planck and table types, Sec. 3.1), the flux reaching radius r is

$$F_E(r) = F_E(a)\, e^{-\tau_E(r)}, \qquad \tau_E(r) = \sum_a \sigma_{a,E} \int_r^{r_{\rm top}} n_a(r')\, dr', \tag{30}$$

the sum running over every photoionizing absorber a and the column starting at zero at the outer face of the domain, $r_{\rm top} = r_{\rm edg}(N)$: no absorber is carried beyond the domain, which is where these expressions differ from Eq. (4) of Caldiroli et al. (2021), whose integral runs to a. The two outer ghost cells are a closure of the reconstruction stencil, not gas on the line of sight: they see the unattenuated beam and add no depth to the cells below (`calc_column_dens`, `utilities.f90`; the same start holds for the Lyα line-center depth of `lya_rt.f90` and the fine-structure escape column of `Cool_coeff.f90`). Until 2026-10-11 the column started at $r(N+N_g)$ and counted the ghosts; past the Roche ridge a hydrostatic ghost of 10^{41} cm^{-3} then shielded the whole domain (update log stage 3 section 138). The photoionization rate of one particle of absorber a and the photoheating rate of the gas are then

$$\Gamma_a(r) = \int_{E_{{\rm th},a}}^{\infty} \frac{F_E(r)}{E}\, \sigma_{a,E}\, dE, \qquad \mathcal{H}(r) = \sum_a n_a(r) \int_{E_{{\rm th},a}}^{\infty} F_E(r) \left(1 - \frac{E_{{\rm th},a}}{E}\right) \sigma_{a,E}\, dE, \tag{31}$$

which are Eq. (3) of Caldiroli et al. (2021) and its rate counterpart, $1 - E_{\rm th}/E$ being the fraction of the photon energy carried off by the photoelectron. Two factors the code adds to (30) are stated where they enter below: the pressure-dependent opacity weight $f(p)$ of the optional model `P`, and the secondary-ionization heat fraction $f_{\rm heat}(E)$ that multiplies the photoelectron share in $\mathcal{H}$ when secondary ionization is on (Sec. 3.5). The rest of this section is the discretization of (30) and (31) on the photon grid and on the radial cells, each line of which was read against `photoionization_field_at_cell_HHe` (`util_ion_eq.f90`), the one routine that forms the field and the rates of a cell.

The transfer is frequency-integrated over the grid of Sec. 3.1 and is pure absorption: there is no scattering term and no diffuse field in the XUV. For cell j the column of each absorber *outside* the cell is accumulated from the top down by `advance_starward_columns` (`util_ion_eq.f90`) on the same rectangle rule the grid uses,

$$N_a^{\rm out}(j) = \sum_{j<j'\le N} n_a(j')\, \Delta r_{j'} R_0\, f(p_{j'}), \tag{32}$$

and the cell's *own* depth is formed from its own density rather than as a difference of two columns, which is the same number in exact arithmetic and loses its digits to cancellation for a thin cell. The two optical depths at photon energy E_k are

$$\tau_k^{\rm out} = \sum_a \sigma_{a,k} N_a^{\rm out}, \qquad \Delta\tau_k = \sum_a \sigma_{a,k}\, n_a(j)\, \Delta r_j R_0 f(p_j), \tag{33}$$

the sum running over H I, the He I ground singlet, He II, He 2^3S when tracked, H_2 when the molecular chemistry is on, and every photoionizable metal stage.

The attenuated flux the cell's *rate* sees is the *mean* of the beam over the cell, not its value at either face (`cell_mean_attenuation` in `utilities.f90`). Since the densities are uniform inside a cell, $\tau(s) = \tau^{\rm out} + s\,\Delta\tau$ and

$$\langle e^{-\tau}\rangle_{\rm cell} = e^{-\tau^{\rm out}}\,\frac{1-e^{-\Delta\tau}}{\Delta\tau}, \tag{34}$$

the second factor being `absorbed_fraction_per_unit_depth`, which the H_2O and OH far-ultraviolet band rates take as well. The quotient $(1-e^{-x})/x$ has one evaluation in the code, `one_minus_exp_over_x` (`utilities.f90`), and every site that needs it goes through it (the recombination radiation below, the ground-capture weights of (60), the fine-structure line escape of Sec. 5., the Sobolev escape of Lyα in Sec. 4.6): its Taylor series through x^8 in Horner form below $x = 10^{-2}$ (the three-term series below 10^{-8}), the closed form above, with an error of at most 2.2×10^{-14}. The closed form alone does not do below the switch: e^{-x} is rounded to a double next to 1, so $(1-e^{-x})/x$ is a staircase in x with relative steps of about ϵ/x, a discontinuous function of the densities $\Delta\tau$ is formed from (2026-10-01, Sec. 9.4). Only this mean makes the photons the cell absorbs equal the photons the beam loses across it; a one-point rule at the inner face is low by 42 percent at $\Delta\tau = 1$, the depth a cell carries at the ionization front. The optional `alpha` form of the 2-D approximation replaces (34) by (23), which has no closed form.

The working flux of the cell is therefore

$$\mathcal{F}_k(j) = F_{{\rm XUV},k}\,\langle e^{-\tau}\rangle_{k,{\rm cell}}\,f(p_j), \tag{35}$$

the model-P factor being folded once into the shared flux weight rather than into each cross section, which is identical because every integrand below is linear in exactly one cross section.

3.4 Photoionization and photoheating rates

Every integral is the rectangle sum over the grid, $\sum_k(\cdot)\,\Delta E_k$, with ΔE_k the exact bin width, so $\sum_k \Delta E_k = E_{\rm top} - E_{\rm floor}$ exactly. For absorber a the photoionization rate per particle and the photoheating rate of *one* particle of that absorber are

$$P_a = \sum_k \mathcal{F}_k\,\frac{\sigma_{a,k}}{E_k}\,\Delta E_k \times 10^{-18}\,{\rm erg^{-1}eV}, \tag{36}$$

$$h_a^{(1)} = \sum_k \mathcal{F}_k\left(1-\frac{E_{{\rm th},a}}{E_k}\right) f_{\rm heat}(E_k)\,\sigma_{a,k}\,\Delta E_k \times 10^{-18}\ \ [{\rm erg\,s^{-1}}], \tag{37}$$

where the factor $1-E_{\rm th}/E$ is the share of the photon left as photoelectron kinetic energy (`photoelectron_share`; the key `photoheat_photon_fraction`, default off, replaces it by a constant fraction), and $f_{\rm heat}$ is the secondary-ionization heat fraction of Sec. 3.5. The heating of the cell is the contraction of these one-particle rates with the composition,

$$\mathcal{H} = \sum_a n_a\,h_a^{(1)}, \tag{38}$$

performed in exactly one place (`photoheating_of_cell`) and split by absorber into the six columns H I, He I, He II, He 2^3S, H_2, metals. The heating efficiency the code forms is

$$\eta_{\rm heat} = \frac{\mathcal{H}}{\displaystyle\sum_k \mathcal{F}_k\Big(\sum_a \sigma_{a,k} n_a\Big)\,\Delta E_k \times 10^{-18}}, \tag{39}$$

the ratio of the energy deposited to the energy absorbed, set to zero in a transparent or unilluminated cell. It is passed to `write_output` (`write_output.f90`) but written to no output file (`Hydro_ioniz.txt` carries r, ρ, v, p, T, the heating and the cooling).

Molecular channels. H_2 absorption is resolved into four mutually exclusive final states whose cross sections are *shares* of σ_{H_2}, so the opacity and the total H_2 destruction rate are unchanged. Each channel is charged its own threshold in (37), and the two deposits that are not a photoelectron, the kinetic energy release of the double channel and the internal excitation the neutral channel radiates away, are stated with the channels themselves in Sec. 6.2.

Self-consistency of the field with the composition. Nothing in (33) depends on any cell inside j, which is what lets the equilibrium sweep march outside in and re-form the field of one cell against its own composition (Sec. 4.4); the cell mean (34) is formed once per pass so that no second attenuation is stacked on the first. The rates are constants of the cell's nonlinear solve, carrying the field dependence inside the residual meaning a re-integration of the photon grid at every residual evaluation.

3.5 Secondary ionization by photoelectrons

A photoelectron faster than $E_{\rm sec} = 30\,\mathrm{eV}$ does not thermalize completely: part of its energy ionizes further neutrals. `electron_energy_degradation.f90` divides the labor between two published determinations. Shull & van Steenberg (1985) supply the energy budget of the ionization channels as a function of the ionized fraction x,

$$\phi_{\rm HI}(x) = 0.3908\left(1-x^{0.4092}\right)^{1.7592}, \qquad \phi_{\rm HeI}(x) = 0.0554\left(1-x^{0.4614}\right)^{1.6660}, \tag{40}$$

whose form vanishes at $x = 1$ where no neutral is left. Dalgarno et al. (1999) supply the heating fraction and everything that depends on molecular hydrogen; their Table 4 mean energies per ion pair, $W(x,E_0) = a(1+cx^b)$, give the energy dependence,

$$f_{\rm HI}(x,E_0) = \phi_{\rm HI}(x)\frac{W_{\rm H^+}(x,1\,\mathrm{keV})}{W_{\rm H^+}(x,E_0)}, \qquad f_{\rm HeI}(x,E_0) = \phi_{\rm HeI}(x)\frac{W_{\rm He^+}(x,1\,\mathrm{keV})}{W_{\rm He^+}(x,E_0)}, \tag{41}$$

so that the high-energy limit is exactly Shull & van Steenberg's. The division of that energy among the targets is recomputed on the cell's own densities, using the collision weight 1.89 that Dalgarno et al.'s eqs. (9) and (10) imply (their coefficients 1.89 and 0.53 are reciprocal to 0.17 percent), with $r_{\rm ref} = n(\mathrm{He\,I})/n(\mathrm{H\,I}) = 0.1$ the reference composition of the fits:

$$w = (n_{\rm HI} + 1.89\,n_{\rm H_2})r_{\rm ref}f_{\rm HI} + n_{\rm HeI}f_{\rm HeI}, \quad c_{\rm HI} = \frac{f_{\rm ion}\,r_{\rm ref}f_{\rm HI}}{w}, \quad c_{\rm H_2} = 1.89\,c_{\rm HI}, \quad c_{\rm HeI} = \frac{f_{\rm ion}f_{\rm HeI}}{w}, \tag{42}$$

with $f_{\rm ion} = f_{\rm HI} + f_{\rm HeI}$. These coefficients are *per target particle* (units cm^3), so $n_{\rm HI}c_{\rm HI} + n_{\rm H_2}c_{\rm H_2} + n_{\rm HeI}c_{\rm HeI} = f_{\rm ion}$ identically and nothing is divided by a vanishing neutral density. The heat fraction is

$$f_{\rm heat}(E_0) = 1 - [1-\eta_{\rm mix}(x,E_0)](1-x) + f_{\rm diss} + f_{\rm vib} + f_{\rm fluor}, \tag{43}$$

with $\eta_{\rm D} = 1 + (\eta_{\rm D,0} - 1)/(1+cx^b)$ their eq. (14) and $\eta_{\rm mix}$ their eq. (12) mixing of the H/He and H_2/He efficiencies. The factor $(1-x)$ is this code's closure and not either source's: the channels that compete with heat need a neutral to act on, and Dalgarno et al.'s form leaves 10 percent of the primary energy unthermalized even at $x = 1$. The three

molecular terms (fragment kinetic energy, direct vibrational excitation of $v = 1, 2$, and the vibrational energy the B and C states return on fluorescing) vanish identically with $n(H_2)$; the last two carry the collisional quench fraction of `h2_vibrational_relaxation.f90`, since a radiated excitation is not heat.

The secondary rates are added to the primary ones of (36): for an absorber a whose photoelectron energy exceeds $E_{\rm sec}$,

$$P_{\rm HI}^{\rm sec} = \sum_k \mathcal{F}_k \, \frac{\sigma_{a,k} n_a}{E_k} \, c_{\rm HI}(E_k) \, \frac{E_k - E_{{\rm th},a}}{E_{\rm th}({\rm H\,I})} \, \Delta E_k \times 10^{-18} \, {\rm erg^{-1}eV}, \tag{44}$$

and likewise for He I and H_2; the H_2 secondaries additionally make one proton per 22 H_2^+ ions (Dalgarno et al., 1999, after their eq. 10), which is an added yield and not a share.

Activation. The key `Secondary_ionization` is *on* by default (`use_sec_ion = .true.`) and, since 2026-10-09, applied from step 0 (`True` or `Immediate`, `sec_ion_immediate = .true.`; update log stage 3 section 127). `Secondary_ionization: Staged` keeps the earlier default: the runtime flag `sec_ion_active` stays false until the marching has settled without the coupling (the state change over a monitor window below `march_change_th`, out of the PLM stage, or the steady gates met; `EXHALE_main.f90`), and it is set again before a JFNK hand-off so that Newton is never asked to solve a system the marched state does not approximately satisfy. Staging was introduced because from a cold initial condition the base feedback had amplified the startup transient into a runaway on high-gravity cases; with the present code that runaway did not occur (MEASURED on `wasp_full` and `wasp_full_newton` from their cold initial conditions with the coupling on from step 0: 20 000 steps without a NaN, and the Newton finish at $\log_{10}\dot{M} = 13.29$ as with staging). `False` disables the coupling entirely.

Photoelectrons below 30 eV. By default a photoelectron below $E_{\rm sec}$ deposits all its energy as heat. Above 10.2 eV that is a limit of the tables of the partition above and not a physical threshold; below 10.2 eV it is physical, no excitation or ionization being open to the electron. The key `Low-energy electron partition: Furlanetto2010` (default `none`; `use_low_energy_partition` in `parameters.f90`, the physics in `low_energy_electron_degradation.f90`) partitions the photoelectrons of 10.2–30 eV by the Monte Carlo deposition tables of Furlanetto & Stoever (2010), their electronic tables at the 14 ionized fractions from 10^{-4} to 0.999, interpolated linearly in E_0 and in $x = n_e/(n_{\rm H} + n_{\rm He})$, the x the partition above is given (`low_energy_electron_tables.f90`). Its heat share $f_{\rm heat} E_0$ is heat; its numbers of H I and He I ionizations per primary are added to the secondary rates of (44), their branching renormalized to the cell's neutral densities by the rule of (42) with the tables' reference composition $r_{\rm ref} = 0.08/0.92$ in place of 0.1; and the energy of the collisionally excited H I lines leaves the cell, as the excitation share of the partition above does. An electron below the 10.2 eV row is heat whole (the authors' interpolator ramps from their 10.0 eV row, which the physics does not), x is clamped into the tabulated range, and the heat is clamped into $[0, 1 - f_{\rm ion}]$. The partition applies only where the secondary-ionization coupling is applied (`use_sec_ion` and, under `Staged`, `sec_ion_active`), on the stellar path (`photoelectron_partition_on_grid`) and the recombination path (`photoelectron_partition_at_energy`) alike, and only in a cell whose *solved* state holds no H_2: one gate, `low_energy_electron_partition`, decides it, and the advection-corrected post-process, whose reconstructed gas carries no molecules, passes the H_2 density of the solved state it corrects. The option is recorded in the state file (token `lowE_partition`), so a state is not restarted under the other setting.

The validity of the option is limited in three ways that the code states and does not repair. (i) The tables describe an atomic H/He gas of He/H = 0.087; only the H I/He I branching is renormalized, the heat and the total ionization share being those of that gas, so in a helium-rich gas, which has fewer H I atoms per electron at the same x, the tables understate the heat below 19.8 eV. (ii) The gate is discontinuous in the H_2 abundance: an arbitrarily small $n(H_2)$ sends the whole 10.2–30 eV power back to heat, no source covering the partition in H_2 at these energies; a continuous treatment of a partly molecular gas is an open design item (`md/TO_BE_DONE.md`, item (AJ)), and the option is off in every stored LHS 1140 b run. (iii) The trapping of the H I line photons in a Lyα-thick cell is not followed, so there the heat below 30 eV is a lower bound. At 30 eV, where the source changes, $f_{\rm heat}$ steps by -0.011 at $x = 10^{-4}$ and by zero at $x \geq 0.5$ (atomic gas, update log stage 3 section 69); the step is not smoothed.

3.6 Reprocessed and non-stellar fields

Three fields in the code are not the direct stellar beam.

Recombination radiation. The ionizing photons emitted when H II, He II and He III recombine are absorbed on the spot (`recombination_radiation_absorbed` in `util_ion_eq.f90`): each cell keeps the fraction $1-e^{-\tau}$ of every channel at its own optical depth (`absorbed_photon_fraction`, formed as τ times the quotient of (34) so that a thin cell keeps its digits) and shares it among H I, He I, He II, H_2 and the metals in the ratio of their absorption coefficients; the part re-absorbed by the recombined species itself is the on-the-spot cancellation of the net coefficients (60). The H II ground captures (`H_rec_escape`, default on) can be taken only by H I and the low-potential metals. With `He_rec_coupling` (default on) each He II cascade exit is represented at one photon energy: the ground-capture continuum starting at $E_{\rm th}$(He I), the $2^1P \rightarrow 1^1S$ resonance line at 21.2 eV, the $2^3S \rightarrow 1^1S$ line at 19.8 eV, and the part of the 2^1S two-photon continuum above the hydrogen edge. The last is not a line: integrating the Drake et al. (1969) shape over 13.598–20.62 eV gives 0.5564 ionizing photons per decay carrying 8.9646 eV, a mean in-band photon energy of 16.110 eV and hence 2.512 eV of photoelectron energy per photon, against 3.511 eV for a flat distribution over the same window. The He III channels are the ground capture at 54.4 eV, the direct capture into He II $n = 2$ (Seaton, 1959), whose continuum starts at 13.605 eV, He II Lyα at 40.81 eV, and the He II $2s$ two-photon continuum, whose spectrum is the fit of Nussbaumer & Schmutz (1984) carried in three bands (13.6–15.4, 15.4–24.6 and above 24.6 eV: 0.112, 0.577 and 0.736 photons per decay). Every case-B capture of He III ends in $2s$ (the fraction of Pengelly 1964) or $2p$, and $2s$ decays by two photons at 526.823 s^{-1} (Drake 1986 eq. 27 at $Z = 2$, with the α-particle mass) unless a proton or He^{2+} collision moves it to $2p$ first (Pengelly & Seaton, 1964). The share H_2 takes ends in the same four final states as the stellar absorption (Sec. 6.2), in the ratio of their cross sections at the photon energy of the channel: it destroys H_2 at the whole absorbed rate, and H_2^+ is made only by what the dissociative, double-ionization and neutral states leave. The dissociative state opens for the photons from 19.8 eV up, the neutral window for He II Lyα and double ionization for the He III ground capture. Each state releases its photoelectron(s) and fragment kinetic energy at the channel energy, plus the mean kinetic energy of the captured electron for a continuum channel; the formation energy the products store and the $n = 2$ excitation the neutral fragments radiate are not heat. Every photoelectron of these channels, of H I, He I, H_2 and the metal ions goes through the partition of Sec. 3.5 at its own energy, the representative energy of its channel and not an average over the capture continuum (`photoelectron_partition_at_energy`, the bracket and interpola-

tion of the stellar path), under the same switches as the stellar ones, the photoelectron and the Auger electron of an inner-shell metal absorption separately: above 30 eV (the He III ground capture on H I, He I, H_2 and the metals below 24.4 eV, He II Lyα on the metals below 10.8 eV) it heats with $f_{\rm heat}$ and adds its secondary ionizations to the same rates; the fragment kinetic energy is heat at once.

Lyman-Werner band. The 912–1201 Å beam is solved once for all of its absorbers together (`fuv_lw_photon_field`), because an H_2O molecule cannot absorb a photon an H_2 line has already taken. Its transmission down to a face is $T = (1-A)e^{-\tau_c}$ with $A = \int \sigma_{\rm pump} f_{\rm shield}(N_{\rm H_2}) n_{\rm H_2} {\rm d}r$ summed along the path, each cell contributing at its own (T, n) the same integral its own line rate takes (the table is built for a uniform slab, and reading A of the whole column at the temperature of the cell the beam has reached would count as taken photons the lines of the cells above have not taken, or the reverse), and across one cell the identity

$$T^{\rm out} - T^{\rm in} = e^{-\tau^{\rm out}} \left[(1 - A^{\rm out}) \left(1 - e^{-\Delta\tau} \right) + \Delta A \, e^{-\Delta\tau} \right] \tag{45}$$

is exact, so the photons the cell removes split into a continuum share and a line share with nothing left over. The continuum factor $e^{-\tau_c}$ is the Draine & Bertoldi (1996) eq. (40) term, identically 1 where the oxygen option is off. Line self-shielding is the tabulated function of $(T, n, N_{\rm H_2})$ of Sec. 6.3. CO is shielded on the same beam by the Visser et al. (2009) function $\Theta(N_{\rm CO}, N_{\rm H_2})$ (`co_self_shielding_table.f90`).

Lyα and the Balmer continuum are described with the excited-hydrogen model in Sec. 4.6.

4. Ionization and chemical state

4.1 The species carried

The composition of a cell is the vector `f_sp`$(j, \cdot)$ of species fractions, laid out once in `species_table.f90`:

column(s)	species	note
1–2	H I, H II	fractions of the hydrogen nuclei
3–5	He I, He II, He III	fractions of the helium nuclei; column 3 *contains* the metastable
6	He 2^3S	the metastable triplet, tracked when `Include He23S` is true (default on)
7–33	the 27 metal ion stages	C, O, N, Mg, Si, Ca, Fe in three stages; Na, K, S in two
34–37	H_2, H_2^+, H_3^+, HeH^+	molecular option (`Molecular chemistry`, `thereis_mol`, default off)
38–40	OH, H_2O, CO	oxygen option (`Oxygen chemistry`, `thereis_oxychem`, default off)

Ten elements carry a nuclear abundance $n_X/n_{\rm H}$, set by `metals.inp` (`metals_input_read.f90`) or by a lower-atmosphere handoff. Masses are in units of the hydrogen *atom*: helium weighs $m_{\rm He}/m_{\rm H} = 3.9715259$ and the metal weights are the CIAAW values converted from the unified mass unit by $1\,{\rm u} = 0.99223573\,m_{\rm H}$.

4.2 The cell system

For each cell the code solves a system of algebraic balance equations at fixed temperature, density and radiation field. The unknowns are stage *fractions*, and one stage of each element is closed as the remainder rather than being an unknown: $n({\rm H\,I}) = (1-x_{\rm H^+})n_{\rm H}$, $n({\rm He\,I}) = (1-x_{\rm He^+}-x_{\rm He^{2+}})n_{\rm He}$, and for a metal element $n(X^0) = (1-x_{X^+}-x_{X^{2+}})n_X$. The layout is

rows	contents
1–3	$x_{\rm H^+}$, $x_{\rm He^+}$, $x_{\rm He^{2+}}$
4	He 2^3S (triplet systems) *or* H_2 (molecular systems)
5–8	H_2^+, H_3^+, HeH^+, He 2^3S (molecular systems)
$m_{\rm base}\ldots$	two rows per metal element (`metal_row_base()`)
$i_{\rm ox}\ldots$	OH and H_2O (oxygen option)

with $m_{\rm base} = 4$, or 5 with the triplet. In the molecular layout $i_{\rm ox} = 8$, or 9 with the triplet, and $m_{\rm base} = i_{\rm ox}$, raised by two when the oxygen chemistry is on (`metal_row_base`, `oxygen_row_base`: one definition, read by the residual and by the driver alike). Seven residual routines cover the configurations: `System_H`, `System_HeH`, `System_HeH_metals`, `System_HeH_TR`, `System_HeH_TR_metals`, `System_HeH_mol` and `System_HeH_mol_metals`. The shared rows live once, in `ion_residual_core.f90`.

The H/He rows. With photoionization rates Γ, recombination coefficients α (case B plus the ground captures whose photons are not re-absorbed on the spot, (60)) and electron-impact ionization coefficients β, `heh_rows` writes

$$0 = n_{\rm HI}\Gamma_{\rm HI} + (n_{\rm HI}\beta_{\rm HI} - \alpha_{\rm HII}n_{\rm HII})\,n_e, \tag{46}$$

$$0 = n_{\rm HeI}\Gamma_{\rm HeI} + (n_{\rm HeI}\beta_{\rm HeI} - \alpha_{\rm HeII}n_{\rm HeII})\,n_e, \tag{47}$$

$$0 = n_{\rm HeII}\Gamma_{\rm HeII} + (n_{\rm HeII}\beta_{\rm HeII} - \alpha_{\rm HeIII}n_{\rm HeIII})\,n_e, \tag{48}$$

with n_e the charge sum of every tracked ion, $n_e = n_{\rm HII} + n_{\rm HeII} + 2n_{\rm HeIII} + \sum_i Z_i n_{{\rm m},i} + n_{H_2^+} + n_{H_3^+} + n_{HeH^+}$ (`calc_ne`), so charge neutrality is imposed identically rather than solved for. Each metal element contributes

$$0 = n_{X^0}\Gamma_{X^0} + (n_{X^0}\beta_{X^0} - \alpha_{X^+}n_{X^+})\,n_e, \qquad 0 = n_{X^+}\Gamma_{X^+} + (n_{X^+}\beta_{X^+} - \alpha_{X^{2+}}n_{X^{2+}})\,n_e, \tag{49}$$

the second row being replaced by $x_{X^{2+}} = 0$ for a two-stage element, and the whole pair by $x_{X^+} = x_{X^{2+}} = 0$ for an element with zero abundance.

The triplet row. With the metastable tracked, the He rows are rewritten in the form of Oklopčić & Hirata (2018) (`heh_tr_rows`): row 2 is the summed He I balance, the sum of their singlet and triplet equations, in which the singlet ground state and the triplet are distinguished, and the fourth row is the 2^3S balance (`tr_triplet_row`, the one

expression of it, read by the atomic and molecular systems and by the transport operator alike),

$$0 = -n_{2^3\mathrm{S}}\Gamma_{2^3\mathrm{S}} + n_e\left[n_{\mathrm{HeII}}\alpha_{2^3\mathrm{S}} + n_{1^1\mathrm{S}}q_{13} - n_{2^3\mathrm{S}}(q_{31g}+q_{31a}+q_{31b}) - n_{2^3\mathrm{S}}\beta_{2^3\mathrm{S}}\right] - n_{2^3\mathrm{S}}\left(A_{31}+n_{\mathrm{HI}}Q_{31}\right), \tag{50}$$

where q_{13} is the collisional feed of 2^3S out of the singlet ground state: the direct excitation $1^1\mathrm{S}\rightarrow 2^3\mathrm{S}$ plus the excitation of every higher triplet term $n \leq 5$ of Bray et al. (2000, Table 2), each of which cascades into 2^3S within the triplets (0.019 of the direct rate at 5000 K, 0.107 at 10^4 K, 0.331 at 2×10^4 K); $q_{31g} = q_{13}^{\mathrm{direct}}(g_1/g_3)e^{E_{13}/kT}$ is the electron-impact (superelastic) de-excitation $2^3\mathrm{S}\rightarrow 1^1\mathrm{S}$, the detailed-balance reverse of the direct excitation, so that the electron collisions between the two levels drive them toward their Boltzmann ratio $3e^{-19.82\,\mathrm{eV}/kT}$ (`deexcitation_rate_HeI_23S_11S`, written with the Boltzmann factor cancelled, so it stays finite where q_{13} underflows, and not negligible in cool gas: $q_{31g}/(q_{31a}+q_{31b}) = 0.054$ at 10^4 K, 0.15 at 5000 K, 0.56 at 3000 K and 2.7 at 2000 K, MEASURED 2026-10-05 from the coded rates); q_{31a} and q_{31b} are the collisional conversions to 2^1S and to 2^1P, the latter together with every singlet level above it (`excitation_rate_HeI_23S_singlets_n3`, 0.019 of $q_{31a}+q_{31b}$ at 10^4 K, MEASURED 2026-10-05), all of the form $C\,\Upsilon\,e^{-E/kT}/(g_i\sqrt{T})$ with the Maxwellian constant $C = h^2/[(2\pi m_e)^{3/2}k^{1/2}] = 8.629132\times10^{-6}$ cm^3 s^{-1} K$^{1/2}$, the effective collision strengths of Bray et al. (2000, Tables 2 and 3) for the network's own transitions and for the higher levels of the sums alike (interpolated inside the tabulated $10^{3.75}$–$10^{5.75}$ K and continued outside it with the temperature dependence of the CHIANTI v11.0.2 strength of the same transition, scaled to the table end) and the NIST level energies, $A_{31} = 1.272\times10^{-4}$ s^{-1} the forbidden radiative decay, $\beta_{2^3\mathrm{S}}$ the electron-impact ionization of the metastable at its own 4.8 eV threshold, and Q_{31} the *total* He(2^3S)+H ionization rate coefficient. Both channels of Q_{31} quench the metastable, so the sink takes the sum; only the Penning branch $f_\mathrm{P} = 0.9$ leaves a lasting proton, and it appears as the source $f_\mathrm{P}Q_{31}n_{2^3\mathrm{S}}n_{\mathrm{HI}}$ in the H^+ row. The remaining 10 percent is associative ionization to HeH^+, which the atomic systems drop (in an atomic gas HeH^+ dissociatively recombines straight back) and the molecular system carries explicitly. In the molecular system the row also loses $k_{\mathrm{H_2}}^{\mathrm{ion}}n_{\mathrm{H_2}}n_{2^3\mathrm{S}}$, the total He($2^3$S)+$\mathrm{H_2}$ ionization, whose two branches both quench the level and neither makes He^+.

The molecular rows. `mol_heh_rows` adds four species balances on the Koskinen et al. (2022) Table 1 network (Sec. 6.1). Schematically, writing g for photorates and k_n for the Table-1 coefficients,

$$\begin{aligned}
\mathrm{H_2}: \quad 0 &= k_6 n_e n_{\mathrm{H_3^+}} + k_9 n_{\mathrm{H_2^+}} n_{\mathrm{HI}} + k_{11} n_{\mathrm{H_3^+}} n_{\mathrm{HI}} + k_{15} n_M n_{\mathrm{HI}}^2 \\
&\quad - \left[g_{\mathrm{H_2}} + g_{\mathrm{LW}} + (k_{10}+k_{13})n_{\mathrm{HII}} + k_{12}n_M + k_{14}n_e + k_8 n_{\mathrm{H_2^+}} + (k_{17}+k_{23})n_{\mathrm{HeII}} + k_{18}n_{\mathrm{HeH^+}} + k_{\mathrm{H_2}}^{\mathrm{ion}}n_{2^3\mathrm{S}}\right] n_{\mathrm{H_2}}, \qquad (51) \\
\mathrm{H_2^+}: \quad 0 &= \left(g_{\mathrm{H_2}} - g^{\mathrm{di}} - g^{\mathrm{dd}} - g^{\mathrm{nd}}\right) n_{\mathrm{H_2}} + k_{10} n_{\mathrm{HII}} n_{\mathrm{H_2}} + k_{11} n_{\mathrm{H_3^+}} n_{\mathrm{HI}} + k_{19} n_{\mathrm{HeH^+}} n_{\mathrm{HI}} + k_{23} n_{\mathrm{HeII}} n_{\mathrm{H_2}} \\
&\quad + f_\mathrm{P} k_{\mathrm{H_2}}^{\mathrm{ion}} n_{2^3\mathrm{S}} n_{\mathrm{H_2}} - \left(k_5 n_e + k_8 n_{\mathrm{H_2}} + k_9 n_{\mathrm{HI}} + k_{\mathrm{He}} n_{1^1\mathrm{S}}\right) n_{\mathrm{H_2^+}}, \qquad (52) \\
\mathrm{H_3^+}: \quad 0 &= k_8 n_{\mathrm{H_2^+}} n_{\mathrm{H_2}} + k_{13} n_{\mathrm{HII}} n_{\mathrm{H_2}} + k_{18} n_{\mathrm{HeH^+}} n_{\mathrm{H_2}} - [(k_6+k_7)n_e + k_{11} n_{\mathrm{HI}}]\, n_{\mathrm{H_3^+}}, \qquad (53) \\
\mathrm{HeH^+}: \quad 0 &= k_{\mathrm{He}} n_{1^1\mathrm{S}} n_{\mathrm{H_2^+}} + (1-f_\mathrm{P}) n_{2^3\mathrm{S}} \left(Q_{31} n_{\mathrm{HI}} + k_{\mathrm{H_2}}^{\mathrm{ion}} n_{\mathrm{H_2}}\right) - \left(k_{16} n_e + k_{18} n_{\mathrm{H_2}} + k_{19} n_{\mathrm{HI}}\right) n_{\mathrm{HeH^+}}, \qquad (54)
\end{aligned}$$

and the H^+ row gains $(g^{\mathrm{di}}+2g^{\mathrm{dd}})n_{\mathrm{H_2}} + k_9 n_{\mathrm{H_2^+}} n_{\mathrm{HI}} + k_{17} n_{\mathrm{HeII}} n_{\mathrm{H_2}} - (k_{10}+k_{13}) n_{\mathrm{HII}} n_{\mathrm{H_2}}$. The factor 2 on the double photoionization branch is the stoichiometry, two protons out of one event, while the $\mathrm{H_2}$ destroyed is still one per event, which is why (51) is not doubled with it. n_M is the $\mathrm{H_2}$-equivalent third-body density of the association R15 and of its reverse R12, the same density in both directions (`h2_third_body_density`; the total particle density with `EXHALE_REACTION_HEAT_RECIPIENTS=0`, Sec. 6.1); k_{He} is $\mathrm{H_2^+ + He \rightarrow HeH^+ + H}$ (Black, 1978) on the ground singlet

n_{1^1S}, the HeH^+ source that replaces the retired Table 1 entry R20 (Sec. 6.1); and the He^+ row loses $(k_{17}+k_{23})n_{HeII}n_{H_2}$ (with the oxygen option also the He^++CO destruction below).

The oxygen rows. With the oxygen option (Sec. 6.6; `thereis_oxychem`, solved only by `System_HeH_mol_metals` since oxygen is a metal element) two more unknowns join the cell system, $x_{OH} = n_{OH}/n_O$ and $x_{H_2O} = n_{H_2O}/n_O$ with n_O the oxygen family of the cell, and `oxygen_carrier_rows` adds their balances on the reactions O1 ($OH+H_2 \rightarrow H_2O+H$), O2 ($O+H_2 \rightarrow OH+H$), their reverses O1r and O2r, the three H_2O photolysis channels O3 ($\rightarrow OH+H$), O4 ($\rightarrow H_2+O(^1D)$) and O5 ($\rightarrow O+H+H$), and the OH photolysis O7 ($\rightarrow O+H$):

$$\mathrm{OH}: \quad 0 = k_{1r}n_{H_2O}n_{HI} + k_2 n_{O^0} n_{H_2} + (J_3+J_4)n_{H_2O} - k_1 n_{OH} n_{H_2} - k_{2r} n_{OH} n_{HI} - J_7 n_{OH}, \tag{55}$$

$$\mathrm{H_2O}: \quad 0 = k_1 n_{OH} n_{H_2} - k_{1r} n_{H_2O} n_{HI} - (J_3+J_4+J_5) n_{H_2O}, \tag{56}$$

and the H_2 row (51) gains the exchange with the water cycle, $-(k_1 n_{OH} + k_2 n_{O^0})n_{H_2} + (k_{1r} n_{H_2O} + k_{2r} n_{OH}) n_{HI}$. Here n_{O^0} is the *free* atomic oxygen, the remainder of the family once the nuclei bound in OH, H_2O and CO are removed, $n_{O^0} = n_O - n_{OH} - n_{H_2O} - n_{CO}$, which the metal block then splits over O I/O II/O III from its own balance: the O I column of the output means free atomic oxygen when the option is on. The hydrogen closure gains the nuclei the carriers hold, $n_{OH} + 2n_{H_2O}$. The forward coefficients k_1 and k_2 are the Baulch et al. (2005) evaluations ($k_1 = 3.6\times10^{-16}\,T^{1.52}e^{-1740/T}$, $k_2 = 6.34\times10^{-12}e^{-4000/T} + 1.46\times10^{-9}e^{-9650/T}$ $\mathrm{cm^3\,s^{-1}}$); the reverses are not transcribed but generated by detailed balance, $k_r = k_f/K_c(T)$ with K_c from the NIST-JANAF Shomate Gibbs energies (Chase, 1998) of H, H_2, O, OH, H_2O, CO and C (`oxygen_rates.f90`), so that the hot limit of the network is the chemical equilibrium of the same thermochemistry. J_3, J_4, J_5 and J_7 are the band-resolved photolysis rates of Sec. 6.6 summed over the four far-ultraviolet bands with their quantum yields. The O4 branch appears in the OH row as if it made OH directly: its product $O(^1D)$ has exactly one sink in the network, O6 ($O(^1D)+H_2 \rightarrow OH+H$), with a lifetime of 1.6×10^{-3} s at the HD 189733 b base, so its local steady state is exact, the flux through O6 equals the $O(^1D)$ production $J_4 n_{H_2O}$, and the pair O4+O6 is one channel whose net effect on H_2 cancels (O4 makes one, O6 consumes one), which is why only O1 and O2 enter the H_2 row; the $O(^1D)$ density is evaluated for the output only. CO is *not* a row of this system: it is a transported carrier (Sec. 6.5) with no formation channel in the network, destroyed one-sidedly by $He^+ + CO \rightarrow C^+ + O + He$ at the Langevin value $1.6\times10^{-9}\,\mathrm{cm^3\,s^{-1}}$ (UMIST RATE22 entry 4068; Millar et al., 2024) and by photodissociation on the Lyman-Werner beam with the Visser et al. (2009) shielding of the star-ward CO and H_2 columns, both inside the carrier step; the cell system sees the CO of the cell as a frozen background and takes the He^+ it consumes, $-k_{D1}n_{CO}n_{HeII}$, in the He^+ row, so the one reaction is complete over the pair of operators and counted once in each.

Normalization of the rows. Before the Newton solve every row of the molecular system, the oxygen, metal and charge-exchange terms included, is divided by the chemical turnover of its element at a composition x_0,

$$r_i = \frac{P_i - L_i}{G_E(x_0)}, \qquad G_E = \sum_{j\in E,\ j\ \text{kinetic}} (P_j + L_j), \tag{57}$$

with $P_j + L_j$ the sum of the magnitudes of the terms of row j and E the element the row's species carries (hydrogen: H^+, H_2, H_2^+, H_3^+, HeH^+; helium: He^+, He^{++}, $He(2^3S)$; each metal its two stage rows, oxygen together with OH and H_2O: the acceptance groups of `chemical_row_layout.f90`; `turnover_or_bound_reciprocal`, `System_HeH_mol.f90`).

x_0 is the starting point of each solve and the scale is held through it (`set_element_turnover_scales`, `System_HeH_mol_metals.f90`): a scale that followed the iterate would bound every row by one and leave the solve no gradient back from an iterate far outside the simplex, which on the cold start of the `mol_metals` regression gate left 214 cells without a root. The groups and the kinetic sum are those of the acceptance measure below: a constraint row of the cell (a transported fraction or He(2^3S) level, the carried or reservoir H_2 partition, the carried oxygen carriers, an identity row) replaces a kinetic balance that is not an equation of the cell, so its gross rate enters no G_E, and the row itself is written $x - x_{\rm fix}$ after the scaling (`mark_constraint_rows_of_cell`, `chemical_row_layout.f90`; until 2026-10-07 the solve scale counted the replaced balances, update log stage 3 section 116). The judgment of a candidate uses the same turnover at the candidate itself. A row that carries no reaction at the evaluated state (an absent element's identity row) keeps the reciprocal of the upper bound with every species at the whole of its element, which is 1 there. The blocks of one residual vector differ by the element abundances, and the dogleg step of `hybrd1` weights rows by their size, so a row far below the others would not determine its unknown. The upper bound itself is not the scale: where a trace ion of one element reacts with another element it exceeds the turnover by the inverse of the ion's fraction (the H^+ row carries He^++H_2 and He(2^3S)+H with He^+ and the metastable at all of the helium, 3.9×10^{12} against a gross rate of 1.3×10^3 cm^{-3} s^{-1} at the root of the LHS 1140 b He/H = 2×10^4 base ghost, MEASURED), which left the H^+ row at 3×10^{-10} of its weight in the solve and in the acceptance test: ten percent of H^+ off the root moved it by 2.7×10^{-11}, against 5.4×10^{-2} in the normalization of Eq. (57). The row's own $P+L$ is not the scale either: it asks a species at 10^{-22} of its element to balance to one part in 10^6 of itself, which neither matters to the gas nor is resolved by a solve whose unknowns are fractions of order one. Against G_E the error of a species counts as the nucleus flux it misplaces, which is what the acceptance tolerance `ieq_res_tol` = 10^{-6} is stated in (`normalized_reaction_residual`, below). The zeros of the rows are unchanged. The constrained continuation below keeps the bound on its own path; the candidate it returns is judged by the same measure.

4.3 The rate coefficients

All of the following live in `Cool_coeff.f90` unless stated. The default set is selected by `Legacy_HHe_rates: False` (default) and `Atomic rate set: default`; the two alternatives are the Hui & Gnedin (1997) fits and the Koskinen et al. (2022) Table 1 fits, and the three meet at the accessor functions at the end of that module.

Recombination. Radiative recombination follows the Badnell (2006, 2023 update) total ground-level fit

$$\alpha_{\rm RR}(T) = \frac{A}{\sqrt{T/T_0}\left(1+\sqrt{T/T_0}\right)^{1-B'}\left(1+\sqrt{T/T_1}\right)^{1+B'}}, \qquad B' = B + C e^{-T_2/T}, \tag{58}$$

implemented as `rr_badnell` (its dummy `T0` keeps the published notation and is not the code's temperature scale). Dielectronic recombination of He II adds the adf09 three-term sum $T^{-3/2}\sum_i c_i e^{-E_i/T}$. Case B is formed by subtracting the capture into the ground level, α_1, from the Milne relation on the ground-state photoionization cross section the radiative transfer itself absorbs with (hydrogenic for H I and He II, the Verner et al. (1996) fit for He I),

$$\alpha_1(T) = \frac{g_n}{g_+}\sqrt{\frac{2}{\pi}}\frac{1}{c^2 (m_e k T)^{3/2}}\int_0^\infty (I+E)^2\,\sigma(I+E)\,e^{-E/kT}\,{\rm d}E, \tag{59}$$

($g_n/g_+ = 2, 1/2, 2$ for H I, He I, He II; `ground_capture_milne`, a 24-point Gauss–Legendre rule in $\ln(1+E/I)$ accurate to 10^{-10}), so that the ground captures and the absorption of the photons they emit are one process and $\alpha_B = \alpha_A - \alpha_1$ for H II, He II and He III. α_1 is 0.998 of Hummer (1994) for H I, 0.999 of his hydrogenic scaling for He II and 1.006–1.011 of Hummer & Storey (1998) for He I. With the metastable tracked, the case-B captures of He II are split between the triplets (all ending in 2^3S) and the excited singlets (ending in 1^1S) in the ratio of the two sums of Hummer & Storey (1998, Table 5), a triplet share of 0.755 at 10 K, 0.771 at 10^4 K and 0.781 at 2.5×10^4 K.

The coefficients the balance uses. A ground capture emits a photon above the edge of the recombined species; the part of those photons the same species re-absorbs in the cell is a capture and a re-ionization that cancel. The coefficients of the rows are therefore

$$\alpha_{\rm HII} = \alpha_B + y_{\rm HI}\,\alpha_1^{\rm H}, \qquad \alpha_{\rm HeII} = \alpha_B + y_{\rm gnd}\,\alpha_1^{\rm He}, \qquad \alpha_{\rm HeIII} = \alpha_B + y_{\rm HeII}\,\alpha_1^{\rm He^+}, \tag{60}$$

$y_r = 1 - n_r\sigma_r\,\Delta r(1-e^{-\tau})/\tau$ the fraction of the photons not re-absorbed by r in the cell, τ the cell optical depth at the edge of every absorber (`ground_capture_escape_weights`), with $(1-e^{-\tau})/\tau$ from the one evaluation of that quotient (Sec. 3.3, at (34)). $y\to0$ is case B, where the gas is thick at the edge; $y\to1$ case A, in the thin outer wind. `H_rec_escape` (default on) governs $y_{\rm HI}$ and `He_rec_coupling` (default on) the two helium weights; a switch that is off sets its weight to zero. The photons taken by other absorbers ionize them (Sec. 3.6, `recombination_radiation_absorbed`), and the photoelectrons of those ionizations go through the same partition as the stellar photoelectrons, at the representative energy of each channel and under the same switches (Sec. 3.5): their secondary ionizations of H I, He I and H_2 are added to the photoionization rates Γ of the rows. The metal stages take Badnell RR + adf09 DR through the same two forms, with the one gap (Ca II, K-like) handled separately and noted at the code site.

Collisional ionization. Voronov (1997) Table I,

$$\beta(T) = A\,\frac{\left(1+P\sqrt{U}\right)U^K e^{-U}}{X+U}, \qquad U = \frac{\Delta E}{k_B T}, \tag{61}$$

with $(\Delta E, P, A, X, K)$ equal to $(13.6, 0, 2.91\times10^{-8}, 0.232, 0.39)$ for H I, $(24.6, 0, 1.75\times10^{-8}, 0.180, 0.35)$ for He I and $(54.4, 1, 2.05\times10^{-9}, 0.265, 0.25)$ for He II, and the metal stages take their own rows of the same table. The ΔE here is Voronov's own fit coefficient, not the measured potential (Mg II 15.2 against 15.035 eV), for the metals as for H and He; the energy *removed* per ionization of H and He is the measured one, $E_{\rm th}$ in erg. Voronov's fits cover 1 eV–20 keV (3 eV for He II, 2 eV for N II and O II); below that, in the 2000–10^4 K of a photoionized wind, the same expression is evaluated, which keeps the exact threshold factor e^{-U} but continues the fitted prefactor where no data constrain it, and no accuracy is claimed there.

Charge exchange. `charge_exchange.f90` carries the reaction list of Huang et al. (2023) Table 4, 63 rows, plus two rows that are not in it, but not all of that table's rate coefficients. A rate that a source calculated or measured is carried as the source gives it, from the original where it could be read (among others Kingdon & Ferland 1996 with Butler & Dalgarno 1979 for N+H^+, Dutta et al. 2001 and Watanabe et al. 2002 for Na+H^+, Watanabe et al. for K+H^+, Rutherford & Vroom 1972 for Fe+X^+, Zhao et al. 2005 for S+H^+, Satta et al. 2013 for Si+He^+, Chenel et al. 2010 for S+C^+, Kwolek

et al. 2019 for Na+Ca$^+$); where the source gives one direction only, the other is computed from it by detailed balance, $k_r = k_f\,[Q(X^s)Q(Y^{t+1})]/[Q(X^{s+1})Q(Y^t)]\,e^{[I(X^s)-I(Y^t)]/kT}$ with NIST internal partition functions and ionization energies, in place of the fitted reverses Table 4 prints. Ten reverses of Table 4 are not carried (A2, A6, A10, A22, C2, C4, C6, D18, D28, D30), because their forward channel ends in an excited or radiative product and so has no collisional reverse from the populated states. One pair, D1/D2 (C+Si$^+$), stays as printed: the paper Table 4 cites for it contains no such rate. `src/tests/charge_exchange_detailed_balance` checks every carried pair against detailed balance and every carried rate against the threshold law. The default active set is Group A, the 23 metal+H/H$^+$ reactions; `cx_full` adds the metal+He (C) and metal+metal (D) groups. The He↔H pair (Group B), $\mathrm{He}^0 + \mathrm{H}^+ \rightleftharpoons \mathrm{He}^+ + \mathrm{H}^0$, is applied separately by `he_h_cx_fvec` so that it is available in *every* system containing helium and is never counted twice; it is gated by `He_H_charge_exchange` (default on). Group E is the two reactions outside Table 4, $\mathrm{O}^{2+} + \mathrm{H}^0 \rightarrow \mathrm{O}^+ + \mathrm{H}^+$ and $\mathrm{N}^{2+} + \mathrm{H}^0 \rightarrow \mathrm{N}^+ + \mathrm{H}^+$ (Barragán et al. 2006, Tables 1 and 2, held at the ends of 10^2–10^5 K), on by default at the published rates because the absence of the first leaves the O III profile wrong by decades in the region a transit probes; `cx_O2p_H 0` and `cx_N2p_H 0` in `metals.inp` restore the Table-4-only set exactly. Each reaction enters as $R = k\,n_{\mathrm{donor}} n_{\mathrm{acceptor}}$, so the zero-abundance guard is automatic. The energy defect of each applied reaction, less the excitation of a product state that leaves as radiation, is heating channel 20 (Sec. 5.1).

The helium-hydrogen pair. $\mathrm{He}^+ + \mathrm{H} \rightarrow \mathrm{He} + \mathrm{H}^+$ has a radiative and a non-radiative channel, and both are carried (`charge_exchange.f90`). The radiative one is row (19) of Stancil et al. (1998) Table 1, $k = 1.20 \times 10^{-15}(T/300)^{0.25}$ cm^3 s^{-1}, the direct radiative charge transfer of Zygelman et al. (1989) weighted by the 1/4 of the singlet approach. The non-radiative one is the Maxwellian average of the $\mathrm{H}^+ + \mathrm{He}(1s^2)$ cross section of Loreau et al. (2014) turned into the exothermic direction by microscopic reversibility, $E_r = E_f + 10.989$ eV in the center-of-mass frame with statistical weights 4 and 1, so that the reverse threshold lies at $x_{\mathrm{th}} = 13.655$ eV/u. Their calculation starts at 15 eV/u and their fit stays finite below the threshold, where the channel is closed, so since 2026-09-26 the reverse cross section between the threshold and 15 eV/u is

$$\sigma_r(x) = \sigma_r(15)\sqrt{\frac{x - x_{\mathrm{th}}}{15 - x_{\mathrm{th}}}}, \qquad x_{\mathrm{th}} \le x \le 15\ \mathrm{eV/u}, \tag{62}$$

and zero below x_{th}: by reciprocity $\sigma_f v$ is constant at low forward energy, capture on the ion-induced-dipole potential with a transfer probability independent of energy, an assumption over forward energies of 0–1.08 eV that no calculation tests. Before, σ_r was zero below 15 eV/u, which closed a channel that is open: the forward rate then fell as $e^{-E/kT}$ and was 1.5×10^{-21} cm^3 s^{-1} at 10^3 K, where it is now 9.56×10^{-17} (9.64×10^{-17} at 10^4 K, against 4.41×10^{-17}). The rate is tabulated from 300 K to 10^5 K (generator `cooling_data/heii_h_charge_exchange_rate.py`), interpolated linearly in $\ln k$ against $1/T$, the first chord continued below 300 K and the 10^5 K value held above. The non-radiative channel is 3.3 percent of the radiative one at 10^4 K and 5.9 percent at 10^3 K (MEASURED 2026-10-05 from `he_h_cx_rates`), so He^+ + H is removed almost entirely by radiative charge transfer; the reverse $\mathrm{He} + \mathrm{H}^+ \rightarrow \mathrm{He}^+ + \mathrm{H}$ is the detailed-balance reverse of the non-radiative channel alone, and its helium reactant is the ground singlet $n(1^1\mathrm{S})$, because the rate carries the barrier $e^{-10.989\,\mathrm{eV}/kT}$ of ground-state helium (the metastable's own charge exchange is a different reaction, not carried).

The third reaction of the pair is the radiative capture $\mathrm{He}^{2+} + \mathrm{H} \rightarrow \mathrm{He}^+(1s) + \mathrm{H}^+ + h\nu$ (`radiative_electron_capture_Hepp_from_H`), the Maxwellian average of the radiative charge-transfer cross

section to $He^+(1s)$ of West et al. (1982, Figs. 4–6, digitized, resolved resonances included; the paper gives no rate coefficient), tabulated over 10^2–10^5 K and held at the ends: 2.0×10^{-13} cm^3 s^{-1} at 10^3 K and 1.65×10^{-13} at 10^4 K (MEASURED 2026-10-05); over 10^3–3×10^4 K it is 16–20 times the constant 1.0×10^{-14} of Kingdon & Ferland (1996) carried before 2026-09-26, which is the default value their section 2 assigns to every reaction without a significant non-radiative channel and not an average of these cross sections. The table is a calculation above 10^3 K and an extrapolation below. The photon carries the whole 40.81 eV defect, so the reaction has no collisional reverse and deposits no heat. It removes He^{2+} from the He^{2+} row, adds a proton to the H^+ row and, where row 2 is the He^+ stage source (the molecular systems), adds He^+ there. All three are applied by `he_h_cx_fvec` under `He_H_charge_exchange`, and by `he_h_cx_fvec_adv` in the advection systems.

Penning ionization of the metastable. The total He(2^3S)+H ionization rate coefficient (`ioniz_HeI23S_H`) is the Maxwell-Boltzmann average of the Movre & Meyer (1997, Fig. 10) total ionization cross sections in the closed form García Muñoz (2025, Fig. 5 and network rows 198–199) publishes,

$$Q_{31}(T) = 10^{-9} \exp\left[-\frac{86.4804}{T} - 0.286766 \ln T + 0.0868445 (\ln T)^2 - 5.73001 \times 10^{-3} (\ln T)^3\right] \text{ cm}^3\,\text{s}^{-1}, \qquad (63)$$

smooth, positive and single-peaked at 3.8×10^3 K, within 2 percent of the cross-section average over 200–10^4 K by the paper's statement and evaluated unclamped outside that range, where no accuracy is claimed. The code notes at that site why the two-branch power law of Taylor et al. (2025) Table 2 is *not* used: it jumps by a factor 1.5724 at 4000 K, which a rate coefficient cannot do, and it disagrees with that paper's own Figure 19, the average of the Cohen & Lane (1971) and Morgner & Niehaus (1979) cross sections; the expression above agrees with Figure 19 to 10–20 percent, so the two independent calculations agree and the published fit is the outlier. The corresponding He(2^3S)+H_2 coefficient is $k = aT^b e^{c/T}$ with the Cohen & Lane (1977) amplitudes of the García Muñoz (2025) network file, reproducing its Table A.5 to 0.13 percent. Both split 0.9 : 0.1 between the Penning and the associative branch.

4.4 Solving one cell

The solver. `solve_ieq` (`newton_solver.f90`) tries a damped Newton iteration with an *analytic* Jacobian first (`newton_dense`, at most 60 iterations and 20 halvings of the step), and on failure restores the entry guess, calls the in-tree MINPACK `hybrd1` from the same guess and continues the Newton from `hybrd1`'s result. The systems without an analytic Jacobian (the molecular and the He 2^3S layouts) are solved by `hybrd1` followed by the same Newton with a forward-difference Jacobian (`hybrd1_resolving_every_unknown`). The solve has converged when the Newton correction $\delta x = -J^{-1} f$ at the returned state resolves *every* unknown relative to its own magnitude,

$$|\delta x_i| \le \tau\,|x_i| + \epsilon \qquad \text{for every } i, \qquad (64)$$

with $\tau = \sqrt{\epsilon_{\text{mach}}} = 1.5 \times 10^{-8}$ the tolerance of the sweep (`EXHALE_IEQ_TOL` names another) and $\epsilon = 2.2 \times 10^{-16}$ the resolution of a fraction at its element total, below which no change of a fraction moves a sum of its element (the resolution floor of the acceptance measure below). The correction is the Newton estimate of the remaining error of each unknown, so a state that meets Eq. (64) is resolved to about τ^2 of each unknown, far inside the acceptance of every kinetic row. A trial step $x + \lambda\,\delta x$ (λ halved from 1) is taken when it lowers $\|f\|_2$ (Armijo) or when the simplified

correction at the trial point, $J^{-1} f(x+\lambda\,\delta x)$ with the Jacobian of x and measured relative to each unknown, falls below $(1-\lambda/4)$ of that of the step; the second test is what lets a trace unknown be resolved once the large rows sit at their rounding and the residual norm, which they dominate, can no longer decrease. Neither stop that stood before was a statement about each unknown: an absolute step of $10^{-11}(1+\max_i |x_i|)$ left a stage whose root is of that order at a relative error of order one, and `hybrd1`'s relative step on the scaled norm of all unknowns, which the $O(1)$ fractions dominate, left H_3^+ in molecular gas out of balance by up to a few percent of its own gross rate (update log stage 3 sections 116 and 120: the H_3^+ non-roots of `hp_zero_seed`, 11992 over the run, are all gone). Each residual routine ships its own exact Jacobian; the photoionization rates are constants of the solve, so those Jacobians are exact for the system actually solved (Sec. 3.4). Run-wide counters record how many cells each route handled, and `EXHALE_FORCE_HYBRD1` forces the MINPACK path for an A/B comparison. `use_newton_ieq` is on by default.

The sweep. `ioniz_eq` (`ionization_equilibrium.f90`) takes the temperature and the total density as *inputs* and overwrites the species fractions in place; chemistry rearranges nucleons and electrons and creates no mass, so the mass density is never corrected by this step and the mass sum of the returned composition is a check against it (`rho_recon`), never a correction. A cell's photoionization rates depend on the columns (32) outside it and on its own depth. The sweep runs from the top down in field blocks of `xuv_field_block_cells` cells, a compile-time constant that is zero, one block of the whole grid: every cell is solved in the field of the composition the sweep was handed, and a positive width would carry the columns of the cells already solved from one block to the next. Inside a cell the pair (own attenuation, composition) is solved `xuv_self_field_passes` times, one by default, the loop stopping early when the accepted stage fractions move by less than `xuv_self_field_tol` = 10^{-9}. Neither setting changes a fixed point: at the fixed point of the coupled source step the composition a cell enters with and the one it returns are the same state, so every width and pass count builds the same field, and only the number of passes of the coupled loop differs (the measured costs that decided both defaults are at their declarations in `ionization_equilibrium.f90`). Cells within a block are independent and the sweep is OpenMP-parallel with the cell scratch declared `threadprivate`; the rate coefficients are evaluated in blocks of a fixed even length (`xuv_rate_block`, `util_ion_eq.f90`), so that the vectorized transcendental calls pair the same lanes at any thread count.

The acceptance measure. A candidate composition is a root when it lies inside the element simplex and the normalized reaction residual $\max_i m_i$ of the system the cell solve used, evaluated at the candidate, is at or below `ieq_res_tol` = 10^{-6} (`normalized_reaction_residual`, `ionization_equilibrium.f90`). Every row of every layout (H only, H/He, H/He with He(2^3S), each with or without metals, molecular, molecular with metals and the oxygen chemistry) has a declared kind and acceptance group (`chemical_row_layout.f90`, which also carries the nuclear stoichiometry of every species, kept apart from the grouping: HeH^+ carries H and He but is measured against hydrogen, OH and H_2O carry H and O and are measured against oxygen). A *kinetic* row (a species balance, a stage-boundary flow, the summed He I balance of the He(2^3S) systems) is measured against the chemical turnover of its group at the candidate,

$$m_i = \frac{|P_i - L_i|}{G_E}, \qquad G_E = \sum_{j\in E,\ j\ \text{kinetic}} (P_j + L_j), \tag{65}$$

with the gross rates $P_j + L_j$ written by the row routines themselves from the same factors as the rows, at the candidate's own composition and electron density, charge exchange included (`ionization_row_gross`). The groups are hydrogen,

helium and each metal element; in the atomic layouts H^+ is the only hydrogen row, so its measure is its imbalance over its own gross rate, and the helium group holds the He I↔He II (or summed He I), He II↔He III and He(2^3S) rows. A *constraint* row (an absent element's identity row, the unused X^{++} of a two-stage element, a transported fraction $x-x_{\rm fix}$, the imposed reservoir partition of a lower ghost) is dimensionless and has no chemical turnover: it is judged in its own units against `constraint_row_tol` $= 10^{-6}$ of the element's nuclei and never enters a G_E. A row or a turnover that is not a finite number makes the measure NaN, and a kinetic row of a group with no turnover and a nonzero imbalance makes it `huge()` (a finite non-root above `ieq_nonroot_res_cap`, so class 6); a group with no turnover and no imbalance is balanced.

Two further statements complete the measure (update log stage 3 section 116). The H_3^+ row of the molecular layouts is also judged against its own gross rate,

$$m_{H_3^+} = \max\left(\frac{|f|}{G_{\rm H}},\ \frac{|f|}{P+L}\,\frac{\texttt{ieq_res_tol}}{\texttt{h3p_row_balance_tol}}\right), \qquad \texttt{h3p_row_balance_tol} = 10^{-3}, \tag{66}$$

because the group norm admits a relative error of a species up to about `ieq_res_tol` over its share of the group turnover, and H_3^+ is the one species measured to carry a share small enough (down to 2×10^{-11}) for that to reach a tenth of its density where it carries 1 to 100 percent of the cooling (`oxygen_chemistry`, r = 1.017–1.040 R_p; `mol_lyman_werner`). And a kinetic row above the tolerance is accepted when its remaining imbalance is below what the composition resolves (the *resolution floor*): its Newton correction along its own unknown, $\delta x_i = |f_i|/|\partial f_i/\partial x_i|$ (a forward difference), is at most $\epsilon = 2.2\times10^{-16}$, the spacing of the floating-point numbers at the element total (every unknown is a fraction of the nuclei of an element, and a smaller change moves none of the element's sums; a correction below $\epsilon|x_i|$ is the special case of it), or the species and its root both stand below `absent_element_density` in nuclei, $(|x_i|+\delta x_i)\,n_E \le 10^{-30}\,\mathrm{cm}^{-3}$ (`fraction_resolution`, `chemical_row_layout.f90`). On the column without ionizing radiation of the regression case `hydrostatic_column` the H^+ row above 1.0097 R_p has $x(H^+) \le 4\times10^{-17}$ and an imbalance of 10^{-6} to 10^{-3} of its gross rate whose correction is of the order of $x(H^+)$ itself; the floor accepts it and refuses nothing a representable composition could improve. Until 2026-10-07 the atomic layouts were measured against an upper bound of each row's rates with every reactant at the whole of its element; on the LHS 1140 b atomic state `ew_fid_atomic_nocond` that bound exceeded the gross H^+ rate by 3.9×10^{12} at the base and 2.1×10^2 at 26 R_p, so a 1 per cent move of H^+ was admitted inside about 1.35 R_p (MEASURED, `src/tests/acceptance_scale`). A slow row inside a fast group (a small share $(P_i+L_i)/G_E$) is admitted with a relative error of its species up to about `ieq_res_tol`$/((P_i+L_i)/G_E)$; the measurement of those shares on the molecular states and the decision taken on it (the group norm for every species, with the own-row criterion of Eq. (66) for H_3^+) are in update log stage 3 sections 112 and 116.

The element simplex. Every unknown is the fraction of one element's nuclei in one species, so a physical state has every fraction $x_i \ge 0$ and, for each element, the tracked species summing to at most its nuclei, the neutral remainder $1-s$ being nonnegative (`ionization_fractions_physical`, `ionization_equilibrium.f90`). The rows are polynomials in the fractions and have roots outside this set. One is the mirror root of a stage whose electrons are its own ions: with $n_e = x\,n_{\rm H}$ the H^+ row of an H-only cell, $n_{\rm H}g(1-x)-n_{\rm H}^2\alpha_B x^2$, has beside x^* the root $-x^*$ to $O(x^{*2})$, with $n(H^+) = n_e < 0$; it solves the row and passes the acceptance measure (its imbalance is 1.7×10^{-16} of its gross rate in the fixture of

`src/tests/acceptance_scale`, section viii), so only the simplex can refuse it. The bound is the resolution of a fraction at its element total: the rounding of a number keeps its sign, so a fraction whose root is zero or positive comes out negative only through a sum or difference formed at the scale of the element total, whose spacing is $\epsilon = 2.2 \times 10^{-16}$ (`fraction_resolution`). A candidate is therefore outside the simplex when a fraction is below $-\epsilon$, or when the k fractions of one element sum above $1 + k\epsilon$ (the neutral remainder has the same resolution, and forming the sum of k terms each at most one adds at most $(k-1)\epsilon/2$; a fraction converted into another element's budget, HeH^+ into helium, OH and H_2O into hydrogen, counts as one term each). A state admitted inside that band is on the face of the simplex to the resolution of the composition, and its densities are written on the face: no negative density leaves the sweep. A candidate outside the simplex is not a root, whatever its residual. It is projected onto the simplex (negative fractions raised to zero, an element's species scaled to its nuclei) and the measure is taken again at the projected state: a root on a face, which the solve resolves to ϵ of its element and can return that far outside it, is a root once projected (class 3; with `hybrd1` alone, excursions of 4×10^{-14} to 2×10^{-13} of the element on `wasp_full_newton`). The mirror root projects to $x = 0$, where the imbalance of the row is its whole production, and is no root. A projected root is taken only when no starting point (the guess built from the previous state, the uncoupled ionization balance at the incoming n_e, all nuclei neutral; in the molecular layouts the dissociation equilibrium in place of the last) gives a root without projection. Failing that, the atomic layouts make a fourth solve, the first bounded non-root of the three re-solved by `hybrd1` and the Newton after it, and the molecular layouts take the constrained continuation; then the bounded non-root under the relaxation amnesty, or the projection of the closest candidate with its residual rechecked (molecular) or the uncoupled balance (atomic) when no attempt stayed inside the simplex (classes 3 to 6 below). On the regression case `hydrostatic_column` a band of 10^{-10} admitted the mirror root as a converged root in 70 cells ($x = -8.0 \times 10^{-12}$, $n(H^+) = -8.9 \times 10^{-2}\,\mathrm{cm}^{-3}$ at $1.31\,R_p$); with the bound above and the fourth attempt every cell ends on the positive root, the non-roots of the whole run fall from 2695 to none, and the H^+ profile agrees with the one the case had before the acceptance measure of update log stage 3 section 112 to 8.8×10^{-11} (update log stage 3 section 118).

Acceptance classes. Every cell's returned state carries a class, and the contract is that classes 1, 2, 3 and 5 are *roots* of the requested equations (each passed the same two tests, a state inside the element simplex and a normalized reaction residual at or below `ieq_res_tol`) and differ only in how the root was found:

class	meaning
1	root of the last attempt made
2	root of the best earlier attempt
3	root after projecting an attempt outside the element simplex onto it
4	not a root; accepted under the relaxation amnesty and reported
5	root of the constrained element-conserving continuation solve
6	not a root, residual above `ieq_nonroot_res_cap`: the cell *keeps the composition it entered the sweep with*

A residual that is not a finite number is a non-root by construction (the sweep tests it explicitly, since NaN fails every comparison), and it falls in class 6: the cell keeps its entry composition. A consumer counting states its chemistry does not describe counts classes 4 and 6 and nothing else.

The hydrogen-only layout (`System_H`, a mixture without helium) faces the same judge: its one row is measured against its own gross rate (the H^+ row is the whole hydrogen group), the candidate must lie in $[-\epsilon, 1+\epsilon]$, and a cell is solved from the guess built from the previous state and, failing a root, from the positive root of its quadratic balance in closed form (`hydrogen_ionization_balance_root`); the classes, the ledger and the non-root streak are those of the other layouts.

The constrained continuation. `constrained_chemical_equilibrium.f90` is the second formulation of the molecular network, and it removes the possibility of an unphysical iterate instead of repairing one. The unknowns are $u_k = \ln n_k$, so $n_k > 0$ for every iterate, converged or not (the logarithmic composition variables of the CEA method, Gordon & McBride 1994); element conservation is an explicit residual row per element rather than a closure, so a species carrying nuclei of two elements (HeH^+, OH, H_2O) satisfies both rows at the accepted root; and n_e is the charge-neutrality sum, never an unknown. Because the network is bistable in its starting point (a molecular basin in the shielded base, an atomic basin above the $H_2 \rightarrow$H front), the solve is a natural-parameter continuation (Allgower & Georg, 2003) in the radiation field: λ scales the field from six decades below the cell's own, where the composition is known in closed form from the H_2 dissociation equilibrium of the local p, T, up to $\lambda = 1$ on the rungs $\lambda = 10^{-6}$, 10^{-4}, 10^{-2}, 0.1, 0.3, 1, each rung starting from the previous rung's root and a failed rung being bisected geometrically toward the last root (down to a field step of 10^{-12}, at most 40 solves for one cell). $\lambda = 0$ is not a starting point: the trace ionization stages of a shielded base are linear in λ, their logarithms diverge there and the Jacobian columns they own vanish. The rungs use the general MINPACK `hybrd` and not `hybrd1`, whose fixed initial trust radius $100\|x\|$ would open a first step of e^{10^4} in every density when $\|u\| \sim 130$. A species whose fraction is imposed from outside the cell (H_2 by the lower-boundary reservoir, the oxygen carriers by carrier transport, H^+, He^+, He^{2+} and He 2^3S by the transported ionization state of Sec. 4.5) is marked fixed: it appears in every reaction and conservation row it belongs to, but it is not an unknown and its own balance row is removed, so the system stays square.

4.5 Ionization carried by the flow

Where the recombination time is not short compared with the flow time, the local balance of Sec. 4.2 is not the answer. Two places in the code carry the ionization state along the flow instead.

The implicit advection systems. `System_implicit_adv_H`, `_HeH` and `_HeH_TR` each take one step of $v_{\rm el}\,\mathrm{d}x/\mathrm{d}r = R(x)$ into cell j, with the variable-step second-order backward differentiation formula (BDF2): $x^{\rm up} \equiv a_1 x_{j-1} - a_2 x_{j-2}$ is the history of the step and $c_1 = g\,h_j/v_{\mathrm{H},j}$ (hydrogen row) and $c_1^{\rm He} = g\,h_j/v_{\mathrm{He},j}$ (helium rows) its rate weights, $h_j = r_j - r_{j-1}$, $w = h_j/h_{j-1}$, $a_1 = (1+w)^2/(1+2w)$, $a_2 = w^2/(1+2w)$, $g = (1+w)/(1+2w)$, so that the rates and the velocity are both those of cell j. Each fraction is a fraction of one element's nuclei, which move with that element's velocity: with element continuity $\nabla\cdot(n_{\rm el}v_{\rm el}) = 0$ the steady stage balance $\nabla\cdot(x\,n_{\rm el}v_{\rm el}) = S$ is $v_{\rm el}\,\mathrm{d}x/\mathrm{d}r = S/n_{\rm el}$, with $v_{\rm He} = v + J/(\rho Y)$ and $v_{\rm H} = v - J/[\rho(1-Y)]$, J the diffusive helium mass flux of the state and Y the helium mass fraction (both equal v without element diffusion). The stage's own eddy term of the transported stages is not in the step; its size against the advective stage flux, $K_{zz}|\mathrm{d}\ln x/\mathrm{d}r|/v_{\rm el}$, is 0.035 at $1.3\,R_p$ and below 10^{-3} beyond $1.75\,R_p$ on the LHS 1140 b reference (MEASURED, update log stage 3 section 82). The first step of the recursion, and every step after a cell it does not

integrate, is backward Euler ($a_1 = g = 1$, $a_2 = 0$; $x^{\rm up} = x_{j-1}$), as is a BDF2 step whose populations come out negative. BDF2 is L-stable, so a population that relaxes within the cell settles on its local equilibrium (the trapezoid rule, A- but not L-stable, would alternate about it); `post_process_adv` states the formula, its derivation and its zero-stability bound $w \le 1+\sqrt{2}$. The systems solve

$$0 = x_{\rm HI}^{\rm up} - x_{\rm HI} + c_1\left[-(\Gamma_{\rm HI} + \beta_{\rm HI} x_e n_{\rm H})\, x_{\rm HI} + \alpha_{\rm HII} x_{\rm HII} x_e n_{\rm H}\right], \tag{67}$$

$$0 = x_{\rm HeI}^{\rm up} - x_{\rm HeI} + c_1^{\rm He}\left[-(\Gamma_{\rm HeI} + \beta_{\rm HeI} x_e n_{\rm H})\, x_{\rm HeI} + \alpha_{\rm HeII} x_{\rm HeII} x_e n_{\rm H}\right], \tag{68}$$

$$0 = x_{\rm HeIII}^{\rm up} - x_{\rm HeIII} + c_1^{\rm He}\left[(\Gamma_{\rm HeII} + \beta_{\rm HeII} x_e n_{\rm H})\, x_{\rm HeII} - \alpha_{\rm HeIII} x_{\rm HeIII} x_e n_{\rm H}\right], \tag{69}$$

with $x_e = x_{\rm HII} + ({\rm He/H})_{\rm local}(x_{\rm HeII} + 2x_{\rm HeIII}) + x_e^{\rm metal}$, and the same He↔H charge-exchange terms on rows 1, 2 and 3. The helium ratio is the global one on the legacy path and the *local*, radius-dependent one when the element diffusion is on, since the global value would misstate n_e by the separation factor. In `System_implicit_adv_HeH_TR` the neutral helium unknown of row 2 is the ground singlet $x(1^1{\rm S})$, whose row is the summed He I balance less the metastable's, and the fourth row is the step of the $2^3{\rm S}$ fraction with the terms of (50) on $c_1^{\rm He}$; the singlet is an unknown of its own rather than a difference, which in a strongly ionized cell whose neutral helium sits mostly in the metastable loses every digit. The He↔H terms, the He^{2+}+H capture included, are `he_h_cx_fvec_adv`. These are the systems of the `_adv` post-process (Sec. 11.1).

Transported ionization stages. With `Ionization transport: True` (default off) the ionization state of hydrogen and helium is carried by the flow instead of being re-solved in every cell as a local equilibrium. The carried fractions are fractions of the element's *nuclei*, $x_k = n_k/n_{\rm el}$: x(H II) of hydrogen and x(He II), x(He III) of helium, carriers `H+`, `He+` and `He++` of the transport-chemistry operator (Sec. 6.5, `diffusive_photochemistry.f90`); the neutral stage of each element is the remainder and is not a row. The two elements are carried together, because the electrons of a transported hydrogen partition would otherwise meet a helium partition that never saw the flow. The key needs helium in the mixture, and in a molecular gas it needs `Molecular carrier transport: True` (the stage sources are then rows (1)–(3) of the molecular network, whose molecular sinks are written at the transported molecular densities); it is refused with `Coupled carrier solve` (and its `On stall` form), whose coupled unknowns are mass fractions and not fractions of the element's nuclei. It applies to an atomic gas and to a gas with metals (`input_read.f90`).

Stage k of an element crosses the face f between cells j and $j+1$ with (`ionization_stage_transport.f90`, equation 1)

$$F_k(f) = x_k(f)\, N_{\rm el}(f) - n_{\rm el}(f) K(f)\, \frac{x_k(j+1) - x_k(j)}{\Delta r(f)}, \tag{70}$$

with $N_{\rm el}$ the nucleus face flux of the element operator (`element_nucleus_face_flux`: advection, gradient, eddy and settling drift in one object, Sec. 6.9), $x_k(f)$ the face fraction of the reconstruction, limiter and donor rule the element mass fractions take, with the donor side chosen by the sign of $N_{\rm el}$ for every stage alike, and the second term the stage's own eddy term relative to its element (the element's eddy flux is already in $N_{\rm el}$). The closing stage takes one minus the carried face fractions and minus the sum of their eddy terms, so that $\sum_k F_k = N_{\rm el}$ at every face to rounding: moving charge between stages moves no nucleus. The eddy term is zero at faces 0 and N, where the element crosses only with the gas. Every stage of an element moves with the element's velocity; the drift of an ion against its neutral in the ambipolar field is not carried, which the code bounds at the routine: exact inside about $1.2\,R_p$ on the certified

LHS 1140 b atomic state and of order 0.02–0.6 of the advective stage flux outward of it (MEASURED, READ from the module header). The source of each row is the balance the local sweep solves for the same gas, evaluated at the trial stage densities on the frozen background of the last sweep: the rows of (46)–(48) in their triplet form or the molecular rows (1)–(3), the He↔H pair and the metal charge exchange of the same reaction set and coefficients (`charge_exchange_stage_sources`).

The admissible set of the carried fractions is the simplex and not a box (`stage_simplex_projection`): each $x_k \geq 0$, and their sum is at most the share of the element the stages may hold, one for hydrogen (whose molecular nuclei the molecular carriers hold separately) and $1-(n_{\mathrm{HeH^+}}+n_{2^3\mathrm{S}})/n_{\mathrm{He}}$ for helium, the $\mathrm{HeH^+}$ bond and a metastable the step holds frozen being no nuclei of the stage rows (`carrier_helium_available_to_stages`). A carrier step that leaves the set is returned to it by raising negatives to zero and then scaling the carried fractions by the bound over their sum, which keeps their ratios and so the charge sum they set; the amounts moved are measured and reported (`stage_simplex_sum_over_limit`, `stage_fraction_under_zero`), never hidden.

In the sweep, every physical cell $j \geq 1$ of a run with the key, once a transport background exists or on a restart, has its x(H II), x(He II) and x(He III) rows replaced by the constraints $x - x^{\mathrm{carried}} = 0$, and the solve starts at the imposed values (`impose_transported_ionization_fractions`, `impose_transported_fractions`); the two lower ghosts are the inflow reservoir, for which no key states an ionized fraction, and keep the sweep's local root. In the marching loop the advection of the carried stages is the divergence of the hydrodynamic face mass flux taken with the mass row inside the Runge-Kutta stages (123), and the operator-split carrier step then carries the element's diffusive part, the stage eddy term and the chemistry; in the fixed-wind relaxation of the stationary partitioned route (Sec. 9.6) the rows carry the whole nucleus flux (70), and every sweep of the hydrodynamic solve is handed the transported fractions. The certification (Sec. 9.3) carries the entries `carrier balance H+, He+, He++`, gated at 10^{-5} in the wind cells, each row's residual against the sum of its own terms (the face-flux divergence and the gross production and loss; a stage row that hands out only its net reads $|S|$ there), and `ionization stage nucleus sum H` and `He`, the identity $\sum_k F_k = N_{\mathrm{el}}$ gated at its floating-point bound. The enthalpy the stage eddy terms carry is (105) of Sec. 6.10.

The transported metastable. With `He 2^3S transport: True` (default off; it requires `Include He23S` and `Ionization transport`, and is refused otherwise) the population of 2^3S is carried as a fourth helium state of the same operator, carrier `He2^3S`, $x_3 = n_{2^3\mathrm{S}}/n_{\mathrm{He}}$ on the helium nucleus flux with the face flux (70),

$$\frac{\partial(n_{\mathrm{He}}x_3)}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\left(x_3 N_{\mathrm{He}} - n_{\mathrm{He}}K_{zz}\frac{\partial x_3}{\partial r}\right)\right] = S_3, \tag{71}$$

with S_3 the right-hand side of (50) (and the He(2^3S)+H_2 quench in a molecular gas), evaluated at the trial population with the singlet $n_{1^1\mathrm{S}} = n_{\mathrm{He}} - n_{\mathrm{HeII}} - n_{\mathrm{HeIII}} - n_{2^3\mathrm{S}} - n_{\mathrm{HeH^+}}$ closing the three carried helium states; the helium simplex is then three-dimensional and reserves the $\mathrm{HeH^+}$ bond alone. In a steady state with $N_{\mathrm{He}} = n_{\mathrm{He}}v$ (no element diffusion) and $K_{zz} = 0$ this is the advection balance of Oklopčić & Hirata (2018, eq. 15); the sweep's row (50) becomes the constraint $x_3 = x_3^{\mathrm{carried}}$, in row 4 of the atomic systems and row 8 of the molecular one (`ieq_triplet_row`). A drift of the metastable against ground-state He I is not carried: the level moves at the velocity of the helium nuclei. Its 19.82 eV are chemical energy, booked by the reactions of the level where they run (the q_{13} cooling, the q_{31g} heating, the Penning heat), so no excitation-energy flux enters the energy equation; the certification reports that flux relative to He I without

gating on it (3×10^{-7} of the energy row's largest term at most on the LHS 1140 b validation state, MEASURED, update log stage 3 section 81). The level's certification entry is `carrier balance He2^3S` at the carrier tolerance, and `level balance He 2^3S` is not applicable under the key. Its row is one closed expression of sign-definite channels, so the scale of its measure is the gross production $n_e(n_{\mathrm{HeII}}\alpha_{2^3\mathrm{S}}+n_{1^1\mathrm{S}}q_{13})$ plus the gross loss (`tr_triplet_row_channels`) and the transport divergence, not the net source: in the wind the two halves cancel to 10^{-8} at 1.2–1.4 R_p, and a measure on the net asked the coupled state for 4×10^{-13} of the channel rates (MEASURED, update log stage 3 section 94). For the same reason the carrier relaxation judges the fixed point of each carrier against that carrier's own largest fraction (`carrier_displacement_on_own_scale`) and asks in addition that the steady carrier residual stand at or below `carrier_accept_tol`, since an implicit step's displacement bounds its residual only by $n/\Delta t$. The option is verified on the stationary route in a molecular gas; the marching route with it and an atomic gas end to end were not exercised (update log stage 3 section 81).

4.6 Excited hydrogen and the Lyα field

`excited_hydrogen.f90` solves the $2s/2p$ statistical equilibrium of Christie et al. (2013, their eqs. (12)–(13)) as a 2×2 linear system,

$$\begin{pmatrix} L_{2p} & -M_{12} \\ -M_{21} & L_{2s} \end{pmatrix}\begin{pmatrix} n_{2p} \\ n_{2s} \end{pmatrix}=\begin{pmatrix} S_{2p} \\ S_{2s} \end{pmatrix}, \qquad n_{2p}=\frac{S_{2p}L_{2s}+M_{12}S_{2s}}{L_{2p}L_{2s}-M_{12}M_{21}}, \tag{72}$$

whose sources are the recombination cascade, collisional excitation from $1s$, radiative pumping by the Lyα mean intensity and, in a molecular gas, the dissociative recombinations $H_2^+ + e$ and $HeH^+ + e$ that leave one hydrogen atom in $n=2$, $k_5 n_{H_2^+} n_e + k_{16} n_{HeH^+} n_e$ (`dissociative_recombination_n2_source`, the rate coefficients of the chemistry; zero when `EXHALE_REACTION_HEAT_RECIPIENTS=0`, and with the level off the 10.199 eV it carries leave as radiation, Sec. 5.1), and whose sinks are radiative decay, the two-photon continuum, collisional de-excitation, ℓ-mixing and photoionization by the stellar Balmer continuum. The atomic data are in one place (`hydrogen_n2_rates.f90`): $A_{2p1s}=6.2649\times10^8\ \mathrm{s^{-1}}$ (NIST ASD, the nonrelativistic dipole value carrying the reduced-mass factor), $A_{2s1s}=8.22461\ \mathrm{s^{-1}}$ (Drake 1986 eq. 27, with the finite nuclear mass and his relativistic factor), $g_{1s}=g_{2s}=2$, $g_{2p}=6$, $f_{\mathrm{Ly}\alpha}=0.4162$, the $1s\to2s$ and $1s\to2p$ collisional rates $C\,\Upsilon\,e^{-E/kT}/(2\sqrt{T})$ with the CHIANTI v11 effective collision strengths (the rates the H I cooling uses too), their detailed-balance inverses, and the case-B coefficient of the ionization balance (the captures into $n\geq2$) split into $2s$ and $2p$ with the case-B fraction of Pengelly (1964, Table I), $\alpha_{2s}=f_{2s}(T)\,\alpha_{\mathrm{B}}$, $\alpha_{2p}=\alpha_{\mathrm{B}}-\alpha_{2s}$ ($f_{2s}=0.322$ at 10^4 K). The $2s\to2p$ ℓ-mixing rate is

$$M_{2s2p}=n_e C^{e}_{2s2p}+n_{\mathrm{HII}}q^{\mathrm{p}}_{2s2p}+n_{\mathrm{HeII}}q^{\mathrm{He^+}}_{2s2p}+n_{\mathrm{HeIII}}q^{\mathrm{He^{2+}}}_{2s2p}, \tag{73}$$

C^e the electron rate of Seaton (1955) eq. (55), his approximation V, Maxwell-averaged in closed form with the same two final states and separations ($5.78\times10^{-5}\ \mathrm{cm^3\,s^{-1}}$ at 10^4 K; the Christie et al. Table 2 rate, 5.35×10^{-6}, averages a cross section whose constant is one tenth of Seaton's first-order dipole constant, and is not used), and the ion rates the impact-parameter cross section of Pengelly & Seaton (1964), eqs. (34)–(41), Maxwell-averaged in closed form with their cut-off $R_c=\min(1.12\hbar v/\Delta E, 0.72v\tau, R_D)$; the Lamb shift and the fine structure set ΔE ($2s_{1/2}$–$2p_{1/2}$ $0.035\ \mathrm{cm^{-1}}$, –$2p_{3/2}$ $0.331\ \mathrm{cm^{-1}}$), which is the smallest cut-off below $n_e\sim4\times10^{12}\ \mathrm{cm^{-3}}$. The closed form holds the lower incomplete gamma functions $\gamma(2,x)=1-e^{-x}(1+x)$ and $\gamma(3,x)=2-e^{-x}(x^2+2x+2)$, which cancel where they are small ($\gamma(n,x)\sim$

x^n/n), so below $x = 0.25$ and 0.5 respectively they are evaluated by their power series in Horner form (`lower_gamma_2`, `lower_gamma_3` in `Cool_coeff.f90`; error at most 1.1×10^{-14} on either side of the switch). The proton coefficient is 4.8×10^{-4} cm^3 s^{-1} at 10^4 K, 8.3 times the electron one, so $2s$ is mixed into $2p$ faster than it decays by two photons wherever $n_e = n_\mathrm{p} \gtrsim 1.5 \times 10^4$ cm^{-3}. The removal of a $2p$ atom without emission of a line photon is

$$D_{2p} = n_e q_{2p1s} + \Gamma_{2p} + M_{2p2s} \frac{A_{2s1s}}{A_{2s1s} + n_e q_{2s1s} + M_{2s2p} + \Gamma_{2s}}, \tag{74}$$

$M_{2p2s} = (g_{2s}/g_{2p}) M_{2s2p}$.

Balmer continuum. The $n = 2$ photoionization rate and the photoelectric heating it delivers are integrals over $[E_\mathrm{th}(\mathrm{H}, n{=}2), E_\mathrm{th}(\mathrm{H})]$ of the run's own spectrum with the hydrogenic cross section $\sigma_2(E) = \sigma_2^\mathrm{thr}(E_2/E)^3$, $\sigma_2^\mathrm{thr} = 1.4 \times 10^{-17}$ cm^2 for $2s$ and $2p$ alike. The code states no source for that value; it is, to its two figures, the $n = 2$ threshold value of the Kramers cross section with the bound-free Gaunt factor of Seaton (1959, eqs. 3 and 10), 1.378×10^{-17} cm^2, and the exact hydrogenic threshold values are 1.478×10^{-17} ($2s$) and 1.355×10^{-17} cm^2 ($2p$), statistical mean 1.386×10^{-17} cm^2 (derived for this document); the $(E_2/E)^3$ form with a constant threshold value leaves out the rise of the Gaunt factor above the edge:

$$\Gamma_2 = \frac{\sigma_2^\mathrm{thr} E_2^3}{h\nu_\mathrm{eV}} \int F_E E^{-4}\,\mathrm{d}E, \qquad h_2^{(1)} = \sigma_2^\mathrm{thr} E_2^3 \int F_E \left(E^{-3} - E_2 E^{-4}\right) \mathrm{d}E. \tag{75}$$

The quadrature resolves the field it integrates, which a fixed rule cannot: for a loaded table the nodes are the table's *own* rows, and each segment is integrated in closed form because the interpolation is a power law there; for the power law the whole band is closed form; only for the smooth Planck field is a composite five-point Gauss-Legendre rule on a geometric subdivision used. The feedback is lagged-explicit: it is computed once per timestep before the ionization and energy solve, and the frozen arrays are read inside `ioniz_eq` and never inside its Newton iteration. What that lag leaves is measured, not assumed: the certification entry `level balance H(n=2)` is the production less the destruction of each level for the populations held, at the composition of the certified state (`excited_hydrogen_level_residual`). The whole module is gated by `use_excited_H`, which is armed by supplying `Stellar Teff [K]:` and `Stellar radius [R_sun]:`; it is *off* by default.

The Lyα field. The mean intensity that pumps $n = 2$ has three forms, selected by `jlya_mode` (`excited_H_update`). The default, mode 0, takes the estimate $J_{\mathrm{Ly}\alpha} = 0.1\, F_\mathrm{LyC}/\Delta\nu_\mathrm{D}$ that Huang et al. (2017, section 2, in the text after their eq. 6) make near the peak of the H photoionization, where F_LyC is the deposited Lyman continuum flux and each ionization is assumed to be balanced by a recombination producing a Lyα photon; EXHALE evaluates it at every cell with $F_\mathrm{LyC} = \xi_\mathrm{day} F_\mathrm{inc}(1 - e^{-\sigma_\mathrm{LyC} N_\mathrm{HI}})$, $F_\mathrm{inc} = L_\mathrm{EUV}/(4\pi a_\mathrm{orb}^2)$, the ionizing flux the whole vertical H I column absorbs, times the cell mean of $1/(1+\tau)$ over the cell's own top-down line-center depth (`lya_photosphere_attenuation_cell_mean`), so the pumping vanishes below the Lyα photosphere. The column-integrated F_LyC and the factor $1/(1+\tau)$ are EXHALE's own choices, not taken from Huang et al. (2017), who use the estimate only for an order of magnitude ($J_{\mathrm{Ly}\alpha} \sim 10^{-8}$ erg cm^{-2} s^{-1} Hz^{-1}) and compute the field itself by Monte Carlo transfer; mode 0 is an approximate estimate. Mode 1, selected by `Jlya RT file:`, reads an externally computed $J_{\mathrm{Ly}\alpha}(r)$, for instance from the LaRT Monte Carlo code, and interpolates it onto the grid. Mode 2, selected by `Jlya escape-prob: True` and refused without `Stellar Lya flux [erg/cm2/s]:` > 0, is the in-line transfer of `lya_rt.f90`, which closes the line with an escape probability

rather than a Monte Carlo, because the columns involved reach $\tau_0 \sim 10^8$ at line center. With $\Delta\nu_\mathrm{D} = \nu_{\mathrm{Ly}\alpha}\sqrt{2kT/m_\mathrm{H}}/c$, the Voigt parameter $a = (A_{2p1s}/4\pi)/\Delta\nu_\mathrm{D}$ and the line-center depth accumulated cell by cell from $n_\mathrm{HI}C_{\mathrm{Ly}\alpha}/\Delta\nu_\mathrm{D}$, the escape probability of a photon emitted at the cell center is

$$\beta = \frac{1}{1+\langle N\rangle}, \qquad \langle N\rangle = \frac{4\sqrt{6}}{\pi^2}\,\frac{\tau_\mathrm{up}+\tau_\mathrm{dn}}{2}\,\Psi(\xi), \qquad \xi = \frac{\tau_\mathrm{dn}}{\tau_\mathrm{up}+\tau_\mathrm{dn}}, \qquad \Psi(\xi) = \sum_{n\ \mathrm{odd}} \frac{\sin(n\pi\xi)}{n^2}, \tag{76}$$

which is Neufeld (1990) eq. (3.27) at zero continuum destruction, reducing for a mid-plane source to Harrington (1973) eq. (40), $\langle N\rangle = 0.909316\,B$. τ_up is the depth of the cell center from the top. By default the planet-ward face is a mirror, $\tau_\mathrm{dn} = \tau_\mathrm{up}$ (Harrington's $B = \tau_\mathrm{up}$); with `Lya absorbing bottom: True` (default off, mode 2 only) it is the rest of the column down to the inner boundary, the molecular layer below acting as a sink rather than a mirror. The form $1/(1+\langle N\rangle)$ and not $1/\langle N\rangle$ is used because $\langle N\rangle$ counts absorptions per photon generated, so a photon is emitted $1+\langle N\rangle$ times before it leaves; this is exact at both ends and needs no join between the thin and thick limits. Under R_II-A redistribution $\langle N\rangle$ is independent of a, so the Voigt parameter survives only in the one-flight penetration of the stellar beam. A second channel runs beside the static one: the photon Doppler-shifts out of resonance over the Sobolev length of the radial velocity gradient, with the escape probability $\beta_\mathrm{S} = (1-e^{-\tau_\mathrm{S}})/\tau_\mathrm{S}$, $\tau_\mathrm{S} = (\pi e^2/m_e c) f_{\mathrm{Ly}\alpha}\lambda_{\mathrm{Ly}\alpha} n_\mathrm{HI}/|\mathrm{d}v/\mathrm{d}r|$, on the radial ray (the angle average over the direction dependence of the Sobolev depth is not taken). Its quotient comes from the one evaluation of $(1-e^{-x})/x$ at (34), so $\beta_\mathrm{S} \to 1$ as $\tau_\mathrm{S} \to 0$ (the closed form used until 2026-10-01 returned 0 below $\tau_\mathrm{S} \approx 10^{-16}$). The two combine as $\beta_\mathrm{tot} = \beta + \beta_\mathrm{S}(1-\beta)$, and β_tot is the β of (77). The trapped internal field and the stellar term are

$$J_\mathrm{int} = \frac{2h\nu^3}{c^2}\,\frac{g_{1s}}{g_{2p}}\,\frac{P(1-\beta)}{\left(A_{2p1s}\beta + D_{2p}\right)n_\mathrm{HI}}, \qquad J_\star = \frac{\xi_\mathrm{day} F_{\mathrm{Ly}\alpha}}{4\pi\Delta\nu_\star}\,\langle T_s E\rangle\,, \qquad J_{\mathrm{Ly}\alpha} = J_\mathrm{int} + J_\star, \tag{77}$$

with $P = \alpha_\mathrm{B} n_e n_\mathrm{HII} + C_{1s2p} n_e n_\mathrm{HI}$ the internal production, $T_s = \mathrm{erfc}\left(x_1/\sqrt{2}X_s\right)$ the fraction of the broad stellar profile (Gaussian of Doppler width X_s set by `Lya stellar halfwidth [km/s]:`, default 70) whose wings reach depth τ, $E = 1+(b-1)(1-\beta_\mathrm{tot})T_s$ the bounded trapping buildup of the penetrating stellar photons, with $b =$ `Lya stellar boost [-]:` (default 5, an empirical factor the code states as tuned to Huang et al. 2023 Fig. 11, not a computed quantity), and $\langle\cdot\rangle$ the mean over the cell's own optical depth. The published solution requires $(a\tau)^{1/3} > 10$ (Neufeld section Va); cells outside that are counted by `lya_wing_domain_record` and are carried there by the thin limit $\beta \to 1$, which (76) reaches smoothly.

5. The thermal source: heating and cooling

5.1 The heating channels

The volumetric heating rate of a cell is a sum of physically distinct deposits, and there is exactly one list of them (`util_ion_eq.f90`): `heating_of_composition` fills every column of `heat_chan`, the total it returns is the sum of those columns in that order, and the header of `output/Heating_breakdown.txt` is built from the same names, so a deposit cannot exist in the energy equation and be missing from the breakdown.

#	name	what it is
1–6	`heat_HI ... heat_metals`	photoionization of each absorber, $n_a h_a^{(1)}$ of (37)–(38)
7	`heat_Hpe[excitedH]`	photoelectric heating of H(n=2) by the Balmer continuum, (75)
8	`heat_Hdx[Lya-deexc]`	collisional de-excitation of the Lyα-pumped $n = 2$ atoms (`incl_deexc_heat`, default on)
9	`heat_He_recomb`	the heat share of the photoelectrons of the H I, He I, H_2 and metal ionizations, and the fragment kinetic energy of the H_2 final states, driven by the recombination radiation of H II, He II and He III absorbed on the spot (`recombination_radiation_absorbed`)
10–12	`heat_He23S_*`	He(2^3S)+H Penning and associative branches, and He(2^3S)+H_2 Penning
13–14	`heat_H2_LW_*`	kinetic energy of the Lyman-Werner dissociation fragments, and collisional de-excitation of the H_2 fluorescence
15	`heat_mol_chem`	net chemical heat of the collisional reactions of the H_2/He network
16–19	`heat_FUV_photolysis`, `heat_oxygen_collisional`, `heat_CO_Hep_transfer`, `heat_CO_photodissoc`	the oxygen option's deposits
20	`heat_charge_exchange`	energy defects of the applied charge-exchange reactions (metal+H/He/metal and He↔H), less the excitation of a product state that leaves as radiation; negative for an endothermic reaction (`charge_exchange_heating`)

Channels 1–6 are the photoheating integrals of Sec. 3.4, already carrying the heating efficiency $f_{\rm heat}$ of (43). Channel 9 carries the same partition: each photoelectron of the recombination radiation is split at the energy of its channel by the partition of Sec. 3.5, under the switches of the stellar path (the secondary-ionization coupling as applied, and below 30 eV the low-energy partition where `Low-energy electron partition` selects it), and only $f_{\rm heat}$ of it is heat; the fragment kinetic energy is heat at once. Neither partition adds a channel: a photoelectron's secondary ionizations go to the rates, and its heat to the channel of the photon that freed it. Channel 15 exists because the return path that makes ionization energy a non-channel in an atomic gas is different in a molecular one (`molecular_reaction_heat.f90`).

Follow one closed cycle:

$$H_2 + h\nu \rightarrow H_2^+ + e \quad \text{(the photon pays } I(H_2) = 15.43\,\text{eV}), \qquad H_2^+ + H_2 \rightarrow H_3^+ + H \quad (+1.70\,\text{eV}),$$

$$H_3^+ + e \rightarrow H_2 + H \quad (+9.25\,\text{eV}, \text{ dissociative: the energy goes into the fragments, not a photon}).$$

The cycle returns to H_2 having turned one H_2 into H+H and having left $I(H_2) - D_0(H_2) = 15.43 - 4.48 = 10.95$ eV in the gas, with no photon anywhere in it to carry that away. Measured on the converged He/H = 0.0793 hot Uranus, that is 81 percent of the total heating rate at $1.02\,r_{\rm base}$. `Molecular reaction heat` is therefore *on* by default; it is unreachable in an atomic run, so no non-molecular result moves.

Where each channel's energy goes (corrected 2026-09-17). That the energy stays in the gas does not make all of it translational, and until item L7g the ledger deposited each dissociative recombination's whole enthalpy as heat. Each channel now deposits what its measured product state leaves to translation:

- $H_2^+ + e$ and $HeH^+ + e$ leave one H atom in $n = 2$ at the collision energies of this layer (Takagi, 2002; Giusti-Suzor et al., 1983; Guberman, 1994), so 10.199 eV goes to the H($n = 2$) population (or radiates, when that population is not tracked) and the gas receives 0.749 and 1.554 eV instead of 10.948 and 11.753.
- $H_3^+ + e \rightarrow H_2 + H$ leaves the H_2 fragment vibrationally excited around $v = 5$–6 (Kokoouline et al., 2001; Strasser et al., 2001), so 2.4795 eV of the 9.250 is internal energy and branches through the vibrational quench fraction rather than being deposited promptly.
- The three-body association $H + H + M$ releases 4.478 eV into a nascent vibrational distribution, and the same quench fraction decides how much of it is collisionally thermalized before it radiates.

The quench fraction itself (`h2_vibrational_relaxation.f90`) is the collisional rate against the radiative one. The radiative side is the *all-level* maximum spontaneous rate over the code's own 302-level H_2 ladder, $5.5939 \times 10^{-6}\,\text{s}^{-1}$, which is 6.56 times the $v = 1$ total decay rate the earlier form used and agrees with the independent Wolniewicz et al. (1998) set to 1.1×10^{-4}; the collisional side is one published quantum calculation per collider, Lique (2015) for H, Jóźwiak et al. (2024) for He and Le Bourlot et al. (1999) for H_2, in place of two 1979 fits with helium absent. In the LHS 1140 b layer the fraction is one to a part in 10^6–10^7, so the layer is in the thermalized limit and these corrections act mainly through the recipients and the association rate; a shallower or cooler layer is not, and the form is stated with that validity where it is used. `EXHALE_REACTION_HEAT_RECIPIENTS=0` restores the prompt-deposit behavior for comparison.

What the list does not hold. The terms of the energy equation that move energy rather than deposit it are not heating channels and are not in `Heating_breakdown.txt`: the viscous dissipation and the conduction divergence of Sec. 2.8, and the divergence of the interdiffusion enthalpy flux q_d of Sec. 6.10. They enter the stationary energy row beside $\mathcal{H} - \Lambda$ (`steady_residual.f90`), and the advected-profile energy solve writes each of them as its own term (Sec. 11.1). The heat a conducting wind gives to the lower atmosphere through the base face, (22), is the one boundary term of that budget and is written to `EXHALE_resolved.out`.

5.2 The cooling channels

`eval_cool` (`util_ion_eq.f90`, through `eval_cool_cells`) assembles the cooling rate of every cell, the atomic, ionic and metal channels through the one routine `radiative_cooling_of_cell` (`Cool_coeff.f90`) that the post-process temperature root also calls, as

$$\Lambda = n_e\,(\Lambda_{\rm ff} + \Lambda_{\rm cx} + \Lambda_{\rm rec} + \Lambda_{\rm ci}) + \Lambda_{\rm metal} + \Lambda_{\rm H_3^+} + \Lambda_{\rm H_2} + \Lambda_{\rm H_2O} + \Lambda_{\rm CO}, \tag{78}$$

in erg cm^{-3} s^{-1}, with n_e the same charge sum the ionization balance uses (`calc_ne`), molecular donors included, since inside a deep molecular base H_3^+ is the dominant ion. The breakdown written to `output/Cooling_breakdown.txt` is read straight off the arrays (78) is built from, so the columns sum to Λ exactly: $10+n_{\rm ion}$ channels (`n_cool_chan`), recombination, collisional ionization, collisional excitation of H I, He I and He II, free-free, the four molecular channels H_3^+, H_2, H_2O and CO, and one line channel for each metal ion.

Recombination. $\Lambda_{\rm rec} = \lambda_{\rm HII} n_{\rm HII} + \lambda_{\rm HeII} n_{\rm HeII} + \lambda_{\rm HeIII} n_{\rm HeIII} + \sum_i \lambda_i n_{{\rm m},i}$, each the kinetic energy the captured electrons carry off, the last sum over the recombining metal ions. For captures of rate coefficient $\alpha(T) = \langle \sigma v \rangle$ in a Maxwellian gas it is exactly

$$\lambda(T) = kT\,\alpha(T)\left(\frac{3}{2} + \frac{{\rm d}\ln\alpha}{{\rm d}\ln T}\right), \tag{79}$$

which Table 1 of Hummer (1994) obeys to 0.1 percent, and it is applied to the *same* coefficient the ionization balance removes the ions with: the net coefficients (60) of H II, He II and He III (case B plus the escaping ground captures, whichever `Atomic rate set` gives $\alpha_{\rm B}$), and for each metal ion the Badnell radiative plus dielectronic (or Huang et al. 2023 iron) coefficient of its balance; the relation holds for dielectronic capture as well, where it charges the resonance energy. The escape weights depend on the densities and the cell width only, so they are frozen while T varies. The logarithmic slope is a centered difference of half step 10^{-3} in $\ln T$ for H II, He II and He III, and is written analytically from the published forms for the metal ions (`metal_recombination_energy_rate`). The kinetic energy of a ground capture whose photon is absorbed by another species is returned with the photoelectron (`recombination_radiation_absorbed`).

Collisional ionization. $\Lambda_{\rm ci}$ charges each ionization its measured potential in erg,

$$\Lambda_{\rm ci} = E_{\rm th}^{\rm HI}\beta_{\rm HI} n_{\rm HI} + E_{\rm th}^{\rm HeI}\beta_{\rm HeI} n_{1^1{\rm S}} + E_{\rm th}^{\rm HeII}\beta_{\rm HeII} n_{\rm HeII} + E_{\rm th}^{2^3{\rm S}}\beta_{2^3{\rm S}} n_{2^3{\rm S}} + \sum_i E_{\rm th}^{i}\beta_i n_{{\rm m},i}, \tag{80}$$

the He I term acting on the ground *singlet* $n(1^1{\rm S}) = n({\rm He\,I}) - n(2^3{\rm S})$, because the metastable sits 19.8 eV up and carries its own 4.8 eV threshold ($\beta_{2^3{\rm S}} = 8.38\times10^{-10}\,T^{1/2}e^{-55338\,{\rm K}/T}$ cm^3 s^{-1}, Black 1981, eq. 12, stated for 10^3–10^5 K and evaluated unclamped outside; `ionization_rate_HeI_23S`, the energy charged being the potential of the level). The metal term charges each collisional ionization the measured potential of the ion; with it and the metal recombination term above, the electron gas, the ionization energy and the recombination radiation close for the metals as for H and He. The same subtraction is made in the photoheating and in the collisional excitation below, so the metastable is never charged the ground-state coefficients as well.

Collisional excitation. $\Lambda_{\rm cx}$ is the coronal (every excitation radiated) sum over the upper levels of the CHIANTI v11.0.2 models of H I and He II and over the 28 upper terms $n \leq 5$ of Bray et al. (2000, Table 2) for He I (the strengths the metastable network uses, Sec. 4.2), with observed level energies, contracted with $n_{\rm HI}$, $n(1^1{\rm S})$ and $n_{\rm HeII}$. The CHIANTI strengths are descaled with the natural cubic spline CHIANTI's own `descale_scups` prescribes. For H I the $n = 2$ part is $10.2\,{\rm eV}\,(q_{1s2s}+q_{1s2p})$ with the same rates the H(n=2) model uses (Sec. 4.6); the $n \geq 3$ part and the He sums are fits $C\,T^{-1/2}\left[a+\ln(1+T/T_0)\right]e^{-T_{\rm ex}/T}$ printed by `cooling_data/fit_hydrogen_helium_excitation.py`, accurate to 1–2 percent. The H I term is the Lyα-dominated hydrogen line cooling, and it is the optically thin coefficient as written: *no escape probability multiplies the Lyα cooling channel.* $\lambda_{\rm HI}$ is `lambda_coex_HI` of `Cool_coeff.f90` and `eval_cool` contracts it with $n_{\rm HI}$ and n_e with no such factor anywhere. The two escape probabilities the code does compute enter elsewhere: β of `lya_rt.f90` (76) enters the H(n=2) populations and the pumping field, and β_{ul} of `fine_structure_line_transfer` enters the metal ground-term statistical equilibrium (81); neither is read by any cooling coefficient. The asymmetry is deliberate: multiplying $\lambda_{\rm HI}$ by an escape probability would leave the trapped fraction with no owner while keeping the de-excitation heating channel 8 as an independent source, that is it would double count. A ledger carrying both is the stated target and is not built.

With the metastable tracked, the He I sum leaves out the $1^1{\rm S}\rightarrow 2^3{\rm S}$ transition, which is the network's own rate q_{13}, and the exchange between the two levels is charged with that rate and its detailed-balance reverse, the electron de-excitation $2^3{\rm S}\rightarrow 1^1{\rm S}$, $q_{31g} = q_{13}(g_1/g_3)e^{E(2^3{\rm S})/kT}$ formed with the Boltzmann factor cancelled from the same Bray et al. (2000) strength (`deexcitation_rate_HeI_23S_11S`, `Cool_coeff.f90`): $E(2^3{\rm S})\left[q_{13}n(1^1{\rm S})-q_{31g}n(2^3{\rm S})\right]$, negative (a heating, the superelastic collisions giving the level's energy to the electrons) wherever the metastable is overpopulated against its Boltzmann ratio to the ground. The same q_{31g} is a loss of the triplet row (`HeITR_coeffs`, Sec. 4.2) and one of the exits of the level in the on-the-spot recombination coupling. Oklopčić & Hirata (2018) leave it out as less probable than the excitations to the singlets; that holds at 10^4 K, where $q_{31g}/(q_{31a}+q_{31b}) = 0.055$, and not in cooler gas (0.56 at 3000 K, 2.7 at 2000 K; MEASURED with these strengths, stated at the rates in `Cool_coeff.f90`). The excitations $1^1{\rm S}\rightarrow n^3L$ into the higher triplets, which feed $2^3{\rm S}$ through their cascade (in q_{13} of the triplet row), are charged their full energy by the He I sum, of which $E(n^3L)-E(2^3{\rm S})$ is radiated in the cascade and $E(2^3{\rm S})$ is carried by the metastable they make. Further terms scale with $n(2^3{\rm S})$: the excitation of $2^3{\rm S}$ within the triplets, radiated back to $2^3{\rm S}$ (10830 Å and the higher triplet lines; a CHIANTI v11 coronal sum, of which the Black (1981) coefficient it replaces is 0.81 at 10^4 K and 0.79 at 5000 K), and the $2^3{\rm S}\rightarrow 2^1{\rm S}$ (0.796 eV), $2^3{\rm S}\rightarrow 2^1{\rm P}$ (1.398 eV) and $2^3{\rm S}\rightarrow n^1L$ ($n \geq 3$) collisional conversions, each removing its excitation energy from the electron gas.

Free-free. $\Lambda_{\rm ff} = 1.426\times10^{-27}\sqrt{T}\sum_{\rm ions}Z_{\rm ion}^2\,\bar{g}(T,Z_{\rm ion})\,n_{\rm ion}$, with $\bar{g}$ the thermally averaged Gaunt factor of van Hoof et al. (2014) as a function of $\gamma^2 = Z^2\,{\rm Ry}/kT$ (their eq. 21), interpolated linearly in $x = \log_{10}\gamma^2$ and clamped to the tabulated range. Ry is the infinite-mass Rydberg unit of energy, as their section 2 defines it, hcR_∞ with the CODATA 2018 R_∞ and the exact SI h, c, k the code carries, so ${\rm Ry}/k = 157887.5$ K (`Cool_coeff.f90`, `Ry_over_kB`). Until 2026-09-26 the code used 157807 K, the constant of the Hui & Gnedin (1997) recombination fits, which moved the regression outputs by at most 1.4×10^{-4} relative. Only $Z_{\rm ion} = 1$ and 2 occur. The molecular ions H_2^+, H_3^+ and HeH^+ do donate to n_e but are deliberately left out of this charge sum: they exist only in the $T\sim10^3$ K molecular base, where this hot-plasma expression is an extrapolation whose emission comes out at frequencies the atmosphere is not thin to, and where H_3^+

and metal lines dominate by far.

Metal lines. $\Lambda_{\rm metal} = n_e \sum_i n_{{\rm m},i} c_i(T, n_e)$, every c_i built from CHIANTI v11.0.2 (collision strengths descaled with the natural cubic spline) where the ion is in it. The C/N/O coefficients are the default (`cno_chianti = .true.`; `cno_cool 0` in `metals.inp` restores the legacy set ported from AIOLOS). Mg II is a closed-form fit to the coronal sum of every excitation out of $3s^2$S with the observed level energies. Ca II and Mg I are (T, n_e) statistical-equilibrium tables with every channel (the Ca II $3d^2$D metastables and the Mg I $3s3p^3$P term besides the resonance lines; at $n_e = 10^9$ cm^{-3} they are 1.7–4 and 5.8–480 times the resonance line alone over 10^4–3000 K). Two entries are not in CHIANTI v11 and are built otherwise: the Na I D doublet with a $\bar{g}$ factor 0.2 calibrated against the CHIANTI Mg II and Ca II values, and the Fe I permitted multiplets from NIST oscillator strengths with the lower levels in Boltzmann equilibrium over the even-parity metastable manifold, the high-density limit, which is an upper limit at most 2.0 times the low-density value over 2000–3×10^4 K (the data for a density-resolved Fe I solve are not available). Above the top density of a (T, n_e) table (10^{14} cm^{-3}) the coefficient continues with the slope of the last interval clipped to $[-1, 0]$, the bounds of a thermalized and a coronal channel.

Two further density-dependent replacements are made on top of coronal curves, because a coronal fit assumes $\Lambda \propto n_e$ and no level saturation while the base of an irradiated atmosphere runs at $n_e \sim 10^9$, $n_{\rm HI} \sim 10^{14}$ cm^{-3}, decades above the critical densities of the lines involved.

- *Fe* II takes a two-dimensional statistical-equilibrium table $\Lambda_{\rm eff}(T, n_e) = \left(\sum_u n_u A_{ul} \Delta E_{ul}\right) / n_e$, whose $1/n_e$ cancels the n_e of the assembly and so leaves the correct saturated emission; at low n_e it reduces to the coronal rate. Without it the forbidden a^6D lines overestimate the base cooling by about four decades.

- *Ground-term fine structure* of C I ([C I] 609.1, 370.4 μm), C II (157.7 μm), N II (205.3, 121.8 μm) and O I (63.2, 145.5, 44.1 μm) is replaced by the exact statistical equilibrium of the ground term,

$$\sum_j f_j R_{ji} = f_i \sum_j R_{ij}, \quad \sum_i f_i = 1, \quad R_{ul} = C_{ul} + \beta_{ul} A_{ul}, \quad R_{lu} = C_{ul} \frac{g_u}{g_l} e^{-E_{ul}/kT}, \tag{81}$$

  with $C_{ul} = n_e k_{e,ul}(T) + n_{\rm HI} k_{{\rm H},ul}(T)$ and $k_{e,ul} = C\,\Upsilon / (g_u \sqrt{T})$ ($C = 8.629132 \times 10^{-6}$ cm^3 s^{-1} K$^{1/2}$, the Maxwellian constant of the triplet row (50)), the effective collision strengths Υ being quartic fits in $x = \log_{10}(T/10^4)$ to the CHIANTI `.scups` values over 10^3–10^5 K (largest error 8.2 percent, at the 10^3 K end on C I) and the H-atom rates taken from Yan & Babb (2023) for C I and N II, Abrahamsson et al. (2007) for O I and Barinovs et al. (2005) for C II. The emitted power per ion is $W_{\rm FS} = \sum_{u>l} f_u \beta_{ul} A_{ul} k_{\rm B} E_{ul}$ and the coefficient returned is $\Lambda_{\rm eff} = W_{\rm FS}/n_e + \Lambda_{\rm rem}$, $\Lambda_{\rm rem}(T, n_e)$ being the statistical-equilibrium table of the channels that leave the ground term (`cool_logLrem_*`; the metastable upper levels of those channels have critical densities of 10^4–10^9 cm^{-3}).

The line trapping β_{ul} of these fine-structure lines is the line-center escape probability formed from the columns above and below the cell (`fine_structure_line_transfer`): the sum over the two faces of the one-face probability of a thermally broadened slab (`line_escape_probability_one_face`), whose thin branch $(1 - e^{-a\tau})/(2a\tau)$, $a = 2.34$, takes its quotient from the one evaluation of $(1 - e^{-x})/x$ at (34). The line-center opacity carries the stimulated-emission factor $1 - e^{-y}$, $y = E_{ul}/T$, formed as y times the same quotient (`line_center_opacity_lte`), so that it tends to y in hot gas for a far-infrared line instead of cancelling. β_{ul} enters *inside* the statistical equilibrium as $A_{ul} \to \beta_{ul} A_{ul}$ rather than as a

factor on the result: in the subcritical limit the cooling is set by the collisional excitation rate and is independent of β, and in the saturated limit the escaping flux is proportional to it. $\beta = 1$ reproduces the optically thin result exactly, and that is the value every other metal ion keeps. With `Base IR field: True` (default off) the same lines also *absorb* the thermal infrared of the lower atmosphere, a blackbody at T_0 covering the sky fraction the base subtends, through the photon occupation number $\bar{n}_{ul}$; this gives the layer a radiative equilibrium floor instead of letting it cool into vacuum.

Molecular infrared. $\Lambda_{H_3^+}$ is built on the emission $W(T)$ of one H_3^+ molecule in LTE into one steradian,

$$\Lambda_{H_3^+} = 4\pi\, n_{H_3^+}\, W(T)\, s(T, n_{H_2}) \quad [\mathrm{W\,molecule^{-1}\,sr^{-1}\ for}\ W], \tag{82}$$

where $W(T)$ is the cooling function of the MiZATeP line list (Mizus et al., 2017, section 5 and Table 7), tabulated at 1 K steps from 20 to 5000 K in the ExoMol file `1H3_p__MiZATeP.cf` and carried as `h3p_mizatep_cooling_table` (`h3p_lte_emission_mizatep`). Between the nodes $\ln W$ is linear in $1/T$, a local Boltzmann factor, so the interpolant is monotone; below 20 K the first interval's Boltzmann form is continued ($W \to 0$ as $T \to 0$) and above 5000 K the 5000 K value is held. The LTE fit of Miller et al. (2013) (their Table 5), used before, is removed: it lies above MiZATeP by factors 1.10, 1.48, 1.89 and 2.1 at 900, 1500, 2000 and 3000 K, and the NMT line list it was stated to sum, summed in LTE, agrees with MiZATeP to 1% over 300–2000 K (header of `h3p_cooling.f90`; `md/h3p_cooling_cause_20261010.md`). The non-LTE departure factor s is that of Miller et al. (2013) Table 6, a ratio of two vibrational-level sums normalized by its own LTE sum, so it multiplies the MiZATeP W unchanged. Its only collider is H_2 through the proton-hopping reaction with the Oka & Epp (2004) coefficient 2×10^{-15} m^3 s^{-1}. Table 6 covers $n(H_2) = 10^6$–10^{14} cm^{-3} in steps of two decades. Between its columns s is interpolated as a power law in $n(H_2)$: the emission per H_3^+ ion can grow at most as fast as the collision rate that populates the emitting levels, so the local index $\mathrm{d}\ln s/\mathrm{d}\ln n$ lies between 0 and 1, as the node-to-node indices of the table do (0.00–0.89), whereas an interpolant linear in s against $\log n$ reached 12.6 just above the 10^8 cm^{-3} column at 2000 K. Between the temperature rows the interpolation stays linear, a form the paper does not constrain. Below the table the code adopts the linear low-collider limit $s(T, n) = s(T, 10^6)\, n/10^6$ and records the cell as evaluated outside the tabulated range. This is the dominant coolant of a molecular base: the atomic channels above are all exponentially suppressed at $T \sim 10^3$ K, so leaving it out leaves that gas with no radiative loss at all. With `Base IR field: True` the rate is the net exchange with the thermal infrared of the lower atmosphere (`h3p_net_cooling_rate`), $\Lambda^{\mathrm{net}}_{H_3^+} = 4\pi\, n_{H_3^+}\, s\,[W(T) - W_{\mathrm{dil}}\, W(T_{\mathrm{rad}})]$: by Kirchhoff's law the absorption of a blackbody at T_{rad} is the LTE emission at T_{rad}, so the one cooling function serves both terms, and the rate vanishes at $T = T_{\mathrm{rad}}$, $W_{\mathrm{dil}} = 1$. Its radiative equilibrium temperature on the MiZATeP table is 958.5 K for $T_{\mathrm{rad}} = 1140$ K, $W_{\mathrm{dil}} = 1/2$ (header of `h3p_cooling.f90`).

With `Molecular IR bands: True` (default off) three further net exchanges are added (`molecular_infrared_cooling.f90`): the H_2 quadrupole and magnetic-dipole line spectrum, and the H_2O and CO vibration-rotation bands. Each is a *net* rate,

$$\Lambda^{\mathrm{net}}_X = n_X \left[E_X(T) - W_{\mathrm{dil}}\, E^{\mathrm{abs}}_X(T; T_{\mathrm{rad}})\right], \qquad E_X(T) = 4\pi \sum_b \sigma_b(T) \int_b B_\nu(T)\, \mathrm{d}\nu, \tag{83}$$

so each vanishes at its own radiative equilibrium temperature instead of running the layer down to nothing; a column is negative wherever that channel heats. The dilution W_{dil} is the same $0.5\, f_{\mathrm{sky}}$ the H_3^+ closure uses and is zero when `Base IR field` is off, which reduces these channels to pure emitters. Without them a converged molecular layer radiates itself down to 190–400 K against an equilibrium temperature of 1100–1400 K.

5.3 Advancing the temperature

In the marching loop. The temperature of a cell is advanced by `solve_energy_semi_implicit` (`energy_semi_implicit.f90`), the inner step of the coupled temperature-composition source step: the marching loop alternates it with the composition sweep until the pair stops moving, so the composition is fixed for one pass and not for the whole step. With the composition and the photoheating frozen over the step,

$$\rho e(T) - \rho e_{\rm old} = \Delta t\,[\mathcal{H} - \Lambda(T)], \qquad \text{i.e.} \qquad R(T) = \varepsilon(T) - \varepsilon_{\rm old} - a\,[\mathcal{H} - \Lambda(T)] = 0, \quad a = \frac{\Delta t}{n_{\rm tot} + n_e}, \tag{84}$$

with $\varepsilon = \rho e/(n_{\rm tot} + n_e)$ the internal energy per particle, which is what `internal_energy_per_particle` returns. The caloric internal energy ρe is that of (14). The root is found by a safeguarded Newton-bisection hybrid on a physical bracket whose ends are the floor of the equation-of-state validity and a ceiling above the heating-only update: the search starts from the explicit-Euler point, brackets the root as it goes, and once both ends are known takes a Newton step whenever that step stays strictly inside the bracket and the bracket keeps halving, bisecting otherwise. The Newton derivative is exact in the equation-of-state term (`heat_capacity_per_particle`) and a secant estimate in $\mathrm{d}\Lambda/\mathrm{d}T$, which affects only the path, the safeguard owning the convergence.

What is tested before the update is returned is the residual at the *returned* temperature T, with the cooling evaluated there:

$$|R(T)| \le 10^{-9} N(T), \qquad N = |\varepsilon(T_{\rm old}) - \varepsilon_{\rm old}| + a\left(|\mathcal{H}| + |\Lambda(T)|\right) + E_{\rm arith}(T)/10^{-9}, \tag{85}$$

(`thermal_balance_norm`). The first term is the caloric offset of the composition at the starting temperature, the second the gross radiative exchange of the step, and $E_{\rm arith}$ an allowance for the rounding of R (a bound under the cooling and equation-of-state evaluation assumptions stated in `energy_semi_implicit.f90`) as evaluated (the rounding of $\varepsilon(T)$, of the difference of two stored energies, of $a(\mathcal{H} - \Lambda)$, and of one unit in the last place of T). The stored thermal energy $\varepsilon_{\rm old}$ itself is *not* in N: a tolerance on it admits the whole net change of a nearly balanced step, which an earlier rule did by accepting $T_{\rm old}$ unsolved, so that the output jumped where a cell crossed that threshold. The same N is used at $T_{\rm old}$, at every trial and in the reported `res_scaled` $= |R|/N$. Where heating and cooling cancel to a fraction f of their sum the admitted defect is $10^{-9}/f$ of the net change, so N is a declared approximation of the size of the step, not a relative bound of a strongly canceled net increment. A cell with a non-finite residual or norm, a zero norm, or an incoming temperature above the ceiling is refused; the iteration cap is 100 cooling evaluations. The cooling written back is $\Lambda(T_{\rm returned})$, evaluated there and not at a previous iterate. There is no clamp: a cell that does not meet the tolerance, that has no bracket, or that reaches the equation-of-state floor is a named failure, nothing is assembled from it, and the step controller retakes the step at half Δt as in Sec. 7.5. The floor is not a reservoir: reaching it means the balance asks for a temperature below the range in which the equation of state and the rate fits are defined, and a floor representing real external heating would have to be a specified reservoir with its own budget. `use_semi_implicit_energy` is on by default.

In the advection post-process. `T_equation.f90` is *not* the marching loop's temperature solve. Its module `equation_T` is used by `post_process_adv.f90` (Sec. 11.1) and by test drivers, and by nothing else; what it solves is the *steady* internal-energy balance of the profile, differenced by the same variable-step BDF2 step as the ionization

systems (Sec. 4.5): every r-derivative at r_j is $(f_j - a_1 f_{j-1} + a_2 f_{j-2})/(g h_j)$ and every other factor is the value at r_j,

$$\nabla\cdot(u\mathbf{v}) + p\,\nabla\cdot\mathbf{v} = \rho v \frac{\mathrm{d}e}{\mathrm{d}r} - p\,v\frac{\mathrm{d}\ln\rho}{\mathrm{d}r} + h\,\nabla\cdot(\rho\mathbf{v}) = \mathcal{H} - \Lambda, \tag{86}$$

with e the internal energy per unit mass and $h = e + p/\rho$ the enthalpy per unit mass. In a run that solved heat conduction or the enthalpy flux of the element, carrier and stage fluxes, $\mathcal{H}$ holds those terms too, as heating (`transport_terms_of_profile`, `set_transport_terms_of_cell`): the conduction of the two neighbors and the enthalpy divergences as a fixed part, the cell's own conduction coefficient times its trial temperature as the part that varies. Conduction then couples the cells, so the cell-by-cell root below is the first sweep of a Newton iteration on the whole column, which Sec. 11.1 states with its acceptance on the energy residual. The history of the step carries each upstream cell's specific internal energy formed at that cell's composition and temperature, not this cell's caloric state evaluated at T_{up}: the rovibrational heat capacity belongs to the gas that holds the molecules, so across a dissociation front two cells at the same temperature store different energy. With the metals on, the root is bracketed rather than guessed (`solve_T_brent`, on by default through `use_brent_tsolve`): 80 logarithmically spaced probes from $\max(0.05T_{\mathrm{guess}}, 1\,\mathrm{K})$ to $4T_{\mathrm{guess}}$ locate the *first* sign change, i.e. the lowest root, and Brent's method closes the bracket to $2\epsilon|b| + \mathrm{tol}/2$ with $\mathrm{tol} = 10^{-10}$ in T/T_0, which is what keeps the solve off the hot spurious root; the cooling it balances is `radiative_cooling_of_cell`, the assembly `eval_cool` reports. A BDF2 step that finds no admissible root is retaken by backward Euler; a cell with no bracket then keeps the run's own temperature, is written `failed` and is counted as rejected, except that in a run with conduction or the enthalpy fluxes it is an unknown of the column solve (Sec. 11.1), which may place its root outside the bracket of this one-cell solve. Without the metals, or with `Brent solver: False`, the root is MINPACK `hybrd1` on the same residual (`post_process_adv.f90`). The metal densities and the fine-structure escape probabilities are held fixed while the root finder varies T, the latter because the optical depth is a column over the whole atmosphere and not a function of this cell's trial temperature.

The heating efficiency. The diagnostic η_{heat} of (39), which the sweep `ioniz_eq` returns beside the heating and cooling, is a property of the composition the sweep returned and is not a free parameter anywhere in the code. It is passed to `write_output.f90` but written to no output file (the columns of `Hydro_ioniz.txt` are r, ρ, v, p, T, heat and cool); the closed-form parameterizations of the same quantity that the literature uses are reproduced offline by `eta_approx.py` for comparison only.

6. The molecular layer, the trace metals and element transport

Everything in this section is *off* by default. A run acquires the molecular layer by asking for it (`Molecular chemistry: True`, which also switches `Molecular base` on so that the base particle count and the chemistry describe one gas), the metals by the presence of `metals.inp` or of elemental abundances in a handoff file, and the element transport by `He_diffusion`. An atomic, metal-free run never enters any of the code described here.

6.1 The H_2/He reaction network

`mol_rates.f90` carries the reaction rate coefficients of Koskinen et al. (2022) Table 1 as published, in cgs ($cm^3\ s^{-1}$, and $cm^6\ s^{-1}$ for the two three-body entries R13 and R15; R13 is multiplied by the total heavy-particle density and R15 by a collider-resolved one, below). EXHALE carries one temperature, so the heavy-particle and the electron temperature of that table are the same variable. The photorates P1–P5 of the paper are marked "SC" there, computed from cross sections, stellar flux and columns; in EXHALE they are the integrals of Sec. 3.4 and are not part of this module.

One entry is deliberately *not* transcribed. R12, the thermal dissociation of H_2, is built by detailed balance from the R15 recombination,

$$k_{12}(T) = \frac{k_{3b}(T)}{K_{\rm eq}(T)}, \tag{87}$$

with $K_{\rm eq}$ the $H+H \rightleftharpoons H_2$ equilibrium constant built from the same internal partition function $Q_{H_2}(T)$ the caloric equation of state uses (Sec. 6.4), because the two Table-1 entries are the same channel in opposite directions and their source measurements do not overlap in temperature. The third body of R13 is the total gas-particle density excluding electrons, its source (Miller et al. 1968) being a single coefficient with no collider dependence measured. R12 and R15 are collider-resolved (corrected 2026-09-17, item L7g): the association rate is the sum $\sum_M k_1(M) n_M$ over $M = H_2$, H and He with the published coefficients of Cohen & Westberg (1983), carried in the code as an H_2-equivalent third-body density so that R12 keeps following R15 by (87). Applying the $M = H_2$ coefficient to the total density, which is what the code did before, over-counts the association: at the base of the LHS 1140 b hot-Uranus column it made the R15 rate 22 percent too large and the molecular chemical heat 21.6 percent too large. No measurement of He as a third body for this reaction exists, so He is given argon's coefficient, and that entry remains an upper bound: a monatomic third body has no internal modes to take up the released energy. Two further limits in a helium-dominated gas, the R16 and R18 differences of order unity to ten with García Muñoz (2025), are recorded at the code site rather than patched.

R19, $HeH^+ + H \rightarrow H_2^+ + He$, is not the Table 1 constant $9.1 \times 10^{-10}\ cm^3\ s^{-1}$, a measurement near 300 K that held at every temperature of the wind is an extrapolation. The code carries the thermal rate of Esposito et al. (2015), quantum close-coupling cross sections for the lowest states and quasi-classical trajectories for the rest, Boltzmann-averaged over all 178 rovibrational states of HeH^+, in the authors' fit $k = \sum_{i=0}^{6} a_i (\log_{10} T)^i$ (maximum error 2.5%; read in the accepted manuscript). It is 1.09×10^{-9} at 300 K, 1.52×10^{-9} at 1000 K and 2.00×10^{-9} near 6000 K, close to the Langevin value 2.09×10^{-9}, and lies within 1–6% of the exact quantum reactive rate of De Fazio (2014) from 100 to 2000 K; above 1000 K it is 1.7–2.2 times the Table 1 constant. The fit is held at its 10 and 15000 K ends outside that range, the authors call the rate questionable above about 10^4 K, and the thermal average assumes HeH^+ internal states in equilibrium at the gas temperature. On the hot-Uranus fixtures the change lowers $n(HeH^+)$ by up to 46% and leaves the gas temperature, velocity and density within 4×10^{-5} (Update log stage 3, section 64).

Two further entries of Table 1 are not transcribed as printed. The first is R17, $He^+ + H_2 \rightarrow H^+ + H + He$, the dissociative charge transfer. Table 1 gives it as $10^{-9} \exp(-5700/T)$ (Moses & Bass, 2000), an Arrhenius form with a 0.49 eV barrier, and that form alone is 2×10^4 below the directly measured rate at 300 K and 200 below it at 400 K. The reaction has two mechanisms: at thermal energy it proceeds by tunnelling out of a long-lived He^+–H_2 complex, because reactants and products do not correlate adiabatically, and only above ~ 400 K does an over-barrier branch take over (Böhringer & Arnold, 1986). The drift-tube measurements give the two-body total as $k_2 = 1.1 \times$

$10^{-13}(300/T)^{0.24}$ cm^3 s^{-1} over 18–408 K (Böhringer & Arnold, 1986), consistent with $(1.1\pm0.1)\times10^{-13}$ at 330 K and $(1.5\pm0.15)\times10^{-13}$ at 78 K (Johnsen et al., 1980). The code therefore carries the sum of the two mechanisms, with the tunnelling branch taken as the measured total less the radiative branch the network carries separately as R23, so that the coded channels sum to the measured total:

$$k_{\mathrm{R17}}(T) = \left[1.1\times10^{-13}\,(300/T)^{0.24} - k_{\mathrm{R23}}\right] + 10^{-9}\exp(-5700/T).$$

Against the only measurement of the rising branch (Johnsen et al., 1980, their Fig. 3, carried to an effective temperature of 700 K) the sum is inside a factor two from 300 to 700 K. The three-body channel both papers measure, $k_3 = 1.6\times10^{-30}(100/T)^{1.27}$ cm^6 s^{-1}, is omitted because $k_3 n = 6\times10^{-19}$ cm^3 s^{-1} at 1000 K and $n = 7\times10^{12}$ cm^{-3}, five decades below the two-body rate.

The second is R20. Table 1 gives $\mathrm{He^+ + H_2 \rightarrow HeH^+ + H}$ as 4.2×10^{-13} cm^3 s^{-1}, citing Schauer et al. (1989). That paper measures the radiative ($\mathrm{He^+ + H_2 \rightarrow He + H_2^+ + }h\nu$, Table 1's R23) and the dissociative ($\mathrm{He^+ + H_2 \rightarrow He + H + H^+}$, R17) charge transfer over $15 \le T \le 40$ K and no $\mathrm{HeH^+}$ channel; it states that the $\mathrm{HeH^+}$ channel "apparently does not become allowed until the collision energy approaches 9 eV", and its $\mathrm{H_3^+}$ signal is the *sum* of the radiative and the $\mathrm{HeH^+}$ channels, which bounds the latter at $\le 1.0\times10^{-14}$ – 42 times below the value Table 1 assigns it. R20 is therefore not carried. The $\mathrm{HeH^+}$ source the network uses instead is the one Table 1 omits, $\mathrm{H_2^+ + He \rightarrow HeH^+ + H}$ at $3.0\times10^{-10}\exp(-6717/T)$ (Black, 1978), whose coefficient is a Maxwellian average over a thermal population of $\mathrm{H_2^+}$ vibrational levels.

Measured on LHS 1140 b at $\mathrm{He/H} = 2.13$ with the $\mathrm{H_2}$ row written out reaction by reaction, the repair rewires the base layer's $\mathrm{HeH^+}$ and $\mathrm{H^+}$ budgets – at the first cell $n(\mathrm{HeH^+})$ falls by a factor 487 at 1023 K and 8×10^4 at 629 K, and $n(\mathrm{He^+})$ rises by 1.11 and 4.4 – and leaves the net $\mathrm{H_2}$ destruction there within a percent, because the helium ions of that layer are sink-limited. The same measurement shows what sets the $\mathrm{H_2}$ of a helium-rich base: the helium-ion and proton channels, with the thermal dissociation R12 nine decades below the first cell's total $\mathrm{H_2}$ loss at 1023 K and twenty-two decades below it at 629 K, and nowhere in that layer reaching 0.4 percent of the loss. The helium-ion channels, not the temperature and not the photons, set the $\mathrm{H_2}$ content of a helium-rich base in this network.

The eight balance rows the network makes are (51)–(54) together with the $\mathrm{H^+}$, $\mathrm{He^+}$, $\mathrm{He^{2+}}$ and He 2^3S rows of Sec. 4.2.

6.2 $\mathrm{H_2}$ photoabsorption resolved into channels

`h2_photo_channels.f90` splits the one measured cross section $\sigma_{\mathrm{H_2}}(E)$ into four mutually exclusive final states whose sum is identically $\sigma_{\mathrm{H_2}}$, so no consumer of the opacity changes and the total $\mathrm{H_2}$ destruction rate is what it was; what changes is which fragments the source terms are told to make.

$$\sigma_N = f_n(E)\,\sigma_{\mathrm{H_2}}(E) \qquad \mathrm{H_2} + h\nu \rightarrow \mathrm{H + H}\ \text{(33–41 eV window)}, \tag{88}$$

$$\sigma_M + \sigma_S + \sigma_D = \left[1 - f_n(E)\right]\sigma_{\mathrm{H_2}}(E) \qquad \text{the three ionizing channels}, \tag{89}$$

with f_n the neutral fraction of Chung et al. (1993) Table 1, carried as a *fraction* so that it rides on this code's $\sigma_{\mathrm{H_2}}$ normalization rather than importing theirs. Inside the ionizing part the split follows the Chung et al. Table II ratio column $r(E) = \sigma(\mathrm{H^+})/\sigma(\mathrm{H_2^+})$, a ratio of time-of-flight detector signals and therefore a ratio of *events*, one count per event and not one per proton; `frac_H2_dissociative_ionization` is that column converted to $\sigma(\mathrm{H^+})/[\sigma(\mathrm{H^+}) + \sigma(\mathrm{H_2^+})]$ and tabulated at 69 energies from 18.076 to 124 eV, linearly interpolated, zero at and below threshold and held flat above

124 eV. The thresholds are 15.4 eV (M, leaves H_2^+), 18.076 eV (S, leaves $H + H^+$), 51.4 eV (D, two protons), and none of its own for N. The D channel is resolved by default (`H2 double ionization: chung80`, the 20 percent of the H^+ events at and above 80 eV that Chung et al. (1993) state, ramped linearly from the 51.4 eV threshold, the ramp not being in the source); `yan_rho` builds it from the double-to-single ratio of Yan et al. (1998) instead (at 110 eV `chung80` gives 15 percent more double ionization than `yan_rho`), and `off` folds it back into S, which reproduces the code before the channel existed. The N window is resolved by default as well (`H2 neutral dissociation: True`).

The energy of one absorbed photon is accounted channel by channel by `h2_channel_energy_recipients`, the single statement of what each channel charges, and the recipients of each sum to $h\nu$ once. Two entries are fixed with no free parameter: the double channel's vertical threshold 51.4 eV (Yan et al., 1998) exceeds the asymptotic product energy $D_0 + 2I(\mathrm{H}) = 31.675$ eV by exactly the kinetic energy release of the two receding protons, 19.725 eV; and the neutral window leaves both fragments in $n = 2$ (the Q2 $^1\Pi_u$ state whose fluorescence curve overlaps this cross section), so $2E(n{=}1 \to 2)$ departs as prompt Lyα and two-photon continuum and is not heat.

6.3 Lyman-Werner photodissociation

`lyman_werner.f90` adds the Solomon process, $H_2(X) + h\nu(11.2\text{–}13.6\,\mathrm{eV}) \to H_2(B,C) \to H + H$, which the Koskinen et al. Table-1 network leaves out (their photorates start at the 15.4 eV edge). The band is 912–1201 Å: 912 Å is the H Lyman edge, and 1201 Å is both the blue edge of band B2 and the red end of the Lyman and Werner line list the shielding table is built from. Draine & Bertoldi's own 912–1110 Å is not used, because a planetary base runs at 700–3200 K where the vibrationally excited levels pump in lines longward of 1110 Å that carry about a third of the dissociations at saturation. One band, one line list, one normalization.

The optically thin rate is formed from the band-integrated *energy* flux at the planet and the mean photon energy of a flat F_λ band, $\langle h\nu \rangle = 2hc/(912 + 1201\,\text{Å}) = 11.7354$ eV:

$$k_{\mathrm{LW}}^{\mathrm{thin}} = \sigma_{\mathrm{LW}} \frac{\xi_{\mathrm{day}} F_{\mathrm{LW}}}{\langle h\nu \rangle}. \tag{90}$$

The band-mean σ_{LW} is not the effective cross section $2.263 \times 10^{-18}\,\mathrm{cm}^2$ that Draine & Bertoldi (1996) Table 1 and 2 imply for a flat-F_λ field restated per 912–1201 Å photon: the rate is the level-resolved table of `h2_self_shielding_table.f90`, normalized per photon of the same band, and that number is only the limit it approaches in optically thin interstellar conditions.

That table is a line-by-line calculation with *overlapping* lines, and it stores three different counts because three are needed: σ_{diss}, dissociations per unit incident band photon fluence, which carries the self-shielding, the line overlap and the trapping of the fluorescent photons and sets the rate and the fragment heating; $p_{\mathrm{single}} = D/(D + \sum A_{ul})$, dissociations per pump with every fluorescent photon free to escape, which sets the vibrational heat return because a trapped decay plus its re-absorption moves one molecule down and another up and deposits nothing; and $p_{\mathrm{eff}} = D/(D + \sum A_{ul} P_{\mathrm{loss}})$, dissociations per photon removed from the beam, which is the band photon ledger. They are related by $\sigma_{\mathrm{diss}} = p_{\mathrm{eff}}\,\sigma_{\mathrm{pump}}$, and the beam's own loss is served from the same table, so the photons the beam loses and the dissociations the rate spends are one absorption with one normalization. Line overlap is not a correction at these columns: at $n_{\mathrm{H}} = 10^{13}\,\mathrm{cm}^{-3}$ the surviving pumping relative to a one-line-at-a-time calculation is 0.45 at $N_{\mathrm{H_2}} = 10^{20}\,\mathrm{cm}^{-2}$ and 900 K, and 4.8×10^{-5}

at 4.4×10^{21} cm^{-2} and 2700 K, which is where the base of a molecular layer sits. The rate enters (51) as $g_{\rm LW}$ and is zero when no band flux is supplied.

6.4 The caloric equation of state

The caloric equation of state that gives the bound H_2 its rovibrational heat capacity is (14) of Sec. 2.5, where it is stated with its validity and its `Caloric EOS: monatomic` switch. `caloric_eos.f90` supplies the four maps the hydrodynamics needs, $(\rho, \rho e) \to T$, $(\rho, \rho e) \to p$, $(\rho, p) \to \rho e$ and $(\rho, p) \to \gamma_{\rm eff}$, and $\gamma_{\rm eff}$ itself for the wave speeds.

6.5 Carrier transport

Everywhere else in the code the composition of a cell is the root of a local algebraic system. A *carrier* is a species the code does not fix that way: one whose chemical time can exceed the flow time, so that its density in a cell is what the flow brought there and not what the cell's own equilibrium would give. `diffusive_photochemistry.f90` solves, for the carriers, the continuity equation of the neutral molecular carriers H_2, OH, H_2O and CO,

$$\frac{\partial n_i}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2 (n_i v + J_i)\right] = P_i - L_i, \qquad J_i = -n_{\rm tot}\,(D_i + K_{zz})\,\frac{\partial f_i}{\partial r} - n_i D_i \left(\frac{1}{H_i} - \frac{1}{H_{\rm atm}}\right), \tag{91}$$

with $f_i = n_i / n_{\rm tot}$, H_2 alone without the oxygen option. With `Ionization transport` the same operator carries the ionization stages as further rows, x(H II) per hydrogen nucleus and x(He II), x(He III) per helium nucleus, and with `He 2^3S transport` (which requires it) the He 2^3S level as $x_3 = n(2^3\mathrm{S})/n_{\rm He}$, a fourth state of the helium partition with the charge of He I (Sec. 4.5). These rows are written on the element's own nucleus face flux and not as (91): across a face $F_k = x_k(f) N_{\rm el}(f) - n_{\rm el}(f) K_{zz}(f)\,\Delta x_k/\Delta r$ (`ionization_stage_transport.f90`), with $N_{\rm el}$ the element nucleus flux of the element operator (advection, gradient, eddy and settling drift in one object, `element_nucleus_face_flux`), the face fraction upwinded on the sign of $N_{\rm el}$, and the closing stage (H I; the He I ground singlet, or He I without the carried level) one minus the carried ones and never a row, so that the stage fluxes of an element sum to its nucleus flux at every face. The eddy term is zero at faces 0 and N, through which the element itself crosses only with the gas. Every stage of an element moves with that element's velocity: the ion–neutral drift in the ambipolar field is not carried, which is exact only inside about $1.2\,R_p$ on LHS 1140 b and a systematic of 0.02–0.6 of the advective stage flux outward of it (MEASURED, stated at the module). The solved set is fixed once by `carrier_set_init`: H_2; H_2, OH, H_2O, CO with the oxygen option; then H^+, He^+, He^{2+} and He 2^3S as the keys add them, at most eight rows.

Between them the four molecular carriers carry every H nucleus that is not atomic, ionized or in a molecular ion and every O and C nucleus that is not a free atom. H_2^+, H_3^+, HeH^+ and O(^{1}D) are left local; on the HD 209458 b molecular example the three molecular ions together carry at most 1.2×10^{-9} of the H nuclei, and the H-nucleus budget is closed against the element total rather than against a sum of transported species, so whatever they hold is simply not available to the carriers. The four molecular carriers are neutral, so the ambipolar field term of the general drift-diffusion equation is absent from (91) rather than neglected; for the carried ionization stages it is the drift stated above, and it is not carried. D_i is Blanc's law over the background carriers (H I, H II, H_2, H_2^+, H_3^+, He I, He II, He III, HeH^+), each pair from the rigid-sphere coefficient of Banks & Kockarts (1973). The ion-neutral polarization enhancement is not added: the transported species only exist where the gas is neutral (electron fraction per H nucleus at most 10^{-7} over the cells that hold H_2), so

the ion pairs carry that share of the Blanc friction sum, and for H_2O the induced-dipole formula would be the wrong one anyway, a permanent dipole of 1.85 D dominating the capture rate. K_{zz} is read from `kzz_cell` and nowhere else; that array is filled once, from the scalar `He_Kzz` or from the lower-atmosphere profile's own K_{zz} column. Settling is $G_i = (m_i - \bar{m}) g_{\rm eff}/(kT)$ with $\bar{m} = \rho/n_{\rm tot}$ and $g_{\rm eff} = -\rho^{-1}{\rm d}p/{\rm d}r$ the settling acceleration of the mixture, the one the element operator uses (eq. 99); thermal diffusion is not carried ($\alpha_T = 0$), there being no evaluated α_T for H_2O, OH or CO in an H_2/He bath.

$P_i - L_i$ is not written in this module: it is obtained by calling the very rows the local equilibrium solve uses, `mol_heh_rows` row 4 for H_2 and `oxygen_carrier_rows` for OH and H_2O, at the trial densities with the background frozen, and the Jacobian is a forward difference of those same calls, so a rate that changes there changes here. CO has no chemical source and only its transport moves it, with a ceiling from `limit_to_element_budget` that keeps it from carrying the oxygen and the carbon out of the wind, where the molecule cannot survive and where the O I and C II lines are the coolant.

The discretization is cell-centered f_i, face-centered fluxes on `r_edg`, one backward-Euler step per call, Newton with a backtracking line search, block tridiagonal in space with blocks of the carrier count (up to 8×8; a carrier the run does not solve keeps an identity row, so the block shape is fixed). Face N carries zero diffusive flux; the drift term switches from central to donor cell on a Péclet test. Whether the base face is an inflow is read from the wind's face mass flux and not from the velocity of cell 1, which carries a collocated odd-even mode. The element reservoirs at the base are Dirichlet, imposed through `melem_ab`, the base density and the equation of state. The *partition* of an element among its carriers is imposed only where the gas below states it: without a handoff the ghost carries the base cell's own partition, face 0 carries no diffusive flux and the base cell derives its partition, which is what makes the comparison against a photochemical handoff a test rather than a tautology. Where a handoff states the H_2 partition (`q_H2_base`, or the profile at its matching level), the inner ghosts hold it as a Dirichlet value and the base face carries, besides the advective inflow of that partition, the eddy, molecular and settling flux against it,

$$J_0 = -\bar{n}_0 (\bar{D}_0 + \bar{K}_0) \frac{X_1 - X_0}{r_1 - r_0} - \bar{n}_0 \bar{D}_0 \bar{G}_0 X_{\rm f}, \tag{92}$$

formed as an interior face is (face means of ghost 0 and cell 1, K_{zz} continuous across the face), with the atomic hydrogen recoiling against it as at every face. The gas below the level is mixed by the same eddies as the gas above it: with the face closed the reservoir partition reached cell 1 by the base mass flux alone, at $0.2\,{\rm cm\,s^{-1}}$ against an eddy speed $K_{zz}/H \approx 250\,{\rm cm\,s^{-1}}$ on LHS 1140 b, and the atomic hydrogen the wind makes aloft was recombined in cell 1 instead of being mixed down into the column below (MEASURED, `md/AN_matching_surface_diagnosis_20261007.md`). The H_2 flux through the base face of the certified state is written to `EXHALE_resolved.out` (`base_face_H2_flux_cgs`, its advective and diffusive halves and `base_face_radius_cm`) for the elemental-flux closure.

The material advection of a carrier is *not* in these rows. It is the divergence of the hydrodynamic face mass flux, $F_\rho(j) Y_c(j)/m_c$, taken with the mass row itself inside the Runge-Kutta stages (123) on the same faces, areas, volumes and time step (`species_face_flux.f90`), so a cell can only lose the carrier the mass row says it loses and a uniform partition is preserved to round-off; the face composition is reconstructed with the limiter of the primitive variables, so it is second order where the profile is smooth. The advective half is in the rows only where the balance is the whole equation rather than the remainder of a split: in the stationary carrier residual, and in the fixed-wind relaxation,

which takes many transport steps at a wind that does not move (`carrier_rows_advect`); there the stage rows take their advective part as $x_k(f)N_{\rm el}(f)$, the element's advection being inside $N_{\rm el}$, and in the marching loop $N_{\rm el}$ carries the element's diffusive half only. That this half must ride on the *face* mass flux is a numerical statement about the front: a first-order donor-cell difference on the cell velocity carries a numerical diffusivity $|v|\Delta r/2$, which on the 500-cell hot Uranus grid is 8.5, 20 and 22 times the molecular D of H_2 at 1.15, 1.30 and 1.50 $R_{\rm p}$. The key is `Molecular carrier transport`, default off.

6.6 The oxygen option

With `thereis_oxychem` (default off) three neutral carriers OH, H_2O and CO join the network. `oxygen_rates.f90` carries the rate coefficients, each transcribed from a named publication and each checked against two independent ports of the same literature that are on this machine (the Photochem `zahnle_earth.yaml` network and the VULCAN `NCHO_photo_network.txt`), every disagreement being recorded at the coefficient. `water_photolysis.f90` supplies the photolysis,

$$\mathrm{H_2O}+h\nu \to \mathrm{OH}+\mathrm{H}, \quad \to \mathrm{H_2}+\mathrm{O}(^1\mathrm{D}), \quad \to \mathrm{O}+\mathrm{H}+\mathrm{H}; \qquad \mathrm{OH}+h\nu \to \mathrm{O}+\mathrm{H},$$

in the four far-ultraviolet bands of Sec. 3.1 (the balance rows these rates enter are (55) and (56)), whose edges are fixed by the branching ratios rather than chosen: the three-body branch opens only in the interval containing Lyα and is 12 percent there, so Lyα needs its own band on branching grounds alone. The first branch alone carries 54.5 percent of the H_2O loss at the HD 189733 b base, which makes the OH/H_2O cycle catalytic rather than a one-way sink. `co_photodissociation.f90` destroys CO on the same Lyman-Werner beam, using a band-mean dissociation cross section times the band photon flux and the Visser et al. (2009) shielding function $\Theta(N_{\rm CO}, N_{\rm H_2})$ rather than their $k_0\Theta$ form, because EXHALE has a band flux and not an interstellar field and k_0 is a statement about a particular field. The other CO destruction channel is the He^+ charge transfer $\mathrm{He^+ + CO \to C^+ + O + He}$ at the Langevin value 1.6×10^{-9} cm^3 s^{-1}, which appears in the He^+ row of (52)'s system and whose +2.2117 eV appears in heating channel 18, one event counted once.

6.7 The lower-atmosphere handoff

The base of the escape model sits at a stated pressure level, and three routes supply the state there.

The analytic column. `lower_column.f90` integrates the hypsometric relation from the 1-bar level up to the base pressure (default 1 μbar), isothermal at $T_{\rm eq}$ with spherical gravity, using the mean molecular weight of the chemical-equilibrium H_2/H/He partition of Visscher et al. (2006) as fitted by Koskinen et al. (2022) eqs. (11)–(13),

$$q_{\rm H_2} = \frac{1.9845 + 10^u - \sqrt{10^u\,(3.9690 + 10^u)}}{2.3670}, \qquad u = -\frac{23672}{T} - \log_{10} p\,[\mathrm{bar}] + 6.2645, \tag{93}$$

with $q_{\rm He}$ from the elemental He/H and $q_{\rm H} = 1 - q_{\rm H_2} - q_{\rm He}$. It returns the radius of the base level and the base mixing ratios, and also a fully atomic bracket, because in chemical equilibrium $q_{\rm H_2}(1\,\mu\mathrm{bar})$ stays large up to $T \sim 2000$ K while photochemistry dissociates H_2 there: a large $q_{\rm H_2}$ on a hot planet is the signal that a photochemical handoff is needed. The validation case is the Koskinen et al. (2022) Model A Uranus-mass planet at 0.05 au, for which the column gives $r_0(1\,\mu\mathrm{bar}) = 1.34\,R_{\rm 1bar}$ with $q_{\rm H_2} \simeq 0.84$.

`base.inp`, the scalar handoff. An optional file of "key value" lines, written by `src/utils/run_lower.py` or `src/utils/vulcan_to_base.py`. Every key belongs to one of five categories and the category states what the value may do to the wind: *provenance* (comments only); *EOS boundary* (`T_base`→ T_0, `r_base`→ R_0, `p_base`, and `q_H2_base`, the H_2 volume mixing ratio of the base gas); *elemental reservoir* (`HeH_base` and `<El>_H_base` for the ten elements, overriding `metals.inp` for that element); *initial guess* (no key today); and *boundary constraint* (`Kzz_base`→`he_kzz`). `q_H2_base` is the q_{H_2} of (15) and is imposed on the H_2 partition of the inflowing ghost species as well, as the partition of the *non-ionized* hydrogen: with $x_2 = 2q(1+\mathrm{He/H})/(1+q)$ (`base_h2_nuclei_fraction`) the H_2 row of a lower ghost is $x_{H_2} = x_2(1-x_{\mathrm{ion}})$, where x_{ion} is the hydrogen in H^+, H_2^+, H_3^+ and HeH^+ and is an unknown of the same solve, so the neutral hydrogen left, $(1-x_2)(1-x_{\mathrm{ion}})$, is never negative (`x_h2_neutral_partition_fixed`, `ion_cell_state.f90`); it is refused above $0.5/(0.5+\mathrm{He/H})$. Species mixing ratios other than `q_H2_base` stay comments: no part of the code consumes them, so presenting them as physics would be false. A `base.inp` that states `p_base` fixes the base level and n_0 is derived from it as in Sec. 1.2, instead of being read from `input.inp`.

`Lower atmosphere profile`, the tabulated handoff. `lower_atmosphere_profile.f90` reads the lower model's solution over an *interval* of pressure that reaches up to the level where EXHALE places its base (the matching surface, the lower model's top face, as the Photochem adapter writes it; a file reaching above it is read, but nothing above the level is taken from it except K_{zz}). It supplies the state at the matching level p_{match} (T, r, q_{H_2}, and the elemental ratios He/H and El/H), which is what the base equation of state and the elemental reservoirs are built from, and K_{zz} as a *profile*, because the homopause, the only consequential thing K_{zz} does, is where $K_{zz} = D_{12}$, a property of two profiles, and the molecular coefficient rises by nearly five decades between the base and $1.2R_{\mathrm{p}}$. Every intensive quantity is interpolated linearly in $\log p$, the variable both models solve on; extrapolation is never performed, except that above the shallowest level the file carries K_{zz} keeps that level's value (`eddy_diffusion_on_grid`). The values go in through exactly the doors the scalar keys use, so nothing in the base equation of state changes; n and ρ carried by the file are *not* imposed, the base density remaining n_0. The hydrogen partition of the inflowing ghosts is the profile's own census at the match: the atomic share of the hydrogen nuclei is $f_{\mathrm{H}} = q_{\mathrm{H}}\, n_{\mathrm{tot}}/n_{\mathrm{H}}$, with the nuclei per particle $n_{\mathrm{H}}/n_{\mathrm{tot}} = (\rho/n_{\mathrm{tot}}m_{\mathrm{H}})/(1+\sum_{\mathrm{El}} X_{\mathrm{El}}m_{\mathrm{El}}/m_{\mathrm{H}})$ from the profile's density and element ratios, which counts the hydrogen of every species of the lower model, and $x_2 = 1-f_{\mathrm{H}}$ (`h_atomic_share_base`, `base_h2_nuclei_fraction`): the hydrogen of the trace molecules the wind does not carry (CH_4, NH_3, H_2O, 1.3×10^{-3} of the nuclei at the LHS 1140 b match) goes onto H_2, the inert majority carrier, and not onto atomic H, whose abundance the H, H_2, He mixture formula of `base.inp` would put sixty times above the profile's (ghost x_{H} 6.6×10^{-4} against 1.1×10^{-5}). One quantity is read below the level: the temperature of the gas there. The bath temperature that heat conduction holds at the base ghost (`conduction_base_level_T`, Sec. 2.8) and the temperature of the two advective ghost cells are both the profile's temperature at the ghost center (`base_reservoir_temperature_at` through `lap_value_at_radius`, linear in r between the bracketing levels and the nearer end outside them), the ghosts' pressure that of the hydrostatic column at that temperature (Sec. 8.1), because the column below the level is the profile's and not the reservoir's isentrope; without a profile both are the isentrope continued from the level.

The elemental-flux closure. The handoff is one-way in each direction: the photochemical model of the lower atmosphere (`src/utils/photochem_to_lower_profile.py`, which runs Photochem) is told elemental escape fluxes $\Phi_{\rm H}$, $\Phi_{\rm He}$ through its upper boundary and writes the profile and `base.inp`, and the wind built on that profile carries elemental fluxes of its own. *The top of the chemistry column is the matching surface*: its top face is placed at $p_{\rm match}$ on the solution's own grid (Photochem's levels are uniform in altitude; the face is moved by $H\ln(p_{\rm face}/p_{\rm match})$ and the chemistry re-converged until the face is at the match to 10^{-3} in $\ln p$, `steady_state_with_top_face_at_match`; every steady-state pass runs under the step budget `--blocks` $\times 100$, which is also Photochem's own step cap, and a pass that stalls is integrated once more from its stalled state at a relative tolerance of 10^{-4}, accepted only under the unchanged steady-state, face, flux and conservation checks and recorded in the header line `steady_state_retry`; a second stall refuses), the profile ends at $p_{\rm match}$ with the state of that face continued from the two top cell centers (`level_at_top_face`), and above the match the wind alone gives $T(p)$, $r(p)$ and the hydrogen partition. The chemistry runs on the climate column with the rise of the column energy balance up to the face, as a persistent $T(p)$ of Photochem. The climate column (Photochem's clima, a multispecies pseudoadiabat from the deep boundary $p_{\rm deep}$ under an isothermal stratosphere, `radiative_convective_column.py`) takes its deep composition from the Gibbs equilibrium of H, He, C, N and O at $(p_{\rm deep}, T_{\rm deep})$ (Photochem's `ChemEquiAnalysis`, condensates included), solved together with the deep temperature as the fixed point $T_{\rm deep} = T_{\rm clima}\big(T_{\rm deep}; x_{\rm eq}(p_{\rm deep}, T_{\rm deep})\big)$ by a bracketed root, and mapped onto the eight carriers clima has opacities for (H_2, He, H_2O, CH_4, CO_2, CO, N_2, NH_3) by the least relative change that keeps the El/H ratios of the elemental vector (to 10^{-6}); more than 10^{-6} of the mixture outside those carriers (at 1100–1500 K mostly atomic H, then HCN) is refused. At fixed (p, T) the CO/CH_4 and N_2/NH_3 balances depend on the elemental vector, and deep equilibrium at these temperatures puts nitrogen mainly in NH_3 (Moses et al., 2011, Sect. 3.1). The approximation is equilibrium *at* the deep boundary, without quenching from deeper, hotter levels: valid where the N_2/NH_3 conversion is faster than the mixing at $p_{\rm deep}$. The deep partition does not propagate: the chemistry computes its own composition from the same vector (update log stage 3 section 162; before 2026-10-29 the column took a prescribed H_2O/CH_4/N_2 partition and refused deep temperatures above 1000 K). The helium flux is imposed on He. The hydrogen flux is split as the wind splits it: H_2 carries the hydrogen the wind passes through its base face on H_2 ($\Phi_{\rm H_2}$, the closure variable below) and atomic H the rest, $\Phi_{\rm H} - \Phi_{\rm H_2}$, negative (an inflow of atoms) where the wind's eddies mix the atomic hydrogen it makes aloft down through the surface. Without a statement of the wind's split (the first iteration of a closure seeded from a wind that did not write it) the hydrogen flux is imposed on H_2 alone where atomic H carries at most 1 percent of the hydrogen nuclei at the model top (`CARRIER_SHARE_TOL`), and split by the top's own share otherwise, made self-consistent in at most `CARRIER_ROUNDS`= 8 rounds. The driver `src/utils/element_flux_closure.py` seeks the fixed point $\Phi_{\rm El} = F_{\rm El}(\Phi_{\rm El})$, El = H, He, by an under-relaxed Picard iteration of the two solves,

$$\Phi^{(k+1)} = \Phi^{(k)} + \omega\big(F^{(k)} - \Phi^{(k)}\big), \qquad \epsilon_{\rm El} = \frac{|F_{\rm El} - \Phi_{\rm El}|}{|F_{\rm El}|}, \tag{94}$$

$\omega = 0.5$ halved (floor 0.125) on every iteration whose residual did not fall, convergence read on the *undamped* residual $\max \epsilon_{\rm El} \le$ tol (default 0.05), at most eight iterations by default. $F_{\rm El}$ is the median of the element flux the wind writes face by face (`write_element_flux_profile`, `binary_element_diffusion.f90`, the Riemann face mass flux times the upwinded mass fraction plus the diffusive flux), over the overlap window, from the first face of constant mass flux up to the profile top, or, when that window is empty (a profile whose top is the matching level), unmeasured or not flat, over

the steady far-field window $r \geq r_{\rm esc}$, where the flux of a stationary state is the flux through the surface, the substitution being recorded. The hydrogen flux handed to the chemistry is a flux of hydrogen *nuclei*: the windows reduce $F_{\rm H}$, the mass flux of component 1 of the binary split (the hydrogen carriers together with the trace metals and the C and O bound in molecules, so that $F_{\rm H} + F_{\rm He} = \dot{M}$), and beside it $F_{\rm H,nuc} = s_{\rm H} F_{\rm H}$, $s_{\rm H} = m_{\rm H} n_{\rm H}/\rho_1$ the hydrogen-nucleus share of component 1's mass in the upwind cell (`lower_profile_F_Hnuc_*`, `steady_F_Hnuc_*` in `EXHALE_resolved.out`), and the closure uses $F_{\rm H,nuc}$ (`hydrogen_nucleus_flux`; $F_{\rm H}$, with the reason logged, only for an EXHALE that writes no $F_{\rm H,nuc}$; update log stage 3 section 134; before 2026-10-10 the metal mass was counted as hydrogen, 0.44 percent on LHS 1140 b). The hydrogen split is a fixed-point variable of its own: $\Phi_{\rm H_2}$, the hydrogen nuclei H_2 carries [g s^{-1}], takes the damped update toward $F_{\rm H_2} = 2 m_{\rm H}\, 4\pi r_0^2\, \phi_{\rm H_2}$ from the wind's `base_face_H2_flux_cgs` $\phi_{\rm H_2}$, and convergence also requires $|F_{\rm H_2} - \Phi_{\rm H_2}|/|F_{\rm H}| \leq$ tol, normalized on the elemental flux because the atomic share can pass through zero (`--phi0-H2`, by default unstated until the first wind states it). A measurement is accepted only when both elemental fluxes are finite and their radial spreads over the window are measured and at most tol (`flux_window_resolved`): a residual smaller than the spread of its window cannot be decided, and the closure then stops as unresolved rather than converged. A nonzero exit of the chemistry, a chemistry that wrote no profile, a wind solve or post-processing pass that exits with a status other than 0 or 2 (status 2, converged and not certified, is carried as a flag; status 2 without the certification's closing line is a runtime error), a wind solve that exits 3, and a solve whose verdict is not `info = 0` after the one continuation the ending policy authorizes are refusals, not iterates. The chemistry and the climate column take the stellar beam the wind's input states (`--face-geometry wind`, the default since 2026-10-10): $\xi_w F$ at the face, ξ_w the wind's dayside dilution (1, $1/2$ or $1/4$ by its `2D approximate method`), in place of each code's own convention (Photochem $0.5F$ at 48°, the climate column $F/4$; `per_model` keeps those); every wind run writes its ξ_w as `dayside_dilution` to `EXHALE_resolved.out`, and the closure refuses a run whose value differs from the one the column models were given (`check_wind_dilution`; the key is written since update log stage 3 section 135, so before it the comparison was never made). The chemistry's K_{zz} is a tracer-mixing coefficient that carries no heat (`--eddy-heat-transport off`, the default): as a heat diffusivity no reference climate exists above $K_{zz} \approx 1.4 \times 10^6$ cm^2 s^{-1} on LHS 1140 b at $\xi_w = 1$, and with turbulent dissipation heating (Hunten, 1974) a value of 10^9 would need 790–1000 times the absorbed stellar flux (update log stage 3 sections 130 and 133). With the option on, the stratosphere solve (`stratosphere_energy_balance`, `radiative_convective_column.py`) hands the climate object back with its own solution, since clima's `TOA_fluxes_dry` leaves its last trial profile in it and a second solve on the same object held a drifted troposphere (section 134); off, that solve is not made. A resumed closure recomputes the measurement of the last recorded iteration from its saved run and refuses to continue unless it is resolved at the requested tolerance and equals the recorded fluxes to 10^{-9}, and with the energy coupling on it refuses a history that carries no conducted-flux columns.

The energy coupling and the column energy balance. With heat conduction in the wind (Sec. 2.8) the wind conducts heat into the lower atmosphere through its base face, and that heat changes the matching-level temperature $T_{\rm match}$ the wind is built on. The certification measures, on the state it certifies, the conductive flux through the base face exactly as the operator forms it, $F_{\rm cond} = \kappa_{1/2}(T_1 - T_{\rm bath})/(r_1 - r_0)$, positive into the column (`heat_conducted_into_lower_atmosphere`, `viscous_conduction.f90`), and writes it

with the two temperatures to `EXHALE_resolved.out` (`base_conductive_flux_cgs`, `base_conduction_T_bath_K`, `base_conduction_T_cell1_K`; `base_conduction_state unmeasured` before a state has been certified). With the closure key `energy_coupling` (default false) the coupling becomes two-way in energy as well as in the elemental fluxes: $F_{\rm cond}$ is a third fixed-point variable, the next chemistry solve takes the damped value $F^{(k+1)} = F^{(k)} + \omega_E (F^{(k)}_{\rm meas} - F^{(k)})$ ($\omega_E = 0.5$, `energy_coupling_omega`) as `--conducted-heat-flux`, and convergence requires, besides (94), that the conducted flux meet the same tolerance in its own residual $|F_{\rm meas} - F|/|F_{\rm meas}|$ (zero against zero is agreement) and that $T_{\rm match}$ move by less than `T_match_tol` (default 6 K, the numerical He 10830 EW target over the EW sensitivity to $T_{\rm match}$ measured in update log stage 3 section 75). A run that does not write the flux, or has conduction off, stops the coupled closure.

The column below $p_{\rm match}$ takes the flux through `src/utils/conducted_heat_column_balance.py`. Its reference state is the climate column $T_0(p)$ of the profile, an isotherm at the skin temperature in the stratosphere, which has no energy equation that could take the flux; the module solves its steady balance at fixed pressure levels in $s = -\ln p$,

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}z}\left(r^2 \kappa \frac{\mathrm{d}T}{\mathrm{d}z}\right) = C(T,p) - C\big(T_0(p),p\big), \qquad \kappa \frac{\mathrm{d}T}{\mathrm{d}z}\bigg|_{p_{\rm match}} = F_{\rm cond}, \qquad T(p_{\rm bottom}) = T_0(p_{\rm bottom}), \tag{95}$$

with κ the mixture conductivity of (21) (the He, H and H_2 terms, the same law at both sides of the matching level) and C the net band cooling of CH_4, NH_3, H_2O, C_2H_2, CO and CO_2 from the k-distribution tables the climate model reads. Only the difference $C(T) - C(T_0)$ enters, so whatever radiative heating holds the climate state is held at its climate value. The conduction is the total one: $F_{\rm cond}$ is the total conductive flux at the match, and the conduction of the climate's own gradient is a term of the equation (the climate column has none), so T_0 is not the solution at $F_{\rm cond} = 0$ where it has a gradient (the tropopause). The unknown is the excess $\delta T = T - T_0$, for precision only, through an exact discrete rewrite with the perturbation flux in full and the reference source S_0, the reference's own imbalance, explicit (`balance_terms`; update log stage 3 section 128; the zero-flux shortcut that returned T_0 was removed because it returned a non-solution where T_0 has a gradient). On the LHS 1140 b k05 column S_0 moves T by at most 3.1×10^{-5} K. Radius, area and gravity are those of the reference profile, held; moving them with T changes $T_{\rm match}$ by -6.4×10^{-3} K there (MEASURED, section 128). The discrete balance is conservative, its source sum equal to the flux through the two ends. With `--eddy-heat-transport on` the face flux also carries the eddy term $\rho c_p K_{zz}(\mathrm{d}T/\mathrm{d}z + g/c_p)$ and the reference T_0 is the stratosphere solved in radiative, molecular and eddy balance (the closure paragraph above). The band emission is selected by `--column-energy-treatment`: `nlte` (default) treats a vibrational band with published V–T de-excitation rates as a two-level system, $e = \varepsilon\,(e_{\rm LTE}(T) - a)$, $\varepsilon = C_{10}/(C_{10} + A_{10})$, $C_{10} = \sum_M k_M(T) n_M$ (M = He, H_2), $a = e_{\rm LTE}(T_0)/2$ the absorption of the upward field of the optically thick stratosphere, with the CH_4 dyad rates measured for both partners (Siddles et al., 1994; Ménard-Bourcin et al., 2005), the NH_3 ν_2 + He rate measured (Hovis & Moore, 1980) and the NH_3 ν_2 + H_2 rate estimated from a measured analog ratio $k(H_2)/k(\mathrm{He})$ (`--nh3-h2-rate-analog`, CD_4 bending modes by default), and the rotational bands and every band lacking rates in LTE; `lte` (every band in LTE, an upper bound on the cooling), `rotational` (a lower bound) and `nlte_partial` bracket it. The transfer of the band photons is selected by `--column-band-transfer`: `thin` (default) lets all of the emission escape, and is valid while the column above absorbs at most 10 per cent of the upward emission excess of any bin carrying more than 1 percent of it; past that the solution is kept and flagged (`thin_layer_limit_exceeded`), not refused. `cooling_to_space` follows the absorption by the overlying column in the cooling-to-space approximation as generalized to a two-level band by Dickinson

(1972, Eqs. 5, 30 and 33): the downward half is absorbed in the thick column below at its climate source function, the upward half escapes with the layer integral of $E_2(\tau)$ over the k-distribution g-points, $\varepsilon \to C_{10}/(C_{10}+\beta A_{10})$, and the exchange inside the warmed layer is neglected and its first-order effect on $T_{\rm match}$ measured on every solution, the solve being refused when it exceeds `--column-exchange-shift-max` (default 1 K). `cooling_to_space_exchange` solves that exchange instead (fixed point on the exchange heating with exact slab E_3 fractions, at most 100 iterations), and has no such bound. Composition, absorption, $r(p)$ and gravity are held in one balance; the adapter alternates the balance with a Photochem solve on the warmed temperature until $T_{\rm match}$ of two successive balances agrees to `--column-energy-tolerance` (default 1 K, at most 8 alternations, a refusal otherwise), and writes the solution to `conducted_heat_column.txt` beside the profile. The overlying gas the transfer sees is the wind's: above the match the wind carries C, N and O as atoms and none of the emitters with its oxygen chemistry off, so the escape of both transfers starts at $p_{\rm match}$ (`include_gas_above_match` false; with the oxygen chemistry on the wind's H_2O and CO above the match would absorb and are not counted). *Recombination heat.* Atomic hydrogen that the wind mixes down through the surface, $\phi_{\rm in}$ atoms cm^{-2} s^{-1}, leaves the column as H_2 at steady state, so by Hess's law the column receives $F_{\rm chem} = \phi_{\rm in} D_0/2$ ($D_0 = 36118.11\,{\rm cm}^{-1}$, Huber & Herzberg 1979); the wind does not charge it, its atoms leaving before they recombine. It enters (95) as a heating held at fixed pressure with the shape of the column's own $H + H\,(+M) \to H_2$ rate per unit mass, normalized to $F_{\rm chem}$ (`recombination_heating_of_inflowing_atoms`; recorded in `conducted_heat_column.txt` and the profile header, `recombination_heat_flux_cgs`). An outflow of atoms carries no such term: its dissociation was charged in the wind.

6.8 Trace metals

`metals.inp` (`metals_input_read.f90`) sets the elemental abundances $n_X/n_{\rm H}$ by number, one ion per line with the neutral label naming the element; a second statement of the same element stops the run, since two abundances for one element are two answers to one question. The same file also carries `cno_cool`, `eos_metals`, `cx_full` and `cx_O2p_H`. The ionization of each element is solved in the same coupled system as hydrogen and helium (`System_HeH_metals.f90` and its triplet and molecular counterparts), through the rows (49); the code carries no parallel decoupled metal solver.

`eos_metals` (default *on*) puts the metals into the bulk gas budget consistently: their mass in ρ and hence in μ, v_0, p_0; their electrons in n_e; and their nuclei in $n_{\rm tot}$ and in the ghost pressure, matching the charge balance the ionization systems already use. The legacy trace approximation (metals in the ionization and cooling but not in the bulk gas) is `eos_metals 0`. A metals-off run is identical either way, every metal sum vanishing.

6.9 Binary element transport

`binary_element_diffusion.f90` treats the gas as two components moving through each other with the single mass-averaged velocity of the hydrodynamics: component 1 is the hydrogen carriers together with the trace metals and the heavy nuclei bound in the molecular carriers, and component He is helium in all stages. Since $\rho_1 + \rho_{\rm He} = \rho$ exactly, the two diffusive mass fluxes close, $J_1 = -J_{\rm He}$, and the composition update creates no mass. The transported variable is the helium *mass* fraction $X = \rho_{\rm He}/\rho \in [0,1]$, which is bounded at both ends unlike the ratio $n_{\rm He}/n_{\rm H}$ that the earlier trace

kernel carried and that diverges exactly in the helium-rich limit this operator exists for. The equation is

$$\frac{\partial(\rho X)}{\partial t}+\frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\left(\rho X v+J\right)\right]=0, \qquad J=-\rho\left(D_{12}+K_{zz}\right)\frac{\partial X}{\partial r}-\rho D_{12}\,GX(1-X), \tag{96}$$

$$G=\frac{(m_{\rm He}-m_{c1})\,m_{\rm H}g_{\rm eff}-\left(\bar{Z}_{\rm He}-\bar{Z}_{c1}\right)eE}{kT}+\alpha_T\frac{\mathrm{d}\ln T}{\mathrm{d}r}-\frac{\mathrm{d}\ln\psi}{\mathrm{d}r} \qquad [\mathrm{cm}^{-1}]. \tag{97}$$

The masses and charges in G are *carrier* quantities: $\psi=n_{1c}/n_{\rm H}$ is the number of collision partners each hydrogen nucleus is spread over (1 atomic, $1/2$ fully molecular) and $m_{c1}=m_1/\psi$. The gradient coefficient collapses to ρD_{12} for any ψ, so the Fickian term is the same in the molecular and the atomic region; the piece of the mole-fraction gradient the chemistry carries does not cancel and rides in G as $-\mathrm{d}\ln\psi/\mathrm{d}r$, which vanishes identically wherever ψ is constant. The ambipolar field is computed and not assumed, following Koskinen et al. (2013) section 2.1,

$$eE=-\frac{1}{n_e}\frac{\mathrm{d}(n_e kT)}{\mathrm{d}r}=-kT\frac{\mathrm{d}\ln(n_e T)}{\mathrm{d}r}, \tag{98}$$

in the logarithmic form, which is exact for the barometric stratification of the base and the same second-order central difference elsewhere. It is gated by `he_ambipolar` (default on); α_T is 0 by default (`He_alphaT`).

The settling acceleration. The force that separates the elements is set by the pressure gradient, not by gravity. Each species obeys $n_s m_s\,\mathrm{D}u_s/\mathrm{D}t=-\nabla p_s+n_s m_s\mathbf{G}+n_s e_s\mathbf{E}+\sum_t n_s m_s\nu_{st}(\mathbf{u}_t-\mathbf{u}_s)$ (Schunk, 1977, eq. 24b). With the drift inertia neglected the sum over all species, electrons included, is the momentum equation of the mixture, $\rho\,\mathrm{D}\mathbf{u}/\mathrm{D}t=-\nabla p+\rho\mathbf{G}$, and eliminating the bulk acceleration leaves the force that drives species s against the mixture as $-\nabla p_s+(\rho_s/\rho)\nabla p+n_s e_s\mathbf{E}$: gravity, the same per unit mass for every species, cancels, and the pressure gradient carries it together with the acceleration of the wind (the Chapman–Cowling pressure-diffusion form). In eq. (97) this is the gravity term with

$$g_{\rm eff}=-\frac{1}{\rho}\frac{\mathrm{d}p}{\mathrm{d}r}=-\frac{p}{\rho}\frac{\mathrm{d}\ln p}{\mathrm{d}r}=g+\frac{\mathrm{D}u}{\mathrm{D}t}, \qquad p=(n_{\rm tot}+n_e)kT, \tag{99}$$

which equals g in a hydrostatic column, exceeds it where the wind accelerates outward ($g+v\,\mathrm{d}v/\mathrm{d}r$ in a steady wind; $v\,\mathrm{d}v/\mathrm{d}r$ reaches g near a sonic point), and is formed (`mixture_pressure_acceleration`) from the state's own total pressure, with the particle count of the equation of state, by the same centered difference of a logarithm over the two neighbours as the ambipolar field (98), so that an isothermal barometric column returns g exactly. In a column of uniform composition the electron pressure is a fixed share of p, $eE=\bar{m}\,m_{\rm H}\,g_{\rm eff}$, and the relative settling mass $(m_{\rm He}-m_{c1})-(\bar{Z}_{\rm He}-\bar{Z}_{c1})eE/(m_{\rm H}g_{\rm eff})$ reaches its plasma limits (2.47 for an H^+ plasma, 1.65 for He^{++}) whatever the density profile. The same $g_{\rm eff}$ drives the trace metals and the molecular carriers. Viscous stress, which is not part of this first-order closure, is left out of the mixture balance. On the LHS 1140 b atomic states $v\,\mathrm{d}v/\mathrm{d}r$ reaches at most 5–6% of g, at 19 R_p (MEASURED, `md/AK_ambipolar_stage_drift_derivation_20261006.md` section 2.1).

The binary coefficient. D_{12} is built pair by pair over the ionization stages and carriers of the two components, each pair in the Chapman–Enskog first approximation $D_{st}=3kT/(16\,n\mu_{st}\Omega^{(1,1)}_{st})$, and combined by Blanc's law, $1/D_{12}=\sum_{s\in\mathrm{He}}\sum_{t\in 1}y_s y_t/D_{st}$ with y the carrier fractions. Neutral–neutral pairs take the rigid-sphere coefficient of Banks & Kockarts (1973); ion–ion pairs the screened Coulomb coefficient (Schunk, 1977, eq. C2); non-resonant ion–neutral pairs the induced-dipole (polarization) form alone,

$$D_{\rm in}=\frac{kT}{2.21\,\pi Z_i e\,n\,(\alpha_n\mu_{\rm in})^{1/2}}, \tag{100}$$

which is eq. 17 of Schunk & Nagy (1980) with the ion charge $Z_i e$ entering the potential $-\alpha_n Z_i^2 e^2/(2r^4)$, α_n the static dipole polarizability of the neutral. Schunk & Nagy (1980) describe the non-resonant interaction as a long-range polarization attraction coupled with a short-range repulsion and still give its momentum-transfer frequency in this form alone (their Table 6); no rigid-core term is added. The momentum-transfer rate coefficient $K = kT/(nD\mu)$ it gives is 1.306×10^{-9} (H^+/He) and $2.355 \times 10^{-9}\,\mathrm{cm^3\,s^{-1}}$ (He^+/H) at every T, 1.6% and 0.6% from their Table 6 (MEASURED, acceptance test T12e). Against the quantum-mechanical H^+/He coefficient of Kimura et al. (1993), 1.05×10^{-9} at 1000–5000 K and 1.17×10^{-9} at 10^4 K, it is 12–24% high; that residual is not explained by the published sources, which do not define the thermal average of the table. Resonant pairs (H^+/H, He^+/He) never enter: both partners carry the same element, and D_{12} is the friction between the elements.

The equation is split. The advection of the element mass fraction is a divergence of the same face mass flux the density rides on and is carried inside the Runge-Kutta stages (123), so each element's nucleus total over the domain changes only by its boundary fluxes to round-off. What is left for the implicit step here is $\rho(X^{\mathrm{new}} - X^{\mathrm{adv}})/\Delta t = -r^{-2}\partial_r(r^2 J)$, taken at the composition the stages advected; the operator advects only when it is *given* the face mass flux, which happens in the fixed-wind relaxation that has no stages to ride on.

The discrete operator is one backward-Euler step with frozen coefficients, faces carrying r^2 areas, Newton on a tridiagonal Jacobian (the step is nonlinear in X through the product $X(1-X)$). The gradient term is central; the drift term keeps a Péclet hybrid, central where $|B|\Delta r \le 2(A+E)$ and donor cell otherwise. The two factors of $X(1-X)$ come from *opposite* sides of the face, because the drift is a counter-flow and a donor-cell rule must take each element's mass fraction from the cell that element leaves:

$$J_{\mathrm{drift}}(f) = \begin{cases} -B_f\, X(j{+}1)\left[1 - X(j)\right], & B_f \ge 0 \ \text{(helium inward)}, \\ -B_f\, X(j)\left[1 - X(j{+}1)\right], & B_f < 0. \end{cases} \tag{101}$$

A donor at $X = 0$ then sends nothing and an acceptor at $X = 1$ receives nothing, so a cell on either end of the axis cannot be pushed past it; with the Péclet switch the same holds in the central branch, and the row argument (time term, gradient flux, upwind advection and drift all of one sign at a cell that first touches an end) gives $0 \le X \le 1$ on the residual itself rather than through a comparison principle. Measured, the worst excursion outside $[0, 1]$ before the assertion clip is -1.05×10^{-1}.

`project_elements` then rebuilds the species vector on the new element totals: helium-bearing species are scaled by $r_{\mathrm{He}} = n_{\mathrm{He}}^{\mathrm{new}}/n_{\mathrm{He}}^{\mathrm{old}}$, component-1 species by r_1, and a species carrying a nucleus of each (HeH^+) by $\min(r_1, r_{\mathrm{He}})$, the nuclei this undercounts for the element with the larger factor being deposited into that element's neutral ground species. Every operation is a multiplication by a nonnegative factor or an addition, so no negative intermediate can arise and both element totals are met exactly. The invariant is that the species handed back carry the mass the species handed in carried, to round-off, which requires r_1 to be the ratio of the two component-1 *masses* read from the cell. The result is the initial guess for `ioniz_eq`, which owns the split *within* an element while this step owns the element totals.

The base of the operator, and what crosses it. The gradient and drift coefficients are built on the faces 1 to $N-1$, and faces 0 and N are left at zero, so **the diffusive flux at the base face is zero by construction and not by the state**: an element crosses either end of the column with the gas alone. The flux the Dirichlet reservoir composition drives sits

at the first solved face, which is face 1 while the operator solves rows 2 to N against the reservoir, and face 0 where a closed base makes cell 1 an unknown. The record `element_base_flux` carries both faces rather than leaving either to be assumed: the advective and the diffusive half at the base face and at the first solved face, in $\mathrm{g\,cm^{-2}\,s^{-1}}$ positive outward (toward increasing r), together with the index of the first solved face and a count of the operator evaluations that produced a composition. A report reads the evaluation the numbers belong to, and a count of zero means no operator has run, which is not a flux of zero. The evaluation matters because the base entry is not steady while the base is: MEASURED on the `lower_profile` configuration, the advective half at the base face reads $+4.47 \times 10^{-8}$ at evaluation 1079, -3.35×10^{-10} at 11220 and $+6.29 \times 10^{-10}$ at 12000, as the base sound wave turns the face flux over in sign (item P6e).

The whole operator is gated by `He_diffusion`, default off. With `he_metal_diffusion` (default off) each trace element additionally carries its own mass fraction and its own binary coefficient; otherwise the metals are frozen to hydrogen, i.e. they ride in component 1 at the metal/H ratio the cell holds, and what `project_elements` preserves for them is that ratio and not the density.

The Dirichlet composition sits at the *center* of cell 1, half a cell above the base level, which since 2026-09-27 is the lower face of the grid (Sec. 8.1). That offset is a first-order term in the cell width which is recorded and not changed; the He/H change near the base is 1.5×10^{-4} (`md/Update_EXHALE_stage3.md` section 29.11).

6.10 The interdiffusion enthalpy flux

Particles that move relative to the mass-averaged velocity carry their enthalpy with them. In a mixture whose species move with velocities $v + w_s$, the energy equation is

$$\frac{\partial \mathcal{E}}{\partial t} + \nabla \cdot \left[(\mathcal{E} + p)\mathbf{v} \right] = \ldots - \nabla \cdot q_d, \qquad q_d = \sum_s h_s J_s, \quad J_s = \rho_s w_s, \quad \sum_s J_s = 0, \tag{102}$$

(Cook, 2009, eqs. 11–13, the Dufour flux and the kinetic energy of the relative motion left out there and here). Because J_s is a flux relative to the *mass*-weighted velocity, h_s is the specific enthalpy and not the internal energy. With the binary element model of Sec. 6.9 every species moves with its element and the electrons with their ions, so $J_1 = -J_{\mathrm{He}}$ and

$$q_d = J_{\mathrm{He}} \left(h_{\mathrm{He}} - h_1 \right) + \sum_X J_X \left(h_X - h_{\mathrm{H}} \right), \tag{103}$$

the metal sum present only where `He_metal_diffusion` moves each metal against hydrogen (taken against h_{H}, component 1 without its metals, the group that recoils) and the metals carry mass in the mixture. The h_c are *sensible* specific enthalpies built from the caloric equation of state (`component_specific_enthalpies` in `binary_element_diffusion.f90`): per particle $h = e + kT = kT\,\gamma/(\gamma - 1)$ for every atom, ion and molecular ion, each ion counted with the Z electrons the ambipolar field makes follow it, and H_2 adding its rovibrational energy of Sec. 6.4; HeH^+ is shared between the two components in the proportion of its nuclei. Formation energy is not in them: the energy update books chemical energy through the formation-energy density, whose reference in the source step is the composition the element transport has just written, so the formation energy of the moved nuclei rides with them already, and the stationary energy row holds chemical energy only as the reaction heating and cooling inside $\mathcal{H} - \Lambda$. J is the operator's whole diffusive flux, gradient, eddy and settling drift together, since eddy mixing carries enthalpy as it

carries the elements. Face values of h_c are the arithmetic means of the two cells, the face rule of every coefficient of the operator, and the flux is zero wherever J is, in particular at faces 0 and N.

The divergence is taken on the face areas and volumes of the element rows, $[A_+ q_d(j) - A_- q_d(j-1)]/V_j$ for $j = 2 \ldots N$, so an interior face cancels between its two cells and the energy the term moves telescopes over the column; cell 1, the Dirichlet reservoir of the element operator, carries no element equation and no interdiffusion energy. Both routes carry it: the marching step subtracts $\Delta t \, \nabla \cdot q_d$, formed from the element fluxes the accepted element step itself carried, from the thermal energy before the coupled source step starts (`EXHALE_main.f90`); the stationary energy row adds $\nabla \cdot q_d$ formed from the element fluxes of the state itself (140).

The transported carriers. A molecular carrier moved by the carrier operator of Sec. 6.5 (H_2, and OH, H_2O, CO with the oxygen chemistry) moves relative to the rest of its own elements as well: its diffusive, eddy and settling number flux Φ_c moves nuclei that the element fluxes do not, and the nuclei of the same elements held by the species that are not carriers move the other way, nucleus for nucleus, so the element fluxes are left as they are. The same identity then gives

$$q_{\mathrm{car}} = \sum_c \Phi_c \left[h_c - \sum_{\mathrm{el}} \nu_{c,\mathrm{el}} \, \bar{h}_{\mathrm{el}} \right], \tag{104}$$

with h_c the enthalpy of one carrier particle and $\bar{h}_{\mathrm{el}}$ the enthalpy per nucleus of el of the species that take up the recoil, each particle with the electrons its charge gave up. These are exactly the species whose densities the composition update rescales (`carrier_write_back`), so the energy row moves the enthalpy of the nuclei the composition rows move: for hydrogen the species of hydrogen nuclei alone that the flow does not carry, H I, H_2^+, H_3^+, and H^+ unless ionization transport carries it; HeH^+ never recoils, since rescaling it would move a helium nucleus with the hydrogen one; for O and C their atomic stages where the metals are in the equation of state. An element with no recoiling species in a cell recoils with its neutral atom. An H_2 molecule rising through atomic hydrogen at zero hydrogen flux thus carries $h_{H_2} - 2h_H = k(u_{\mathrm{rv}} - 5T/2)$, about $-1.5\,kT$ at 1000 K. Φ_c is the carrier operator's own face flux for the state in hand (`molecular_carrier_enthalpy_face_flux` in `diffusive_photochemistry.f90`), with the temperature and particle count of that state; the outer end of the carrier column is closed, so $q_{\mathrm{car}}(N) = 0$, and so is the base unless a handoff states the H_2 partition there, when $q_{\mathrm{car}}(0)$ is the enthalpy the base face exchanges with the lower atmosphere; the divergence, taken over cells $1 \ldots N$ on the carrier rows' geometry, sums to $-A_0 q_{\mathrm{car}}(0)$ over the column. The stationary energy row adds $\nabla \cdot q_{\mathrm{car}}$ beside $\nabla \cdot q_d$; the marching step subtracts $\Delta t \, \nabla \cdot q_{\mathrm{car}}$ after an accepted carrier step, from the fluxes of the composition that step returned (its own fluxes to first order in Δt, exactly at a fixed point). MEASURED on the certified well-mixed He/H = 0.19 state of LHS 1140 b before the term existed: $|\nabla \cdot q_{\mathrm{car}}|$ is a median 1.8 percent of the heating over $1.001 < r/R_p < 1.03$, up to 10 percent, comparable to the net heating where heating and cooling cancel, and that state is no longer stationary with the term (the carrier row stands at 7×10^{-3} on the first pass). The key `Interdiffusion enthalpy flux:` is **on by default** whenever `He_diffusion` moves helium, a molecular carrier is transported or `Ionization transport` carries the stages, and controls all three terms (the stage term is below), because the term belongs to the energy equation of any mixture whose components move relative to one another; `False` exists to reproduce published escape models that omit it (Koskinen et al., 2013, eq. 3), (Koskinen et al., 2022, eq. B3). With `He_diffusion` off the element term is identically zero with no arithmetic, and so is the carrier term without transported carriers. The environment variable `EXHALE_INTERDIFF_ENTH_SCALE` = s, $0 \le s \le 1$ (1 when unset),

multiplies the element term $\nabla\cdot q_d$ (`interdiffusion_enthalpy_scale` in `binary_element_diffusion.f90`); it is a continuation factor and not physics, used to reach the state of the full equation ($s=1$) in steps from one solved without the term, and since it changes the energy equation the state token `interdiff_enth` carries the value of s wherever $s\neq 1$, so a state of a continuation step is never read as one of the model. MEASURED on the LHS 1140 b `He_diffusion` states: the divergence reaches 13 to 26 percent of the local energy budget at 1.1–5 R_p, and on the molecular He/H = 1.8 state (24 outer passes with the term, not yet certified) it lowers T by a median 166 K over 1.3–2 R_p and He/H by 3.1 percent, and moves $\log_{10}\dot{M}$ by +0.002.

The carried ionization stages. With `Ionization transport` the stage k of an element crosses a face with $F_k = x_k(f)\,N_{\rm el}(f)+E_k$, $E_k=-n_{\rm el}K\,\partial x_k/\partial r$ (`ionization_stage_transport.f90`, equation 1): the first term moves the stage with its element, whose enthalpy is in $(\mathcal{E}+p)v$ and in (103), and the stage's own eddy term E_k moves it relative to the element, the neutral stage carrying $-\sum_k E_k$. By (102) that relative motion carries

$$q_{\rm st}=\sum_{\rm el}\sum_{k}E_k\left[h_k-\bar{h}_{\rm el}\right], \tag{105}$$

with h_k the enthalpy of one particle of stage k with its Z_k electrons and $\bar{h}_{\rm el}$ the enthalpy per nucleus of the recoiling species, the same set as for the molecular carriers: hydrogen in H I, H_2^+, H_3^+ (H^+ is the carried stage, H_2 a carrier with its own term, so no nucleus is counted twice), helium in He I with the He 2^3S level inside it (where `He 2^3S transport` carries the level, it has charge 0 and the sensible enthalpy of He I, so its own eddy term adds nothing to $q_{\rm st}$, and its excitation energy is chemical energy booked by its reactions; MEASURED on the certified LHS 1140 b validation state, that excitation-energy flux relative to He I would be at most 3×10^{-7} of the energy row's largest term). Since every heavy particle carries the same $kT\,\gamma/(\gamma-1)$, a stage exchange at fixed element flux moves only the electrons' enthalpy: with the recoil in the neutral atom $q_{\rm st}=h_e\sum_{\rm el}\sum_k Z_kE_k$, $h_e=kT\,\gamma/(\gamma-1)$, exactly so for helium; H_2^+ and H_3^+ in the hydrogen recoil lower $\bar{h}_{\rm H}$ by their share of one electron per two and three nuclei. Ionization energy is not in $q_{\rm st}$, for the reason given above. The term holds under zero current (the electrons follow the ions), one temperature for electrons, ions and neutrals, and no stage drift beyond E_k: the ion–neutral drift that the one-velocity stage transport leaves out (0.02–0.6 of the advective stage flux outward of $\approx 1.2\,R_p$ on LHS 1140 b) is absent together with its enthalpy. E_k is the stage rows' own (`ionization_stage_face_flux`), zero at faces 0 and N, so the column sum of the divergence vanishes (`ionization_stage_enthalpy_face_flux`); the divergence enters the stationary energy row, the marching update and the advected-profile energy solve together with the molecular term, under the same key, and the state token `stage_enth` records it. MEASURED on the certified He/H = 2.09 conduction state with carried stages (`.P2/ew_fit/phase3`): the term reaches 1.7×10^{-3} of the energy-row scale at 1.16 R_p and exceeds the 10^{-6} tolerance in 369 of 500 cells, so that state is not stationary with it.

7. Numerical scheme

7.1 The radial grid

The domain above the base level $r=1$ is discretized into N finite-volume cells (`Grid cells:`, default 500) with $N_g=2$ ghost cells at each end; all grid arrays run over $1-N_g\le j\le N+N_g$. `define_grid.f90` builds the cell centers

r_j, the interfaces $r_{j+1/2}$ as their midpoints, and the widths $\Delta r_j = r_{j+1/2} - r_{j-1/2}$. The last is an exact identity of the discretization and not an approximation: it is the same face pair that forms ΔV_j in (5), the same pair the gravitational source differences, and the same width the column integrals $n_j \Delta r_j$ and the cell optical depths use.

Three grid types are available, and since 2026-09-27 every one of them places the first face exactly on the base level, $r_{\rm edg}(0) = 1$, so that the reservoir stated there is the state of that face (Sec. 8.1); the two lower ghost cells lie below the level. `Uniform` takes $\Delta r = (r_{\rm max} - 1)/N$ and $r_j = 1 + (j - \frac{1}{2})\Delta r$, so its faces are $1 + j\,\Delta r$ and the outer face of cell N is $r_{\rm max}$. `Stretched` places the centers in geometric progression, $r_j = 2x^j/(1+x)$, so that the face between r_0 and r_1 is the level, with x the root of $(N+N_g)\ln x - \ln[(1+x)/2] - \ln r_{\rm max} = 0$ (the center of the outermost ghost at $r_{\rm max}$), found by Newton's method from $x = r_{\rm max}^{1/(N+N_g)}$, which climbs to it monotonically because the function is increasing and concave. `Mixed`, the type every shipped configuration uses, lays $N_{\rm low}$ uniform cells of width $\Delta r_{\rm base}$ on the base and then stretches the remaining $N_{\rm up} = N - N_{\rm low}$ cells by a factor k that solves

$$\frac{1 - k^{N_{\rm up}}}{1 - k} = \frac{r_{\rm max} - r_{N_{\rm low}}}{\Delta r_{\rm base}} \tag{106}$$

by Newton-Raphson to 10^{-8}, which is Eq. (10) of Caldiroli et al. (2021). The two numbers are the key `Base grid [dr,cells]:` and default to $\Delta r_{\rm base} = 2 \times 10^{-4} R_p$ and $N_{\rm low} = 50$, the values ATES hard-codes. The widths are then smoothed once by a 1-2-1 pass running downward from the junction to the base, the centers are rebuilt from the smoothed widths, and the grid is mapped by the affine map that sends its first face to 1 and the center of its outermost ghost to $r_{\rm max}$, which multiplies every width by one factor and so keeps the smoothed shape. For all three types the first ghost center is then restated as the mirror of the first cell center, $r_0 = 2 - r_1$, which moves it by rounding only and makes the first face exactly 1.0; the run stops if that face is not 1 or if a ghost cell reaches $r \le 0$. Until 2026-09-27 the level was a ghost *center* ($r_0 = 1$ on the `Mixed` grid, $r_{1-N_g} = 1$ on the other two), so every cell center has since moved (by half a cell on the `Mixed` grid, by one and a half on the other two) and a state written before then does not load on the present grid (Sec. 8.4). The requirement behind $\Delta r_{\rm base}$ is that the pressure scale height $H = k_B T/(\bar{m} g)$, with $\bar{m} = \rho/n_{\rm tot}$ the mean particle mass, be resolved by several cells at the base; a configuration that violates it is the one that produces a base-layer momentum imbalance no solver removes.

Two windows are located once, at grid construction: $j_{\rm min}$, the first cell with $r \ge r_{\rm esc}$ (`Escape radius [R_p]:`), which the marching convergence measure of Sec. 7.7 is taken over, and $j_{\rm flux}$, the first cell with $r \ge r_{\rm flux}$ (default $1.2\,R_p$), which the stationary flux gate uses. Both are clamped to mid-domain, with a message, where the requested radius lies outside the domain: a deep Roche-lobe-filling case can have $r_{\rm max}$ smaller than the default escape radius, and an empty window would make the measure $-\infty$ and the state converged at step zero.

7.2 Reconstruction

Both schemes reconstruct the *primitive* variables $W = (\rho, v, p)$ to the two sides of every face, and both write the left state of face j from cell j and the right state of face j from cell $j+1$.

PLM. `PLM_rec.f90` builds a limited slope from the three cell averages $j-1, j, j+1$ and evaluates it at the two faces,

$$W_{L,j+1/2} = W_j + \tfrac{1}{2}\sigma\,(r_{j+1} - r_j), \qquad W_{R,j-1/2} = W_j - \tfrac{1}{2}\sigma\,(r_j - r_{j-1}), \tag{107}$$

with σ the generalized minmod (monotonized central) slope of the right, left and central difference quotients at $\theta = 2$,

$$\sigma = \text{minmod}\left(\theta \frac{\Delta W_{j+1/2}}{r_{j+1} - r_j},\ \theta \frac{\Delta W_{j-1/2}}{r_j - r_{j-1}},\ \frac{W_{j+1} - W_{j-1}}{r_{j+1} - r_{j-1}} \right), \tag{108}$$

the slope being zero whenever the three arguments do not share a sign. This is the generalized minmod limiter as Käppeli & Mishra (2016) state it, at the least diffusive value for which the scheme stays total-variation diminishing; ATES uses the same limiter and attributes it to Kurganov & Tadmor (2000). The difference from ATES is that the quotients here are formed on the actual center spacing of a stretched grid. The same scalar limiter is exported as `PLM_rec_scalar`, so that a transported mass fraction rides on the same discretization as the density whose face flux carries it.

WENO3. `Reconstruction.f90` uses the third-order WENO reconstruction with the nonlinear weights of Yamaleev & Carpenter (2009). With $\Delta W_{j\pm1/2}$ the two interface jumps bounding cell j,

$$\tau = \Delta W_{j+1/2} - \Delta W_{j-1/2}, \qquad S_0 = 1 + \frac{\tau^2}{\Delta W_{j+1/2}^2 + \epsilon_j}, \qquad S_1 = 1 + \frac{\tau^2}{\Delta W_{j-1/2}^2 + \epsilon_j}, \tag{109}$$

for each primitive component $W = \rho, v, p$, which is their $1 + \tau/(\epsilon + \beta_r)$ with their already-squared τ (their Eq. 22) and the floor folded in. The floor carries the solution scale in the units of the component: $\epsilon_j = (\Delta r_j W_{s,j})^2$ with $W_{s,j}$ the largest $|W|$ of cells $j-1$, j, $j+1$ of the cell-average primitive (for the pressure the cell pressure, also under the well-balanced option), the floor of Eq. (111) below, and $S = 1$ where the stencil is all zeros. The face states are

$$\begin{aligned} W_{L,j+1/2} &= W_j + \frac{S_0 C_2(j) \Delta W_{j+1/2} + D_1(j) S_1 C_1(j-1) \Delta W_{j-1/2}}{S_0 + D_1(j) S_1}, \\ W_{R,j-1/2} &= W_j - \frac{D_2(j) S_0 C_2(j) \Delta W_{j+1/2} + S_1 C_1(j-1) \Delta W_{j-1/2}}{D_2(j) S_0 + S_1}, \end{aligned} \tag{110}$$

with the purely geometric coefficients $C_1(j) = \Delta V_{j+1}/(\Delta V_j + \Delta V_{j+1})$, $C_2 = 1 - C_1$, $D_1(j) = \Delta V_{j+1}/(\Delta V_j + \Delta V_{j-1})$ and $D_2(j) = \Delta V_{j-1}/(\Delta V_j + \Delta V_{j+1})$. Each candidate is the linear interpolant of two cell averages *in the volume coordinate*, so the jump to a neighbor is weighted by the cell's own volume share; these shares, and not the nonlinear weights, are what set the order on a stretched grid. The ideal weights $d_0 = 2/3$, $d_1 = 1/3$ of the uniform-grid scheme enter only through their ratio, which is what D_1 and D_2 carry: exactly $1/2$ on a grid uniform in the volume coordinate and $1/2 + O(h)$ otherwise.

Two limits of the scheme are stated at the routine. What runs is a third-order WENO reconstruction *with* those weights and not the energy-stable ESWENO scheme of the same paper, whose extra dissipation flux (their Eq. 48) is not computed here. And the floor is their prescription (their Eq. 65, $\epsilon = \max(\|u_0^2\|, \|(u_{0,\xi})^2\|)\,\Delta\xi^2$) read with the local amplitude of the stencil and the local cell width. Until 2026-10-09 the primitive floor was $\epsilon = \Delta r_j^2$, which has the h^2 scaling but not the solution scale: its face values were not invariant under a rescaling $W \to cW$ (relative departure 2×10^{-4} to 6×10^{-3} on smooth and stored LHS 1140 b states, against $\leq 4 \times 10^{-16}$ with the present floor), and in the outer domain, where ρ and p are 10^{-10} in code units against Δr_j^2 of 10^{-2} to 0.2, it made the weights linear, so a jump there was reconstructed with an overshoot of $1/6$ of the jump and a negative face that only the positivity limiter repaired (MEASURED, the suite `src/tests/weno3_primitive_floor`). Both floors keep the third order of the face values on smooth data (rate 3.00–3.01). With the local floor the marching regression case `wasp_full` no longer reached the old marching stop, a cell-centered spread $du < 10^{-3}$ of $\rho_j v_j r_j^2$; that was a property of the stop and not of the floor: a certified

stationary state of the same case has a face mass-flux spread of 5.3×10^{-13} and a cell-centered du of 3.6×10^{-3} on 500 cells (1.7×10^{-3} on 1000), set by the two cells next to the outer ghost, and the domain-scale floor tried meanwhile has a stationary du of 1.8×10^{-3} as well (`md/weno3_local_floor_marching_stall_diagnosis_20261009.md`). The marching stop now reads the face mass flux and the change of the state (Sec. 7.7; update log stage 3 sections 125 and 127). *With* `Well balanced: True` *the pressure face values are second order* (rate 1.99–2.06 on a heated and on a hydrostatic profile, for every floor tested): the reconstruction of the departure from each cell's own hydrostatic equilibrium is the second-order well-balanced construction of Käppeli & Mishra (2016), and the floor does not set that. No floor keeps a face positive across a jump of ratio above about $1/\Delta r_j^2$, which is what the positivity limiter below is for.

WENO3 of a transported scalar. A mass fraction or an element fraction is reconstructed by `Reconstruct_scalar` with the same face formula, Eq. (110), and the floor carries the square of the scalar's own scale,

$$\epsilon_j = \left(\Delta r_j \, q_{s,j}\right)^2, \qquad q_{s,j} = \max\left(|q_{j-1}|, |q_j|, |q_{j+1}|\right), \tag{111}$$

so that the face values commute with a scaling of the scalar ($q \to c\,q$ gives c times the faces). The scale is the local one and not a norm over the column as in the published prescription, because a stage fraction spans many decades along one column (He^{++} from 9×10^{-17} to 7×10^{-4} of the helium on the refused LHS 1140 b He/H 2.09 state) and a column norm would set the floor by its largest value. The primitive reconstruction uses the same floor (Eq. 109). With $\epsilon = \Delta r_j^2$ instead, every scalar whose jumps are below Δr_j in code units, which is every trace stage, took the linear weights: the unlimited three-point interpolant, which is not bounded by the two cells of a face. At a local minimum it put the neighbours' value on the minimum's own outflow face (He^{++} at 6×10^{-12} of the helium between 3.4×10^{-6} and 5.6×10^{-5} left its cell at a face fraction of 1.8×10^{-5}), which the carried stages of the stationary route could not balance (update log stage 3 section 121).

Donor face of the carried ionization stages. The scale floor does not remove the defect: on the LHS 1140 b He/H 1.2 state a cell whose He^{++} had been emptied still sent He^{++} out at a reconstructed face fraction of 9.9×10^{-6} (measured at the face between cells 358 and 359), so the semidiscrete rate of that cell pointed out of the nonnegative domain and no carrier interval could be retained. The carried stages (and He 2^3S where it is transported) therefore take the donor cell's fraction at every face, $d = j$ for $N_{\rm el} \geq 0$ and $d = j+1$ otherwise,

$$x_{k,f} = x_{k,d}, \qquad x_{{\rm close},f} = 1 - \sum_k x_{k,d}, \qquad F_k = N_{\rm el}\, x_{k,f} - n_{{\rm el},f} K \frac{x_{k,j+1} - x_{k,j}}{\Delta r}, \tag{112}$$

with $N_{\rm el}$ the element's face flux, so that the stage fluxes still sum to it and an empty donor advects nothing. The rule is first order in space; its leading numerical diffusive flux coefficient is $|N_{\rm el}|\,\Delta r/2$ (a fraction diffusivity $|N_{\rm el}|\Delta r/(2n_{\rm el})$ for locally constant coefficients on a uniform grid). On a He/H 100 state whose carrier rows converge (the wind itself uncertified) it moved the mass-loss rate by –0.005 percent and the width-matched He 10830 equivalent width by –0.17 percent, against –0.35 and –0.20 percent from doubling the cells (update log stage 3 section 163).

Positivity of the face states. Neither scheme has a positivity property: a face value of ρ or p can cross zero while every cell average on the stencil is positive, and at Mach 60 the thermal pressure is 2×10^{-4} of the total

energy density, so a reconstruction across a jump tips it negative and the sound speed takes the square root of it. `positivity_limited_faces` therefore scales each offending face state *continuously* toward its own cell average,

$$W_{\rm face} \leftarrow W_{\rm avg} + \theta\,(W_{\rm rec} - W_{\rm avg}), \qquad \theta = \min_{q\in\{\rho,\,p\}} \frac{q_{\rm avg} - \varepsilon}{q_{\rm avg} - q_{\rm rec}}, \qquad \varepsilon = \epsilon_{\rm mach}\, q_{\rm avg}, \tag{113}$$

which is the linear scaling idea of Zhang & Shu (2010a) applied to the primitive vector, and not their limiter: they scale the conserved vector in two steps with θ the root of a quadratic, one θ per cell against one per face state here, and their floor is absolute against one unit in the last place of the cell average here. What carries over is their Lemma 2.5, that a convex combination of an admissible average with the reconstruction is admissible; their Theorem 2.1 does not, its hypotheses being a Gauss-Lobatto quadrature set and a CFL condition nothing here imposes. Continuity is the point: a hard switch at the crossing is a step discontinuity of the residual and no stationary solve can descend across one. A face whose reconstruction is admissible has $\theta = 1$ and is not rewritten.

The PLM to WENO3 continuation. The optional two-stage choice `Reconstruction scheme: PLM+WENO3` marches with PLM until the convergence measure falls below the first `du_th` value (or its plateau is detected) and then continues with WENO3. Replacing one discretization by another between two time steps is a jump of the right-hand side, so the key `Reconstruction continuation:` interpolates instead. `reconstruction_continuation_rhs` (`steady_residual.f90`) evaluates the homotopy

$$R_\lambda(u) = (1-\lambda)\,R_{\rm PLM}(u) + \lambda\,R_{\rm WENO3}(u), \tag{114}$$

returning its flux-difference and source parts separately, and blends the stored interface fluxes, the face pressures and the three momentum terms on the same λ. Everything the two schemes disagree about is inside the two evaluations: the face reconstruction, the extrapolated reconstructed-face boundary states, and the pressure split of Sec. 2.3. At $\lambda \le 0$ and $\lambda \ge 1$ a single evaluation is taken and the two pure operators are reproduced exactly; only the open interval costs two right-hand sides. Because the admissible set $\rho > 0$, $\rho e > 0$ is convex and each stage is then a convex combination of two forward Euler stages, the continuation cannot manufacture a positivity failure that neither scheme has. The ramp is adaptive: λ advances by $\Delta\lambda$ where the relative state change of the step stays below a tolerance and $\Delta\lambda$ is halved where it does not.

7.3 Approximate Riemann solvers

`Num_Fluxes.f90` carries three fluxes, selected by the mandatory key `Numerical flux:`. Each side's total energy and sound speed are built with its *own* heat capacity, which is what the caloric equation of state makes different across a face inside the H_2 front; the physical flux is

$$F(W) = \big(\rho v,\ \rho v^2[+p],\ v(\mathcal{E}+p)\big), \tag{115}$$

the bracketed pressure being present under PLM without the well-balanced option, as Sec. 2.3 requires.

HLLC. The three-wave solver of Toro et al. (1994), with the Davis/Einfeldt speed bounds of Batten et al. (1997),

$$S_L = \min\big(0,\, v_L - a_L,\, v_R - a_R\big), \qquad S_R = \max\big(0,\, v_L + a_L,\, v_R + a_R\big), \tag{116}$$

$$S_* = \frac{p_R - p_L + \varphi_L v_L - \varphi_R v_R}{\varphi_L - \varphi_R}, \qquad \varphi_K = \rho_K (S_K - v_K), \tag{117}$$

and the star states

$$u_K^* = \frac{\rho_K (S_K - v_K)}{S_K - S_*} \left(1,\ S_*,\ \frac{\mathcal{E}_K}{\rho_K} + (S_* - v_K) \left[S_* + \frac{p_K}{\varphi_K} \right] \right)^\top, \tag{118}$$

the flux being F_K, $F_K + S_K(u_K^* - u_K)$ or the mirror of those according to the signs of S_L, S_*, S_R. Because S_L and S_R are clamped at zero, a face with every physical wave moving outward returns $F(W_L)$. In a star branch the pressure needed by (7) is the effective numerical pressure $p_K + S_K \rho_K^* (S_* - v_K)$, not just the side pressure or the usual contact pressure. A supersonic branch uses the upstream pressure without a star increment. The momentum flux carries the complementary term, preserving the complete Cartesian HLLC flux when pressure and ram are added. `Numerical flux:` is mandatory and has no default; HLLC, ROE and LLF are separate choices. A stationary residual result from one flux must not be treated as a certificate of a state computed with another.

Roe. The linearized solver of Roe (1981), with the averages

$$\tilde{\rho} = \sqrt{\rho_L \rho_R}, \qquad \tilde{v} = s v_L + (1-s) v_R, \qquad \tilde{H} = s H_L + (1-s) H_R, \qquad s = \frac{\sqrt{\rho_L}}{\sqrt{\rho_L} + \sqrt{\rho_R}}, \tag{119}$$

$\tilde{a} = \sqrt{(\bar{\gamma} - 1)(\tilde{H} - \tilde{v}^2/2)}$, and the flux

$$F_{j+1/2} = \tfrac{1}{2} \left(F_L + F_R \right) - \tfrac{1}{2} \sum_{i=1}^{3} \tilde{\alpha}_i \, |\tilde{\lambda}_i| \, \tilde{K}^{(i)}, \tag{120}$$

with the wave strengths $\tilde{\alpha}_{1,3} = (\Delta p \mp \tilde{\rho} \tilde{a} \, \Delta v)/2\tilde{a}^2$ and $\tilde{\alpha}_2 = \Delta \rho - \Delta p / \tilde{a}^2$. The acoustic strengths $\tilde{\alpha}_{1,3}$ may be formed with the velocity jump scaled by $\min(\mathrm{Ma}, 1)$ after Rieper (2011), which is the published low-Mach cure for a flux whose dissipation does not vanish with the Mach number; the option is `Low Mach velocity jump:`, it is off by default, and it is not recommended on a near-hydrostatic column. The factor is exactly zero where two neighboring velocities cancel in the Roe average, so it removes the damping of the odd-even ($2\Delta r$) velocity mode at precisely the faces that mode lives on; measured on the LHS 1140 b low-XUV base it turned a solve unmodified Roe completes into a nearly pure odd-even mode. Rieper's premise, that the local and the global Mach number are comparable, does not hold on a column running from $M \sim 10^{-8}$ at the base to 10^{-2} at the top (item L25, 2026-09-17). The single index $\bar{\gamma}$ is the arithmetic mean of the two face gammas, exact wherever the two sides agree; a run whose caloric equation of state makes the index vary across the front, that is molecular chemistry with the ladder heat capacity, is refused this flux at startup rather than being given a wave speed with no bound property. A Harten-Hyman entropy correction is applied to the acoustic eigenvalues from the star state that `speed_estimate_ROE.f90` builds (the primitive-variable, two-rarefaction and two-shock estimates of Toro (2009) chapter 9, selected by the pressure ratio). Where that estimate has no admissible star state, two rarefactions separating fast enough leave vacuum, or no branch returns a positive star pressure and positive star densities, the face is given the HLLE flux instead,

$$F_{j+1/2} = \frac{S_R F_L - S_L F_R + S_L S_R (u_R - u_L)}{S_R - S_L}, \qquad \begin{aligned} S_L &= \min(v_L - a_L,\ \tilde{v} - \tilde{a}), \\ S_R &= \max(v_R + a_R,\ \tilde{v} + \tilde{a}), \end{aligned} \tag{121}$$

which is positively conservative under those bounds (Einfeldt et al., 1991), where the Roe flux is not. The number of such faces is counted and reported.

Local Lax-Friedrichs.

$$F_{j+1/2} = \tfrac{1}{2}\left[F_L + F_R - \alpha\,(u_R - u_L)\right], \qquad \alpha = \max\left(|v_L| + a_L,\ |v_R| + a_R\right), \tag{122}$$

the coefficient being the spectral radius of the flux Jacobian (Rusanov, 1961). It is the most diffusive of the three and resolves no contact wave; it is also the flux the positivity repair below substitutes at a single face, because with cell-average input states it maps positive density and positive internal energy to positive density and positive internal energy under $\Delta t(|v| + c)/\Delta r \leq 1$ (Perthame & Shu, 1996; Zhang & Shu, 2010b).

7.4 Positivity repair of a stage

A Runge-Kutta stage can leave the admissible set $\rho > 0$, $\rho e > 0$ even with admissible face states, because the source terms are bounded by no flux choice. `positivity_limited_fluxes` (`RK_rhs.f90`) repairs such a stage by replacing, at the offending cells only, the high-order interface fluxes with the first-order Lax-Friedrichs flux of the neighboring cell averages and rebuilding those cells' update from the mixture; the replacement is swept until no cell is inadmissible or until both faces of a cell are already first order, in which case no flux choice repairs it and the caller halves the time step. Replacing one face rather than all of them is the flux correction of Hu et al. (2013). The stored face fluxes are written back after the repair. During PLM–WENO3 continuation, the pressure contributions receive their reconstruction weights once; a previously blended face pressure is not weighted a second time. This preserves the same pressure/source pairing as the unrepaired update. The transported species rows ride on F^{ρ} and a species flux built on a discarded high-order flux would not sum to the mass update the density took. Every count is reported at the end of a run, separately for repaired faces belonging to accepted steps, so that a run in which nothing fired is identifiable as the run an unguarded build gives.

7.5 Time integration and the time step

The marching loop of `EXHALE_main.f90` advances the hydrodynamics with the three-stage strong-stability-preserving Runge-Kutta method (Gottlieb & Shu, 1998),

$$\begin{aligned} u^{(1)} &= u^n - \Delta t\,\left(dF - S\right)\big|_{u^n}, \\ u^{(2)} &= \tfrac{1}{4}\left[3u^n + u^{(1)} - \Delta t\,\left(dF - S\right)\big|_{u^{(1)}}\right], \\ u^{n+1} &= \tfrac{1}{3}\left[u^n + 2u^{(2)} - 2\Delta t\,\left(dF - S\right)\big|_{u^{(2)}}\right], \end{aligned} \tag{123}$$

each stage followed by the boundary condition, by the admissibility test of Sec. 7.4, and by the species advection sub-step that rides on the face mass fluxes that stage built. A stage that cannot be repaired causes the whole step to be discarded and retaken from u^n at half Δt, up to a bounded number of bisections.

The time step is the explicit-stable interval of the scheme actually in use, derived at `eval_dt.f90` for the spherical finite-volume divergence the right-hand side assembles,

$$dF_j = \frac{A_{j+1/2}F_{j+1/2} - A_{j-1/2}F_{j-1/2}}{V_j}, \qquad A = r^2, \qquad V_j = \frac{r_{j+1/2}^3 - r_{j-1/2}^3}{3}. \tag{124}$$

Every stage of (123) is a convex combination of forward-Euler steps of the same length (the SSP coefficient is 1), so the restriction is the forward-Euler one of (124). The time-step calculation estimates each face signal by $S_{j+1/2} =$

$\max(|v|+c_s)$ over the two adjacent cell averages. With face area $A_f = r_f^2$ and cell volume $V_j = (r_{j+1/2}^3 - r_{j-1/2}^3)/3$ (both without the common 4π), the code selects

$$\Delta t = C_{\rm CFL} \min_{1\le j\le N} \frac{V_j}{\max\left(A_{j-1/2}S_{j-1/2},\, A_{j+1/2}S_{j+1/2}\right)}, \qquad C_{\rm CFL} = 0.6 \text{ by default}, \tag{125}$$

with $c_s = \sqrt{\gamma_{\rm eff} p/\rho}$ from the cell's own composition. In the Cartesian uniform limit $A_{j-1/2} = A_{j+1/2}$ and $V_j = A\,\Delta r$, and (125) returns $C_{\rm CFL}\Delta r/(|v|+c_s)$, so the number the `CFL:` key carries keeps its usual meaning and nothing is recalibrated. The stricter condition that the two fans of one cell may not meet inside it replaces the maximum by the sum of the two face terms, which corresponds to $C_{\rm CFL}/2$ on a uniform grid with equal face signal speeds.

The minimum is taken over evolved cells only. Ghost states supply the signals at the two boundary faces; their own cell widths do not enter the minimum. In spherical geometry $V_j/A_{j+1/2} < \Delta r_j < V_j/A_{j-1/2}$, so replacing V_j/A_f by the center spacing changes the interval.

S_f is formed from the cell averages bounding the face and not from the reconstructed face states, which inside a steep cell can carry a larger $|v|+c_s$; that margin is the one a Godunov number below unity leaves. During initialization, `Time stepping: Local` gives each cell its own Δt_j, the same expression without the minimum. This is pseudo-time relaxation, not a physical time history: unequal cell intervals do not advance shared face transfers over one common elapsed time. Physical mode therefore refuses local time stepping. In physical-time marching the local array is uniformly the global step, which keeps the update arithmetic identical to the scalar form. A ghost row, which is not evolved, is given the interval of the evolved cell it borders so that a stage formed over the padded range is arithmetic on a physical interval.

(125) is not only a marching quantity: it supplies the stationary route's initial pseudo-time and the floor of the JFNK pseudo-transient continuation (Sec. 9.5), so its value moves solver behavior as well as a time step. The no-step evaluation route does not call it.

In physical-time mode, the default integration-error diagnostic compares one complete step with two half steps from the same starting state every 20 accepted steps. If the scaled difference exceeds its limit, the entire attempt is retried with a shorter interval. The complete-step result is retained when accepted; the two half steps provide the estimate. This sampled estimate does not assert a bound on every unsampled step. It is inactive by default during initialization, whose local time steps do not define a physical-time trajectory.

7.6 Operator splitting of one step

One accepted marching step applies the following operators, in this order, with the state after each one carried into the next (`EXHALE_main.f90`):

1. the three Runge-Kutta stages (123), each with its reconstruction, flux assembly, boundary fill, positivity test and species advection sub-step, and the advection of the adiabat $K = p/\rho^\gamma$ on the same face mass fluxes (`adiabat_advection_stage`), after which the internal energy of a cold, highly supersonic expanding cell is taken from K (`internal_energy_from_adiabat`; the dual-energy method, below);

2. the conversion to primitive variables, the composition-dependent particle and electron counts, and the temperature (13);

3. the advected element composition written into the species vector (`species_advection_project`), followed by the implicit diffusive half of the element transport (`element_diffusion_step` in `binary_element_diffusion.f90`, Sec. 6.9), active only with `He_diffusion`, and, on an accepted element step, the energy update $\mathcal{E} \leftarrow \mathcal{E} - \Delta t\, \nabla \cdot q_d$ from the element face fluxes of that step (Sec. 6.10);

4. the vertical transport of the carriers with their chemistry (`photochemical_transport_step` in `diffusive_photochemistry.f90`, Sec. 6.5): the molecular carriers with `Molecular carrier transport`, the ionization stages with `Ionization transport` and He 2^3S with `He 2^3S transport`, followed, where that interval was covered, by $\mathcal{E} \leftarrow \mathcal{E} - \Delta t\, \nabla \cdot q_c$ from the composition the step returned (`carrier_enthalpy_divergence_of_state`);

5. the refresh of the lagged H(n=2) populations and their Balmer source (`excited_hydrogen.f90`, Sec. 4.6);

6. the coupled source step: a bounded fixed-point iteration alternating the composition equilibrium at the current temperature (`ioniz_eq`, Sec. 4.4) with the semi-implicit energy update (84) at the returned composition, exiting when the temperature and the composition both stop moving;

7. the boundary condition;

8. the Crank-Nicolson viscous and conductive stage (20), where those options are on, and the boundary condition again;

9. the periodic Shapiro filter, where it is on, and the boundary condition again.

Steps 3, 4, 8 and 9 are no-ops unless their key is present, and the two enthalpy updates also need `Interdiffusion enthalpy flux` (on by default). In a physical-mode run (a trajectory with a clock) a refused element step or a carrier interval the transport operator could not cover refuses the whole attempted step, which is restored from its checkpoint and retaken with a reduced step (`attempted_step_reduced_dt`) up to a retry budget whose exhaustion stops the run; while the run is reaching a state the refusal is counted and the step goes on. A step whose energy update found no temperature, or whose Crank-Nicolson stage failed (a temperature below the floor of the equation of state, a non-finite solution, a pivot breakdown; `viscous_conduction_step` prints the cell, the solved value and the energy a floor would have injected), has no state to march on and is refused in both modes: it is restored and retaken at half the interval within the same budget, whose exhaustion stops the run with exit status 2 (Sec. 2.8). The Shapiro filter is the 1-2-1 low-pass pass $u_j \leftarrow u_j + \frac{1}{4}\epsilon(u_{j-1} - 2u_j + u_{j+1})$ applied every `shapiro_every` steps, the way CETIMB damps the base sound wave (Koskinen et al., 2013). It is off by default and is recorded as a validity state of every step it runs on, because unlike the low-Mach term of Sec. 2.7 it alters the marched state with no source term behind it and therefore has no counterpart in the stationary residual.

The internal energy of a cold, highly supersonic flow. The scheme carries $\mathcal{E} = \rho e + \frac{1}{2}\rho v^2$ and recovers ρe as the difference, whose ratio to the kinetic term is $2/[\gamma(\gamma-1)M^2]$, 1.1×10^{-3} at $M = 40$. There the difference carries the truncation error of the kinetic-energy flux (not round-off), amplified by that ratio. MEASURED on the Yan et al. (2022) WASP-52 b atomic run (He/H 98/2, 10 R_p): between 9.03 and 9.14 R_p the adiabat $p/\rho^{5/3}$ fell 7.2-fold where

the net heating is positive, which an adiabatic flow with heating cannot do; the temperature reached the floor and the energy update refused the step. The adiabat is therefore also carried, $\partial_t(\rho K)+\nabla\cdot(\rho K v)=0$, on the face mass fluxes, volumes and SSP-RK3 weights of the mass row with the donor-cell face value of K, taken fresh from the state at the start of every step (`adiabatic_entropy_transport.f90`; the dual-energy method of Ryu et al. 1993 and Bryan et al. 1995, whose threshold values have not been compared). After the stages, in a cell where $\rho e < 10^{-2}\,\mathcal{E}$ ($M \gtrsim 13$), $\nabla\cdot v \geq 0$ (no shock, across which K is not conserved) and the gas has the constant γ of an atomic mixture (the caloric equation of state of molecular gas has no single adiabat), $\rho e = K\rho^{\gamma}/(\gamma-1)$ and $\mathcal{E}$ is reset to it plus the kinetic energy. Heating, cooling, conduction and viscosity follow as before, operator split. Total energy is not conserved in the reset cells, by less than 10^{-2} of their $\mathcal{E}$, the size of the error it replaces; the run log prints the number of reset cell-steps. A run whose every cell is below that Mach number or compressive is unchanged (update log stage 3 section 139).

7.7 The marching convergence measure and the stop conditions

In initialization and relaxation runs, a numerical marching stop uses the conserved mass flux through the faces and the change of the state. A stationary finite-volume wind carries one numerical mass flux, $\Phi_{j+1/2} = r_{j+1/2}^2\,\widehat{F}_{\rho,j+1/2}$, through every face; its spread over the faces at $r \geq r_{\rm flux}$ is the flux gate of the steady solvers (Sec. 9.2), and the stop reads the same quantity against its own threshold `march_flux_spread_th` (4×10^{-4}), not the steady gate `flux_spread_th` (2×10^{-5}). This stop records a numerical relaxation snapshot. Physical-time profile acceptance is assessed over a specified window, separately from this stop. The second measure is the state change over one monitor window of $N_{\rm resid} = 500$ steps,

$$\chi = \max_{1\leq j\leq N} \max\left(\left|\frac{\rho_j}{\rho_j^{\rm w}}-1\right|, \left|\frac{E_j}{E_j^{\rm w}}-1\right|, \frac{|(\rho v)_j-(\rho v)_j^{\rm w}|}{\sqrt{\rho_j^{\rm w}E_j^{\rm w}}} \right), \tag{126}$$

with the superscript w denoting the state $N_{\rm resid}$ steps before; the momentum density passes through zero near the base, so its change is measured against the momentum scale $\sqrt{\rho E}$ of the cell. An initialization or relaxation run reaches its numerical marching stop when the face spread is below `march_flux_spread_th` and $\chi <$ `march_change_th` $= 10^{-4}$ at a monitor step (out of the PLM stage), and stops *without* a claim (the stall net) when $\chi < 10^{-4}$ has held over $N_{\rm stall} = 2000$ steps while the face flux is not flat and has stopped falling (less than 1 percent drop from one monitor window to the next; a face spread still falling is a march still converging). The step cap `EXHALE_MAXSTEPS`, the hard cap `count_max` (10^6), and the steady gates where `Resid tol:` was stated can also stop those runs. In a physical trajectory, the settled, stall, and residual flags are reported at monitor steps and do not terminate integration. A trajectory ends at a step cap (`Max steps`, `count_max` or `EXHALE_MAXSTEPS`), or on a hard failure such as a nonfinite state or retry exhaustion. In v1.10, an enabled regional stabilization test also ends it on a pass or exhaustion. In an earlier `wasp_full` calculation, χ fell by a factor of 0.88 every 1000 steps with either WENO3 floor, corresponding to an e-folding of 16 windows and an estimated remaining change of about 16χ, or 2×10^{-3} at the threshold. In the same earlier calculation, marching `wasp_full` on from a face spread of 4.0×10^{-4} to 1.9×10^{-5} (27 500 further steps, an e-folding of about 9000 steps) changes ρ by at most 1.45×10^{-3}, v by 1.24×10^{-3}, T by 3.3×10^{-4} and $\log_{10}\dot{M}$ by less than 0.005. The environment variables `EXHALE_MARCH_FLUX_TH`, `EXHALE_MARCH_CHANGE_TH` and `EXHALE_MONITOR_STEPS` replace the two thresholds and $N_{\rm resid}$ for measurement. The marching stop is *not* a stationarity claim or a physical-time profile acceptance test: its face threshold is twenty times the steady gate, so the state it writes

is a relaxation snapshot. That state is certified like any other written state; when the certification refuses it, it is written `certified=F cert_reason=no_stationary_claim` and the run exits 0. Only the residual gate met while marching and a Newton finish (`steady_gates_converged` or `newton_finished`, `certification_note_stationarity_claim` in `EXHALE_main.f90`) claim a stationary state, and only they lead to status 2 on a refusal (Sec. 9.2; update log stage 3 section 135).

The default acceptance of physical-time marching in v1.00 and v1.10 is checked across every output snapshot spanning the final $K = 3$ flow-crossing times. The crossing time is calculated from the last snapshot, using the larger of the flow and acoustic times and, when the adiabatic index is not recorded, the more restrictive of the stated adiabatic-index bounds. Every physical cell must have a full-window range of at most 0.004 dex in each of T and ρ. In a cell with Mach number at least 0.01 throughout the window and no velocity sign change, the velocity range divided by the minimum $|v|$ must be at most 0.01. In the quasi-hydrostatic cells and near sign changes, the range is divided by the maximum speed in that region. The checker in `EXHALE_v1.10/LHS1140b/tools/marching_acceptance.py` also requires coverage and sufficient snapshot density; it reports the measured worst band and failing cells. These are finite-duration profile conditions, not a stationary residual test.

The cell-centered product of the averages,

$$\mathrm{du} = \frac{\max_{j\ge j_{\min}} F_j - \min_{j\ge j_{\min}} F_j}{|\overline{F}|}, \qquad F_j = \rho_j v_j r_j^2, \tag{127}$$

which the marching stopped on until then (`du_th`, 10^{-3}, the ATES criterion $\Delta\dot{M}/\dot{M} < 10^{-3}$), is still formed every step and printed as a diagnostic. It is not constant on a stationary discrete state: a certified stationary `wasp_full` state with a face spread of 5.3×10^{-13} has du = 3.6×10^{-3} on 500 cells and 1.7×10^{-3} on 1000, set by the two cells next to the outer ghost (`md/weno3_local_floor_marching_stall_diagnosis_20261009.md`; `md/outer_boundary_subsonic_review_20261009.md` reviews that boundary). Conversely, the old stop accepted the control march of `wasp_full` at step 12 627 with a face spread of 0.11, and a march of the same case with the same floor (secondary ionization held off) was still changing by 10 percent per 500 steps at step 12 000. The du-type criteria are kept where they decide a transition and not a claim: the PLM→WENO3 switch at the first `du_th` value (0.5, far above any stationary du), with its stall alternative. The JFNK hand-off read du $< 10^{-2}$ (`newton_du_switch`) until 2026-10-10; with the outer face at $2R_p$ the du of `wasp_full_newton` never fell below its floor of 1.04×10^{-2} and the hand-off came through its stall clause after 45 118 steps, so it now reads the face spread (`newton_flux_switch`, default 0.1, update log stage 3 section 131; written to `EXHALE_resolved.out` under that name). The second `du_th` value is read and no longer stops a run. A march from a certified stationary state leaves it at about 1.2×10^{-5} in face spread per step, beginning at the base cells, and settles near a face spread of 3–4 $\times 10^{-4}$ (MEASURED on `wasp_full`, 7000 steps): the fixed point of the operator-split march is not the root of the stationary residual to that level, so the stationary residual and the marching profile criterion must be reported as distinct measurements.

The largest one-step change of the conserved state over the escape window,

$$\mathrm{dtu} = \max_{k\in\{1,2,3\}} \max_{j_{\min}\le j\le N} \left| 1 - \frac{u_{k,j}^{n+1}}{u_{k,j}^{n}} \right|, \tag{128}$$

is printed as a diagnostic; it no longer stops a run. With `Solver: Newton [switch]` the marching is a warm-up only: the hand-off is armed the first time a monitor step sees the face spread at or above the switch, and fires out

of the PLM stage at a monitor step where it is below it, or where the spread has settled (the stall net) within five times the switch. The optional handoff is independent of physical-time profile acceptance. In physical mode, `Marching snapshot interval` selects an accepted-step count for separately archived intermediate snapshots; zero disables them. A snapshot has no stationary claim and carries `certified=F` with a `marching_snapshot_s<step>` reason. The final cap endpoint is written once as the final state, rather than duplicated in the archive. The older periodic write to the current output files remains a separate overwriting write path; it does not supply the immutable snapshot series used for profile acceptance.

8. Boundary and initial conditions

8.1 The lower boundary

The lower boundary is a characteristic condition imposed *at the base face* $r_{-1/2}$, with the number of conditions set by the sign and magnitude of the Mach number there (`base_boundary.f90`). At that face the eigenvalues are $v-c$, v and $v+c$, and a wave enters the domain when its eigenvalue is positive, the radial coordinate increasing outward:

$$\begin{array}{lll} 0 < v < c & \text{subsonic inflow} & 2 \text{ reservoir conditions, } 1 \text{ interior} \\ -c < v < 0 & \text{flow reversal} & 1 \text{ reservoir condition, } 2 \text{ interior} \\ v > c & \text{supersonic inflow} & 3 \text{ reservoir conditions, } 0 \text{ interior} \\ v < -c & \text{supersonic outflow} & 0 \text{ reservoir conditions, } 3 \text{ interior.} \end{array} \tag{129}$$

The interior relation is the linearized $v-c$ compatibility condition in its LODI form (Thompson, 1987; Poinsot & Lele, 1992),

$$p_b - \rho_i c_i v_b = p_i - \rho_i c_i v_i, \tag{130}$$

with the subscript i denoting the interior state continued to the face. The closed-form Riemann invariant $v - 2c/(\gamma-1)$ is deliberately not used: it assumes a single constant γ, and the caloric equation of state gives every cell its own $\gamma_{\rm eff}(x_{\rm H_2}, T)$, so that invariant is wrong wherever the H_2 fraction or the temperature varies across the boundary stencil, which is the whole molecular base. The sound speed c_i is the interior one, because the outgoing wave is an interior wave.

The reservoir is (p,s). The two reservoir conditions are the pressure and the specific entropy of the lower atmosphere at the base *level*, not the pressure and the density. The distinction matters as soon as the base composition changes: with (p,s) the temperature follows the composition, with (p,T) the entropy does. The entropy is carried implicitly, as the isentrope through $(p_{\rm base}, T_{\rm base})$ at the base composition, which is what `base.inp` and the lower-atmosphere profile reader state.

The face is the level. The level is $r = 1$, and every grid type places its first face exactly there, $r_{-1/2} = r_{\rm edg}(0) = 1$ (Sec. 7.1; `base_boundary_states` stops the run if it is not). The reservoir's (p,s) is therefore the state of the face side it owns, with no transport along any isentrope, and the contact between the reservoir and the domain neither moves with the cell width nor encloses a slab of reservoir gas; the two lower ghost cells lie below the level. Until 2026-09-27

the level was the center of a ghost cell, the face lay half a cell above it on the `Mixed` grid, and the reservoir was carried up to the face along its own hydrostatic isentrope. That put a slab of reservoir gas of thickness $\Delta r/2$ under the domain and made the base pressure of the domain $p_0[1-\Delta r/(2H_{\rm res})]$, a first-order error in Δr: MEASURED, the molecular lower layer of the LHS 1140 b He/H = 1.8 state sat 2.2 and 1.15 percent low in density on 500 and 1000 cells, and the pressure gap between the interior at the level and the reservoir, $2.93/1.44/0.70$ per cent on 500/1000/2000 cells, is $0.114/0.067/0.038$ percent with the face at the level in that historical v1.00 comparison (boundary model v4). These measurements do not establish the current v1.10 result. The wind quantities stay first order in Δr with errors 1.6–1.9 times smaller; the remaining first-order error is likely the thermal layer at the contact, which the grid does not resolve (cell-1 temperature 800/695/604 K against the 226 K reservoir), and two first-order terms were found and left: the element-diffusion Dirichlet He/H at the cell-1 center (Sec. 6.9) and the conduction bath at the ghost center $r(0)$ (`viscous_conduction.f90`).

The interior state at the face. The interior state (ρ_i, v_i, p_i) that (130) reads is built from the cell average of cell 1 alone (`interior_state_at_level_face`), never from the reconstruction stencil, because that stencil reaches into the ghost and a boundary condition that reads its own output is circular. How cell 1 is carried to the face depends on the reconstruction the boundary has to agree with.

With the well-balanced reconstruction (`Well balanced: True`, Sec. 2.6; boundary model `face_at_level_discrete_equilibrium_v7`) it is cell 1's own constant-density equilibrium at the face,

$$\rho_i = \rho_1, \qquad p_i = P_{\rm dn}(1) = p_1 + \rho_1\left[\Phi_c(1) - \Phi_i(0)\right], \tag{131}$$

the pressure the reconstruction itself extrapolates cell 1 to at the same face, formed by the same expression from the same cell averages and potential arrays, with $T_i = p_i/(\hat{n}_1\rho_1)$ at cell 1's particle count. The boundary and the reconstruction then measure the pressure of cell 1 at the face from one discrete equilibrium. That equilibrium is mechanical, the discrete form of $dp/dr = -\rho\, d\Phi/dr$, and says nothing about the temperature or the entropy (Käppeli & Mishra, 2016, Sect. 2.1); it holds for a column of any entropy stratification, and the reservoir is not changed by it: it still states the pressure and the entropy of the gas it supplies. The isentropic continuation below, which this replaces under the well-balanced option, differs from $P_{\rm dn}(1)$ by about $(1/2\gamma)(\Delta r/2H)^2$ of p, a truncation difference between two second-order constructions with no physics in it. MEASURED (`src/tests/grid_and_gates`, `base_reservoir_equilibrium` and `low_mach_face_flux`): on a static column built in the reconstruction's discrete equilibrium and meeting the reservoir's pressure at the level, that difference left a face-0 flux of 1.46×10^{-4} of $(\rho c)_{\rm res}$, $v_b = -2.6\times10^{-4}\,c_{\rm res}$ and a cell-1 momentum row of 3.1×10^{-3} of the weight, 1.47×10^{4} times the wind flux against a manufactured wind at Mach 10^{-8}; with (131) and the ghosts below, all three are at the rounding level (below 10^{-14}) on the isothermal column, on a column warming upward and on a column with a pressure-balanced contact at the level, and the face flux of the manufactured wind is within 2.1×10^{-3} (isothermal) and 2.5×10^{-3} (warming) of the wind at every Mach number from 10^{-3} to 10^{-9}, falling at first order with the base cell width.

Without it the reconstruction carries no discrete equilibrium to share, and cell 1 is continued along its own hydrostatic isentrope by `continue_hydrostatic_isentrope` (boundary model `face_at_level_v6`), which uses the fact that at

rest and at constant entropy the Bernoulli function is uniform,

$$u(T)+T-\frac{\Phi(r)}{\hat{n}}=\text{const} \quad\Longrightarrow\quad u(T_b)+T_b=u(T_a)+T_a-\frac{\Phi(r_b)-\Phi(r_a)}{\hat{n}}, \tag{132}$$

with u the internal energy per particle of the caloric equation of state and $\hat{n}$ the particles per unit mass, solved by Newton with $dh/dT=c_v+1$; the density then follows from $d\ln\rho=c_v\,d\ln T$ as $\rho_b=\rho_a(T_b/T_a)^{c_v(T_{\rm mid})}$. The midpoint heat capacity makes this exact to third order in $\ln(T_b/T_a)$, and the routine is only ever used over half a cell and over the two ghost cells, where that logarithm is below 0.1 and the density error is below 10^{-7} relative. It is not a general isentrope integrator over a scale height and is not used as one. The same routine gives, under both models, the temperature of the reservoir continued below the level, which the ghost cells carry. The conduction operator reads its bath at the ghost center $r(0)$ from `base_reservoir_temperature_at`: where a lower-atmosphere profile is handed over, the bath is the profile's own temperature at that radius (`lap_value_at_radius` in `lower_atmosphere_profile.f90`, linear in r between the bracketing levels), because the gas below the level is the profile's column and not an isentrope: on the energy-coupled LHS 1140 b profiles the column energy balance warms the 185 K stratosphere to $T_{\rm match}$ within about 0.1 scale height below the level (Sec. 6.7), and the isentrope through (p_b,T_b) stands about 100 K above that in the two ghost cells (417–422 against 253–327 K, MEASURED, `md/AN_matching_surface_diagnosis_20261007.md`); without a profile it is the isentropic continuation, and a reservoir that returns no positive temperature leaves the ghost's own (`conduction_base_level_T` in `viscous_conduction.f90`). With a profile the ghost cells of the advection carry the same temperature (next paragraph), so the boundary states one ghost temperature; without one they keep their own isentrope. The bath is the reservoir's at rest, at inflow and during a reverse flow alike, so the conduction boundary does not follow the direction the contact is upwinded on. The isentropic continuation balances an atmosphere that is isentropic between the level and the center of cell 1 and no other rest state exactly: hydrostatic balance does not imply constant entropy, and the continuation starts from a point state while cell 1 holds an average.
In both models the interior velocity is carried to the face at constant $\rho v r^2$, $v_i=\rho_1 v_1 r_1^2/(\rho_i r_b^2)$, rather than copied.

What is held, and the order of the three steps. The acoustic matching comes first: the pressure is a reservoir condition in every subsonic branch, $p_b=p_{\rm res}$, and the face velocity then follows from (130) as $v_b=v_i+(p_b-p_i)/(\rho_i c_i)$, so v_b is a *result* of the matching and does not depend on ρ_b. The direction of the contact is read off the matched state next, and only then is the contact upwinded: the entropy and composition of the face are the reservoir's where gas enters the domain and the interior's where it leaves,

$$\rho_b=(1-w_{\rm rev})\,\rho_{\rm res}+w_{\rm rev}\,\rho\big(s_i,p_b\big),\qquad w_{\rm rev}\in\{0,1\}. \tag{133}$$

The upwind choice is discrete and has no width. A contact carrying different entropies on its two sides has direction-dependent traces; a face density between them is an interpolation and not a law of thermal contact, and the average of the two isentropes that a symmetric handover returns at zero has no physics behind it. This is what distinguishes boundary model v2 from v1, whose entropy source was handed over by a cubic smoothstep in the face Mach number.
The interior's mass flux is carried at the face density (well-balanced model). Through a stationary face the mass flux is continuous: a contact cannot stand at a face gas flows through, so in a steady state the gas just inside the level has the density ρ_b of the face state the contact selects. Under the well-balanced model $\rho_i=\rho_1$ is the density of a mechanical

model of cell 1 and not of the gas at the face, so once ρ_b is chosen the advective part of the face velocity is the interior's mass flux at that density,

$$v_b = \frac{\rho_1 v_1 r_1^2}{\rho_b r_b^2} + \frac{p_b - p_i}{\rho_i c_i}, \tag{134}$$

and the face state carries the interior's own mass flux wherever the pressures balance. MEASURED on the manufactured winds above, dividing by ρ_1 instead made the base face carry 3.9 (isothermal) and 7.8 (warming) percent more than the wind at every Mach number, and the ghosts, which continue $\rho_b v_b r^2$, the same excess. The direction of the contact is read from the velocity at ρ_i; $\rho_1 v_1$ has the same sign at any density, so the two differ only where the two terms of v_b cancel to $O(\Delta r/H)$, a face that carries no resolved flow either way.

At rest and at inflow the reservoir owns the level. At a pressure-balanced contact at rest the two sides may carry two entropies and characteristic theory alone does not choose between them; this model chooses the reservoir, and the physical assumption is stated rather than carried implicitly: the lower atmosphere below the base level is dense and radiatively controlled, so on the time scales of a stationary solution it is a heat bath and a column standing at rest above it takes the level's own temperature. Its validity is a subsonic base at a level lying inside the radiatively controlled lower atmosphere, and it says nothing about a supersonic base or about a level above that region. Only a *reverse* flow hands the trace to the interior, and then the reservoir keeps exactly the one condition its single entering characteristic allows, the pressure.

Which quantity the direction of the contact is read from. The contact moves with the gas, so the side that owns the face is the side the level's mass flux comes from. The first interior cell's own product $\rho_1 v_1 r_1^2$ is not that flux at a base carrying a collocated odd-even velocity mode: MEASURED on the certified LHS 1140 b atomic state, the Riemann face flux is $+1.00000\,F_{\rm wind}$ at every face of the grid while the cell-centered product reads $-2.00\,F_{\rm wind}$ at cell 1, and the hot-Uranus base reads $-196\,F_{\rm wind}$ against a face flux of +0.98. The matched face velocity v_b is the face's own velocity and is what characteristic theory names, but it carries $(p_{\rm res} - p_i)/(\rho_i c_i)$, and a converged state leaves that pressure difference at about 10^{-4} of p, which at a base Mach number of 3×10^{-7} is a velocity two orders above the flow's own. The direction is therefore taken from the mass flux the state carries, read where the cell-centered product *is* a flux: the wind window $r \geq r_{\rm flux}$ the convergence gate is already defined on. The face flux itself is not available here, being produced by the Riemann solve this boundary feeds.

The window has standing only when it carries a flux at all and carries *one* flux, both properties of its own profile:

$$M_{\rm wind} = \frac{F_{\rm wind}}{\rho_i r_b^2 c_i}, \qquad d_{\rm window} = \frac{\sqrt{\langle F^2\rangle - \langle F\rangle^2}}{|\langle F\rangle|}, \qquad F = \rho v r^2, \tag{135}$$

the moments taken over the physical cells of the window. Where $|M_{\rm wind}| \geq 10^{-12}$ and $d_{\rm window} \leq 2 \times 10^{-3}$ the window states the direction; otherwise the matched face velocity does, and then a velocity below the rounding of the cancellation it is formed from, $4\epsilon[(|p_i| + |p_b|)/(\rho_i c_i) + |v_i|]$, counts as zero and the reservoir owns the level. A relative standard deviation is used rather than a range because the range is not a differentiable function of the state: its extrema are carried by particular cells, and MEASURED with one window cell swept through the argmax the two one-sided derivatives of the face density are −242.2 and +0.0128, a slope jump of 1.9×10^4 at a point where the value itself is continuous. The spread threshold is the geometric middle of the measured gap between the two populations, to one digit: 302 certified stationary states run from 4.06×10^{-5} to 1.82×10^{-4} and a stalled transient reads 1.90×10^{-4}, while the tightest state

that is not a wind reads 1.39×10^{-2}. The amplitude floor acts on no state the code meets: the smallest window Mach number measured anywhere is 3.7×10^{-9}, at a cold start.

The direction is held through each steady solve. The separation above is one of *written* states. The iterates of a steady solve are not written states, and they do cross the threshold. MEASURED on three stalled LHS 1140 b molecular continuations (the well-mixed He/H = 0.3169 and 0.3347 rungs and the $K_{zz} = 10^9$ He/H = 1.09 rung, stage exports of every pass): every solve that converged returned $d_{\mathrm{window}} = 8.4 \times 10^{-5}$ to 1.4×10^{-4} with the reservoir owning the level, and every solve that failed returned $d_{\mathrm{window}} = 2.000 \times 10^{-3}$ to 4.1×10^{-3} with the interior owning it, several at 2.000×10^{-3} to four digits. A solve entered at a written state is pushed across the threshold by its first steps, and to reach the root it has to cross it again through a residual that steps there; the outer passes then alternated between a converged and a failed solve for 60 to 80 passes. The upwind side of the contact is a property of a solution, so while a steady solve runs the side is the one the rule selects at the solve's first evaluation, its entry state, and it does not change inside the solve. The rule is applied again to the state the solve returns; where a converged solve returns a state whose rule selects the other side, one more solve from that state holds that side, so a root is accepted only on the side its own rule selects, the condition every written state was judged by before. Only the path to the root changes. The marching route and every evaluation outside a steady solve apply the rule at every evaluation, and `EXHALE_BASE_CONTACT_PER_EVALUATION=1` does so inside the solves as well, for a control experiment. On the well-mixed He/H = 0.3169 rung, which had alternated for 80 passes, holding the side certified the state in six passes.

Holding the incoming acoustic invariant $p_b + \rho c v_b = p_{\mathrm{res}}$ instead of the static pressure is present in the source as a weight that is zero. Both forms are exactly well balanced at rest; on a moving solution the invariant makes the static face pressure fall below the reservoir's by a fraction M of itself where a real reservoir's Bernoulli drop is a fraction M^2, which is why the weight is zero.

The supersonic branches. A supersonic *outflow* takes every characteristic out of the domain, so nothing may be prescribed and the whole face state passes to the interior. That transition is a change in the number of outgoing characteristics and its width is a declared numerical regularization, a C^1 cubic smoothstep in M_i of half width `base_face_mach_blend` $= 10^{-8}$ about $M_i = -1$; it is the module's one designed-in width and it does not touch the entropy source. A supersonic *inflow* would need a third reservoir condition which a (p, s) reservoir does not have, so the state is not a solution of a well-posed problem there; the run reports the crossing and does not invent a velocity. A compatibility relation is linear and far from a solution can ask for any velocity, so the face state is capped at Mach 5 and the events are counted.

The ghost cells. `base_ghost_averages` fills the two lower ghosts from the face state downward, with the velocity carrying a constant $\rho v r^2$ so the reconstruction sees no spurious velocity gradient. Without the well-balanced reconstruction they are the *volume averages* of the hydrostatic isentrope through the face state, continued below it, evaluated by 8-point Gauss-Legendre quadrature weighted by r^2. Where a lower-atmosphere profile is handed over, the gas below the level is the profile's column: each ghost carries the profile's temperature at its center, the temperature the conduction bath reads (`base_reservoir_temperature_at`), and the volume average of the pressure of the column at rest at that $T(r)$, $d \ln p / dr = -(d\Phi/dr)/(\hat{n} T(r))$, continued down from the face pressure (Simpson's rule in $1/T$ over 32 panels,

`hydrostatic_pressure_on_profile`), the density being $p/(\hat{n}T)$. With it they continue the reconstruction's discrete equilibrium below the level,

$$P_{\rm up}(0) = p_b, \qquad P_{\rm up}(-1) = P_{\rm dn}(0), \qquad p_j = \frac{P_{\rm up}(j)}{1-[\Phi_i(j)-\Phi_c(j)]/\chi_j}, \quad \rho_j = \frac{p_j}{\chi_j}, \tag{136}$$

each ghost keeping the ratio $\chi_j = p_j/\rho_j$ of the construction above, that is the temperature of the gas below the level (the reservoir-side isentrope's, or the profile's) at the particle count the reservoir states, so that only its pressure and density move, together, by a factor $1+O[(\Delta r/H)^2]$. The equilibrium mismatch of face 0 is then $P_{\rm dn}(1)-p_b$, the same pressure difference (130) closes, and it vanishes with the WENO3 departures of faces 0 and 1 (whose stencils read the two ghosts) and the jump of the face below the ghosts, whose state is $P_{\rm dn}(0)$ on both sides, on any column at rest in discrete equilibrium with the reservoir. A ghost wider than two scale heights of the gas it holds, $[\Phi_i(j)-\Phi_c(j)]/\chi_j \geq 1$, has no such equilibrium and keeps the isentropic average; the events are counted. Writing the face value into a ghost cell average instead of either construction is a half-cell error: the two differ by $\Delta r/2H$, which is first order in Δr in the state and *zeroth* order in the momentum residual it produces, an imbalance that no resolution removes. `Rec_BC` then puts the face state itself into the left slot of the base face; the right slot is the interior's reconstruction of cell 1 down to that face and is deliberately left alone, the Riemann solver being what mediates the two.

A column sampled from a continuum hydrostatic profile (an analytic isothermal or isentropic column as cell averages) is not a rest state of the discrete scheme: its cell averages lie $O[(\Delta r/H)^2]$ of p from the discrete equilibrium through the same reservoir pressure, and under the well-balanced model that distance appears at the base face as $P_{\rm dn}(1)-p_{\rm res}$ (MEASURED on the analytic isothermal column, face-0 flux 3.1×10^{-4} of $(\rho c)_{\rm res}$ on the coarsest base grid, falling at order 2.0). A seed has to be projected onto the discrete equilibrium for its lower layer to be at rest. The measurement key `EXHALE_SEED_HYDROSTATIC_CELLS=`K (off unless set; it requires `Well balanced: True`) does that for a seed at load (`seed_layer_in_discrete_equilibrium`, `init.f90`): holding p/ρ and the mass fractions of every cell, it sets cell 1 so that its constant-density equilibrium reaches the reservoir pressure at the level face, cells $2\ldots K$ so that the equilibrium mismatch of faces $1\ldots K-1$ vanishes, and scales every cell above K by the one factor that balances face K; a seed so projected is a new state, and its exact code-unit file is not read.

The base Mach number is recorded every step and a first crossing into supersonic inflow is warned about, as a validity check on the boundary that is independent of the chemistry (`Apply_BC.f90`).

The ghost composition is a fixed point of the sweep that produced it (boundary model v3). The boundary is built on the two lower ghost cells, whose composition is not prescribed but solved: the composition sweep is a map from the composition it is entered at to the composition it returns, the entry state setting the columns, the cell depths, the electron and particle counts, the secondary-ionization partition, the shielding and the ionization the base handoff's partition is stated against. A ghost that is merely one application of that map is therefore a boundary that depends on the composition the run happened to be entered at, and MEASURED it does: on one frozen physical column, reservoir, spectrum, options and executable, two entry compositions returned ghosts 4.7×10^{-3} to 5.5×10^{-2} apart in their trace ions, both accepted by every test the sweep applies at reaction residuals of about 3×10^{-26}, and the base continuity row of the first physical cell read 8.335×10^{-8} in one and 2.923×10^{-9} in the other, against the same tolerance. Neither state is a fixed point, and the difference does not close with the inner solve: over six decades of the cell solver's stopping accuracy the two states stay put to 10^{-15} while remaining that far apart.

Two statements close it, and together they are what distinguishes boundary model v3 from v2. *The seed is part of the model*: the ghost solve starts from the composition of the gas the reservoir holds at the base level, whose partition within each element is the first physical cell's, at the first sweep of a run, and from the ghost the previous sweep returned at every sweep after it. The first is the only seed that is a function of the physical column and the declared inputs alone, which is why the loader installs it in place of the ghost rows a restart pair carries. *The returned ghost is a fixed point of the map that produced it*: `ioniz_eq` applies its own map again at the ghost it returned, restoring the cells above the ghosts to the composition they were handed at each application, until the largest absolute change of any ghost species density, in the cell's own units, is at most 10^{-11} and the ghost's heavy-particle plus electron count moves by at most 10^{-6} of itself. A boundary that cannot reach that in twenty applications stops the run by name, with both moves and the accuracy, exactly as the ghost's imposed H_2 partition closure does. What the routine returns for the physical cells is still one application of one map, so the stationary residual is still one application of one operator.

The accuracy is set by the base continuity row, which is a near-cancellation of two face fluxes and therefore reads the ghost composition with a large gain: MEASURED, a ghost difference of 4.7×10^{-3} takes that row from 3.90 to 111.3 of its own rounding floors, so one floor of the row is bought by 4.4×10^{-5} of ghost composition and 10^{-11} is a quarter of one floor. **The norm is absolute and not per species.** The sweep pins the composition absolutely, in the cell's own units, and not stage by stage, and no quantity the boundary builds reads a trace stage against its own value: MEASURED on a molecular carrier configuration, a ghost settled into an exact two-cycle whose H I moved by 6.5×10^{-12} and whose H_3^+ moved by 2.2×10^{-12} of the cell's composition, worth 0.06 of one rounding floor of the base row, while against their own values those same moves read 4.1×10^{-9} and 5.1×10^{-3}. A relative statement cannot be made for a stage sitting at 4×10^{-10} of the cell. Species below 10^{-18} of the cell's own density are left out of the comparison. The bracket is stated rather than tuned: 10^{-11} sits a factor 1.5 above the lowest plateau measured, so a configuration whose composition solve plateaus higher stops the run, and the answer there is to measure that plateau.

The temperature leg closes with the composition. The ghost's heavy-particle plus electron count is what turns its temperature into the pressure `base_ghost_averages` builds and into the internal energy and sound speed the caloric map returns, so a ghost whose composition has stopped moving but whose count has not is a ghost whose rates belong to one gas and whose thermodynamic state to another; the count is measured across every application and the rates are rebuilt from the composition each application is entered at. Because the applications are *one* sweep, nothing that separates one sweep from the next may turn over between them; the frozen background's readiness is held at the value the sweep was entered with and takes the sweep's own value only when the sweep ends.

The contract covers the lower ghosts of a run whose network is molecular and carries helium, which is where the boundary's composition holds the trace ions the base row reads. A run outside it is marked reached before the first traversal and leaves after it, so not one number of such a run moves. The atomic ghost is not under the contract and its seed dependence is not measured. On the state that motivated the contract, one refused molecular generation, the evaluation now gives one answer whatever the ghost solve is started from: three seeds 9.7×10^{-1} apart certify with the same cell-1 row 2.923×10^{-9} and stand at most 8.0×10^{-11} apart over every column of both state files (MEASURED, item P1 step 2).

The model identity, the seed and what the fixed point reached are printed in the boundary report and the identity travels in every state file (Sec. 8.4).

8.2 The upper boundary

At an outflowing outer face the characteristics $v+c$ and v leave the domain; $v-c$ leaves too only where the outflow is supersonic. Where it is subsonic, $v-c$ enters and one condition has to be supplied from outside, and the ghost, a continuation of the interior, supplies it implicitly: at a subsonic outer edge the continuation is a physical outer condition that can set the interior mass flux, and its validity is a property of where the outer face is placed. The rule is stated once, for every reconstruction: which characteristics cross a face is a property of the flow and not of the scheme that reads the ghost (`Apply_BC.f90`). The run therefore reports the Mach number of the outermost cell with every $\dot{M}$ and, below 1.1, states that the outer ghost can set it (`EXHALE_main.f90`). MEASURED on `wasp_full_newton` (`md/outer_boundary_subsonic_review_20261009.md` section 5): with the default L_1 face at $1.574\,R_p$ (Mach 0.95, sonic point near $1.6\,R_p$), ρ and v differ from the solution with the face at $2\,R_p$ (Mach 1.50) by up to 15 and 17 percent near $1.31\,R_p$ and $\log_{10}\dot{M}$ by 0.01, five times the grid effect between the 2 and $3\,R_p$ domains, both supersonic. For a transonic wind the outer face is put past the sonic point (`Outer radius [R_p]:`); the WASP-121 b regression cases run at $2\,R_p$ (update log stage 3 section 127.6). The continuation (`free_outflow_ghost`) is the isothermal hydrostatic one,

$$\rho_g = \rho_N \exp\left[-\frac{\Phi_c(N+k)-\Phi_c(N)}{p_N/\rho_N}\right], \qquad p_g = p_N \exp[\cdots], \qquad v_g = v_N, \tag{137}$$

the state of rest the last physical cell carries outward under the code's own potential with the isothermal sound speed squared p/ρ held at cell N's value. Since ρ and p are scaled by the same factor, the ghost is at the *same* temperature as cell N, so the ionization sweep and the sound speed of the time step see the temperature they saw before, at the density the stratification gives. The absorbing columns do not read the ghosts at all: they start at the outer face $r_{\rm edg}(N)$ (Sec. 3.3).

A zero-gradient copy would assert $dp/dr = 0$ at the outer face, a hydrostatic imbalance of the order of the local weight that no resolution removes, and under PLM it is worse: the forward difference entering the limiter is then exactly zero, so the limited slope of cell N vanishes and that cell carries no pressure gradient at all. On an analytic hydrostatic column the momentum residual of cell N in units of its own weight converges at order 1.80 for (137), 1.42 for a linear extrapolation and not at all (0.006) for the zero-gradient copy. The velocity stays a copy: the outgoing characteristics carry it out of the domain and the boundary has nothing of its own to say about it; a gradient-carrying ghost velocity either drives the run into a base-breathing limit cycle or divides by a ghost density the stratification can make arbitrarily small. A ghost that leaves the admissible set falls back on the zero-gradient copy and is counted. The continuation neglects the inertial term in the ghost cells, which holds where $M^2 H/L_v \ll 1$ with L_v the length over which the velocity varies; a subsonic Mach number alone is not sufficient, and the incoming acoustic condition of a subsonic edge is not justified by this approximation, which is why the outer face is placed beyond the sonic point where a run can place it. No other treatment of a subsonic face is implemented (decision 2026-10-12): the escape models the code is compared with (Salz et al., 2016; Caldiroli et al., 2021) put the face past the sonic point and state no incoming condition, and a relaxed characteristic condition would make $\dot{M}$ depend on an exterior pressure that none of them states. Below Mach 1.1 the run labels its $\dot{M}$ as not quantitative.

A supersonic cell N: the steady-wind ghost. Where cell N flows out faster than its adiabatic sound speed, all three characteristics leave and the ghost is the gas the wind carries beyond the face, not a state at rest, and (137) is outside

its own validity condition. Past the ridge of the Roche potential it is also unbounded: Φ falls outward, the factor $\exp[-(\Phi_g-\Phi_N)/c^2]$ exceeds one, and the ghost becomes denser than cell N without bound (MEASURED with the binary of before 2026-10-11: 1.7×10^6 cm^{-3} in cell N against 3.4×10^{34} and 4.9×10^{63} in the ghosts, WASP-121 b at $12\,R_p$, step 0). The ghost of a supersonic cell N is therefore the steady isothermal wind through cell N at $c^2=p_N/\rho_N$: v_g is the supersonic root of (139) through (r_N,v_N), evaluated at r_g (`isothermal_wind_speed`, `grav_func`, the root solver the transonic initial condition uses), $\rho_g=\rho_N v_N r_N^2/(v_g r_g^2)$ and $p_g/\rho_g=c^2$, valid in any potential; where a steep rise of Φ leaves no supersonic root the ghost takes $v_g=c$. The face is supersonic, so the ghost enters only cell N's reconstruction slope. A subsonic cell N keeps (137) (update log stage 3 sections 127.4 and 138).

For the reconstructed face states, `Rec_BC` extrapolates outward with a zero-gradient copy under PLM and a linear extrapolation under WENO3. The diffusive operators carry zero flux through the outer face: heat conduction (`thermal_conduction_coeffs` forms no right-face term in cell N, Sec. 2.8), the binary element operator and the eddy part of the carrier and stage fluxes; the gas crosses that face only with the advective flux.

8.3 Initial conditions

`IC mode`: selects among `cold`, `transonic`, `hot_parker`, `auto` and `windae` (`input_read.f90`; an unknown word is warned about and taken as `cold`); the older `Transonic IC` and `Hot Parker IC` keys select the same families and take precedence over `auto`. `set_IC.f90` builds the first three, `auto` chooses between the first two, and `windae` is built in `init.f90`.

Cold hydrostatic (`cold`, the default). An isothermal hydrostatic atmosphere at the base temperature,

$$\rho(r)=\rho_{\rm bc}\exp\!\left[-b_{\rm eff}\big(\Phi_c(r)-\Phi(1)\big)\right],\qquad p=\frac{\hat{n}_{\rm bc}+\delta_{\rm bc}}{\rho_{\rm bc}}\,\rho,\qquad T=1, \tag{138}$$

with $\rho_{\rm bc}$ the density at the base level $r=1$ (the ghosts take the same stratification continued downward), $\delta_{\rm bc}=10^{-10}$ the hydrogen ionization fraction of the seed, whose electrons it adds, a small linear velocity seed $v=\frac{1}{2}(r-1)$ that vanishes at the level, and a nearly neutral composition. The factor $b_{\rm eff}$ starts at 1 and is raised in steps of 0.2 until the density at the cell nearest the middle of the domain falls below 10^{-8}, so that a low-gravity planet does not start with a domain at uniform density; the value used is reported, and densities below 10^{-8} are then floored there. This is the ATES initial condition. In a Roche domain carried past L1 the potential in (138) is its running maximum from the base outward: inside the ridge this is Φ_c itself, and past it the density is held at its value on the ridge (in practice the 10^{-8} floor) instead of rising as $\exp[+|\Delta\Phi|]$, which reached 5×10^{43} cm^{-3} at $15\,R_p$ for HD 209458 b before 2026-10-11 (update log stage 3 section 138).

Transonic (`transonic`). The steady isothermal-wind profile obtained from the algebraic Bernoulli integral of the isothermal wind equation in the code's own potential,

$$\tfrac{1}{2}v^2-c^2\ln v-2c^2\ln r+\Phi(r)=B, \tag{139}$$

with B fixed at the sonic point, where $v=c$ and $d\Phi/dr=2c^2/r$. The subsonic root is taken inside the sonic radius and the supersonic root outside it, and the density follows from $\rho v r^2=\dot{M}$ with $\dot{M}$ pinned to the base density. It exists only

where the potential admits an interior sonic point; where it does not, the cold hydrostatic profile is kept. It is needed for deep Roche-lobe-filling cases whose sonic point sits near L1 and which launch no wind from a hydrostatic start.

Hot Parker warm seed (`hot_parker`). Seeding the full isothermal Parker wind, its density as well as its velocity, destabilizes the run: the hot, low-density Parker column is globally inconsistent with the code's non-isothermal equilibrium and drains onto the base. The warm seed therefore keeps the stable cold hydrostatic *density* and overlays only a warm temperature and ionization ramp and the Parker *velocity* as a head start, both anchored to the cold static base through a cubic smoothstep window that is zero at $r = 1$ and unity above a stated transition radius. This removes the inverted base pressure gradient that otherwise drives an inflow and a thermal runaway.

Automatic selection (`auto`). One exact probe decides between the cold and the transonic family with no tunable threshold: does the *cold* base sound speed admit an interior sonic point in the actual potential? If it does the transonic profile is taken, which catches both the deep Roche-lobe cases, where the Roche $d\Phi/dr \to 0$ toward L1 always crosses the critical condition, and the low-gravity cases whose sonic radius falls inside the domain; if it does not, the cold hydrostatic profile is taken. The hot Parker seed is deliberately not auto-selected.

Wind-AE (`windae`). The ported Wind-AE solver in `src/modules/wind_ae/` is run in process to build a warm-start profile on the code's own grid, which is then loaded through the restart path.
Every family fills the whole species vector, and a molecular run seeds the H_2 column from the same two functions the inflowing ghost uses: the handoff value where a lower atmosphere states one (Sec. 6.7), and the chemical-equilibrium mixing ratio (93) at the local pressure of each cell of the seed and T_0 otherwise, clipped at the element-ratio ceiling $q_{\max} = 0.5/(0.5+y)$. The reason is that the optically thick base equilibrium is strongly molecular, and a zero-H_2 start makes the base composition solve land on the spurious atomic root.
Two of the families set `force_start`, which suppresses the convergence test for the first iterations: a transonic profile already carries $\rho v r^2 = \text{const}$, so the marching measure of (127) would be near zero at step 0 and the run would exit before the cold wind had heated to its steady state.

8.4 Restart

`Load IC? True` restores the state from `Hydro_ioniz_IC.txt` and `Ion_species_IC.txt` (`load_IC.f90`). Files carrying the `# columns` schema header are read by label, so every species the file holds, metal ionization stages included, is restored; a headerless legacy file takes the original fixed seven-column read and the metals are rebuilt from the abundances, which is reported. An element the file does not carry, or carries as identically zero, is initialized from its abundance exactly as a cold start does. An element whose reservoir a lower-atmosphere handoff states has its loaded column renormalized onto that reservoir by one factor common to all its ionization stages (a file with the metadata block arrives within the He/H tolerance below, its `# reservoir` line being compared first, so the factor is 1 for it; a state bound for another reservoir is carried there by `map_state_to_grid.py --reservoir` first). The file's own He/H ratio is compared against the input's to 10^{-6} relative (`heh_dev_tol`), and a state written at a different element ratio is rescaled or refused rather than silently adopted, the input file being the authority on composition.

The exact state. Beside the dimensional pair every written state has `output/conserved_state.txt` (`conserved_state_restart.f90`): the conserved variables u and every species fraction of every cell in code units, written to 17 significant digits, which read back as the same binary64 values. Read as `output/conserved_state_IC.txt` beside the two `_IC` files, it restores the physical cells and the upper ghosts at the bits the run ended on, where the dimensional route rebuilds $(\rho, \rho v, \mathcal{E})$ from 16-digit ρ, v, p with the energy reformed from p through the caloric equation of state. It is used only when every recorded condition matches the present run (format, N, N_g and the species count, the checksums of the two dimensional files it was written with, the reservoir and option fields, the boundary model, the species order, the normalization constants, the input files, the gravity and irradiation fields, the cell and face radii, and a checksum of its own data); any mismatch, a load that changed an option or migrated a legacy token, or a seed whose lower layer is projected leaves it unused with a printed notice, and the restart proceeds from the dimensional pair. The lower ghosts are derived again by the boundary in either case. A state whose bits matter, the input of a stationary evaluation in particular, is to be loaded with it: MEASURED, a certified state reloaded from the text pair alone moved its rounding-level mass row past its tolerance at one cell (3.316×10^{-12} against 3.0×10^{-12}), and reloaded exactly it certified (update log stage 3 section 83).

The intent of a restart. `Restart intent:` states what the loaded state is continued as, and is refused when no state is being loaded. `trajectory` continues the history, including its elapsed time, and is refused with `Run mode: init`; `relaxation` treats the state as a numerical iterate and marches toward stationarity; `stationary` measures the loaded state and enters the stationary solve from it at once (pseudo-time 1.0, no CFL step), and requires `Solver: Newton`. Absent the key, a physical run continues the trajectory and any other run relaxes. `stationary` takes a second word: `evaluate` certifies the stored pair, writes the products from it and stops without a step (Sec. 11.3), and `equilibrate` puts the loaded composition on its own fixed point before the solve. The stationary solve of a loaded state writes no `_adv` products (Sec. 11.4).

`Restart option change:` names the entries of the state file's option line that are allowed to differ from this run's (`load_IC.f90`, `opt_name`). A token that decides which species the state files carry (`metals`, `mol`, `oxychem`, `carrier`) is refused: naming it cannot make the file's rows this run's rows. `iontrans` and `he23strans` are not of that kind, since the stage and the 2^3S columns exist either way and the key decides whether they are a local root or a carried population, so they may be named. `carrier_newton` is a token of the route and not of the equations and may differ unnamed (the difference is reported and recorded as a `route_change` line). A token absent from a file written before it existed (`interdiff_enth`, `carrier_enth`, `lowE_partition`, `stage_enth`, `he23strans`) reads as `F`, the equations of the code that wrote it. The two tokens of a continuation factor, `cond` and `interdiff_enth`, carry a factor $s \neq 1$ as `ES23.16E3` (`T` at $s = 1$), so two factors are two states. A five-decimal (F7.5) token, written from 2026-09-29 until 2026-10-01, stands for the interval $[v - 5 \times 10^{-6}, v + 5 \times 10^{-6}]$ and not for one factor: a requested factor inside it loads the state as a seed with its certification claim dropped (`legacy_factor_token`), and a factor outside it is a change of the equations under the rule of every other token.

Every state file carries the lower boundary as well as the state: one line names the boundary model that built it and one states the prescribed reservoir with its version, $(p, T, \hat{n}, r_{\mathrm{level}})$ at round-trip precision. The reservoir is the only boundary input the physical column cannot reconstruct, which is why it travels; the ghost rows the file holds are the boundary's

own output and are rebuilt at the read from the column and this run's reservoir rather than adopted. A state written under an older boundary model is *reported* and loaded, not refused: the loader names the model that wrote it, names the model this run solves, and says that the boundary is rebuilt from the physical column and this run's reservoir and that its ghost composition is solved to this model's own seed and fixed point, so the state's residual is not the residual it was written with. A file written before the identity was recorded says so as well. A state written before 2026-09-27 (boundary model v3 or older) is nevertheless refused, by the grid check and not by the boundary model: every cell center has moved since the face was put on the level (by half a cell on the `Mixed` grid), and the loader compares the file's centers with the run's over the physical cells to 10^{-10} relative (`verify_restart_radii`, `load_IC.f90`); a non-finite center is a mismatch and is refused, not a match. A state on another grid enters a run only as an initialization seed mapped onto the new grid by `src/utils/map_state_to_grid.py` (`--ic` writes the `_IC` names), which writes `mode=init t_phys=0 certified=F cert_reason=mapped_seed` into its `# coupling` line, and the run has to solve it again. The mapper interpolates in $\ln r$ (ρ, p and the source rates in $\log_{10}$, v and T linearly) and carries each species as its ratio n_s/n_{H} to the hydrogen-nucleus density, so a ratio uniform in the source stays uniform to rounding; it is not a conservative remap. The first physical cells are mapped from physical cells: the source's lower ghost rows are the reservoir, boundary data that the loader does not read, so a target center between the base face and the source's first center (the first cell of a refined base block) takes the source's cell 1 values instead of an interpolation across the face against the reservoir row; the upper ghosts are the free-outflow continuation and stay interpolation partners. `--reservoir El/H v` multiplies every stage column of the element by one factor, the ratio of the new value to the source's `# reservoir` line, at fixed pressure (the density and the temperature follow the added nuclei). The hydrogen-nucleus density of every row is held: for helium the factor k multiplies $\mathrm{HeH^+}$ too, and the $(k-1)\,n(\mathrm{HeH^+})$ hydrogen nuclei that brings are taken out of H I, H II, $\mathrm{H_2}$, $\mathrm{H_2^+}$ and $\mathrm{H_3^+}$ by one factor $g = 1-(k-1)\,n(\mathrm{HeH^+})/n_{\mathrm{H}}^{\text{H-only}}$, so that every row carries k times its source He/H and its source ratio of every other element exactly. Left in, that hydrogen moves each element ratio of the row by $-(k-1)\,n(\mathrm{HeH^+})/n_{\mathrm{H}}$, 1.5×10^{-6} for $n(\mathrm{HeH^+})/n_{\mathrm{H}} = 2.0\times10^{-6}$ and $k = 1.75$ (the LHS 1140 b He/H ascent). The map is refused when the base rows do not come out at the new ratio within `BASE_ROW_RATIO_TOL` $= 10^{-6}$ relative, the tolerance of the loader that reads the file, which catches a species holding nuclei of two elements that the factor does not move (the oxygen carriers OH, $\mathrm{H_2O}$ and CO the tool does not multiply). `--extrapolate-beyond` continues a state above a stated radius at constant temperature, isothermal hydrostatic density, constant mass flux and frozen composition, a seed and not a solution of the steady equations.

A stationary run that alternates a wind solve with a composition relaxation also leaves a restartable pair of its last completed outer pass, the generation `output/pass_state/<id>/` that `output/pass_state/current` names (Sec. 9.6); copied to the `_IC` names it is loaded like any state file, and it is never a certified state. The periodic write of the marching loop (every 1000 steps) is written `certified=F` with `cert_reason=marching_snapshot_s<step>` for the same reason.

9. The stationary state

The time-dependent marching solution is the standard result. A stationary residual solve is an optional calculation that seeks a root of the discrete equations. A marching solution can satisfy the finite-duration profile criterion without a

stationary certificate; a JFNK solution that is not obtained does not by itself make that marching result fail. This section defines the stationary residual, its certificate, and the optional solvers.

9.1 The stationary residual

`assemble_residual` (`steady_residual.f90`) evaluates the right-hand side of one forward Euler step of the marching operator and subtracts the radiative source from its third row,

$$\begin{aligned} R_1 &= dF_1 - S_1, \\ R_2 &= dF_2 - S_2 - S_\mu, \\ R_3 &= dF_3 - S_3 - (\mathcal{H} - \Lambda) - S_\mathcal{E} + \nabla\cdot q_d + \nabla\cdot q_c, \end{aligned} \tag{140}$$

with $S_\mathcal{E}$ the viscous and conductive source (20) (the conductivity carrying its continuation factor `EXHALE_CONDUCTION_SCALE`), $\nabla\cdot q_d$ the divergence of the interdiffusion enthalpy flux of the diffusing elements and $\nabla\cdot q_c$ that of the transported carriers relative to their elements, the molecular carriers (104) and the carried ionization stages (105) together (`carrier_enthalpy_divergence`, registered by `carrier_set_init` in `diffusive_photochemistry.f90`), all of Sec. 6.10. Both are formed from the element, carrier and stage fluxes of the state itself and are identically zero, with no arithmetic, unless `Interdiffusion enthalpy flux` is on (its default) and `He_diffusion` moves the elements, a molecular carrier is transported, or `Ionization transport` carries the stages; the composition is therefore an argument of every assembly. dF and S are those of (6) to (11), evaluated through the same `reconstruction_continuation_rhs` used for the hydrodynamic stages. Sharing those spatial terms does not establish a common fixed point of the complete split march and the stationary system: composition elimination, transport and boundary updates must also agree, and a finite time step introduces a splitting error. A stationary certificate and a finite-duration marching profile test therefore measure different properties. The lowest ghost cell has no lower face and therefore no balance to state; its row is explicitly zero, so that no reader of R finds a heating rate standing alone in a row that has no fluxes.

A residual row is a dimensional quantity and has to be divided by something before it can be compared with a tolerance. Every row is divided by the largest term *that row itself contains* (`residual_row_scale`):

$$\begin{gathered} s_1(j) = \frac{\max\left(|F_{j-1/2}|A_-,\ |F_{j+1/2}|A_+\right)}{\Delta V_j}, \quad s_2(j) = \max\left(|\mathrm{ram}|,\ |\partial_r p|,\ \rho|\partial_r\Phi|,\ |S_\mu|\right), \\ s_3(j) = \max\left(|dF_3|,\ |S_3|,\ \mathcal{H},\ \Lambda,\ |S_\mathcal{E}|,\ |\nabla\cdot q_d + \nabla\cdot q_c|\right). \end{gathered} \tag{141}$$

Since ΔV_j is exactly the volume the flux difference was divided by, R_1/s_1 is the fractional change of the mass flux across the cell, and the mass row having no source, that fraction is the whole of what the row says. The momentum scale uses the three gathered physical terms of Sec. 2.3 and not dF_2 and S_2, for the reason given there.

A signal-speed bound $D_k(|v|+c_s)/\Delta r$ is not usable as the scale of these rows: nothing in any of them moves at c_s, so in a quasi-hydrostatic layer that bound stands far above the terms the row holds and R_1 divided by it is the fractional flux error per cell *times* the local Mach number, which at Mach 5×10^{-5} is a blindness of four decades.

These are the scales of the *measure* only. The Newton system and its merit keep the state scales D_k of `state_scales_of_cell`, so the merit and the acceptance test are deliberately not one expression apart.

9.2 The acceptance test

`steady_gates_met` is the only definition of an accepted steady state in the code: every route that may stop on "the equations are satisfied", the two stationary solvers and the marching loop under `Resid tol:`, calls it. Four gates:

1. the *residual gate*, $\|R\| < $ `resid_tol`, with $\|R\|$ the maximum over physical cells and rows of $|R_k(j)|/s_k(j)$;

2. the *flux gate*, the spread of the Riemann face mass flux $F_{f+1/2}r^2_{f+1/2}$ over $r \geq r_{\rm flux}$ below `flux_spread_th` (default 2×10^{-5} at $r_{\rm flux} = 1.2\,R_p$). A residual norm does not say this: a slowly varying flux has a small local derivative, and two states that differ by a factor 25 in the spread differ by only 1.9 to 5.7 in every residual norm tried. It is taken on the *face* flux and not on the cell-centered $\rho v r^2$, because the finite-volume update moves the face flux and the centered product is a reconstruction of the state;

3. the *chemistry gate*, no cell carrying a composition that is not a root of the chemical network, that is no cell in acceptance class 4 or 6 of Sec. 4.4: every kinetic row balanced to `ieq_res_tol` of the chemical turnover of its element group at the state, H_3^+ also to 10^{-3} of its own gross rate, a row whose Newton correction is below the resolution of the composition counted as balanced, and every constraint row met to its own tolerance (Eqs. 65 and 66). Heat and cool in the energy row are evaluated on the accepted composition, so on such a cell the row is a number computed on a state the chemistry does not describe;

4. the *carrier gate*, where $n(H_2)$ is among the unknowns: the carrier row balanced to `carrier_resid_th` of its own largest terms.

A caller that cannot supply the third makes no statement about the chemistry and the test rests on the others; the marching stop is such a caller.

9.3 Certification

The four gates above judge the hydrodynamic rows and at most one species row. A configuration that also carries an elemental transport balance, a level population or a set of eliminated species would reach its answer with nothing said about those equations. `certification.f90` closes that: it lists every equation the configuration makes independent and returns, for each, one of three statuses, *not applicable*, *evaluated* with its residual measured on *this* state, or *unavailable*. The third is the point of the construction: a routine that returns zeros when its frozen background is not ready is indistinguishable at the call site from a balanced equation, and an active equation with no evaluator returns nothing at all. Both now say so, and a state carrying either is written and not certified.

The inventory covers the three hydrodynamic rows, one balance for each transported carrier (the molecular carriers H_2, OH, H_2O and CO, and with `Ionization transport` the carried H II, He II and He III, with `He 2^3S transport` also He 2^3S), the ionization stage nucleus sums of H and He, the He/H partition and the trace elemental transport balances, the He 2^3S and H(n=2) level balances, and the eliminated-species closure of the coupled cell system. Where the metastable is carried its local level balance is *not applicable*: the sweep's row is then the constraint $x_3 = x_3^{\rm carried}$, and the level's equation is its transported balance, the entry `carrier balance He2^3S`. A carrier balance is measured as its steady residual over the sum of the row's own terms, the transport flux divergences and the reaction production

and loss (`carrier_residual`, `diffusive_photochemistry.f90`). The ionized stages (the proton and the two helium ions, in atomic and in molecular gas) carry their gross production and loss, formed from the gross rate G of their rows as $P = (G+f)/2$, $L = (G-f)/2$ (every term of those rows is sign-definite; the He↔H charge exchange is inside both G and f, the metal charge exchange added on its own sides): with the net alone the scale of the He^+ row held only the transport where He^+ is made by photoionization and destroyed by $He^+ + H_2$ at about $330\,cm^{-3}\,s^{-1}$ each (net 6×10^{-3}, LHS 1140 b route B, $1.2\,R_p$) and read a state balanced to 5×10^{-8} of its channels as 4.2×10^{-4} (MEASURED, update log stage 3 section 116). The He 2^3S row records the production $n_e(n_{\mathrm{HeII}}\alpha_{2^3\mathrm{S}} + n_{1^1\mathrm{S}}q_{13})$ and the loss $n_{2^3\mathrm{S}}[\Gamma_{2^3\mathrm{S}} + n_e(q_{31g}+q_{31a}+q_{31b}+\beta_{2^3\mathrm{S}}) + A_{31} + n_{\mathrm{HI}}Q_{31}]$ of (50) separately (in a molecular gas the loss adds the He(2^3S)+H_2 quench) (`tr_triplet_row_channels`, `ion_residual_core.f90`), because there the net is the difference of two channels that cancel to about 10^{-8} of themselves at 1.2–1.4 R_p on the LHS 1140 b states and is not one of the row's terms; scaled by the net, the row was asked for 4×10^{-13} of its channel rates and its measure wandered across 10^{-5} from pass to pass (MEASURED, update log stage 3 section 94). The molecular rows (H_2, the proton of a molecular gas, the oxygen carriers) are scaled by their gross channels by the same rule. The stage sums are the identity $\sum_k F_k(f) + F_{\mathrm{close}}(f) = N_{\mathrm{el}}(f)$ at every face, summed over all stages of one element, read from the stationary evaluation that formed the stage fluxes rather than re-formed, so the number the report carries is the one the rows were built with; one entry per element, because the identity is a statement about one element's own nucleus flux. It gates, at the floating-point bound of the identity: the identity is algebraic and has no truncation error, so the whole of what a tolerance on it can clear is the rounding of the sums, $2(5n_k+2)\epsilon g$ with g the ratio of magnitude sums measured at the face the measure was taken at, and at its algebraic ceiling ($3/2$ for one carried stage, 2 for two or more) where the operator reports no g. Constructions that break the identity on purpose stand ten decades above that bound and the states this code produces at 0.07 to 0.08 of it. Hydrogen carries one stage ($n_k = 1$), helium two, and three where `He 2^3S transport` carries the metastable as a further helium state. Every row measure is the *maximum* over physical cells of $|\mathrm{res}|/\max(\mathrm{scale}, \mathrm{floor})$, never a volume average: a front or a thin layer must not be averaged away by the wind that carries most of the volume. The volume average is reported beside it and is never decisive. The measurement has no side effects; every module array the carrier evaluator writes is held aside around the call, allocation status included, because a certification is a measurement of a state and not an operation on it.

Some entries are recorded beside the inventory and gate nothing: the mass closure $\sum_i f_i A_i - 1$ of the composition with its worst cell; with `Conduction` the conductive heat flux through the base face of the certified state (`heat_conducted_into_lower_atmosphere`, `viscous_conduction.f90`), written to `EXHALE_resolved.out` as `base_conductive_flux_cgs` with the bath and cell-1 temperatures it is formed from; with a lower-atmosphere profile the elemental face fluxes of the state and their medians and spreads over the overlap and steady-flux windows, written to the same record (`write_element_flux_profile`, `binary_element_diffusion.f90`); and with `He 2^3S transport` the divergence of the 19.82 eV the carried level moves relative to He I, against the energy row's largest term, which is chemical energy booked by the level's reactions and not a term of the energy row. These are the numbers the energy coupling of the elemental-flux closure reads, and they are measured on the state about to be written for that reason.

Exit status and acceptance. The run's exit status reports the certification of the state it wrote (`certification_stop_uncertified`, `certification.f90`): 0 when that state is certified or when the run made

no stationary claim (the marching stop of Sec. 7.7, whose state is a relaxation snapshot, a step cap, a stall, a NaN; the refused state is then written `certified=F cert_reason=no_stationary_claim`); 2 when the run declared a stationary state (the residual gate met while marching, or a Newton or steady finish with info = 0; `certification_note_stationarity_claim`) and the certification refuses it, after the state is written in full and the refusing rows named, the last line of the log being the certification's closing line (a gfortran run-time error also exits 2, without that line); 3 when the stationary solve of a steady route (a stationary restart or the direct steady route) returned info $\neq 0$, its last iterate written with `cert_reason=stationary_solve_failed` (`certification_note_stationary_solve_failed`; until 2026-10-10 that iterate was written as a relaxation snapshot and the run exited 0, update log stage 3 section 127.5). Certification is a diagnostic beside the default physical-time marching profile criterion of Sec. 7.7. The default criterion in both trees adds no stationary-root, drift-trend, slow-chemistry or observable-change gate. A stationary route still reports its own exit status; it does not decide whether a separately measured marching profile window passes. A caller must require complete outputs and distinguish that route's numerical failure from a completed diagnostic record.

A certificate belongs to the state written. What a certificate measures includes quantities *derived* from the conserved state and the composition: the temperature, the heating and cooling of that composition in the field it attenuates, and the composition of the two lower ghost cells, which no file stores and which the boundary derives. Taken from the iteration that produced the state, they depend on its history: the base continuity row reads the lower ghost with a gain of order $1/\mathrm{Mach}$, and on an LHS 1140 b state the ghost the last sweep of a solve had left (taken at the temperature of an earlier iterate) and the ghost the reload solved at the state's own temperature differed by 1.8×10^{-12} in temperature and moved the cell-1 mass row by 13 per cent of its anchor, across it (MEASURED, update log stage 3 section 94). `sources_and_ghosts_of_the_held_composition` (`EXHALE_main.f90`) therefore rebuilds them from the pair alone: the temperature from u through the caloric map of f_{sp}, and one sweep that holds the composition of every cell above the lower ghosts (`ioniz_eq_sources_of_held_composition`, `ionization_equilibrium.f90`), builds every rate from that composition, contracts heat and cool with its densities, hands the interior back bit for bit (asserted; the run stops otherwise) and solves the two lower ghosts. The certification of the state a steady route is about to write (`certify_the_state_to_be_written`), the joint test of every stationary outer pass (Sec. 9.6) and the certification of a loaded restart all stand on it, so a pass is accepted, and a file is certified, on the numbers its reload measures.

`Restart intent: stationary evaluate` certifies the *stored pair* in this way, before any composition sweep, and that certification is the verdict: the `certified=` field of every file the route writes, its exit status (0; 2 when the file claimed to be stationary and the stored pair is refused; 7 when the input is not refused and the advection-derived product is) and the `stored pair verdict` of the log. The route writes the stored pair and its derived products, then takes one composition sweep from the loaded state and reports its movement and its certification beside the verdict (`certification_reports_side_by_side`); the swept composition is never written and sets nothing. The verdict was a function of the iteration before this: a sweep taken before the certification moved the physical composition by 5×10^{-12} at most, and the mass row of a wind cell judged against its rounding anchor read 0.987 of its tolerance on the stored state and 1.054 after the sweep (MEASURED, LHS 1140 b He/H = 3.0, update log stage 3 section 88). `Restart intent: stationary` makes the same stored-pair certification as the answer to the file's claim and starts its solve from the swept

state.

The capacity of a report is the size of the inventory. A report holds `cert_max_entries` entries, and that number is not a round constant: it is formed from the counts that set it, three hydrodynamic rows plus one entry for each transported carrier, two stage nucleus sums, the He/H partition, one for each trace element, two level balances and one for each coupled cell system, which is $3+8+2+1+10+2+7 = 33$ today (eight carriers, ten trace elements, seven cell systems; `cert_max_entries` in `certification.f90`). It follows the inventory when a carrier, a trace element or a cell system is added; the eighth carrier, He 2^3S, raised it from 32. The capacity is exactly consumed, every entry being written for every configuration with the status *not applicable* where it does not apply: MEASURED over the regression logs of the 32-entry binary, before the metastable was a carrier, a configuration with the ionization stages transported and metals on printed `active equations 11 of 32 in the inventory`, so no free slot is left. The entry writers therefore address only the entry their own call created, and an entry that does not fit stops the run before any field is written, naming the equation that did not fit, the capacity and the entry that stands last and unchanged. Both halves matter. What an overfilled report costs is that a new equation's measure, tolerance and verdict are attached to the name of the equation before it, and a certification then states a verdict about a set of equations it does not name; raising the constant would leave that available one equation later.

The tolerances are one per kind of row, so that the report says which number refused a state: 3×10^{-12} for mass, 10^{-8} for momentum, 10^{-6} for energy, 10^{-5} for an elemental or carrier balance in the wind, and the composition solve's own root tolerance for the closure and the level balances. One number for the three hydrodynamic rows is not possible: on one converged state the three reach values seven decades apart and their one-ulp arithmetic floors two decades apart.

The v1.10 element and carrier verdict uses 10^{-5} in every applicable physical cell, retaining each row's absence mask. The radius windows and former restricted-cell comparisons are diagnostic only. A prescribed reservoir cell with a zero element residual does not independently test the boundary condition. The mass row's tolerance is a function of the cell,

$$\mathrm{tol}_1(j) = \max\Big[3 \times 10^{-12},\ \min\big(1,\ c_{\mathrm{round}}\, f(j)\big)\Big], \qquad c_{\mathrm{round}} = 10, \tag{142}$$

anchored on the rounding floor of that cell's own continuity row,

$$f(j) = \epsilon_{\mathrm{mach}}\, \frac{\max_{\mathrm{faces}}\big[\rho(|v|+c_s)A\big]}{\max_{\mathrm{faces}}|AF|}. \tag{143}$$

The reasoning is that each interface flux is assembled from two reconstructed states and from their jumps, and where the column is nearly hydrostatic those jumps stand many decades below the states, so a correctly rounded interface flux carries the last bit of the $O(\rho c_s)$ quantities the Riemann solve is built from and not the last bit of its own value ρv. The smallest flux difference the arithmetic can resolve is then $\epsilon\,\rho(|v|+c_s)A/\Delta V$, which is one ulp in the wind and about ϵ/Mach in a layer whose flux is a tiny residue of the momentum. Measured at the base of three reference configurations that floor spans four decades, 5.7×10^{-14} to 3.2×10^{-11}, so a single fixed number would hold one state to eight ulps of its own base layer and another to a tenth of one ulp. The margin $c_{\mathrm{round}} = 10$ stands a factor 3.7 above the largest measured ratio between the step the row takes when one ulp is added to every density and the estimate (143). The ceiling of unity is not an operative branch on any state measured: the row measure is the fractional change of the mass flux across a cell, so a state whose estimated floor reaches 1 is one whose mass row cannot be judged at all.

9.4 The Jacobian-free Newton-Krylov solve

The unknowns of the direct solve are the conserved state on the physical cells,

$$Y = \left(\rho_j,\ (\rho v)_j,\ \mathcal{E}_j,\ \{y_{s,j}\}\right)_{j=1}^{N}, \qquad n_{\rm var} = 3 + n_{\rm spec}, \tag{144}$$

where the optional species rows $y_{s,j}$ exist only where `Coupled carrier solve` puts the transported balances into the Newton system (Sec. 9.6): the elemental mass fractions of the diffusing elements and the transported molecular carriers, registered by `set_transported_species_rows` from the configuration. Every other solve has $n_{\rm var} = 3$. The carried ionization stages and He 2^3S are never Newton unknowns: a stage unknown is a fraction of one element's nuclei and not a species mass fraction, so `input_read.f90` refuses `Ionization transport` with a coupled carrier solve, and the direct steady route (`EXHALE_PTC=1`), whose last iteration would put the stages back on their local root, refuses it as well. The ghost cells are not unknowns: the boundary condition fills them from the interior at every residual evaluation. The ionization state and the temperature are eliminated *locally* inside the residual, that is, given Y the equilibrium sweep of Sec. 4.4 re-solves the composition and returns the consistent heating and cooling of Sec. 5., so that $F(Y)$ is a residual of size $n_{\rm var}N$ and not of the full chemical system. That makes F a function of Y only if the elimination has one fixed point for one Y, and it needs the rates of a cell to be smooth in the cell's own densities: the absorbed fraction $(1-e^{-\Delta\tau})/\Delta\tau$ of (34) in closed form is a rounding staircase at small $\Delta\tau$, and on the LHS 1140 b conduction states (cell depth $\Delta\tau \approx 1.5\times10^{-7}$ at the H I edge where $x(\mathrm{H}^+)$ crosses 0.5) the fixed point of the elimination lay on a step, the elimination alternated between two compositions 1.8×10^{-11} apart in $x(\mathrm{H}^+)$, and the energy rows stalled at 10^{-5}–10^{-4} against the tolerance 10^{-6}. With the series evaluation of Sec. 3.3 (one `one_minus_exp_over_x` in `utilities.f90`, a Taylor series below $x = 10^{-2}$, the code's one definition of the quotient) the same state certifies in one outer pass (MEASURED 2026-10-01, `md/Update_EXHALE_stage3.md` section 58).

The solve is a Jacobian-free Newton-Krylov iteration in the sense of Knoll & Keyes (2004): the Krylov method needs only the action of the Jacobian on a vector, which is taken as a forward difference of the residual, so the nonlocal radiative coupling never has to be assembled.

The outer iteration of `solve_steady_jfnk` is, per step: evaluate the full residual F at the iterate; build the banded finite-difference Jacobian of the *frozen-radiation* residual as a preconditioner; solve $(I/\Delta\tau + J)\,\delta Y = -F$ by right-preconditioned GMRES with the matrix-vector product taken matrix free, so that the nonlocal radiative coupling is captured; take a step along a controlled direction; and ramp $\Delta\tau$.

The banded preconditioner. WENO3 reaches two cells on either side and within a cell every variable couples, so in the flat ordering $Y(n_{\rm var}(j-1)+k)$ a column couples to rows within $|\mathrm{row}-\mathrm{col}| \le 3n_{\rm var}-1$; the half-bandwidths are that number and a graph coloring with stride $n_{\rm color} = k_l + k_u + 1 = 6n_{\rm var}-1$ makes the row supports of same-color columns disjoint, so that the frozen-radiation residual, which is strictly local, is captured exactly by that many finite-difference probes. For $n_{\rm var} = 3$ this is 8 and 17. The band is factored by LAPACK `dgbtrf` and applied as the right preconditioner of the Krylov cycle. The finite-difference and Krylov probe steps are taken in the diagonally scaled variables of `cell_state_scales`, the characteristic magnitude of each unknown in its own cell; a probe point that leaves the set of states the code can describe, a chemical balance row going to `NaN` or overflowing, is answered by halving the step and sampling closer to the iterate, at most six times, and never by accepting a non-finite residual or

inventing a derivative.

The step control. Where the system carries no species row the step is the Krylov direction with a backtracking line search on the merit $\|F/D\|_2$, with a Grippo non-monotone reference, the worst true merit of the last five iterates, so that sideways steps are admissible and the solve can leave a bad neighborhood. Where the system carries a species row the step control is a scaled trust region with a Powell dogleg: the path runs from the origin to the Cauchy point s_U, the minimizer of the linear model along the steepest descent direction, and from there to the Krylov point s_N, and the step is where that path first meets the ball of radius Δ,

$$s = a_U s_U + a_N s_N, \qquad \begin{cases} a_U = 0,\ a_N = 1 & \|s_N\| \le \Delta, \\ a_U = \Delta/\|s_U\|,\ a_N = 0 & \|s_U\| \ge \Delta, \\ a_U = 1-\tau,\ a_N = \tau & \text{otherwise}, \end{cases} \tag{145}$$

with τ the root of $\|s_U + \tau(s_N - s_U)\| = \Delta$ in $[0,1]$. Returning the two coefficients rather than the vector is deliberate: the image As follows from As_U and As_N by linearity and costs no further matrix-free product. The Krylov leg is solved without the pseudo-transient shift, $As_N = -r_0$ rather than $(I/\Delta\tau + A)s_N = -r_0$, because the dogleg falls monotonically along its path only when that leg is the model's own minimizer. A species unknown also carries a box, the fraction-to-the-boundary rule holding the step inside the physical range of a mass fraction, and an element with no reservoir is excluded from the registry altogether: its unknown would sit exactly on its own lower bound with no room below, and the boundary rule then returns zero for any direction with a component pointing there, so the whole Krylov direction would be refused.

A trial that is not a gas is refused before anything is evaluated. Every trial, probe and Jacobian column the solve evaluates is first screened cell by cell on its conserved state: finite, $\rho > 0$ and $\rho e = E - (\rho v)^2/(2\rho) > 0$, with the one definition the marching route and the restart reader use (`admissible_conserved_cell`, which also requires the internal energy formed exactly as the conversion to primitives forms it to be positive, so that the two cannot disagree within rounding). A cell that fails has no temperature, so `eval_residual` refuses the trial right after unpacking it, before the equation of state, the base boundary state or any rate coefficient is evaluated, as the named refusal "conserved state that is not a gas" with its own counter in the solve's refusal census; the temperature is never clipped. A refused trial has no residual: the residual vector is a quiet NaN, the radiative pair and the particle count are zero, the composition is the seed, and the products a discarded evaluation would restore are restored; every caller that asks for admissibility reads the flag before any of these, and the merit of a refused trial is not formed. An evaluation that does not ask is an evaluation of the iterate, which is admissible by construction: each solve screens the state it is entered at and returns at once as not converged when that state is not a gas, and adopts only admitted trials, so meeting such an iterate stops the run with the condition named. A finite-difference column whose forward probe is refused at every length (the base point within the shortest probe of the boundary of the gas, e.g. a cell whose internal energy is below about $2^{-6}\sqrt{\epsilon_{\text{mach}}}$ of its kinetic energy, probed along its momentum) is sampled on its backward side, $[F(Y) - F(Y - h\,e)]/h$, which is the same derivative to first order, exactly as the Jacobian-vector products already were; a column refused on both sides stays unresolved and is counted. MEASURED, the open-face LHS 1140 b route B closure before this screen: a

trial reached the composition sweep with cell 2 at $T = -1.4 \times 10^{-5}$ K and stopped the run inside the charge-exchange turnover, whose "thread holds no cell" state was a temperature sentinel (`md/osc_diagnosis_20261008.md` section 5); that state is now its own flag, set with the temperature on every thread, every caller hands the turnover bound its cell's temperature, and a cell at a non-positive temperature is stopped where its rate coefficients would be formed (update log stage 3 section 125).

The ledger and the stop. The best iterate reached so far is kept, with its composition, its flux spread and whether it met the gates. The non-monotone search accepts sideways steps, which is what lets the solve leave a bad neighborhood and also what lets it circle, so when the best iterate has not improved for 20 outer iterations the solve returns to it and switches to a monotone Armijo acceptance, and if that buys nothing in another 20 it stops and reports that it stagnated, a different statement from "no descent direction exists". The loop-top stop requires the certified rows of Sec. 9.3 and not only the merit: the two are different functionals, and a window on the merit alone bounds nothing about the quantity the state is judged by.

A measurement option, off by default, changes what accepts a pseudo-transient step on the line-search branch: with `EXHALE_PTC_SER=1|2` a trial is accepted while $\|R\| \geq 10^{-3}$ (`EXHALE_PTC_SER_RNORM`) when it is admissible and no cell's density or total energy moves by more than 0.3 of itself (`EXHALE_PTC_SER_MAX_CHANGE`), no merit descent being asked, and $\Delta\tau$ follows the switched-evolution-relaxation ratio (`=1`) or doubles on a full step (`=2`); the returns to the best iterate and the stagnation stop are not taken, and the solve hands back its last iterate unless the best one met the gates. MEASURED on two LHS 1140 b atomic He/H ≈ 9.7 continuation rungs that the merit line search returned unmoved, `=2` reached info = 0 in 22 and 28 iterations (update log stage 3 section 42); it is what the continuation chains of that planet run with. The return flag is 0 when the returned state met the acceptance gate and every judged row, 1 at the iteration limit, 2 when the solve ended without a state meeting the gate, and 3 at an iterate where no feasible descent direction exists. A flag of 2 is not a solver failure: the solve converged and the certification refused the state it hands back, naming the row, and that state is written with the refusal recorded rather than marched on.

9.5 Pseudo-transient continuation

`solve_steady_ptc` is the same residual solved with the banded Jacobian used as the *operator* and not only as a preconditioner: per iteration $(I/\Delta\tau + J)\delta Y = -F$ is solved by LAPACK banded LU, the step is taken with a backtracking line search and a positivity test, and $\Delta\tau$ follows the switched-evolution-relaxation ramp $\Delta\tau \leftarrow \Delta\tau\, \|R\|_{\mathrm{old}}/\|R\|_{\mathrm{new}}$, cut on a failed step. As $\Delta\tau \to \infty$ this is Newton's method; at small $\Delta\tau$ it behaves like explicit relaxation, which is the robust startup the continuation exists for. The route is not available with a carrier unknown: it builds its Jacobian from the frozen residual, which by construction performs no equilibrium sweep and therefore has no carrier row, and giving it one would mean assembling that row from a background nothing in the routine refreshes. The caller is told so rather than being handed a silently three-row answer.

The starting $\Delta\tau$ matters: the marching handoff and the stationary restart (`Restart intent: stationary`) start the continuation from $\Delta\tau_0 = 1$, a reloaded state being already relaxed, and the direct steady route (`EXHALE_PTC=1`) from the CFL step; `EXHALE_PTC_DTAU0` replaces the start for measurement.

9.6 The partitioned outer iteration

A species the run moves by an operator-split transport step instead of by a Newton row, which is the diffusing elements and the molecular carriers whenever `Coupled carrier solve` is off and the carried ionization stages and He 2^3S always, has a stationary balance no hydrodynamic solve touches; a single steady solve would freeze it at whatever state it was handed. `steady_wind_with_element_diffusion` in `EXHALE_main.f90` alternates the two: solve the wind at fixed composition, relax the composition to its own steady state in that wind, solve again. "Fixed composition" means the *transported* part of it: the element fractions the diffusion operator moves, the carrier densities and the carried stage fractions, which every equilibrium sweep of the hydrodynamic solve is handed as imposed values. The local part, the ionization state of every cell and the molecular reaction balance at the carriers' densities, is resolved inside every evaluation of the hydrodynamic residual (Sec. 9.1), so the wind solve carries its own chemistry and only the transport of composition is alternated. A run with neither element diffusion nor carrier transport therefore has one outer pass: the wind solve alone. The composition relaxations are the fixed-wind element relaxation of `binary_element_diffusion.f90` (Sec. 6.9), under-relaxed with a factor ω starting at 0.5, and the fixed-wind carrier relaxation of `diffusive_photochemistry.f90` (Sec. 6.5), which transports the carried stages as well, bounded by a movement trust rather than damped. That relaxation stops at the fixed point of the carrier block at the held wind, and the fixed point is judged carrier by carrier: the largest change of each carrier over one full-length step against the largest fraction *that* carrier holds (`carrier_displacement_on_own_scale`), below 10^{-10} (`relax_tol`), and the steady carrier residual of the present rows on the certification's measure at or below 10^{-8} (`carrier_accept_tol`). The carriers span ten decades, He 2^3S being 5×10^{-11} of the helium nuclei at $1.2\,R_p$ on the LHS 1140 b states, and against the largest fraction of all carriers a step that moved the level by 20 percent of itself read 10^{-11} and closed the block after one step with its row at 9×10^{-5} of its own terms; a displacement alone does not bound the steady residual either, an implicit step's residual being bounded only by $n/\Delta t$ (MEASURED, update log stage 3 section 94). The advecting flux handed to the element operator is the face mass flux of the state just solved, read from the row terms that state assembled, so the operator is given its wind and does not reach into the residual for one.

What is accepted is only a state on which the certification of the *full* set of active equations passes, evaluated on the refreshed state the routine would hand back. A hydrodynamic solve returning 0 while an alternated species row refuses is not an accepted state: that flag is a statement about the rows in the Newton registry, and the alternated rows stand outside it by construction.

A failed hydrodynamic solve hands on nothing. Its returned state is an iterate and not a wind, so neither the next composition update nor the next solve is taken from it. The last converged wind of the alternation is defined as the conserved state (ghosts included) and the species fractions of the most recent pass whose solve converged, taken where that pass's joint test takes them, with the frozen carrier background and the acceptance record of that solve (its residual norm and flux spread) beside them; every other quantity the loop reads (primitive state, temperature, excited hydrogen, the radiative heating and cooling and the sweep ledger, the ghosts) is rebuilt from that pair at held composition. The wind is a solution *for its composition*, so the two are restored together: an old wind is never given a newer composition and called its solution. The boundary inputs do not change inside one call of the alternation, and the record does not outlive it. After a failed solve the pair is restored, the pass keeps its failure flag, and the next

composition update is taken from the restored wind with a shorter step; where no step length of that update can be shortened the alternation ends, because the next pass would repeat the failed one. Where no solve of the alternation has converged yet there is no wind to restore and no composition update has been consumed, so the solve is repeated once from its own iterate (the same equations; nothing is relabeled), and if that fails as well the alternation ends with that iterate and its flag. MEASURED before this rule, on the open-face LHS 1140 b route B closure: the failed pass-3 iterate, handed on, carried base cells at 2929, 46 and 1181 K into the pass-4 solve, which evaluated the trial at -1.4×10^{-5} K (`md/osc_diagnosis_20261008.md` sections 2 and 5).

An open base face starts from a seed that runs continuously from its ghost. With the H_2 partition of the lower ghosts stated (the open face of the matching surface), a seed produced under another boundary carries another hydrogen partition in its base cells (route B: $n(\mathrm{HI})/n_{\mathrm{tot}} = 0.11$ in cell 1 against 10^{-5} in the ghost). A step in the partition, wherever it stands, drives an H_2 diffusive flux whose relative-transport enthalpy is an energy sink no stationary composition carries, and a hydrodynamic solve at that frozen composition balances it by cooling and compressing a cell. MEASURED both ways: with no seed rule the step is at face 0 and cell 1 falls to 57 K (route B, sink 19.67 and heat 19.31 code units; `md/cold_base_cell_diagnosis_20261009.md`) or 83 K (LHS 1140 b He/H 1.2); with the ghost's ratio given to the whole uniform base region (1..N_{low}, the rule of update log stage 3 section 126 until 2026-10-28) the step is at cells $N_{\mathrm{low}}/N_{\mathrm{low}}+1$ and cell 51 falls to 20–55 K. So on the stationary restart route the seed's ratio is a geometric blend over the base region (section 161):

$$\ln r_j^{\mathrm{seed}} = \ln r_g + w_j(\ln r_j^{\mathrm{loaded}} - \ln r_g), \qquad w_j = \frac{j}{N_{\mathrm{low}}+1},$$

r_g the $n(\mathrm{HI})/n(\mathrm{H_2})$ of ghost 0, each cell keeping its hydrogen nuclei and its mass ($f(\mathrm{HI})+2f(\mathrm{H_2})$ held, f a number per unit mass); a cell with no H I or no H_2 keeps its loaded values, and a marching run is not changed, since it evolves the seed in time. MEASURED (section 161): the cold cell at the top of the base region is gone (pass 1: route B 20.2 K $\rightarrow$ 207.5 K, He/H 1.2 29.6 K $\rightarrow$ 229.9 K), but a broad cool layer remains below it (He/H 1.2 final state 360–400 K over cells 45–56, against a seed whose coolest cell is 501 K), because the blend still changes the ratio by many decades over the 50 cells. A carrier relaxation before the first solve, added with the section 126 seed, was removed (section 127.5): that seed put all the hydrogen of the seeded cells into H_2, so a carrier relaxation at frozen element densities asked for up to 1.31 times a cell's hydrogen and the element limiter refused every entry.

The progress control. A pass makes progress when the joint distance of the state to the certification falls, or, where that distance is carried by a row that is not hydrodynamic, when the composition's own residual falls; three consecutive passes without progress end the loop rather than spend the budget (20 passes, `EXHALE_OUTER_PASSES`) on a fixed point the alternation is not approaching. The endings are named: the joint certification passes; a row of the state is not finite; the element composition update is refused; the joint measure stops falling; the pass budget ends uncertified; no solve has converged (twice from the entry composition); a failed update cannot be shortened.

The two step lengths of the composition half, the element under-relaxation ω (start 0.5, `EXHALE_DIFF_OMEGA`; floor 0.125; it doubles back toward its start and never above it) and the carrier movement bound, the largest relative change of one cell's particle count $n_{\mathrm{tot}}+n_e$ one pass may make (start 10^{-2}, `EXHALE_CARRIER_TRUST`; floor 10^{-3}; ceiling 0.5), answer to the hydrodynamic solve that consumes the update (revised 2026-09-27). *A pass whose carrier update ended*

on its movement bound is not a pass without progress as long as the solve that consumed it converged and the joint distance rose by no more than a factor 1.1: the state moved by the whole bound, and the relative measures cannot show it because residual and scale fall together. MEASURED on the LHS 1140 b He/H = 1.8 state with transported ionization stages, every pass ended on the bound while the gated measure stood at 2.17 to 2.20×10^{-2}; the earlier control read that as stagnation, halved the bound to its floor and ended the loop, while the same state with the bound held at 0.5 certified in 14 passes. Such a pass is neither counted toward the ending nor allowed to shorten a step, and the bound doubles, up to 0.5. The bound halves when a bound-limited update raised the joint distance of a solved wind by more than 1.1, when it preceded a failed solve (the update is retried from the restored wind, so whether or not it ended on the bound), and when a pass without progress moved the carriers short of the bound on a solved wind; a relaxation that kept no step leaves it where it is. ω halves on a pass without progress whose wind was solved and on a failed solve that consumed an element update, and doubles back toward its starting value after a converged solve on a pass with progress. The element half is often the pace of the alternation: MEASURED on certified LHS 1140 b conduction rungs, the elemental He/H row fell by 0.505 a pass, which is $1-\omega$, while at `EXHALE_DIFF_OMEGA=1.0` it fell by about a decade a pass and three rungs certified in 6 or 7 passes instead of 21 or 22 or a refusal (update log stage 3 section 44); the continuation chains of that planet run at $\omega = 1$, and the default is unchanged. Since a failed solve restores the last converged wind, every failed solve after the first converged one started from an update of a solved wind, and that update is what is shortened. `EXHALE_CARRIER_TRUST_HOLD` freezes the bound for every pass. Recorded as open (`md/TO_BE_DONE.md` item (AL)): the starting bound for transported stages (the state above certifies from 0.5 and not from 10^{-2}), and the molecular He/H = 0.083 continuation, where a growing bound makes the H_2 row of the outermost cells overshoot and neither control certifies.

The state of every pass, kept. Every pass after which the iteration goes on publishes the state the next pass starts from, rebuilt from (u, f_{sp}) in the order of the checkpoint restoration with every side effect on the caloric mixture undone, as one generation `output/pass_state/<id>/` (`Hydro_ioniz.txt`, `Ion_species.txt`, `manifest.txt`) written by the writers of the final state, with `<id>` = `r<run identity>_p<n>`, so a run stopped from outside keeps its last completed pass. The pair carries `certified=F cert_reason=pass_snapshot_p<n>` in both halves, because the certification of a pass is taken before its composition update and the snapshot is the state after it, and a `# pass_snapshot` line with the movement bound and ω of the next pass, which a restart does not inherit. The generation is written in a temporary directory, renamed onto its name, and published by renaming a one-line file onto `output/pass_state/current`; an interruption at any point therefore leaves `current` naming a complete generation. Both halves state `state_id=<id>` on their coupling line, and the loader refuses a pair whose halves state two identities or only one. Nothing the run reads afterwards changes: the final state files of a run are bitwise those of the binary before the snapshot existed.

Two rules keep the state self-consistent. The progress control stands *ahead* of the composition update, so the pass that ends the iteration takes no update and what is handed back is the very state the certification of that pass was evaluated on; and the last pass of the budget takes no update either, an update existing only to be consumed by the next hydrodynamic solve. Taking it would replace a stationary wind by a state whose hydrodynamic rows the update itself spoiled, by eleven decades in the mass row on a molecular configuration.

With `Coupled carrier solve: True` the alternation does not arise for the carriers at all: $n(H_2)$ becomes a fourth

Newton unknown per cell, one solve returns a wind and a carrier partition that are steady states of each other, and the outer loop runs once. A third value, `On stall` (2026-09-17), runs the alternation and hands the state to that block at the pass where the alternation gives up: the joint distance has not fallen in three consecutive passes AND the carrier relaxation of this pass and of the two before it ended on the movement bound. What is known about the two routes where they have been compared: on a certified molecular state the alternation certifies at outer pass 12 while the block moves its residual by 0.6 percent in the same wall clock, every one of its linear solves ending with the Krylov subspace exhausted, so the block as it stands is limited by its preconditioner and not by the species box.

What the alternation cannot do, measured. The movement bound on the carrier relaxation is not what holds the refusing molecular states: relocating the cells that control it only moves the control to the edge of whatever region is exempted, and the refusing row is worse afterwards. What the measurements point at is the splitting itself. The bound exists because the wind is held while the carriers relax, and a composition good enough for the carrier rows is already outside the range in which holding the wind is admissible: the one relaxation that reached the carriers' own fixed point moved the base particle count almost seven times the bound, and the hydrodynamic solve that had to follow it did not converge, its mass rows outside tolerance in every cell over a hundred iterations. With no alternated species at all, which is what an atomic run and a fully coupled run both give, the body is one steady solve followed by the state refresh and the flag is the solve's own.

10. v1.10 regional stabilization and stationary closure

This section describes the regional diagnostic and the lower-boundary thermochemical closure in v1.10. The regional diagnostic, the physical-time profile criterion, and stationary certification answer separate questions.

10.1 Finite-duration regional condition

Let $q_j(t)$ be a physical-cell quantity, assigned to a lower, transition, or upper radial region. The monitor partitions accepted physical time into windows of length W. It forms a time average $\overline{q}_{j,k}$ in window k by linear interpolation between accepted endpoint states, splitting intervals exactly at window boundaries. The first complete window fixes the reference $\overline{q}_{j,1}$ for that candidate. The check includes the largest deviation of a window mean from this reference, the change between neighboring window means, and the span of all window means:

$$D_j = \max\left\{ \max_k \frac{|\overline{q}_{j,k} - \overline{q}_{j,1}|}{s_j}, \max_{k>1} \frac{|\overline{q}_{j,k} - \overline{q}_{j,k-1}|}{s_j}, \frac{\max_k \overline{q}_{j,k} - \min_k \overline{q}_{j,k}}{s_j} \right\}.$$

The separate full-range check is

$$A_j = \frac{\max_{t\in\text{candidate}} q_j(t) - \min_{t\in\text{candidate}} q_j(t)}{s_j}.$$

These tests are applied to density, velocity, pressure, temperature, electron and total particle number density, and the atomic species populations. Positive fluid quantities and regional inventories use their first-window means as scales; velocity uses an initial sound-speed scale, and species populations use an initial hydrogen-nucleus density scale. Therefore the numerical limits are not relative error bounds on a rare population or a near-zero velocity.

The regional inventories are total mass and *internal* energy over the physical cells of each region; internal energy subtracts kinetic energy from the conserved total energy. The mass-flow quantity is the actual numerical face mass flux at the upper face of each region, with the accepted Runge–Kutta stage weights $1/6$, $1/6$, and $2/3$. Its window-mean change and span are checked, while its instantaneous range is recorded but not used as a stop condition. A zero reference flow uses an acoustic-flow scale at machine precision to keep the denominator nonzero.

The optional regional diagnostic uses $D_j \leq 0.01$ and $A_j \leq 0.05$ for upper cell quantities, and $D_j \leq 0.10$, $A_j \leq 0.20$ for lower and transition cell quantities. Both inventory measures and mass-flow means have limits of 0.01 above the upper cut and 0.05 below it. Every active condition must hold for eight consecutive windows containing at least 40 accepted endpoints each. The transition region may be empty, but the lower and upper regions must be nonempty. An unsuccessful candidate resets its fixed reference. A settling interval, if specified, is excluded from the first candidate's extrema.

The automatic window is twice the initial outgoing acoustic travel time. The default maximum duration is the settling interval plus twice the full confirmation duration. The accepted endpoint at a window edge is included in the last range test. Rejected numerical steps contribute neither time nor states. The source reports the effective cuts, limits, sample counts, and duration in `regional_convergence_status.txt`.

This finite-duration result supports the statement that the modeled upper atmosphere changed by less than the stated limits while lower-atmosphere variations remained within their stated limits over the tested duration. It does not prove stationarity, infinite-time boundedness, or observational accuracy. A separate stationary residual calculation is required for a stationary claim. The regional thresholds are not the default marching profile criterion of Sec. 7.7.

10.2 Lower-boundary thermochemical closure in the stationary problem

The stationary residual in v1.10 evaluates a thermochemical state for the lower ghost cells at a fixed physical column. The same boundary calculation is queried when the stored state is certified. The routines `stationary_ghost_state` and `boundary_chemistry_of_state` are defined in `stationary_boundary_chemistry.f90`; the reservoir composition is formed in `reservoir_composition.f90`. A boundary closure that fails its own convergence test cannot be treated as a stationary root, even if a different residual or profile diagnostic is small. The enabled fluid, composition, radiation, and boundary conditions must all be assessed at the same state and under the stated input conditions.

The regional test is evaluated in global physical marching of an atomic atmosphere. It does not evaluate the stationary boundary closure or certify a Newton–Krylov result, and no automatic transition between the two conditions is implemented. Existing Huang and Salz results have not been newly accepted by the definitions or thresholds in this section.

11. Output and post-processing

11.1 The advection-corrected profiles

For the reason given in Sec. 4.5, `post_process_adv.f90` re-solves the profile at the end of a marching run (its Newton finish included) and in a stationary evaluation, in ten outer passes (`n_pp_passes`), with the ionization advected: the

systems (67)–(69) are solved cell by cell from the base outward, each cell taking its history x^{up} from the solved fractions of the two cells below it and each element's stages stepping on the velocity of that element's nuclei (below), and the temperature is then re-solved on the steady internal-energy equation (86) with the Brent bracket of Sec. 5.3. In a run that solved heat conduction or an enthalpy flux of the element, the molecular carrier or the carried stage fluxes (Sec. 6.10), the heating of that equation holds them too, evaluated with the corrected temperature (`transport_terms_of_profile`; the stage term enters with the carrier term, `carrier_enthalpy_divergence_of_state`), and cell 1 conducts against the bath the run's conduction operator holds at the base (`conduction_base_level_T`, Sec. 8.1): conduction (21) couples each cell to both neighbors, so after the marching sweep the corrected cells are solved together by a Newton iteration on the column (banded Jacobian, conductivity and enthalpy flux divergence re-evaluated at each iterate); cells that keep the run's temperature are fixed values of that solve, while a cell whose own marching energy step found no root (outside the Brent bracket $[0.05, 4]\,T_{\mathrm{in}}$, or the $[0.5, 2]\,T_{\mathrm{in}}$ band that guards the metal-cooled residual against its spurious hot root) is an unknown of it, started from the run's temperature and decided by the residual like the rest (`cells_marching_unrooted` in the record; `EXHALE_ADV_COLUMN_EXCLUDE_UNROOTED=1` makes such cells fixed values again, for measurement). That iteration stops when no temperature moves by more than 10^{-10} of itself, or after 100 iterations, which is a stopping rule and decides nothing; the column is then judged on its residual: the energy row R of every unknown, formed again at the final profile with the transport terms of that profile, must satisfy $|R| \le a_E + 10^{-6} S_E$, S_E the largest of its physically grouped terms (the advective energy divergence as one term, the $p\,\mathrm{div}\,v$ work, heating, cooling, the conduction divergence and the two enthalpy divergences, of the element fluxes and of the carrier and stage fluxes), 10^{-6} the energy tolerance of the stationary certification, and $a_E = 10\,\epsilon$ times the sum of the magnitudes of the parts the row is assembled from (the grouped terms, the two halves of the advective difference and the three face terms of the conduction divergence; the stagnated $|R|$ MEASURED on three LHS 1140 b conduction states is 0.73–0.99 ϵ of that sum). The temperature band of the cell-by-cell solve is not applied to the column: it refused a root of the conduction-coupled equation (a cell at 10.35 R_p whose row closes to 9.5×10^{-12} of its largest term at twice the run's temperature). The column is rejected when the banded solve fails, a residual, step or term is not finite, a temperature is not positive, or the residual test fails; a rejected column keeps the marching profile and every one of its unknowns is written `failed`, and the outcome, the reason, the triggering cell, the worst residual and the number of cells without a marching root are recorded in the `# adv_derived_state:` line of both `_adv` headers.

What that acceptance covers is the column subproblem as it is solved, with lagged coefficients: the conductivity and the element, carrier and stage enthalpy divergences are formed on the *input* composition and not on the derived one, and the fine-structure escape probabilities are those of the profile the pass started from, which the cooling written after the last column forms again. Three things are not tested: the full energy residual of the derived state on one composition with its own coefficients, the convergence of the ten outer passes, and the momentum balance of the corrected pressure on the supplied ρ and v. An accepted column is therefore recorded as `converged_on_residual` and the product as `verdict=unverified`, `outer=unverified`, never as more; the verdict is `rejected` when the column of the last pass was rejected or any written row is `failed`, and a file without the line has an unknown derived state, which is not a pass either. After every composition update of a pass (the H/He sweep with its cells restored to equilibrium, and under `pp_metals = 2` the metal re-solve and its failure branch) the particle count, the electron density and the mean molecular weight are formed again from that composition, then the ionized fraction of the photoelectron partition, and only then

the heating, the energy solve and the output pressure (the written p equals $(n_{\rm tot}+n_e)kT$ of the written composition to 7.8×10^{-16}, MEASURED on `wasp_full` with `pp_metals = 2`, against 4.4×10^{-8} before 2026-10-01). Without these terms the corrected temperature of a conduction-dominated layer answers another equation (7900 K against the run's 5500 K at 1.2 R_p on the fiducial LHS 1140 b atomic case).

The step. Both recursions are one-step integrations of an ODE in r, $\mathrm{d}y/\mathrm{d}r = F(y,r)$, with the variable-step second-order backward differentiation formula (BDF2),

$$y_j - a_1 y_{j-1} + a_2 y_{j-2} = g\,h_j F(y_j, r_j), \qquad a_1 = \frac{(1+w)^2}{1+2w}, \quad a_2 = \frac{w^2}{1+2w}, \quad g = \frac{1+w}{1+2w}, \tag{146}$$

$h_j = r_j - r_{j-1}$, $w = h_j/h_{j-1}$: the left side over $g\,h_j$ is $\mathrm{d}P/\mathrm{d}r$ at r_j of the quadratic P through the three points, so the step is exact for quadratics and the recursion second order in the cell width (the formula of Li & Liao 2022, eq. (1.4) of the preprint). Every r-derivative of a step, the known density profile of the energy equation included, is taken at r_j by this rule, and every other factor (rates, heating and cooling, the velocity of the residence time) is the value at r_j. The recursion starts, and restarts after every cell it does not integrate, with backward Euler ($a_1 = g = 1$, $a_2 = 0$); a cell whose step ratio exceeds the zero-stability bound $w \le 1+\sqrt{2}$ is integrated by backward Euler, and a BDF2 step whose populations come out negative or whose temperature has no root is retaken by backward Euler, which keeps the populations of a linear rate system nonnegative; the run log counts all three. BDF2 is L-stable, like backward Euler, so in the stiff limit a population that relaxes within the cell returns its local equilibrium; the trapezoid rule, A- but not L-stable, would alternate about it from cell to cell. Until 2026-09-27 every step was backward Euler with the rates at r_j and the residence time at r_{j-1}, a first-order recursion that was the main source of the first-order grid convergence of the He 10830 equivalent width. MEASURED on nine certified LHS 1140 b states (He/H 1.6, 1.8 and 2.13 on 500, 1000 and 2000 cells) with the state held fixed, the post-process share of the equivalent-width error falls from +3.0/+1.0 percent to below 3×10^{-4}, the `_adv` 2^3S column converges at order 1.98–2.14 (1.00–1.01 before), and a closed-form column test gives 1.99; the equivalent width itself moves by −3.9 percent (500 cells) to −0.97 per cent (2000 cells) at He/H = 1.6 and still converges at first order, what remains being the solved state's own base closure and the transit integration.

Three properties of that pass are worth stating.

- The third term of (86), the enthalpy flux carried by the divergence of the mass flux, $h\,r^{-2}\mathrm{d}(\rho v r^2)/\mathrm{d}r$, vanishes for a stationary mass flux and for nothing else, so a correction built without it would describe a converged wind and no other state. The divergence is the mass row of the state itself, the face mass fluxes of its Riemann solve differenced over the cell, $(A_{j+1/2}F_{j+1/2} - A_{j-1/2}F_{j-1/2})/\mathrm{d}V_j$, so the term is an exact zero wherever the two faces carry the same flux. (Until 2026-09-27 it was the same operator applied to the cell-centered $\rho v r^2$ of two neighboring cells, which at the base of a quasi-hydrostatic layer carries the odd-even pattern of the cell-centered velocity; its difference set the temperature of the lower layer.)

- A cell is corrected only where the correction means something. Above the Damköhler number 10^2, formed with the *slowest* relaxing species of the solved system, the gas reaches local equilibrium many times over while it crosses the cell and the equilibrium solution already solves the advection equation, so the cell is pinned to it.

Below an equilibrium ion fraction of 10^{-6} the residual, which carries the neutral fraction, cannot represent the ion density to better than about a percent. Where the mass row of the supplied state exceeds `adv_conditional_tol` as a fraction of its own largest term, the cell keeps the run's own value; each cell's verdict travels in the output file as `adv_T_status`, `adv_comp_status` and `adv_mass_row`.

- The reconstruction treats the gas as H/He plus trace metals and *excludes* the molecular species: H_2, H_2^+, H_3^+ and HeH^+ are not passed in and are not re-solved, so inside a molecular layer the `_adv` particle count omits the neutral H_2 and the molecular-ion electrons. The output file says so in its own header rather than only in the source. The same holds for the energy balance: the post-process lacks the molecular chemistry heat and the H_3^+ and molecular band cooling, so wherever the heating and cooling it omits, measured on the state handed in as the difference between the run's own channels and its own at the run's temperature and composition, exceed the ones it carries, the row keeps the run's temperature with `adv_T_status` = 3 (`unsupported`); MEASURED on the LHS 1140 b molecular states, 196/389/776 rows on 500/1000/2000 cells, up to 1.134–1.137 R_p, with an equivalent-width change of at most 1.2×10^{-4} of itself. The balance of each population the advection systems solve is judged the same way (the composition class): the systems are molecule-free, so where H^+, He^+ or $He(2^3S)$ loses faster to H_2 (R10 and R13; R17 and R23; the $He(2^3S) + H_2$ ionization) than to the channels the systems carry, on the state handed in, the row keeps the run's composition and temperature with status 3 in both fields, and the metal re-solve of `pp_metals = 2` skips it. MEASURED on the LHS 1140 b molecular photochemical state, the rows it refuses at 1.005–1.08 R_p had carried an $n(\mathrm{He}\,2^3S)$ 70 to 5600 times the run's. The comparison has no free parameter, and passing it does not bound the omitted H_2 loss to a small error: it says only that the omitted loss is not the larger one.

The first of those gates has a fourth companion: wherever $v \leq 0$, or the velocity of the hydrogen or helium nuclei is ≤ 0, in the cell or the one below it, the cell keeps the converged equilibrium ionization and temperature, because the step takes its history from the cells below, which are not upstream when the gas moves inward, and the residence time h/v is then negative as well. The relaxation rate entering the Damköhler number is the total rate at which each solved population is destroyed and re-formed, and the *minimum* over the species the system solves is taken, because $He(2^3S)$ relaxes orders of magnitude more slowly than hydrogen ($A_{31} = 1.272 \times 10^{-4}\ \mathrm{s}^{-1}$ sets the floor) and the vector is only as local as its slowest member. Each element's rates are weighed with the residence time of that element's nuclei, h/v_H or h/v_He, the times its step uses.

The velocity of each element's nuclei. The fractions the advection systems step are fractions of one element's nuclei, and with `He_diffusion` those nuclei do not move with the mass-averaged velocity: with J the diffusive helium mass flux of the state (`helium_diffusive_face_flux`, the flux the element row of the run is the divergence of; the cell value is the mean of its two faces, and faces 0 and N carry none) and Y the helium mass fraction,

$$v_\mathrm{He} = v + \frac{J}{\rho Y}, \qquad v_\mathrm{H} = v - \frac{J}{\rho(1-Y)}, \tag{147}$$

and the hydrogen row of each step takes $c_1 = g\,h_j/v_\mathrm{H}$, the helium rows $c_1 = g\,h_j/v_\mathrm{He}$ (both equal v without element diffusion; the ghost cells keep v). With element continuity $\nabla \cdot (n_\mathrm{el} v_\mathrm{el}) = 0$ the steady stage balance is then $v_\mathrm{el}\,\mathrm{d}x/\mathrm{d}r =$

$S/n_{\rm el}$. MEASURED on the LHS 1140 b He/H = 2.09 reference, $v_{\rm He}/v = 0.82$–0.97 and $v_{\rm H}/v = 2.1$–1.08 over 1.3–4 R_p, and stepping both elements on v had put the frozen-in He^+ fraction 3.5 percent low and H^+ 6 per cent high against the transported stages of the solution (update log stage 3 section 82). The stage's own eddy term $-n_{\rm el}K\,\partial x/\partial r$ is not in the step, which is second order in r; the step is valid where $v_{\rm el} \gg K|{\rm d}\ln x/{\rm d}r|$, a ratio MEASURED on the same state at 0.035 (He^+) at 1.3 R_p and below 10^{-3} beyond 1.75 R_p for $K = 10^9$ cm^2 s^{-1}, and a run with a larger K or a slower wind has to be checked against it.

What a refused row carries. A refused field is written with the run's own numbers, exactly: a composition field that is not `corrected` carries the handed-in species and metal columns, a temperature field that is retained, unsupported or not evaluated the handed-in temperature, and a row whose composition is refused and whose temperature is refused but not failed carries the handed-in pressure, heating and cooling too. The one exception is the temperature of a `failed` row of a rejected column, which is the marching profile of that pass. A row with a corrected temperature and a refused composition is therefore not the solution as a whole.

11.2 The mass-loss rate

The quantity the run reports as `Log10 of steady-state Mdot` is measured at one radius, by `steady_mass_loss_rate` in `EXHALE_main.f90`, on the state each route writes (the stored pair in a stationary evaluation), and appended to `EXHALE_setup.out`:

$$\dot{M} = 4\pi\,\rho(j)\,v(j)\,r(j)^2\,n_0 v_0 m_{\rm H} R_0^2 \qquad \text{at} \qquad j = \max(N-20,\,1), \tag{148}$$

the offset of 20 cells keeping the evaluation far enough from the outer boundary, and the floor at $j = 1$ keeping it out of the base ghosts where the boundary closure and not the wind sets ρ and v. A logarithm is taken only where the flow there is directed outward: $\rho v \le 0$ is an inflow, which is not a mass-loss rate, and the run then reports the signed flux instead of a logarithm. The dayside convention of Table 1 is applied here and only here: $\log_{10}\dot{M} \to \log_{10}\dot{M} - \log_{10} 2$ for `Rate/2 + Mdot/2`, and $-\log_{10} 4$ for `Mdot/4`.

For a quantitative interpretation of the mass-loss rate from a marching solution, report the measured profile band over the stated final physical-time window. A small value of du alone does not establish that band. A stationary solve and its certificate may be reported separately when available.

11.3 Output files

Which state the products are built from. A solved state's advection-corrected products are made by `Restart intent: stationary evaluate` (`stationary_state_of_the_loaded_restart` in `EXHALE_main.f90`), which loads the state (with its exact code-unit file, Sec. 8.4), certifies the stored pair and writes every product from that pair without entering the time integration. The stored pair is the conserved state and the composition exactly as loaded, with the temperature from u through the caloric map of that composition and the heating, the cooling and the two lower ghosts from one sweep that holds the composition of the cells above the ghosts and hands it back bit for bit (`sources_and_ghosts_of_the_held_composition`, `ioniz_eq_sources_of_held_composition`). The same routine rebuilds these quantities for the joint test of every stationary outer pass and for the certification of every state

a steady route writes (certify_the_state_to_be_written), so a certificate is a statement about the state in the file and its reload measures the same numbers: the base continuity row reads the lower ghost with a gain of order the inverse base Mach number, and a ghost left by the last sweep of a solve at the temperature of an earlier iterate had moved a cell-1 mass row by 13 percent of its rounding-anchored tolerance between a solve and its reload (MEASURED, update log stage 3 section 94). The products are Hydro_ioniz.txt, Ion_species.txt, conserved_state.txt, the channel breakdowns and the mass-loss rate of the stored pair, the _adv pair derived from it, and EXHALE_resolved.out. One composition sweep from the loaded state is then taken, and its movement and certification are printed beside the stored pair's, entry by entry; it is neither written nor the verdict. The evaluation exits with status 2 when the loaded state claims to be stationary and the certification of the stored pair refuses it, 7 when the input is not refused but the derived product is (a rejected or absent derived-state record: a rejected column energy solve or a failed row of the _adv files), and 0 otherwise; the record is written into the _adv headers either way, and 7 is used by no other stop of the code. The marching route keeps its exit status for the state it solved and records the derived product only in the _adv headers and the log. The route this replaced took one time step at CFL: 1.0e-12 with Do only PP: True, which suppressed the movement but did not remove it, reconstructed with whatever Reconstruction scheme: stated rather than with the scheme the solution was reached under, and wrote the stepped state back over Hydro_ioniz.txt. The difference is largest where the advection operator is weakest: the operator's own weighting measure adv_mass_row falls from 1.57 to 1.7×10^{-9} at the first cell and from 3.1×10^{-5} to 4.0×10^{-14} at the last, while the He I 10830 equivalent width moves by 5×10^{-6} of itself.

Every file written to output/ carries a schema header, so a reader adapts instead of counting columns. Hydro_ioniz.txt opens with # EXHALE schema 2 and then

```
# columns r[Rp] rho[mH/cm3] v[cm/s] p[cgs] T[K] heat[erg/cm3/s] cool[erg/cm3/s]
```

(with three further status columns on the _adv file). Column 2 is ρn_0, the *mass* density in units of $m_{\rm H}$ per cm^3 with the metals included under the eos_metals policy, not a number density. The first and last Ng rows of every file are ghost cells filled by the boundary conditions, and their $\rho v r^2$ is a boundary extrapolation. Ion_species.txt carries the species fractions in the species_table order of Sec. 4.1; Heating_breakdown.txt and Cooling_breakdown.txt carry the channel decompositions of Sec. 5., each built from the array the energy equation itself used, so the channel sum and the heat or cool column of Hydro_ioniz.txt are the same number to round-off; Excited_H.txt, Lyman_Werner.txt, FUV_bands.txt, Oxygen_chemistry.txt and OI_levels.txt carry the optional physics; conserved_state.txt is the exact code-unit state beside every written pair (Sec. 8.4); the generation pass_state/<id>/ named by pass_state/current is the state of the last completed outer pass of an alternating stationary run, never certified (Sec. 9.6). The two _adv files carry a # derived_from: line rather than the # coupling: header of a state file (item L9, 2026-09-17): they describe an advection-corrected composition, which is not the composition any certification was made on, so the input state's certification pair is carried there as provenance and not as a claim about the numbers in the file. The Python loader examples/exhale_io.py reads all of them through those headers (exhale_io.Run('output/') gives .r, .T, .Mdot and the rest), and it should be used in place of any new column parser.

EXHALE_resolved.out, written in the run directory and not in output/ (write_resolved_config,

write_setup_report.f90), is the machine-readable record of the configuration the wind actually uses after input.inp and the lower-atmosphere handoff are resolved (radius, mass, base temperature, He/H, orbit, the grid and the physics switches), as key value lines to full double precision; EXHALE_transit.py reads it in place of input.inp. It is written at startup and again after the certification of the written state, and then also carries the elemental fluxes over the overlap window of that state and, when conduction is on, the heat the wind conducts into the lower atmosphere through the base face, $q = \kappa_{1/2}(T_1 - T_{\rm bath})/(r_1 - r_0)$ in $\mathrm{erg\,cm^{-2}\,s^{-1}}$, positive leaving the wind (base_conductive_flux_cgs, with base_conduction_T_bath_K and base_conduction_T_cell1_K; base_conduction_state says unmeasured before the wind exists). That flux is the base face's own term of the conduction operator, $T_{\rm bath}$ the bath at $r(0)$ of Sec. 8.1, measured on the state the certification is made on (heat_conducted_into_lower_atmosphere, viscous_conduction.f90) and not on the last residual evaluation, which inside a Newton solve is a perturbed state; it is what the energy coupling of the elemental-flux closure with the lower atmosphere reads (Sec. 6.7).

11.4 The transmission spectrum

EXHALE_transit.py, with the library exhale_transit_lib.py, builds transit transmission spectra from either of the two states a run can leave, selected by EXHALE_TRANSIT_STATE: the _adv profiles by default, or the solved state pair Hydro_ioniz.txt and Ion_species.txt with EXHALE_TRANSIT_STATE=solution, which is how a stationary run is read. The lines are the same either way: the He I 10830 Å triplet, Lyα, Hα and Hβ, and the metal doublets Mg II h&k, Ca II H&K and Na I D, the metal lines being skipped automatically for a metals-off run. It integrates the Voigt line opacity over impact parameter, averages over the stellar disk, and convolves with the instrument profile and the planet's rotation; the Balmer branch builds the non-LTE $2s/2p$ populations with the level balance of Christie et al. (2013) and the rate coefficients of the Fortran model of Sec. 4.6 (transcribed in exhale_transit_lib.py, n2_populations, and equal to them to 3×10^{-13} relative), but with a field of its own, because neither the solver's Lyα field nor its Balmer rate is in the profiles the tool reads: $J_{\mathrm{Ly}\alpha}$ from a file or the estimate of Huang et al. (2017), and a diluted stellar blackbody for the Balmer continuum. An optional triaxial Roche geometry comes from roche_recon.py. Overrides are environment variables (EXHALE_TRANSIT_*: resolving power, He window, rotation period, azimuthal samples) and the output keeps the historical tpm_*.txt names, each with its own # transit_schema 1 metadata block. The physics of that tool is documented separately in docs/transmission_spectrum.pdf and is not repeated here.

Which state the spectrum is built from. The two selections are two answers for one column and not two formats of one answer. The _adv pair is a post-process of a written state (the end state of a marching run or the stored pair of a stationary evaluation): a steady advective correction cell by cell, second order in the cell width (variable-step BDF2 along the flow, each element's stages at the velocity of its own nuclei), with no stage eddy term, refused in a cell where an assumption of the steady equations fails. It is an independent discretization of the same column and no equation of the run was solved for it. The solved pair is the state the solver produced and the certification judged; where the ionization stages are transported its composition is the partition the stage transport itself produced, so an advective correction on top of it would be a second and different transport approximation of the same stages. The stationary solve of a loaded state (Restart intent: stationary) therefore writes no _adv products and none are manufactured for it

by the transit tool; the evaluation of a solved state (`stationary evaluate`) runs the post-process on its stored pair and writes them, and where they are absent the default selection refuses by name and names the other selection rather than reading it. The two files of a selection are read together, the O I ground-term populations included: the temperature of one state with the composition of the other solves neither set of equations. Every saved curve carries the selection, the path, md5 and size of each file read, and the state's own provenance, coupling, boundary model and reservoir lines. The `_adv` selection is refused with exit status 7 when the profile's `# adv_derived_state:` record says `rejected` or when the profile has no record (an unknown derived state), unless `EXHALE_TRANSIT_DIAGNOSTIC=1`, which makes the curves as diagnostic data and writes the refusal or the unknown record into their metadata. Within a profile that passes, the row census refuses a row whose temperature field is not `corrected` or whose composition field is `failed`; a refused row enters the spectrum with the run's own numbers (Sec. 11.1), and the share of the line-center depth it carries is reported as a contribution.

What the choice costs was MEASURED on `wasp_full`, which carries both states, as the peak transit depth relative to itself (before the atomic-data corrections of 2026-09-24/25, which moved both states, and before the post-processor took the solver's $n = 2$ rates; not re-measured since): He I 10830 −0.64 percent, Hα −1.28, Hβ −3.48, Lyα unchanged at its saturated ceiling, and the optically thin metal lines +0.1 to +0.8. The metastable helium column is 12 percent *larger* in the solved state and its line is nonetheless shallower, because the advection correction moves that column outward (column-weighted mean radius 1.2941 against $1.3112\,R_p$) and a saturated line's disk-averaged depth follows how far the absorbing shell reaches; Hβ moves most because it is the line still on the linear part of the curve of growth (item P6f).

11.5 A model transit line against an observed one: the width-matching kernel

A one-dimensional steady model has no line-of-sight velocity field beyond its radial outflow and its thermal motion, and its transit lines come out narrower than the observed ones (on LHS 1140 b He I 10830 the model red pair has a FWHM of about 0.27 Å against the measured 0.841 Å). For every planet and every line compared with an observed transit profile the model is therefore compared with the observation through one convention: the line *width* is matched by a Gaussian velocity convolution, and the model is fitted to the observed *equivalent width* of that width-matched curve. The convention is written once, for LHS 1140 b He I 10830, in `LHS1140b/he10830_equivalent_width.py`, and a comparison on another planet or line generalizes it rather than introducing another.

1. The excess absorption $(\max T - T)/\max T$ is formed from the synthetic transit depth of `tpm_He10830.txt` (column 4, `T_rot+instr`; user decision 2026-10-06, before it the instrument-only column 3), which is convolved with the planet rotation (tidally locked, the azimuthal disk average) and the instrument at the instrument's own resolving power, on its own air-wavelength grid (`transit_file_air_excess`).

2. That excess is convolved with a Gaussian of FWHM f in velocity, in the planet rest frame and with no shift, $\sigma_\lambda = f\lambda_0/(2.35482\,c)$ with $\lambda_0 = 10830$ Å in air (`broaden_excess`, `scipy.ndimage.gaussian_filter1d` with its reflecting boundary, which conserves the absorption summed over the grid). The instrument kernel is already inside the curve and Gaussians commute, so the order of the two convolutions does not matter.

3. f is chosen for each model separately so that the extractor applied to the observation (for He I 10830 the three-Gaussian `he_line_metrics.fit_metrics`, whose width is the FWHM of the blended red pair) returns the observed FWHM of the same feature (`width_matching_fwhm_kms`: Brent's method on f in $[1, 60]\,\mathrm{km\,s^{-1}}$ to $10^{-3}\,\mathrm{km\,s^{-1}}$; on LHS 1140 b $f \approx 22\,\mathrm{km\,s^{-1}}$).

4. The equivalent width of the model is the window equivalent width of the broadened curve through the same operator as the observed one: the curve, converted to vacuum wavelengths, is interpolated linearly onto the observation's own samples inside the window (on LHS 1140 b vacuum 10832.60–10834.20 Å in the planet rest frame) and integrated with their trapezoid weights (`model_equivalent_width`). Any fit of a planet parameter to the observed equivalent width (He/H, $\dot{M}$, the stellar EUV) is judged on this width-matched value.

`he10830_equivalent_width.py --width-matched <tpm_He10830.txt> <output> [f]` writes the broadened excess with its record (the kernel FWHM, the criterion, and the equivalent width and line metrics with and without the kernel; `width_matched_product`). In the equivalent-width record of a state (`LHS1140b/models/.P1/fiducial_conduction/ew_of_state.sh`), `EW_NONTHERMAL_KERNEL=width_match` adds `fwhm_kms=` and `EW_kernel=` to each line of `EW.txt`, and `EW=` stays the unbroadened window equivalent width; a number in place of `width_match` applies a given f.

The kernel stands for a velocity field that the one-dimensional model does not contain. It is not a computed physics term: it carries no energy or momentum budget and moves no absorber, and its width is measured from the data for each model rather than being a parameter of the model. Every result that uses it states f and says so. Because the reflecting convolution conserves the absorption summed over the grid, the window equivalent width falls as f grows only by the absorption the broadened wings carry out of the window.

12. The input keys that select physics

Runtime configuration is by the presence of a key, not by a compile flag (Sec. 1.1). Three companion files are read at startup and their mere presence is itself a switch: `metals.inp` turns the trace metals on, `opacity.inp` the tabulated continuum opacities, and `base.inp` the lower-atmosphere handoff.

Table 2 lists the keys that select physics or the numerical scheme, with the default the code takes when the key is absent. The defaults were read from `parameters.f90` and the parsing from `input_read.f90`; keys that only name a file, a path or an output option are omitted, and so are the planet and stellar parameters of the core block (radius, mass, equilibrium temperature, orbit, He/H, luminosities), which every run states. The core-block lines that select physics or the scheme are mandatory and have no default: they are marked as such. A key stated twice is refused at startup (`refuse_duplicate_keys`), and a line that matches no key is reported with a warning. The retired lower-boundary keys `Valve eps`, `Hydrostatic base`, `Base ghost temperature` and `Base velocity`, and the renamed or merged `Oxygen transport` and `Stellar FUV B1 flux`, stop the run with the replacement named; `Resid norm` is ignored with a warning.

Table 2: Physics and algorithm keys of `input.inp`, with the value the code takes when the key is absent.

Key	Default	What it selects, and where
Geometry, grid and the one-dimensional approximation		
`2D approximate method:`	mandatory	`Mdot/4`, `Rate/2`, `Rate/4` or `alpha`: the dayside dilution of every stellar beam and the factor on $\dot{M}$ (Table 1; `parameters.f90`, `utilities.f90`).
`Grid type:`	mandatory	`Uniform`, `Mixed` or `Stretched`; every type puts the first face on the base level, $r_{\rm edg}(0) = 1$ (`define_grid.f90`).
`Grid cells:`	500	Number of physical cells; ghosts are added on top (`define_grid.f90`).
`Base grid [dr,cells]:`	2×10^{-4}, 50	Width and count of the uniform base region of the `Mixed` grid (`define_grid.f90`).
`Domain mode:`	Roche	`Spherical` drops the tidal and centrifugal terms of (12) (`grav_field.f90`).
`Outer radius [R_p]:`	L1 radius	Outer boundary; required in spherical mode (`input_read.f90`). A transonic wind needs it past the sonic point; the Mach number of the outermost cell is reported with $\dot{M}$ (`EXHALE_main.f90`).
`Outer shells [r_face,cells]:`	none	Cells appended beyond the constructed grid out to the face `r_face`, the cells below unchanged bit for bit (`define_grid.f90`).
Discretization		
`Numerical flux:`	mandatory	`HLLC`, `ROE` or `LLF` (`Num_Fluxes.f90`); `ROE` is refused with `Molecular chemistry` unless `Caloric EOS: monatomic`, because the Roe average assumes one constant adiabatic index.
`Low Mach velocity jump:`	off	On the `ROE` branch only, scales the normal velocity jump of the Roe dissipation by $\min(\lvert U_{\rm Roe}\rvert / a_{\rm Roe}, 1)$ (Rieper, 2011) (`Num_Fluxes.f90`).
`Reconstruction scheme:`	mandatory	`PLM`, `WENO3` or the two-stage `PLM+WENO3` (`Reconstruction.f90`).
`Reconstruction continuation:`	off	$\Delta\lambda$ of the homotopy (114) between the two operators (`steady_residual.f90`).
`CFL:`	0.6	CFL number of (125) (`eval_dt.f90`).
`Time stepping:`	global	`Local` gives every cell its own pseudo-time step (`eval_dt.f90`).

Table 2 continued

Key	Default	What it selects, and where
`Well balanced:`	off	Carries the departure from the cell's own hydrostatic equilibrium instead of the state (`Reconstruction.f90`, `RK_rhs.f90`).
`Low-Mach damping:`	off, $M_{\rm th} = 10^{-3}$	Gated fourth-difference dissipation (19) of the contact mode (`low_mach_dissipation.f90`).
`Shapiro filter:`	off, every 4 steps	1-2-1 low-pass filter of strength $\epsilon > 0$ on the marched state (`Apply_BC.f90`).
`Caloric EOS:`	`ladder`	`monatomic` forces $\gamma = 5/3$ everywhere (`caloric_eos.f90`).
`Energy solver:`	semi-implicit	`Explicit` reverts the source update to forward Euler (`energy_semi_implicit.f90`).
Boundary and initial conditions		
`Base BC:`	`density`	`pressure <microbar>` sets the base level from a pressure instead of a density (`input_read.f90`, `base_boundary.f90`).
`IC mode:`	`cold`	`cold`, `transonic`, `hot_parker`, `auto` or `windae` (`set_IC.f90`).
`Transonic IC:`	off	Older synonym of `IC mode: transonic`.
`Hot Parker IC:`	off, $T_{\rm wind} = 10^4$ K	Older synonym of `IC mode: hot_parker`.
`Run mode:`	`init`	`phys` declares a physical integration (one global Δt, a clock that advances only after an accepted step); `init` permits local pseudo-time, pseudo-transient continuation and non-root chemistry as numerical devices (`parameters.f90`).
`Restart intent:`	`relaxation` (`trajectory` under `Run mode: phys`)	What a loaded state is continued as: `trajectory`, `relaxation`, `stationary`, or `stationary evaluate` (the stored pair certified and its products written without a step, exit 2 on a refused input and 7 on a refused `_adv` product, Sec. 11.3), or `stationary equilibrate` (the loaded composition put on its own fixed point first) (`input_read.f90`, `load_IC.f90`).
`Restart option change:`	none	Names the option tokens of the state file that may differ in this run (among them `cond` and `interdiff_enth`, which carry the continuation factors below); any other difference refuses the restart, and `metals`, `mol`, `oxychem` and `carrier`, which decide the species columns of the state files, can never be named (`load_IC.f90`).

Table 2 continued

Key	Default	What it selects, and where
Convergence and the stationary solvers		
`du_th [PLM,WENO3]:`	10^{-3}	Threshold on the cell-centered spread (127) at which PLM hands over to WENO3; a second, smaller value makes the run two-stage. Stops nothing since 2026-10-09: the marching stops on the face spread below 4×10^{-4} and χ (126) below 10^{-4} (`EXHALE_main.f90`).
`Escape radius [R_p]:`	mandatory	Inner edge of the window (127) is taken over (`define_grid.f90`).
`Resid tol:`	off	Residual convergence instead of the flux spread; the solvers use 10^{-5} when it is absent (`steady_residual.f90`).
`Flux spread tol:`	2×10^{-5}, $r_{\rm flux} = 1.2$	The flux gate of the acceptance test (`steady_residual.f90`).
`Solver:`	marching only	Optional `Newton [<switch>]` hands over to the stationary solve once the face mass-flux spread falls below the switch, default 0.1 (a *du* threshold, default 10^{-2}, until 2026-10-10; `EXHALE_main.f90`).
`Stall:`	10^{-6}, 2000 steps	Plateau detector of the PLM stage; `N_stall` is also the length of the stall net of the marching stop (`EXHALE_main.f90`).
`Level tol:`	off	Mass-flux level stability required beside every marching stop.
`Max steps:`	10^6	Hard cap on marching iterations.
`Newton solver:`	on	`False` reverts the composition solve to MINPACK `hybrd1` (`newton_solver.f90`).
`Brent solver:`	on	`False` replaces the bracketed root of the post-process temperature solve by MINPACK `hybrd1` (`T_equation.f90`).
Atomic hydrogen and helium		
`Include He23S?`	on	The He 2^3S metastable as a solved level (`System_HeH_TR*.f90`).
`ATES_photoionization_rate`	off	Reverts He I photoionization to the legacy ATES fit (`cross_sec.f90`).
`Legacy_HHe_rates`	off	Reverts H/He recombination and collisional ionization to the ATES fits (`Cool_coeff.f90`).
`Atomic rate set:`	`default`	`Koskinen2022` swaps four atomic H/He rate coefficients (`Cool_coeff.f90`, `mol_rates.f90`).

Table 2 continued

Key	Default	What it selects, and where
`Secondary_ionization`	on, from step 0	Secondary ionization by fast photoelectrons above 30 eV; `Staged` (the default until 2026-10-09) applies it once the marching has settled, `False` not at all (`electron_energy_degradation.f90`).
`Low-energy electron partition:`	`none`	`Furlanetto2010` partitions the photoelectrons of 10.2–30 eV, stellar and recombination, into heat, H I and He I ionizations and escaping line photons (Furlanetto & Stoever, 2010), in cells without H_2 and wherever the secondary-ionization coupling is applied; `none` heats with all of their energy (`low_energy_electron_degradation.f90`, Sec. 3.5).
`He_rec_coupling`	on	He II and He III recombination radiation absorbed on the spot; off is case B for both (`util_ion_eq.f90`).
`H_rec_escape`	on	H II ground captures whose photons are not re-absorbed by H I in the cell are net recombinations; off is case B everywhere (`util_ion_eq.f90`).
`He_H_charge_exchange`	on	The He/H charge-exchange pair (`charge_exchange.f90`).
`Photoelectron heating:`	`excess`	Whether a photoionization deposits $h\nu - E_{\rm th}$ (`excess`), $h\nu$ (`full`) or a stated fraction in $(0, 1]$ of $h\nu$ (`util_ion_eq.f90`).
`Deexc heat`	on	Collisional de-excitation heating of H(n=2) (`excited_hydrogen.f90`).
Radiation field		
`Use only EUV`	mandatory	`False` includes the X-ray band of the photon grid (`input_read.f90`).
`Spectrum type:`	mandatory	`Load`, `Power-law`, `Planck` or `Monochromatic`: the spectrum of the whole photon grid (`sed_read.f90`, `J_inc.f90`).
`Stellar Teff:` + `Stellar radius:`	0 (off)	Enable the H(n=2) model and its Balmer-continuum coupling, whose band is read from the run's own spectrum type (`excited_hydrogen.f90`).
`Stellar Lya flux:`	0	Stellar Ly-α at the orbit; also band B2 of the oxygen photolysis (`lya_rt.f90`).
`Jlya escape-prob`	parameterized	`True` replaces the parameterized pumping field $J_{\rm Ly\alpha} = 0.1 F_{\rm LyC}/\Delta\nu_D$ by the Neufeld escape-probability Ly-α transfer (`lya_rt.f90`).
`Jlya RT file:`	none	Import a $J_{\rm Ly\alpha}(r)$ profile computed elsewhere.
`Lya stellar halfwidth:`	70 km/s	Half-width of the stellar Ly-α line.

Table 2 continued

Key	Default	What it selects, and where
`Lya stellar boost:`	5.0	Bounded buildup of the trapped stellar beam.
`Lya absorbing bottom`	reflecting	Closes the Ly-α domain from below with an absorber.
`Stellar LW flux:`	from the spectrum file, else 0	912–1201 Å band flux driving H_2 photodissociation; with molecular chemistry and a loaded spectrum an unstated key is integrated from the file, which must reach the band (`lyman_werner.f90`, `input_read.f90`).
`Stellar FUV B3/B4 flux:`	from the spectrum file, else 0	The two longer FUV bands of the oxygen photolysis, integrated from a loaded spectrum in the same way when the oxygen chemistry is on (`water_photolysis.f90`).
Molecules, metals and the lower atmosphere		
`metals.inp` present	metals off	Trace metals solved in the same coupled system as H and He (`System_HeH_metals.f90`).
`eos_metals`	1	Metals in the bulk mass, electron and particle budget (`composition.f90`).
`cno_cool`	1 (CHIANTI)	0 reverts C/N/O line cooling to the legacy analytic fits (`Cool_coeff.f90`).
`Coronal cutoff width:`	0.1	Width w of the roll-off $\exp(-x^2)$, $x = (T_{\rm floor}/T - 1)/w$, of the coronal-excitation guard below the 10^3 K floor of the CHIANTI fits (`coronal_excitation_cutoff`, `Cool_coeff.f90`).
`Molecular chemistry:`	off	The $H_2/H_2^+/H_3^+/HeH^+$ network (`System_HeH_mol.f90`, `mol_rates.f90`).
`Oxygen chemistry:`	off	OH, H_2O and CO with their photolysis (`oxygen_rates.f90`, `water_photolysis.f90`).
`Molecular carrier transport:`	follows oxygen chemistry	Vertical transport of the molecular carriers, H_2 alone or H_2, OH, H_2O and CO with the oxygen chemistry (`diffusive_photochemistry.f90`).
`Composition update holds:`	energy	What the composition update of the stationary alternation holds beside ρ and v: the conserved energy, or the pressure (`EXHALE_main.f90`).
`Coupled carrier solve:`	off	`True`: $n(H_2)$ (and the diffused element fractions) as Newton unknowns beside the three hydrodynamic ones from the first pass; `On stall`: from the pass at which the alternation stops approaching a joint fixed point; off is the alternation (`steady_newton.f90`).

Table 2 continued

Key	Default	What it selects, and where
`Ionization transport:`	off	x(H II) per hydrogen nucleus and x(He II), x(He III) per helium nucleus carried with the flow on their elements' nucleus fluxes, as rows of the transport-chemistry operator (`ionization_stage_transport.f90`, `diffusive_photochemistry.f90`). Needs helium; in a molecular gas it needs `Molecular carrier transport`; refused with `Coupled carrier solve`.
`He 2^3S transport:`	off	$x_3 = n(2^3\mathrm{S})/n_{\mathrm{He}}$ carried as a fourth helium state of the same operator, eq. (71); needs `Include He23S` and `Ionization transport`.
`Molecular base:`	off (on with `Molecular chemistry`)	Removes the H nuclei bound into H_2 from the base particle count (15) (`composition.f90`); `Molecular chemistry` turns it on.
`H2 double ionization:`	`chung80`	$H_2 + h\nu \rightarrow 2H^+ + 2e$ resolved above 51.4 eV; `yan_rho` the other model, `off` folds it into the single dissociative channel (`h2_photo_channels.f90`).
`H2 neutral dissociation:`	on	$H_2 + h\nu \rightarrow H + H$ over 33–41 eV (`h2_photo_channels.f90`).
`Molecular reaction heat:`	on	Deposits the energy the collisional molecular reactions release (`molecular_reaction_heat.f90`).
`Molecular IR bands:`	off	H_2, H_2O and CO infrared line cooling (`molecular_infrared_cooling.f90`).
`Base IR field:`	off	The infrared coolants see the thermal radiation $B_\nu(T_0)$ of the atmosphere below the base instead of vacuum (`util_ion_eq.f90`, `Cool_coeff.f90`).
`Lower atmosphere:`	none	`analytic` or `vulcan` lower-atmosphere pre-step, with an optional 1-bar radius as its second word (`lower_column.f90`).
`Lower atmosphere profile:`	none	The lower atmosphere handed over as a table over pressure (`lower_atmosphere_profile.f90`).
`Lower column:`	off	1-bar radius of the analytic column.
Diffusion and molecular transport		
`He_diffusion`	off	He/H diffusive separation as a transported element (`binary_element_diffusion.f90`).
`He_metal_diffusion`	off	Extends the same operator to the trace metals.
`He_Kzz`	0	Constant eddy diffusion coefficient where no profile states one.
`He_alphaT`	0	Thermal-diffusion factor.

Table 2 continued

Key	Default	What it selects, and where
`He_ambipolar`	on	Ambipolar term of the element diffusion.
`Interdiffusion enthalpy flux:`	on (with `He_diffusion`, a transported molecular carrier or `Ionization transport`)	The enthalpy flux $\sum_s h_s J_s$ of the diffusing elements, of the transported carriers and of the carried ionization stages relative to their element in the energy equation, Sec. 6.10 (`binary_element_diffusion.f90`, `diffusive_photochemistry.f90`).
`Viscosity:`	off	Radial viscous force and its dissipation (20), with the calibrated $\mu(T)$; a number μ_0 [s] in place of `True` is a diagnostic power law $\mu = \mu_0 T^s$ in code units, $s = 0.7$ by default (`viscous_conduction.f90`).
`Conduction:`	off	Heat conduction with the mixture κ of (21), against a bath at the lower atmosphere's own temperature at the base ghost center (the profile's temperature where a profile is handed over); the heat conducted through the base face of the certified state is written to `EXHALE_resolved.out` (`base_conductive_flux_cgs`, (22)) (`viscous_conduction.f90`, `base_boundary.f90`).

Four environment variables change the equations or the solver a run applies and are therefore stated here. `EXHALE_CONDUCTION_SCALE` and `EXHALE_INTERDIFF_ENTH_SCALE` are continuation factors $0 \leq s \leq 1$ on the conductivity and on the interdiffusion enthalpy divergence (1 when unset, which is the equation; a value outside $[0, 1]$ stops the run for the conduction factor); a state solved at $s < 1$ is a step of that continuation and not a prediction of the model, and the factor is part of the identity of the state it writes (the `cond` and `interdiff_enth` tokens of the options field). `EXHALE_REACTION_HEAT_RECIPIENTS=0` restores the prompt deposit of the molecular reaction heat for comparison (Sec. 5.1). `EXHALE_PTC_SER=1` accepts the pseudo-transient steps of the stationary solve by the pseudo-time control rather than by merit descent, and `=2` also doubles the pseudo-time on an accepted step (`steady_newton.f90`, Sec. 9.). The other variables the source reads (`EXHALE_MAXSTEPS`, a deterministic step cap used to compare runs at one step count, the thread counts, and the diagnostic switches, dumps, fault injections and control experiments, 187 names in all on 2026-10-05) change no equation of a production run and are not listed.

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